ETAS RTA-BSW v12.7.0

RTA-Diag Stack Reference Guide

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1 RTA-DIAG

1.1 Introduction

A vehicle ECU may have sensors and actuators connected to it so that it can calculate output signals based on the incomming information. While the ECU software executes, it may encounter unexpected and undesired situations, such as:

- Invalid input signals due to defect sensors in the system.
- Unacceptable control deviations.
- Corrupt or missing data delivered to it by other ECUs.
- Defective system components (e.g. a broken lamp, a flat tire, or an over-heated turbo charger).

These undesired conditions are expressed as diagnostic events, and if such events occur, it is usually necessary log them for subsequent review during the development cycle, or in the vehicle service station, in order to analyze and fix the problem.

It might also be necessary to trigger protective system actions in the ECU to avoid hardware damage or to initiate substitute reactions in order to retain the complete or partial availability of the system.

The RTA-DIAG stack provides the functionality necessary for -

- Storing events and failures.
- Controlling system reactions.
- Accessing the ECU to extract information relating to diagnostic events.

The sections below provides a brief summary of the modules contained within the RTA-DIAG stack.

1.1.1 RTA-DIAG Modules

The RTA-DIAG stack consists of the following modules, each of which is described in its own sub-section in this Reference Guide.

Diagnostic Communication Manager (Dcm)

- Dcm provides a common API for diagnostic services, and its functionality is used by external diagnostic tools during development, manufacturing or vehicle servicing.

Diagnostic Event Manager (Dem)

- The Dem module is responsible for processing and storing diagnostic events and associated data.

Diagnostic Log and Trace (Dlt)

- Dlt handles and stores log and trace messages, which are generated by

calling APIs provided by the Dlt module.

Function Inhibition Manager (FiM)

- The Fim module provides a control mechanism for the inhibition (enabling/disabling) of functionality within software components.

Rba_DemObdBasic

- Rba_DemObdBasic is an extension to the Dem module. It provides the basic functionality for OBD diagnostic data, as described in the CARB ODB law.

1.1.2 RTA-DIAG Architecture

The diagram below shows the position of the RTA-DIAG modules in the Autosar layered architecture.

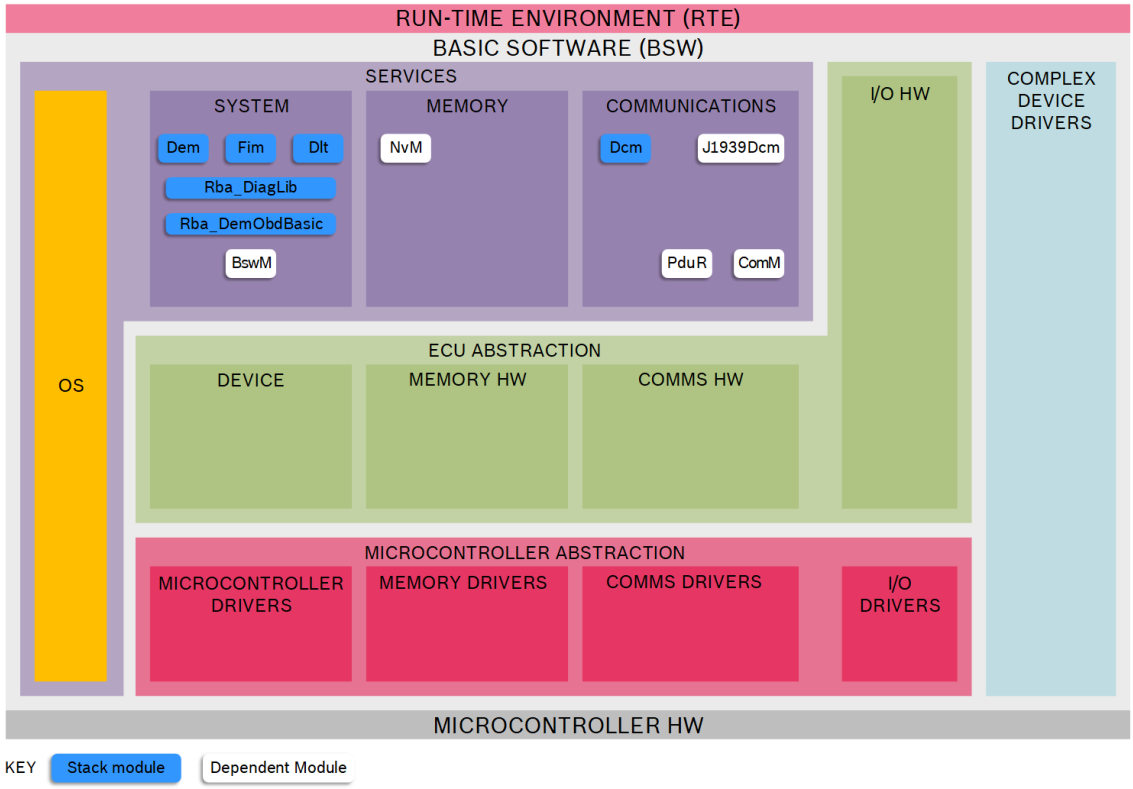

Figure 1.1.RTA-DIAG Architecture

2 Diagnostic Communication Manager

Module name	Dcm
Package	RTA-DIAG
AUTOSAR Module ID	53
AUTOSAR Version	23-11

2.1 Introduction

The Diagnostic Communication Manager (Dcm) is located in the system services abstraction layer, forming part of the diagnostic (RTA-DIAG) stack.

Dcm provides a common API for diagnostic services, and its functionality is used by external diagnostic tools during development, manufacturing or vehicle servicing.

Dcm ensures diagnostic data flow and manages diagnostic states, especially diagnostic sessions and security states. It also checks if the diagnostic service request is supported and if the service may be executed in the current session according to the aforementioned diagnostic states.

The Dcm module is network-independent, which means that all network-specific functionality (i.e. the specifics of networks like CAN, LIN, FlexRay or MOST) is handled outside of Dcm by other BSW modules.

2.1.1 Interaction/Dependencies

The diagram in the RTA-DIAG Introduction section shows the position of Dcm (and the other RTA-DIAG modules) in the Autosar layered architecture. The interaction/dependencies of Dcm with other modules is summarised here.

In processing the requested diagnostic service, Dcm will interact with other BSW modules or with SW-Components (through the RTE) to obtain the requested data, or to execute the requested commands. Dcm has interfaces to, and dependencies on, the following:

Diagnostic Event Manager (Dem)

- The Dem module retrieves information related to fault memory on Dcm's behalf.

Protocol Data Unit Router (PduR)

- The PduR module provides the network-independent interface to Dcm. Dcm receives diagnostic messages from PduR, and it assembles gathered information and sends messages back through PduR. Proper operation of Dcm relies on the PduR interface supporting all service primitives defined for the Service Access Point (SAP) between the diagnostic application layer and the underlying transport layer.

Communication Manager (ComM)

- The ComM module provides functions so enable Dcm to indicate "active"

and “inactive” states for diagnostic communication. Dcm provides functionality to handle the communication requirements of “Full”, “Silent” and “No-Communication”, and also provides the functionality to enable/disable Diagnostic Communication, if requested by ComM.

BswM

- If the initialization of Dcm is the consequence of a jump from the bootloader, Dcm notifies BswM that the application was updated. Dcm also informs BswM of communication mode changes.

rba_DiagLib

- The rba_Diaglib provides common functionality and utils for all RTA-DIAG modules, including Dcm.

Module Configuration

This module can be generated by the RTA-BSW Configuration Generator.

For any unsupported automatic configuration, the module can be manually configured by the user.

2.2 Supported Features

2.2.1 Core Features

- Dcm receives, processes and checks diagnostic messages from PduR.
- It assembles the gathered information and sends a message back through the PduR module.
- Dcm interacts with other BSW modules (or with SW-Components through the RTE) to obtain requested data, or to execute requested commands.

2.2.2 Dcm Sub-Modules

Dcm consists of the following sub-modules:

The Diagnostic Session Layer (DSL)

- The DSL sub-module ensures a smooth data flow of diagnostic requests and responses, it supervises and guarantees diagnostic protocol timing, and it manages diagnostic states (especially diagnostic session and security).

The Diagnostic Service Dispatcher (DSD)

- The DSD sub-module checks the validity of incoming diagnostic requests, and keeps track of the progress of service request executions.

The Diagnostic Service Processing (DSP)

- The DSP sub-module handles the actual diagnostic service requests, which it receives via a function call from the DSD sub-module.

2.2.3 Parallel Diagnostic Processing

Parallel processing of OBD and other requests. Earlier implementations of Dcm supported only one active protocol at a time. With this feature, OBD along with another non-OBD protocol can be handled at the same time.

- Parallel processing is only supported in Default session.
- The parameter DcmRbMaxRequestsInParallel is used to process multiple diagnostic protocol requests in parallel.
- Parallel processing is not supported in Non-Default session and the handling of multiple diagnostic protocol requests would be based on protocol priority.

Currently Parallel protocol processing is supported only for

- ReadDataByIdentifier(0x22)
- Read DTC Information(0x19)
- Diagnostic Session Control(0x10)
- ECUReset(0x11)
- Authentication(0x29)
- RoutineControl(0x31)
- TesterPresent(0x3E)
- ClearDiagnosticInformation(0x14)

Services in a Default session and for other services are not currently supported. It is therefore recommended to request only the services listed here in parallel.

2.2.4 PostBuild

- Currently only PostBuild selectable is supported in Dcm and is limited to postbuild parameters DcmDslProtocolRowUsed, DcmDsdServiceUsed and DcmDspPidUsed.
- If these parameters are not configured then by default these will be considered as supported in all postbuild variants. If any of these parameters are not supported or not needed in any postbuild variant then that parameter should be configured as FALSE for that variant.
- If postbuild is enabled, postbuild variant should be passed to Dcm_Init as the parameter.
- If postbuild is disabled, NULL_PTR should be passed to Dcm_Init as the parameter.

See the Configuration Advice and Limitations for more details.

2.2.5 Additional Features

The Dcm module also includes the following features.

- OBD Services: $01, $02, $03, $04, $06, $07, $08, $09 and $0A.
- Unified Diagnostic Services (UDS) 14229-1: 0x10, 0x11, 0x14, 0x19, 0x22, 0x23, 0x27, 0x28, 0x29, 0x2A, 0x2C, 0x2E, 0x2F, 0x31, 0x34, 0x35, 0x36, 0x37, 0x3D, 0x3E, 0x85, 0x86.
- Multicore.
- Memory mapping.
- Generation of Sender-Receiver elements for Dem.
- Splitting of responses for KWP services.
- Asynchronous Server Point handling for DidData, SecurityAccess and Routine Control Interfaces.
- Flexible length handling for Seed and Key handling in Security Access service. See the Configuration Advice for more details.

2.3 Extensions

2.3.1 Configuration Extensions

The following configuration parameters are available in version v12.7.0 in addition to those defined in AR v22-11

2.3.1.1 Dcm/DcmConfigSet/DcmDsd/DcmDsdServiceTable/DcmDsdService/DcmDsdSubService/DcmRbDsdSubServiceUserModeRuleFnc

Short Name	DcmRbDsdSubServiceUserModeRuleFnc
Description	Function name used by the application to perform user specific checks on the subfunction.In case the checks fails, NRC value set by the application can be returned to the tester. In case the function is not configured for sub service , the DcmAppl API DcmAppl_UserSubServiceModeRuleService shall be called.
Introduction	Function name used by the application to perform user specific checks on the subfunction.In case the checks fails, NRC value set by the application can be returned to the tester. In case the function is not configured for sub service, the DcmAppl API DcmAppl_UserSubServiceModeRuleService shall be called.
Multiplicity	0..1
Default Value	-
Type	ECUC-FUNCTION-NAME-DEF

2.3.1.2 Dcm/DcmConfigSet/DcmDsd/DcmDsdServiceTable/DcmDsdService/DcmRbDsdSidTabSidInitFunction

Short Name	DcmRbDsdSidTabSidInitFunction
Description	Function pointer of the service initializing function.User may see the function name for Dcm specific services by pressing “Ctrl+Enter”
Multiplicity	0..1
Default Value	-
Type	ECUC-STRING-PARAM-DEF

2.3.1.3 Dcm/DcmConfigSet/DcmDsd/DcmDsdServiceTable/DcmDsdService/DcmRbDsdSidTabUserModeRuleFnc

Short Name	DcmRbDsdSidTabUserModeRuleFnc
Description	Function name used by the application to perform user specific checks on the SID.In case the checks fails, NRC value set by the application can be returned to the tester. In case the function is not configured for SID, the DcmAppl API Xxx_UserService ModeRuleService shall be called.
Introduction	Function name used by the application to perform user specific checks on the SID.In case the checks fails, NRC value set by the application can be returned to the tester. In case the function is not configured for SID, the DcmAppl API Xxx_UserService ModeRuleService shall be called.
Multiplicity	0..1
Default Value	-
Type	ECUC-FUNCTION-NAME-DEF

2.3.1.4 Dcm/DcmConfigSet/DcmDsd/DcmDsdServiceTable/DcmRbNRCForService

Short Name	DcmRbNRCForService
Description	The NRC updated in this field shall be returned for any request in the protocol referring this service table if the service is not supported in active session. ISO recommends that the parameter should be set to 0x7F for UDS.Users may configure any NRC value for this parameter, according to project requirements, for other protocols.
Multiplicity	0..1
Value Range	1..254

Default Value	127
Type	ECUC-INTEGER-PARAM-DEF

2.3.1.5 Dcm/DcmConfigSet/DcmDsd/DcmRbDsdRequestASPManufacturerNotificationEnabled

Short Name	DcmRbDsdRequestASPManufacturerNotificationEnabled
Description	Only relevant when dcmDsdRequestManufacturerNotificationEnabled parameter is TRUE.This parameter is used to select the type of Server Handling required. TRUE indicates Asynchronous Server Call Point handling shall be used for every Manufacturer ServiceRequestNotification. FALSE indicates Synchronous server call point handling shall be used.
Multiplicity	0..1
Default Value	-
Type	ECUC-BOOLEAN-PARAM-DEF

2.3.1.6 Dcm/DcmConfigSet/DcmDsd/DcmRbDsdRequestManufacturerNotificationEnabled

Short Name	DcmRbDsdRequestManufacturerNotificationEnabled
Description	Allows to enable or disable the requested notification mechanism for the Manufacturer.
Multiplicity	1..1
Default Value	-
Type	ECUC-BOOLEAN-PARAM-DEF

2.3.1.7 Dcm/DcmConfigSet/DcmDsd/DcmRbDsdRequestSupplierNotificationEnabled

Short Name	DcmRbDsdRequestSupplierNotificationEnabled
Description	Allows to enable or disable the requested notification mechanism for the Supplier.
Multiplicity	1..1
Default Value	-
Type	ECUC-BOOLEAN-PARAM-DEF

2.3.1.8 Dcm/DcmConfigSet/DcmDsl/DcmDslProtocol/DcmDslProtocolRow/DcmDslConnection/DcmDslMainConnection/DcmDslROEConnectionRef

Short Name	DcmDslROEConnectionRef

Description	Reference to a ResponseOnEvent connection which is used for the processing of ResponseOnEvent events.
Multiplicity	0..1
Default Value	-
Type	ECUC-REFERENCE-DEF

2.3.1.9 Dcm/DcmConfigSet/DcmDsl/DcmDslProtocol/DcmDslProtocolRow/DcmDslConnection/DcmDslResponseOnEvent

Short Name	DcmDslResponseOnEvent
Description	This container contains the configuration of a ResponseOnEvent connection.
Introduction	The PDU referenced by this transmission channel can produce meta data items of type TARGET_ADDRESS_16 and SOURCE_ADDRESS_16.
Multiplicity	0..1
Default Value	-
Type	ECUC-PARAM-CONF-CONTAINER-DEF

2.3.1.10 Dcm/DcmConfigSet/DcmDsl/DcmDslProtocol/DcmDslProtocolRow/DcmDslConnection/DcmDslResponseOnEvent/DcmDslRoeTxConfirmationPduId

Short Name	DcmDslRoeTxConfirmationPduId
Description	Identifier of the PDU that is used by the lower level module for transmission confirmation of responses on this connection.
Multiplicity	0..1
Value Range	0..65535
Default Value	-
Type	ECUC-INTEGER-PARAM-DEF

2.3.1.11 Dcm/DcmConfigSet/DcmDsl/DcmDslProtocol/DcmDslProtocolRow/DcmDslConnection/DcmDslResponseOnEvent/DcmDslRoeTxPduRef

Short Name	DcmDslRoeTxPduRef
Description	Reference to a Pdu in EcuC that is used for this ResponseOnEvent transmission connection (only needed for ROE Transmission Type is TYPE2). One PDU required if ROE Transmission Type is TYPE2. No PDU required if ROE Transmission Type is TYPE1.
Multiplicity	0..1
Default Value	-

Type	ECUC-REFERENCE-DEF

2.3.1.12 Dcm/DcmConfigSet/DcmDsl/DcmDslProtocol/DcmDslProtocol-Row/DcmRbDslKWPTimings

Short Name	DcmRbDslKWPTimings
Description	Timings set required for KWP protocol
Multiplicity	0..1
Default Value	-
Type	ECUC-PARAM-CONF-CONTAINER-DEF

2.3.1.13 Dcm/DcmConfigSet/DcmDsl/DcmDslProtocol/DcmDslProtocol-Row/DcmRbDslKWPTimings/DcmRbP2MaxDefaultTime

Short Name	DcmRbP2MaxDefaultTime
Description	Default time of P2 max (This time activated during the start of communication) in seconds
Multiplicity	1..1
Value Range	0..4293
Default Value	-
Type	ECUC-FLOAT-PARAM-DEF

2.3.1.14 Dcm/DcmConfigSet/DcmDsl/DcmDslProtocol/DcmDslProtocol-Row/DcmRbDslKWPTimings/DcmRbP2MaxLimitTime

Short Name	DcmRbP2MaxLimitTime
Description	Limit time of P2 max (Aceess timing parameter service while changing the ECU timngs cross check the limit by refering this configured value) in seconds
Multiplicity	1..1
Value Range	0..4293
Default Value	-
Type	ECUC-FLOAT-PARAM-DEF

2.3.1.15 Dcm/DcmConfigSet/DcmDsl/DcmDslProtocol/DcmDslProtocol-Row/DcmRbDslKWPTimings/DcmRbP3MaxDefaultTime

Short Name	DcmRbP3MaxDefaultTime
Description	Default time of P3 max (This time activated during the start of communication) in seconds
Multiplicity	1..1
Value Range	0..4293

Default Value	-
Type	ECUC-FLOAT-PARAM-DEF

2.3.1.16 Dcm/DcmConfigSet/DcmDsl/DcmDslProtocol/DcmDslProtocol-Row/DcmRbDslKWPTimings/DcmRbP3MaxLimitTime

Short Name	DcmRbP3MaxLimitTime
Description	Limit time of P3 max (Aceess timing parameter service while changing the ECU timngs cross check the limit by refering this configured value) in seconds
Multiplicity	1..1
Value Range	0..4293
Default Value	-
Type	ECUC-FLOAT-PARAM-DEF

2.3.1.17 Dcm/DcmConfigSet/DcmDsl/DcmDslProtocol/DcmDslProtocol-Row/DcmRbDslProtocolNRC21OnPriorityAssessmentReject

Short Name	DcmRbDslProtocolNRC21OnPriorityAssessmentReject
Description	This parameter defines the reaction when a new requests arrives on this protocol, when there is already a request with higher or same priority under processing. TRUE: answer with NRC21 BusyRepeatRequest FALSE: silently ignore the request DcmDslProtocolRows for which DcmRbDslProtocolNRC21OnPriorityAssessmentReject is configured, the Dcm shall consider the parameter DcmRbDslProtocolNRC21OnPriorityAssessmentReject instead of DcmDslDiagRespOnSecondDeclinedRequest for rejection of a second declined request.
Introduction	TRUE: answer with NRC21 BusyRepeatRequest. FALSE: silently ignore the request
Multiplicity	0..1
Default Value	-
Type	ECUC-BOOLEAN-PARAM-DEF

2.3.1.18 Dcm/DcmConfigSet/DcmDsl/DcmDslProtocol/DcmDslProtocol-Row/DcmRbMinimumPageSize

Short Name	DcmRbMinimumPageSize

Description	Configuration of minimum page size for a consecutive page when DcmPagedBufferEnabled == TRUE. The consecutive page will be transmitted as soon as it is filled with the minimum page size. This has should be configured as such to avoid any timeout in the Tps while waiting for the consecutive page to be filled. This should not be greater than the DcmDslBufferSize referenced by the DcmDslProtocol RxBufferRef of this DcmDslProtocolRow. In case the parameter is left unconfigured and DcmPagedBufferEnabled == TRUE, then the DcmDslBufferSize referenced by the DcmDslProtocolRxBufferRef of this DcmDslProtocolRow will be considered as the DcmRbMinimumPageSize
Multiplicity	0..1
Value Range	8..4294967294
Default Value	-
Type	ECUC-INTEGER-PARAM-DEF

2.3.1.19 Dcm/DcmConfigSet/DcmDsl/DcmRbDslKWPSplitResponseFeature

Short Name	DcmRbDslKWPSplitResponseFeature
Description	The container to enable/disable the feature of splitting the responses for KWP services. If configured, the split of responses from any service shall be supported for KWP2000 services. If not configured, the split of responses by the services shall be disabled. Based on the configuration of the above container, the compiler switch DCM_CFG_SPLITRESPSUPPORTEDFORKWP shall be generated. This container shall be effective only if there is a KWP protocol configured.
Multiplicity	0..1
Default Value	-
Type	ECUC-PARAM-CONF-CONTAINER-DEF

2.3.1.20 Dcm/DcmConfigSet/DcmDsl/DcmRbDslKWPSplitResponseFeature/DcmRbDslKWPTimeBetweenSplitResponses

Short Name	DcmRbDslKWPTimeBetweenSplitResponses

Description	The parameter to define the timing between any 2 split responses in seconds. The timing has to be in multiples of DcmTaskTime. This parameter is mandatory if the container DcmRbDslKWPSplitResponseFeature is configured. The value in the parameter shall be generated as DCM_CFG_SPLITRESPONSETIME-FORKWP.
Multiplicity	1..1
Value Range	0..4293.0
Default Value	-
Type	ECUC-FLOAT-PARAM-DEF

2.3.1.21 Dcm/DcmConfigSet/DcmDsp/DcmDspAuthentication/DcmDspAuthenticationConnection/DcmDspAuthenticationTargetIdentificationModeRuleRef

Short Name	DcmDspAuthenticationTargetIdentificationModeRuleRef
Description	Reference to a mode rule that is used to evaluate the target identification.
Multiplicity	0..0
Default Value	-
Type	ECUC-REFERENCE-DEF

2.3.1.22 Dcm/DcmConfigSet/DcmDsp/DcmDspAuthentication/DcmDspAuthenticationConnection/DcmRbDspAuthenticationAlgorithmIndicator

Short Name	DcmRbDspAuthenticationAlgorithmIndicator
Multiplicity	0..1
String Length Range	32..32
Default Value	-
Type	ECUC-STRING-PARAM-DEF

Description: Algorithm Indicator supported by this Authentication connection in hexadecimal format. This shall be configured in the exact same order as the algorithmIndicator parameter sent in the request for Service 0x29, SubFunction 0x05, 0x06 and 0x07. Eg: 06092A864886F70D01010A0000000000, in this case the expected algorithmIndicator parameter in the request shall be [06,09,2A,86,48,86,F7,0D,01,01,0A,00,00,00,00,00] Indicates the algorithm used in the generating and verifying Proof of Ownership (POWN), which further determines the parameters used in the algorithm and possibly the session key creation mode.This field is a 16 byte value containing the BER encoded OID value of the algorithm used. The value is left aligned and right padded with zero up to 16 bytes. Mandatory to be configured if DcmRbDspAuthenticationConcept != DCM_APCE_ASYMMETRIC. The algorithm Indicator has to be consistent with the algorithm configured in the corresponding CsmSignatureVerify Job referenced in

DcmDspAuthenticationVerifyProofOfOwnerShipClientJobRef.

2.3.1.23 Dcm/DcmConfigSet/DcmDsp/DcmDspAuthentication/DcmRbAuthenticationMaxLenCustomChallenge

Short Name	DcmRbAuthenticationMaxLenCustomChallenge
Description	This container contains the configuration of the maximum number of Bytes for the Challenge info passed by application to DCM. This paramter is only valid fo APCE Authentication concept
Multiplicity	0..1
Value Range	1..65535
Default Value	-
Type	ECUC-INTEGER-PARAM-DEF

2.3.1.24 Dcm/DcmConfigSet/DcmDsp/DcmDspAuthentication/DcmRbDspAuthenticationConcept

Short Name	DcmRbDspAuthenticationConcept
Description	The authentication concept that is supported by Dcm.This parameter decides which sub-functions are supported for service 0x29. DCM_APCE_ASYMMETRIC : Sub-functions 0x01, 0x02, 0x03 and 0x04. ACR_ SYMMETRIC : Authentication with Challenge-Response (ACR) and symmetric cryptography. DCM_ACR_ SYMMETRIC : Sub-functions 0x05. 0x06 and 0x07. Sub-functions 0x00 and 0x08 are independent of this parameter. Hint: DCM_ACR_ASYMMETRIC is not supported currently, even though ISO 14229 supports it.
Multiplicity	1..1
Default Value	DCM_APCE_ASYMMETRIC
Enumeration Literals	DCM_APCE_ASYMMETRIC, DCM_ACR_SYMMETRIC
Type	ECUC-ENUMERATION-PARAM-DEF

2.3.1.25 Dcm/DcmConfigSet/DcmDsp/DcmDspAuthentication/DcmRbDspAuthenticationCustomChallengeServer

Short Name	DcmRbDspAuthenticationCustomChallengeServer
Description	This parameter enables the provision of the Callouts DcmAppl_CustomChallengeGen () to the application. DcmAppl_CustomChallengeGen () : The function allows the application to modify or generate a new challenge based on the initial challenge provided by the CSM.
Multiplicity	0..1

Default Value	false
Type	ECUC-BOOLEAN-PARAM-DEF

2.3.1.26 Dcm/DcmConfigSet/DcmDsp/DcmDspAuthentication/DcmRbDspAuthenticationEnaPownApplCallbk

Short Name	DcmRbDspAuthenticationEnaPownApplCallbk
Description	This parameter will enable application callback to encode and decode the POWN server and client data
Multiplicity	0..1
Default Value	false
Type	ECUC-BOOLEAN-PARAM-DEF

2.3.1.27 Dcm/DcmConfigSet/DcmDsp/DcmDspAuthentication/DcmRbDspAuthenticationEnableApplCustomChallenge

Short Name	DcmRbDspAuthenticationEnableApplCustomChallenge
Description	This parameter enables the provision of the Callouts DcmAppl_ChallengeGenComplete(), DcmAppl_GetChallengePOWNServer() and DcmAppl_GetChallengePOWNClient to the application. DcmAppl_ChallengeGenComplete() : Provides the Challenge Server, Challenge Client and their respective length to application. DcmAppl_GetChallengePOWNServer() : Dcm fetches the custom challenge for POWN calculation of server from application which is calculated based on the challenge client and challenge server provided in the first call back. DcmAppl_GetChallengePOWNClient() : Dcm fetches the custom challenge for POWN verification of client from application which is generated based on the challenge client and challenge server provided in the first call back.
Multiplicity	0..1
Default Value	false
Type	ECUC-BOOLEAN-PARAM-DEF

2.3.1.28 Dcm/DcmConfigSet/DcmDsp/DcmDspAuthentication/DcmRbDspAuthenticationEnableCustomRole

Short Name	DcmRbDspAuthenticationEnableCustomRole

Description	This parameter enables the provision of the Callouts DcmAppl_GetCustomRole() to application. DcmAppl_GetCustomRole() : The function allows the application to define and return a custom authentication role that can be used for further processing or decision-making within the DCM module.
Multiplicity	0..1
Default Value	false
Type	ECUC-BOOLEAN-PARAM-DEF

2.3.1.29 Dcm/DcmConfigSet/DcmDsp/DcmDspAuthentication/DcmRbDspAuthenticationEnableFailedCountInterface

Short Name	DcmRbDspAuthenticationEnableFailedCountInterface
Description	This parameter will enable to callout a counter for number of failed attempts and for specific implementation after authentication result is obtained from KeyM.
Multiplicity	0..1
Default Value	false
Type	ECUC-BOOLEAN-PARAM-DEF

2.3.1.30 Dcm/DcmConfigSet/DcmDsp/DcmDspAuthentication/DcmRbDspAuthenticationInvalidAlgorithmIndicatorNRC

Short Name	DcmRbDspAuthenticationInvalidAlgorithmIndicatorNRC
Description	Defines the NRC that shall be sent in case of an invalid algorithm indicator (the algorithm indicator received in the request is not matching the algorithm indicator configured in the corresponding DcmRbDspAuthenticationAlgorithmIndicator). If this is not configured, NRC 0x10 shall be sent.
Multiplicity	0..1
Value Range	0..255
Default Value	-
Type	ECUC-INTEGER-PARAM-DEF

2.3.1.31 Dcm/DcmConfigSet/DcmDsp/DcmDspAuthentication/DcmRbDspAuthenticationPownClientMaxSize

Short Name	DcmRbDspAuthenticationPownClientMaxSize
Description	Parameter to set maximum size of proof of ownership of client, This shall be configured if DcmRbDspAuthenticationEnaPownApplCallbk is set to true.

Multiplicity	0..1
Value Range	1..65535
Default Value	-
Type	ECUC-INTEGER-PARAM-DEF

2.3.1.32 Dcm/DcmConfigSet/DcmDsp/DcmDspClearDTC/DcmRbDspCDI-ISOVersion

Short Name	DcmRbDspCDIISOVersion
Description	This parameter indicates the version of ISO the project needs, the projects can choose between ISO version ISO14229-1 2013 edition and newer edition ISO14229_01_2020. This parameter is only relevant to CDI and is used to choose the Memory Selection byte can be supported or not. If ISO14229-1 2013 edition is chosen, the Memory Selection byte is not supported. If ISO14229-1 2020 edition is chosen, the Memory Selection byte is supported
Multiplicity	0..1
Default Value	ISO14229_01_2020
Enumeration Literals	ISO14229_01_2013, ISO14229_01_2020
Type	ECUC-ENUMERATION-PARAM-DEF

2.3.1.33 Dcm/DcmConfigSet/DcmDsp/DcmDspComControl/DcmDspCom-ControlSetting/DcmRbDspComControlReEnableUserModeRuleFnc

Short Name	DcmRbDspComControlReEnableUserModeRuleFnc
Description	Function name used by the application to perform user specific check to verify if communication modes needs to be re-enabled for the channels.In case the checks fails, communication modes shall be re-enabled. In case the function is not configured , the DcmAppl API DcmAppl_UserCommCtrlReEnableModeRuleService shall be called.
Introduction	Function name used by the application to perform user specific check to verify if communication modes needs to be re-enabled for the channels.In case the checks fails, communication modes shall be re-enabled. In case the function is not configured for sub service, the DcmAppl API DcmAppl_UserCommCtrlReEnable-ModeRuleService shall be called.
Multiplicity	0..1
Default Value	-
Type	ECUC-FUNCTION-NAME-DEF

2.3.1.34 Dcm/DcmConfigSet/DcmDsp/DcmDspControlDTCSetting/DcmRbDspControlDTCSettingReEnableUserModeRuleFnc

Short Name	DcmRbDspControlDTCSettingReEnableUser ModeRuleFnc
Description	Function name used by the application to perform user specific check to verify if DTC setting needs to be re-enabled.In case the checks fails, DTC setting shall be re-enabled. In case the function is not configured , the DcmAppl API DcmAppl_UserDTC-SettingEnableModeRuleService shall be called.
Introduction	Function name used by the application to perform user specific check to verify if DTC setting needs to be re-enabled.In case the checks fails, DTC setting shall be re-enabled. In case the function is not configured , the DcmAppl API DcmAppl_UserDTC-SettingEnableModeRuleService shall be called.
Multiplicity	0..1
Default Value	-
Type	ECUC-FUNCTION-NAME-DEF

2.3.1.35 Dcm/DcmConfigSet/DcmDsp/DcmDspData/DcmDspDataBlockIdRef

Short Name	DcmDspDataBlockIdRef
Description	NRAM blockId to access the data.
Introduction	Only relevant if DcmDspDataUsePort==USE_BLOCK_ ID.
Multiplicity	0..1
Default Value	-
Type	ECUC-REFERENCE-DEF

2.3.1.36 Dcm/DcmConfigSet/DcmDsp/DcmDspData/DcmDspDiagnosisScaling/DcmDspAlternativeDataProps

Short Name	DcmDspAlternativeDataProps
Description	This container contains the configuration (parameters) of an alternative Diagnosis Representation by the means of ECU configuration parameters.

Introduction	The physical unit of the alternative data representation is defined by the DataPrototype referenced by DcmDspExternalSRDataElementClass. Additionally the definition of a text table mapping can be a defined for DcmDspDataTypeCategory TEXTTABLE and SCALE_LINEAR_AND_TEXTTABLE
Multiplicity	0..1
Default Value	-
Type	ECUC-PARAM-CONF-CONTAINER-DEF

2.3.1.37 Dcm/DcmConfigSet/DcmDsp/DcmDspData/DcmDspDiagnosisScaling/DcmDspAlternativeDataProps/DcmDspDataTypeCategory

Short Name	DcmDspDataTypeCategory
Description	Data category of the alternative Diagnosis Representation.
Multiplicity	0..0
Default Value	-
Enumeration Literals	LINEAR, SCALE_LINEAR_AND_TEXTTABLE, TEXTTABLE
Type	ECUC-ENUMERATION-PARAM-DEF

2.3.1.38 Dcm/DcmConfigSet/DcmDsp/DcmDspData/DcmDspDiagnosisScaling/DcmDspAlternativeDataProps/DcmDspLinearScale

Short Name	DcmDspLinearScale
Description	This container contains the configuration (parameters) of an linear scale of the alternative Diagnosis Representation.
Multiplicity	0..1
Default Value	-
Type	ECUC-PARAM-CONF-CONTAINER-DEF

2.3.1.39 Dcm/DcmConfigSet/DcmDsp/DcmDspData/DcmDspDiagnosisScaling/DcmDspAlternativeDataProps/DcmDspLinearScale/DcmDspDiagnosisRepresentationDataLowerRange

Short Name	DcmDspDiagnosisRepresentationDataLowerRange
Description	Lower Range for this scale of the data in the alternative Diagnosis Representation.
Multiplicity	1..1
Value Range	-Inf..Inf
Default Value	-
Type	ECUC-FLOAT-PARAM-DEF

2.3.1.40 Dcm/DcmConfigSet/DcmDsp/DcmDspData/DcmDspDiagnosisScaling/DcmDspAlternativeDataProps/DcmDspLinearScale/DcmDspDiagnosisRepresentationDataOffset

Short Name	DcmDspDiagnosisRepresentationDataOffset
Description	Data offset of the alternative Diagnosis Representation for this scale.
Multiplicity	1..1
Value Range	0..Inf
Default Value	0
Type	ECUC-FLOAT-PARAM-DEF

2.3.1.41 Dcm/DcmConfigSet/DcmDsp/DcmDspData/DcmDspDiagnosisScaling/DcmDspAlternativeDataProps/DcmDspLinearScale/DcmDspDiagnosisRepresentationDataResolution

Short Name	DcmDspDiagnosisRepresentationDataResolution
Description	Data resolution of the alternative Diagnosis Representation for this scale.
Multiplicity	1..1
Value Range	0..Inf
Default Value	-
Type	ECUC-FLOAT-PARAM-DEF

2.3.1.42 Dcm/DcmConfigSet/DcmDsp/DcmDspData/DcmDspDiagnosisScaling/DcmDspAlternativeDataProps/DcmDspLinearScale/DcmDspDiagnosisRepresentationDataUpperRange

Short Name	DcmDspDiagnosisRepresentationDataUpperRange
Description	Upper Range for this scale of the data in the alternative Diagnosis Representation.
Multiplicity	1..1
Value Range	-Inf..Inf
Default Value	-
Type	ECUC-FLOAT-PARAM-DEF

2.3.1.43 Dcm/DcmConfigSet/DcmDsp/DcmDspData/DcmRbDspDataArrayInfoSubElement

Short Name	DcmRbDspDataArrayInfoSubElement

Description	String Input which contains the information Sub Array Index for DcmSubElementInImplDataElementInstance or structure type info for DcmSubElementInDataElementInstance. If DcmSubElement InDataElementInstance is configured then this parameter should the structure Type as well as the as the sub-structure type separated by commas: eg: struct Test { uint8 A, uint32 B }; If B is the SubElement required then this parameter must be filled with - Test,B If DcmSubElementInImplDataElementInstance is configured then this parameter should contain the sub array index to be accessed:
Multiplicity	0..1
Default Value	-
Type	ECUC-STRING-PARAM-DEF

2.3.1.44 Dcm/DcmConfigSet/DcmDsp/DcmDspData/DcmRbDspDataUseAsynchronousServerCallPoint

Short Name	DcmRbDspDataUseAsynchronousServerCallPoint
Description	Only relevant in case DcmDspDataUsePort is set to USE_DATA_ASYNCH_CLIENT_SERVER.This parameter is used to select the type of Server Handling required.Set the value to TRUE for Asynchronous server call point handling else Synchronous server call point handling is carried out
Multiplicity	0..1
Default Value	false
Type	ECUC-BOOLEAN-PARAM-DEF

2.3.1.45 Dcm/DcmConfigSet/DcmDsp/DcmDspDid/DcmDspDidExtRoe

Short Name	DcmDspDidExtRoe
Description	Provide information to manage a responseOnEvent request on this DID externally. If this container doesn't exist, an ROE event on this DID will be managed internally.In RTA-BSW DCM internally managed DIDs are not supported, if the projects want monitoring of DID as per Autosar 4.0.2, then it is mandatory to configure the DcmDspDidExtRoe
Introduction	The R-Port for ROEServices interface is named ROEServices_<SWC> where <SWC> is the name of this container.
Multiplicity	0..1

Default Value	-
Type	ECUC-PARAM-CONF-CONTAINER-DEF

2.3.1.46 Dcm/DcmConfigSet/DcmDsp/DcmDspDid/DcmDspDidExtRoe/DcmDspDidRoeActivateFnc

Short Name	DcmDspDidRoeActivateFnc
Description	Function name to activate/deactivate an ROE event managed externally.This parameter is related to the interface xxx_ActivateEvent.This parameter is to be configured mandatorily if atleast one of the signal of the DID is of the following DcmDspDataUse-Port==USE_DATA_SYNCH_FNC/USE_DATA_ASYNCH_FNC/USE_BLOCK_ID. IF all the signals of the DID are set to ===USE_DATA_SENDER_RECEIVER/USE_DATA_ASYNCH_CLIENT_SERVER/USE_DATA_SYNCH_CLIENT_SERVER
Introduction	Only relevant if DcmDspDataUsePort== USE_DATA_SYNCH_FNC/USE_DATA_ASYNCH_FNC. This parameter is related to the interface xxx_ActivateEvent.
Multiplicity	0..1
Default Value	-
Type	ECUC-FUNCTION-NAME-DEF

2.3.1.47 Dcm/DcmConfigSet/DcmDsp/DcmDspDid/DcmDspDidExtRoe/DcmDspDidRoeEventId

Short Name	DcmDspDidRoeEventId
Description	EventId to be used within Apis Dcm_TriggerOnEvent() and xxx_ActivateEvent()
Multiplicity	1..1
Value Range	0..255
Default Value	-
Type	ECUC-INTEGER-PARAM-DEF

2.3.1.48 Dcm/DcmConfigSet/DcmDsp/DcmDspDid/DcmRbAtomicSender-ReceiverCommunication

Short Name	DcmRbAtomicSenderReceiverCommunication
Description	This parameter is used to select the type of communication required.Set the value to TRUE for Atomic SenderReceiver communication else regular SenderReceiver communication will be carried out
Multiplicity	1..1

Default Value	false
Type	ECUC-BOOLEAN-PARAM-DEF

2.3.1.49 Dcm/DcmConfigSet/DcmDsp/DcmDspDid/DcmRbDspIsDIDAvailableFnc

Short Name	DcmRbDspIsDIDAvailableFnc
Description	Function name to request the application to check if the DID requested is supported in the variant.The user can leave the field empty if the check is not required for this DID
Introduction	Function name to request the application to check if the DID requested is supported in the variant.The user can leave the field empty if the check is not required for this DID
Multiplicity	0..1
Default Value	-
Type	ECUC-FUNCTION-NAME-DEF

2.3.1.50 Dcm/DcmConfigSet/DcmDsp/DcmDspDidInfo/DcmDspDidControl/DcmRbDspDidControlUserModeRuleFnc

Short Name	DcmRbDspDidControlUserModeRuleFnc
Description	Function name used by the application to perform user specific checks before IO Control of a DID.In case the checks fails, NRC value set by the application can be returned to the tester. In case the function is not configured for DID Control, the DcmAppl API DcmAppl_UserDIDModeRuleService shall be called.
Introduction	Function name used by the application to perform user specific checks before IO Control of a DID.In case the checks fails, NRC value set by the application can be returned to the tester. In case the function is not configured for DID Control, the DcmAppl API DcmAppl_UserDIDModeRuleService shall be called.
Multiplicity	0..1
Default Value	-
Type	ECUC-FUNCTION-NAME-DEF

2.3.1.51 Dcm/DcmConfigSet/DcmDsp/DcmDspDidInfo/DcmDspDidControl/DcmRbDspIoCtrlResetOnSessSecurityChange

Short Name	DcmRbDspIoCtrlResetOnSessSecurityChange

Description	Indicates if resetting of IOControl on session / security change is enabled / disabled.
Introduction	True = IOcontrol on session / security change is enabled False = IOcontrol on session / security change is disabled
Multiplicity	0..1
Default Value	true
Type	ECUC-BOOLEAN-PARAM-DEF

2.3.1.52 Dcm/DcmConfigSet/DcmDsp/DcmDspDidInfo/DcmDspDidRead/DcmRbDspDidReadUserModeRuleFnc

Short Name	DcmRbDspDidReadUserModeRuleFnc
Description	Function name used by the application to perform user specific checks before Reading a DID.In case the checks fails, NRC value set by the application can be returned to the tester. In case the function is not configured for DID Read, the DcmAppl API DcmAppl_UserDIDModeRuleService shall be called.
Introduction	Function name used by the application to perform user specific checks before Reading a DID.In case the checks fails, NRC value set by the application can be returned to the tester. In case the function is not configured for DID Read, the DcmAppl API DcmAppl_UserDIDModeRuleService shall be called.
Multiplicity	0..1
Default Value	-
Type	ECUC-FUNCTION-NAME-DEF

2.3.1.53 Dcm/DcmConfigSet/DcmDsp/DcmDspDidInfo/DcmDspDidWrite/DcmRbDspDidWriteUserModeRuleFnc

Short Name	DcmRbDspDidWriteUserModeRuleFnc
Description	Function name used by the application to perform user specific checks before writing a DID.In case the checks fails, NRC value set by the application can be returned to the tester. In case the function is not configured for DID Write, the DcmAppl API DcmAppl_UserDIDModeRuleService shall be called.
Introduction	Function name used by the application to perform user specific checks before writing a DID.In case the checks fails, NRC value set by the application can be returned to the tester. In case the function is not configured for DID Write, the DcmAppl API DcmAppl_UserDIDModeRuleService shall be called.

Multiplicity	0..1
Default Value	-
Type	ECUC-FUNCTION-NAME-DEF

2.3.1.54 Dcm/DcmConfigSet/DcmDsp/DcmDspMemory/DcmDspMemoryIdInfo/DcmDspReadMemoryRangeByLabelInfo/DcmRbDspReadMemoryRangeUserModeRuleFnc

Short Name	DcmRbDspReadMemoryRangeUserModeRuleFnc
Description	Function name used by the application to perform user specific checks before reading a memory range.In case the checks fails, NRC value set by the application can be returned to the tester. In case the function is not configured for memory range, the DcmAppl API DcmAppl_UserMemoryRangeModeRuleService shall be called.
Introduction	Function name used by the application to perform user specific checks before reading a memory range.In case the checks fails, NRC value set by the application can be returned to the tester. In case the function is not configured for memory range, the DcmAppl API DcmAppl_UserMemoryRangeModeRuleService shall be called.
Multiplicity	0..1
Default Value	-
Type	ECUC-FUNCTION-NAME-DEF

2.3.1.55 Dcm/DcmConfigSet/DcmDsp/DcmDspMemory/DcmDspMemoryIdInfo/DcmDspReadMemoryRangeInfo/DcmRbDspReadMemoryRangeUserModeRuleFnc

Short Name	DcmRbDspReadMemoryRangeUserModeRuleFnc
Description	Function name used by the application to perform user specific checks before reading a memory range.In case the checks fails, NRC value set by the application can be returned to the tester. In case the function is not configured for memory range, the DcmAppl API DcmAppl_UserMemoryRangeModeRuleService shall be called.

Introduction	Function name used by the application to perform user specific checks before reading a memory range.In case the checks fails, NRC value set by the application can be returned to the tester. In case the function is not configured for memory range, the DcmAppl API DcmAppl_UserMemoryRangeModeRuleService shall be called.
Multiplicity	0..1
Default Value	-
Type	ECUC-FUNCTION-NAME-DEF

2.3.1.56 Dcm/DcmConfigSet/DcmDsp/DcmDspMemory/DcmDspMemoryId-Info/DcmDspWriteMemoryRangeByLabelInfo/DcmRbDspWrite-MemoryRangeUserModeRuleFnc

Short Name	DcmRbDspWriteMemoryRangeUserModeRuleFnc
Description	Function name used by the application to perform user specific checks before writing to a memory range.In case the checks fails, NRC value set by the application can be returned to the tester. In case the function is not configured for memory range, the DcmAppl API DcmAppl_UserMemoryRangeModeRuleService shall be called.
Introduction	Function name used by the application to perform user specific checks before writing to a memory range.In case the checks fails, NRC value set by the application can be returned to the tester. In case the function is not configured for memory range, the DcmAppl API DcmAppl_UserMemoryRangeModeRuleService shall be called.
Multiplicity	0..1
Default Value	-
Type	ECUC-FUNCTION-NAME-DEF

2.3.1.57 Dcm/DcmConfigSet/DcmDsp/DcmDspMemory/DcmDspMemoryId-Info/DcmDspWriteMemoryRangeInfo/DcmRbDspWriteMemo-ryRangeUserModeRuleFnc

Short Name	DcmRbDspWriteMemoryRangeUserModeRuleFnc

Description	Function name used by the application to perform user specific checks before writing to a memory range.In case the checks fails, NRC value set by the application can be returned to the tester. In case the function is not configured for memory range, the DcmAppl API DcmAppl_UserMemoryRangeModeRuleService shall be called.
Introduction	Function name used by the application to perform user specific checks before writing to a memory range.In case the checks fails, NRC value set by the application can be returned to the tester. In case the function is not configured for memory range, the DcmAppl API DcmAppl_UserMemoryRangeModeRuleService shall be called.
Multiplicity	0..1
Default Value	-
Type	ECUC-FUNCTION-NAME-DEF

2.3.1.58 Dcm/DcmConfigSet/DcmDsp/DcmDspMemory/DcmDspMemoryId-Info/DcmRbDspDownloadMemoryRangeInfo

Short Name	DcmRbDspDownloadMemoryRangeInfo
Description	Provides the range of memory address allowed for download
Multiplicity	0..*
Default Value	-
Type	ECUC-PARAM-CONF-CONTAINER-DEF

2.3.1.59 Dcm/DcmConfigSet/DcmDsp/DcmDspMemory/DcmDspMemoryId-Info/DcmRbDspDownloadMemoryRangeInfo/DcmRbDspDownload-MemoryRangeHigh

Short Name	DcmRbDspDownloadMemoryRangeHigh
Description	High memory address of a range allowed for download
Multiplicity	1..1
Value Range	0..4294967294
Default Value	-
Type	ECUC-INTEGER-PARAM-DEF

2.3.1.60 Dcm/DcmConfigSet/DcmDsp/DcmDspMemory/DcmDspMemoryId-Info/DcmRbDspDownloadMemoryRangeInfo/DcmRbDspDownload-MemoryRangeLow

Short Name	DcmRbDspDownloadMemoryRangeLow

Description	Low memory address of a range allowed for download.
Multiplicity	1..1
Value Range	0..4294967294
Default Value	-
Type	ECUC-INTEGER-PARAM-DEF

2.3.1.61 Dcm/DcmConfigSet/DcmDsp/DcmDspMemory/DcmDspMemoryIdInfo/DcmRbDspDownloadMemoryRangeInfo/DcmRbDspDownloadMemoryRangeSecurityLevelRef

Short Name	DcmRbDspDownloadMemoryRangeSecurityLevelRef
Description	Link to the Security Access Levels needed for download access on this memory address. If there is no reference, no check of security level shall be done.
Multiplicity	0..*
Default Value	-
Type	ECUC-REFERENCE-DEF

2.3.1.62 Dcm/DcmConfigSet/DcmDsp/DcmDspMemory/DcmDspMemoryIdInfo/DcmRbDspUploadMemoryRangeInfo

Short Name	DcmRbDspUploadMemoryRangeInfo
Description	Provides the range of memory address allowed for upload
Multiplicity	0..*
Default Value	-
Type	ECUC-PARAM-CONF-CONTAINER-DEF

2.3.1.63 Dcm/DcmConfigSet/DcmDsp/DcmDspMemory/DcmDspMemoryIdInfo/DcmRbDspUploadMemoryRangeInfo/DcmRbDspUploadMemoryRangeHigh

Short Name	DcmRbDspUploadMemoryRangeHigh
Description	High memory address of a range allowed for upload.
Multiplicity	1..1
Value Range	0..4294967294
Default Value	-
Type	ECUC-INTEGER-PARAM-DEF

2.3.1.64 Dcm/DcmConfigSet/DcmDsp/DcmDspMemory/DcmDspMemoryIdInfo/DcmRbDspUploadMemoryRangeInfo/DcmRbDspUploadMemoryRangeLow

Short Name	DcmRbDspUploadMemoryRangeLow
Description	Low memory address of a range allowed for upload.
Multiplicity	1..1
Value Range	0..4294967294
Default Value	-
Type	ECUC-INTEGER-PARAM-DEF

2.3.1.65 Dcm/DcmConfigSet/DcmDsp/DcmDspMemory/DcmDspMemoryIdInfo/DcmRbDspUploadMemoryRangeInfo/DcmRbDspUploadMemoryRangeSecurityLevelRef

Short Name	DcmRbDspUploadMemoryRangeSecurityLevelRef
Description	Link to the Security Access Levels needed for upload access on this memory address. If there is no reference, no check of security level shall be done.
Multiplicity	0..*
Default Value	-
Type	ECUC-REFERENCE-DEF

2.3.1.66 Dcm/DcmConfigSet/DcmDsp/DcmDspMemoryTransfer/DcmDspMemoryTransferFnc

Short Name	DcmDspMemoryTransferFnc
Description	Function name for memory transfer services.
Introduction	Only relevant if DcmDspMemoryTransferUsePort is set to false.
Multiplicity	0..0
Default Value	-
Type	ECUC-FUNCTION-NAME-DEF

2.3.1.67 Dcm/DcmConfigSet/DcmDsp/DcmDspPeriodicTransmission/DcmRbDspRDPIISOVersion

Short Name	DcmRbDspRDPIISOVersion

Description	This parameter indicates the version of ISO the project needs, the projects can choose between ISO version ISO14229_01_2020_02 edition /ISO 14229-01 2013-03 edition and older edition ISO14229_01_2006_12. This parameter is only relevant to Rdpi and is used to choose the NRC sent when the no. of PDID in a request exceeds the configured DcmDspMaxPeriodicDidToRead. If ISO14229_01_2020_02 edition or ISO 14229-01 2013-03 edition is chosen NRC 0x13 will be sent, else NRC 0x31 will be sent which is according to ISO14229_01_2006_12. ISO14229_01_2020_02 is the latest released ISO version. If the parameter is left unconfigured , ISO14229_01_2020_02 shall be considered as default
Multiplicity	0..1
Default Value	ISO14229_01_2020_02
Enumeration Literals	ISO14229_01_2020_02, ISO14229_01_2013_03, ISO14229_01_2006_12
Type	ECUC-ENUMERATION-PARAM-DEF

2.3.1.68 Dcm/DcmConfigSet/DcmDsp/DcmDspPeriodicTransmission/DcmRbDspRdpiType2FreeTxPduCount

Short Name	DcmRbDspRdpiType2FreeTxPduCount
Description	This parameter indicates the number of main function calls available for Dcm before releasing the TxPdu in case of no confirmation from the lower layers. For example if this value is configured as 5 and say DcmTasktime is 10ms, 50 msecs of time is available for the confirmation to be received, if no confirmation is received the TxPduID is released for later transmission. This parameter has to be configured to a value greater than the timeout value for RDPI available in the Tp Modules
Multiplicity	0..1
Value Range	1..255
Default Value	10
Type	ECUC-INTEGER-PARAM-DEF

2.3.1.69 Dcm/DcmConfigSet/DcmDsp/DcmDspPid/DcmDspPidData/DcmDspPidDataPos

Short Name	DcmDspPidDataPos
Description	This is the position in bit of the PID structure and will not start at position 0 in case a support information is availabe (for bit-mapped PIDs).

Multiplicity	1..1
Value Range	0..2040
Default Value	-
Type	ECUC-INTEGER-PARAM-DEF

2.3.1.70 Dcm/DcmConfigSet/DcmDsp/DcmDspPid/DcmDspPidData/DcmDspPidDataSize

Short Name	DcmDspPidDataSize
Description	Length of data associated to the PID in bit.
Multiplicity	0..1
Value Range	0..2040
Default Value	-
Type	ECUC-INTEGER-PARAM-DEF

2.3.1.71 Dcm/DcmConfigSet/DcmDsp/DcmDspPid/DcmDspPidData/DcmDspPidService02/DcmRbDspPidDemDataElementClassRef

Short Name	DcmRbDspPidDemDataElementClassRef
Description	Reference to DemDataElementClass in DEM configuration.Allows to link the DCM PID and DEM PID configuration for Mode $02.
Multiplicity	0..1
Default Value	-
Type	ECUC-REFERENCE-DEF

2.3.1.72 Dcm/DcmConfigSet/DcmDsp/DcmDspRequestControl/DcmRbDspRequestControlFnc

Short Name	DcmRbDspRequestControlFnc
Description	Function name for reading value by TID (DcmDspRequestControlFnc-function). This is only relevant if DcmDspRequestControlUsePort==FALSE.
Multiplicity	0..1
Default Value	-
Type	ECUC-FUNCTION-NAME-DEF

2.3.1.73 Dcm/DcmConfigSet/DcmDsp/DcmDspRequestControl/DcmRbDspRequestControlUsePort

Short Name	DcmRbDspRequestControlUsePort

Description	If this is false, the DCM will use the function defined in DcmDspRequestControlFnc. If this is true, R-Port requiring the interface. The name of this port is the name of the container.
Multiplicity	1..1
Default Value	-
Type	ECUC-BOOLEAN-PARAM-DEF

2.3.1.74 Dcm/DcmConfigSet/DcmDsp/DcmDspRoe/DcmDspRoeEvent

Short Name	DcmDspRoeEvent
Description	This container contains a list of all supported Roe eventTypeRecords which are accepted by this ECU.
Introduction	At most one DcmDspRoeEvent container is allowed to define a DcmDspRoeEventProperties container with the choice DcmDspRoeOnDTCStatusChange.
Multiplicity	1..255
Default Value	-
Type	ECUC-PARAM-CONF-CONTAINER-DEF

2.3.1.75 Dcm/DcmConfigSet/DcmDsp/DcmDspRoe/DcmDspRoeEvent/DcmDspRoeEventId

Short Name	DcmDspRoeEventId
Description	EventId for a global identification of this ROE event it is used within APIs Dcm_TriggerOnEvent() and the ModeDeclarationGroup.
Introduction	The ratio Ids should be sequentially ordered beginning with 0 and no gaps in between.
Multiplicity	1..1
Value Range	0..254
Default Value	-
Type	ECUC-INTEGER-PARAM-DEF

2.3.1.76 Dcm/DcmConfigSet/DcmDsp/DcmDspRoe/DcmDspRoeEvent/DcmDspRoeEventProperties

Short Name	DcmDspRoeEventProperties
Description	This container contains the properties of Roe eventTypeRecords.
Introduction	In one DcmDspRoeEventProperties container one DcmDspRoeOnDTCStatusChange or DcmDspRoeOnChangeOfDataIdentifier container shall be defined.

Multiplicity	1..1
Default Value	-
Type	ECUC-CHOICE-CONTAINER-DEF

2.3.1.77 Dcm/DcmConfigSet/DcmDsp/DcmDspRoe/DcmDspRoeEvent/DcmDspRoeEventProperties/DcmDspRoeOnChangeOfDataIdentifier

Short Name	DcmDspRoeOnChangeOfDataIdentifier
Description	This container contains the eventTypeRecord supported for onChangeOfDataIdentifier eventType.
Multiplicity	0..1
Default Value	-
Type	ECUC-PARAM-CONF-CONTAINER-DEF

2.3.1.78 Dcm/DcmConfigSet/DcmDsp/DcmDspRoe/DcmDspRoeEvent/DcmDspRoeEventProperties/DcmDspRoeOnChangeOfDataIdentifier/DcmDspRoeDidRef

Short Name	DcmDspRoeDidRef
Description	Reference to a Did which is watched.
Multiplicity	1..1
Default Value	-
Type	ECUC-REFERENCE-DEF

2.3.1.79 Dcm/DcmConfigSet/DcmDsp/DcmDspRoe/DcmDspRoeEvent/DcmDspRoeEventProperties/DcmDspRoeOnDTCStatusChange

Short Name	DcmDspRoeOnDTCStatusChange
Description	This container contains the eventTypeRecord supported for onDTCStatusChange eventType.
Multiplicity	0..1
Default Value	-
Type	ECUC-PARAM-CONF-CONTAINER-DEF

2.3.1.80 Dcm/DcmConfigSet/DcmDsp/DcmDspRoe/DcmDspRoeEvent/DcmDspRoeEventProperties/DcmDspRoeOnDTCStatusChange/DcmDspRoeDTCStatusMask

Short Name	DcmDspRoeDTCStatusMask
Description	Value of the relevant DTCStatusMask
Multiplicity	0..0

Value Range	0..255
Default Value	-
Type	ECUC-INTEGER-PARAM-DEF

2.3.1.81 Dcm/DcmConfigSet/DcmDsp/DcmDspRoe/DcmDspRoeEvent/DcmDspRoeEventProperties/DcmRbDspRoeDTCRecordInfoOnDTCStatusChange

Short Name	DcmRbDspRoeDTCRecordInfoOnDTCStatusChange
Description	Container to support the eventType DcmRbDspRoe DTCRecordInfoOnDTCStatusChange. If this container is configured then the eventType DcmRbDspRoe DTCRecordInfoOnDTCStatusChange is supported otherwise not.
Multiplicity	0..1
Default Value	-
Type	ECUC-PARAM-CONF-CONTAINER-DEF

2.3.1.82 Dcm/DcmConfigSet/DcmDsp/DcmDspRoe/DcmDspRoeEvent/DcmDspRoeEventProperties/DcmRbDspRoeOnComparisonOfValues

Short Name	DcmRbDspRoeOnComparisonOfValues
Description	This container contains the eventTypeRecord supported for onComparisonOfValues eventType.
Multiplicity	0..1
Default Value	-
Type	ECUC-PARAM-CONF-CONTAINER-DEF

2.3.1.83 Dcm/DcmConfigSet/DcmDsp/DcmDspRoe/DcmDspRoeEvent/DcmDspRoeEventProperties/DcmRbDspRoeOnComparisonOfValues/DcmDspRoeDidRef

Short Name	DcmDspRoeDidRef
Description	Reference to a Did which is watched for eventType onComparisonOfValues.
Multiplicity	1..1
Default Value	-
Type	ECUC-REFERENCE-DEF

2.3.1.84 Dcm/DcmConfigSet/DcmDsp/DcmDspRoe/DcmDspRoeEvent/DcmDspRoeInitialEventStatus

Short Name	DcmDspRoeInitialEventStatus

Description	Initial Roe status of this RoeEvent
Multiplicity	1..1
Default Value	DCM_ROE_CLEARED
Enumeration Literals	DCM_ROE_CLEARED, DCM_ROE_STOPPED
Type	ECUC-ENUMERATION-PARAM-DEF

2.3.1.85 Dcm/DcmConfigSet/DcmDsp/DcmDspRoe/DcmDspRoeEventWindowTime

Short Name	DcmDspRoeEventWindowTime
Description	This container configures the available EventWindowTime in this Ecu.
Introduction	This container contains a sub-set of EventWindowTimes supported by the Dcm, to limit the Ecu resources.
Multiplicity	1..*
Default Value	-
Type	ECUC-PARAM-CONF-CONTAINER-DEF

2.3.1.86 Dcm/DcmConfigSet/DcmDsp/DcmDspRoe/DcmDspRoeEventWindowTime/DcmDspRoeEventWindowTime

Short Name	DcmDspRoeEventWindowTime
Description	Value of the EventWindowTime
Multiplicity	1..1
Default Value	-
Enumeration Literals	DCM_ROE_EVENT_WINDOW_CURRENT_AND_FOLLOWING_CYCLE, DCM_ROE_EVENT_WINDOW_CURRENT_CYCLE, DCM_ROE_EVENT_WINDOW_INFINITE
Type	ECUC-ENUMERATION-PARAM-DEF

2.3.1.87 Dcm/DcmConfigSet/DcmDsp/DcmDspRoe/DcmDspRoeEventWindowTime/DcmDspRoeStorageState

Short Name	DcmDspRoeStorageState
Description	If this parameter is set to TRUE the StorageStateBit will be evaluated if this EventWindowTime is requested.
Multiplicity	1..1
Default Value	-
Type	ECUC-BOOLEAN-PARAM-DEF

2.3.1.88 Dcm/DcmConfigSet/DcmDsp/DcmDspRoe/DcmDspRoeInterMessageTime

Short Name	DcmDspRoeInterMessageTime
Description	Provide the minimum time in seconds between two transmissions of ROE event. It is used for the delay between two different consecutive Roe transmissions.
Multiplicity	1..1
Value Range	0..5
Default Value	-
Type	ECUC-FLOAT-PARAM-DEF

2.3.1.89 Dcm/DcmConfigSet/DcmDsp/DcmDspRoe/DcmRbDspWindowTimeInReportEvents

Short Name	DcmRbDspWindowTimeInReportEvents
Description	If set to TRUE the Event Window Time will be the part of Roe request for event type ReportActiveEvents. If set to FALSE the Event Window Time will not be the part of Roe request for event type ReportActiveEvents;
Multiplicity	1..1
Default Value	-
Type	ECUC-BOOLEAN-PARAM-DEF

2.3.1.90 Dcm/DcmConfigSet/DcmDsp/DcmDspRoutine/DcmDspRequestRoutineResults/DcmDspRequestRoutineResultsOut/DcmDspRequestRoutineResultsOutSignal/DcmDspArgumentScaling/DcmDspAlternativeDataProps

Short Name	DcmDspAlternativeDataProps
Description	This container contains the configuration (parameters) of an alternative Diagnosis Representation by the means of ECU configuration parameters.The physical unit of the alternative data representation is defined by the DataPrototype referenced by DcmDspExternal SRDataElementClass. Additionally the definition of a text table mapping can be a defined for DcmDspDataTypeCategory TEXTTABLE and SCALE_LINEAR_AND_TEXTTABLE
Introduction	The physical unit of the alternative data representation is defined by the DataPrototype referenced by DcmDspExternalSRDataElementClass. Additionally the definition of a text table mapping can be a defined for DcmDspDataTypeCategory TEXTTABLE and SCALE_LINEAR_AND_TEXTTABLE

Multiplicity	0..1
Default Value	-
Type	ECUC-PARAM-CONF-CONTAINER-DEF

2.3.1.91 Dcm/DcmConfigSet/DcmDsp/DcmDspRoutine/DcmDspRequestRoutineResults/DcmDspRequestRoutineResultsOut/DcmDspRequestRoutineResultsOutSignal/DcmDspArgumentScaling/DcmDspAlternativeDataProps/DcmDspLinearScale

Short Name	DcmDspLinearScale
Description	This container contains the configuration (parameters) of an linear scale of the alternative Diagnosis Representation.
Multiplicity	1..1
Default Value	-
Type	ECUC-PARAM-CONF-CONTAINER-DEF

2.3.1.92 Dcm/DcmConfigSet/DcmDsp/DcmDspRoutine/DcmDspRequestRoutineResults/DcmDspRequestRoutineResultsOut/DcmDspRequestRoutineResultsOutSignal/DcmDspArgumentScaling/DcmDspAlternativeDataProps/DcmDspLinearScale/DcmDspDiagnosisRepresentationDataLowerRange

Short Name	DcmDspDiagnosisRepresentationDataLowerRange
Description	Lower Range for this scale of the data in the alternative Diagnosis Representation.
Multiplicity	1..1
Value Range	-Inf..Inf
Default Value	-
Type	ECUC-FLOAT-PARAM-DEF

2.3.1.93 Dcm/DcmConfigSet/DcmDsp/DcmDspRoutine/DcmDspRequestRoutineResults/DcmDspRequestRoutineResultsOut/DcmDspRequestRoutineResultsOutSignal/DcmDspArgumentScaling/DcmDspAlternativeDataProps/DcmDspLinearScale/DcmDspDiagnosisRepresentationDataOffset

Short Name	DcmDspDiagnosisRepresentationDataOffset
Description	Data offset of the alternative Diagnosis Representation for this scale.
Multiplicity	1..1
Value Range	0..Inf
Default Value	0
Type	ECUC-FLOAT-PARAM-DEF

2.3.1.94 Dcm/DcmConfigSet/DcmDsp/DcmDspRoutine/DcmDspRequestRoutineResults/DcmDspRequestRoutineResultsOut/DcmDspRequestRoutineResultsOutSignal/DcmDspArgumentScaling/DcmDspAlternativeDataProps/DcmDspLinearScale/DcmDspDiagnosisRepresentationDataResolution

Short Name	DcmDspDiagnosisRepresentationDataResolution
Description	Data resolution of the alternative Diagnosis Representation for this scale.
Multiplicity	1..1
Value Range	0..Inf
Default Value	-
Type	ECUC-FLOAT-PARAM-DEF

2.3.1.95 Dcm/DcmConfigSet/DcmDsp/DcmDspRoutine/DcmDspRequestRoutineResults/DcmDspRequestRoutineResultsOut/DcmDspRequestRoutineResultsOutSignal/DcmDspArgumentScaling/DcmDspAlternativeDataProps/DcmDspLinearScale/DcmDspDiagnosisRepresentationDataUpperRange

Short Name	DcmDspDiagnosisRepresentationDataUpperRange
Description	Upper Range for this scale of the data in the alternative Diagnosis Representation.
Multiplicity	1..1
Value Range	-Inf..Inf
Default Value	-
Type	ECUC-FLOAT-PARAM-DEF

2.3.1.96 Dcm/DcmConfigSet/DcmDsp/DcmDspRoutine/DcmDspRequestRoutineResults/DcmDspRequestRoutineResultsOut/DcmDspRequestRoutineResultsOutSignal/DcmDspRoutineSignalLength

Short Name	DcmDspRoutineSignalLength
Description	Provide the length in bits of the signal in the RoutineControl request/response
Multiplicity	0..1
Value Range	0..65535
Default Value	-
Type	ECUC-INTEGER-PARAM-DEF

2.3.1.97 Dcm/DcmConfigSet/DcmDsp/DcmDspRoutine/DcmDspRequestRoutineResults/DcmRbDspRequestRoutineResultsIn

Short Name	DcmRbDspRequestRoutineResultsIn

Description	Provide description of input parameter of RequestResults subservice for RoutineControl service.
Multiplicity	0..1
Default Value	-
Type	ECUC-PARAM-CONF-CONTAINER-DEF

2.3.1.98 Dcm/DcmConfigSet/DcmDsp/DcmDspRoutine/DcmDspRequestRoutineResults/DcmRbDspRequestRoutineResultsIn/DcmRbDspRequestRoutineResultsInSignal

Short Name	DcmRbDspRequestRoutineResultsInSignal
Description	Provide description of a routine signal used in RoutineControl service.
Multiplicity	1..*
Default Value	-
Type	ECUC-PARAM-CONF-CONTAINER-DEF

2.3.1.99 Dcm/DcmConfigSet/DcmDsp/DcmDspRoutine/DcmDspRequestRoutineResults/DcmRbDspRequestRoutineResultsIn/DcmRbDspRequestRoutineResultsInSignal/DcmRbDspRoutineSignalEndianness

Short Name	DcmRbDspRoutineSignalEndianness
Description	Defines the endianness of the data belonging to a Request Routine Response In Signal.If projects configures OPAQUE no endianness conversion will be supported/applicable
Introduction	If no DcmDspRoutineSignalEndianness is defined the value of DcmDspDataDefaultEndianness is applicable.
Multiplicity	0..1
Default Value	-
Enumeration Literals	BIG_ENDIAN, LITTLE_ENDIAN, OPAQUE
Type	ECUC-ENUMERATION-PARAM-DEF

2.3.1.100 Dcm/DcmConfigSet/DcmDsp/DcmDspRoutine/DcmDspRequestRoutineResults/DcmRbDspRequestRoutineResultsIn/DcmRbDspRequestRoutineResultsInSignal/DcmRbDspRoutineSignalLength

Short Name	DcmRbDspRoutineSignalLength

Description	Provide the length of the Routine signal in bits.If DcmDspRoutineRequestResINSignal is configured as VARIABLE_LENGTH, then Dcm shall update the currentDataLength parameter of the configured DcmAppl API (DcmDspRequestResultsRoutineFnc) in BYTES before calling the API .CurrentDataLength parameter signifies the actual length of the variable length signal positioned at the end in request.
Multiplicity	0..1
Value Range	0..65535
Default Value	-
Type	ECUC-INTEGER-PARAM-DEF

2.3.1.101 Dcm/DcmConfigSet/DcmDsp/DcmDspRoutine/DcmDspRequestRoutineResults/DcmRbDspRequestRoutineResultsIn/DcmRbDspRequestRoutineResultsInSignal/DcmRbDspRoutineSignalPos

Short Name	DcmRbDspRoutineSignalPos
Description	Provide the position of the signal in the RoutineControl request/response.The position is defined in bits.
Introduction	The position is defined in bits.
Multiplicity	1..1
Value Range	0..65535
Default Value	-
Type	ECUC-INTEGER-PARAM-DEF

2.3.1.102 Dcm/DcmConfigSet/DcmDsp/DcmDspRoutine/DcmDspRequestRoutineResults/DcmRbDspRequestRoutineResultsIn/DcmRbDspRequestRoutineResultsInSignal/DcmRbDspRoutineSignalType

Short Name	DcmRbDspRoutineSignalType
Description	Provide the type of the signal in the RoutineControl request/response.
Multiplicity	1..1
Default Value	-
Enumeration Literals	BOOLEAN, SINT16, SINT16_N, SINT32, SINT32_N, SINT8, SINT8_N, UINT16, UINT16_N, UINT32, UINT32_N, UINT8, UINT8_N, VARIABLE_LENGTH
Type	ECUC-ENUMERATION-PARAM-DEF

2.3.1.103 Dcm/DcmConfigSet/DcmDsp/DcmDspRoutine/DcmDspStartRoutine/DcmDspStartRoutineIn/DcmDspStartRoutineInSignal/DcmDspArgumentScaling/DcmDspAlternativeDataProps

Short Name	DcmDspAlternativeDataProps

Description	This container contains the configuration (parameters) of an alternative Diagnosis Representation by the means of ECU configuration parameters.The physical unit of the alternative data representation is defined by the DataPrototype referenced by DcmDspExternal SRDataElementClass. Additionally the definition of a text table mapping can be a defined for DcmDspDataTypeCategory TEXTTABLE and SCALE_LINEAR_AND_TEXTTABLE
Introduction	The physical unit of the alternative data representation is defined by the DataPrototype referenced by DcmDspExternalSRDataElementClass. Additionally the definition of a text table mapping can be a defined for DcmDspDataTypeCategory TEXTTABLE and SCALE_LINEAR_AND_TEXTTABLE
Multiplicity	0..1
Default Value	-
Type	ECUC-PARAM-CONF-CONTAINER-DEF

2.3.1.104 Dcm/DcmConfigSet/DcmDsp/DcmDspRoutine/DcmDspStartRoutine/DcmDspStartRoutineIn/DcmDspStartRoutineInSignal/DcmDspArgumentScaling/DcmDspAlternativeDataProps/DcmDspLinearScale

Short Name	DcmDspLinearScale
Description	This container contains the configuration (parameters) of an linear scale of the alternative Diagnosis Representation.
Multiplicity	1..1
Default Value	-
Type	ECUC-PARAM-CONF-CONTAINER-DEF

2.3.1.105 Dcm/DcmConfigSet/DcmDsp/DcmDspRoutine/DcmDspStartRoutine/DcmDspStartRoutineIn/DcmDspStartRoutineInSignal/DcmDspArgumentScaling/DcmDspAlternativeDataProps/DcmDspLinearScale/DcmDspDiagnosisRepresentationDataLowerRange

Short Name	DcmDspDiagnosisRepresentationDataLowerRange
Description	Lower Range for this scale of the data in the alternative Diagnosis Representation.
Multiplicity	1..1
Value Range	-Inf..Inf
Default Value	-
Type	ECUC-FLOAT-PARAM-DEF

2.3.1.106 Dcm/DcmConfigSet/DcmDsp/DcmDspRoutine/DcmDspStartRou-

tine/DcmDspStartRoutineIn/DcmDspStartRoutineInSignal/DcmDspArgumentScaling/DcmDspAlternativeDataProps/DcmDspLinearScale/DcmDspDiagnosisRepresentationDataOffset

Short Name	DcmDspDiagnosisRepresentationDataOffset
Description	Data offset of the alternative Diagnosis Representation for this scale.
Multiplicity	1..1
Value Range	0..Inf
Default Value	0
Type	ECUC-FLOAT-PARAM-DEF

2.3.1.107 Dcm/DcmConfigSet/DcmDsp/DcmDspRoutine/DcmDspStartRoutine/DcmDspStartRoutineIn/DcmDspStartRoutineInSignal/DcmDspArgumentScaling/DcmDspAlternativeDataProps/DcmDspLinearScale/DcmDspDiagnosisRepresentationDataResolution

Short Name	DcmDspDiagnosisRepresentationDataResolution
Description	Data resolution of the alternative Diagnosis Representation for this scale.
Multiplicity	1..1
Value Range	0..Inf
Default Value	-
Type	ECUC-FLOAT-PARAM-DEF

2.3.1.108 Dcm/DcmConfigSet/DcmDsp/DcmDspRoutine/DcmDspStartRoutine/DcmDspStartRoutineIn/DcmDspStartRoutineInSignal/DcmDspArgumentScaling/DcmDspAlternativeDataProps/DcmDspLinearScale/DcmDspDiagnosisRepresentationDataUpperRange

Short Name	DcmDspDiagnosisRepresentationDataUpperRange
Description	Upper Range for this scale of the data in the alternative Diagnosis Representation.
Multiplicity	1..1
Value Range	-Inf..Inf
Default Value	-
Type	ECUC-FLOAT-PARAM-DEF

2.3.1.109 Dcm/DcmConfigSet/DcmDsp/DcmDspRoutine/DcmDspStartRoutine/DcmDspStartRoutineIn/DcmDspStartRoutineInSignal/DcmDspRoutineSignalLength

Short Name	DcmDspRoutineSignalLength
Description	Provide the length in bits of the signal in the RoutineControl request/response

Multiplicity	0..1
Value Range	0..65535
Default Value	-
Type	ECUC-INTEGER-PARAM-DEF

2.3.1.110 Dcm/DcmConfigSet/DcmDsp/DcmDspRoutine/DcmDspStartRoutine/DcmDspStartRoutineOut/DcmDspStartRoutineOutSignal/DcmDspArgumentScaling/DcmDspAlternativeDataProps

Short Name	DcmDspAlternativeDataProps
Description	This container contains the configuration (parameters) of an alternative Diagnosis Representation by the means of ECU configuration parameters.The physical unit of the alternative data representation is defined by the DataPrototype referenced by DcmDspExternal SRDataElementClass. Additionally the definition of a text table mapping can be a defined for DcmDspDataTypeCategory TEXTTABLE and SCALE_LINEAR_AND_TEXTTABLE
Introduction	The physical unit of the alternative data representation is defined by the DataPrototype referenced by DcmDspExternalSRDataElementClass. Additionally the definition of a text table mapping can be a defined for DcmDspDataTypeCategory TEXTTABLE and SCALE_LINEAR_AND_TEXTTABLE
Multiplicity	0..1
Default Value	-
Type	ECUC-PARAM-CONF-CONTAINER-DEF

2.3.1.111 Dcm/DcmConfigSet/DcmDsp/DcmDspRoutine/DcmDspStartRoutine/DcmDspStartRoutineOut/DcmDspStartRoutineOutSignal/DcmDspArgumentScaling/DcmDspAlternativeDataProps/DcmDspLinearScale

Short Name	DcmDspLinearScale
Description	This container contains the configuration (parameters) of an linear scale of the alternative Diagnosis Representation.
Multiplicity	1..1
Default Value	-
Type	ECUC-PARAM-CONF-CONTAINER-DEF

2.3.1.112 Dcm/DcmConfigSet/DcmDsp/DcmDspRoutine/DcmDspStartRoutine/DcmDspStartRoutineOut/DcmDspStartRoutineOutSignal/DcmDspArgumentScaling/DcmDspAlternativeDataProps/DcmDspLinearScale/DcmDspDiagnosisRepresentationDataLowerRange

Short Name	DcmDspDiagnosisRepresentationDataLowerRange
Description	Lower Range for this scale of the data in the alternative Diagnosis Representation.
Multiplicity	1..1
Value Range	-Inf..Inf
Default Value	-
Type	ECUC-FLOAT-PARAM-DEF

2.3.1.113 Dcm/DcmConfigSet/DcmDsp/DcmDspRoutine/DcmDspStartRoutine/DcmDspStartRoutineOut/DcmDspStartRoutineOutSignal/DcmDspArgumentScaling/DcmDspAlternativeDataProps/DcmDspLinearScale/DcmDspDiagnosisRepresentationDataOffset

Short Name	DcmDspDiagnosisRepresentationDataOffset
Description	Data offset of the alternative Diagnosis Representation for this scale.
Multiplicity	1..1
Value Range	0..Inf
Default Value	0
Type	ECUC-FLOAT-PARAM-DEF

2.3.1.114 Dcm/DcmConfigSet/DcmDsp/DcmDspRoutine/DcmDspStartRoutine/DcmDspStartRoutineOut/DcmDspStartRoutineOutSignal/DcmDspArgumentScaling/DcmDspAlternativeDataProps/DcmDspLinearScale/DcmDspDiagnosisRepresentationDataResolution

Short Name	DcmDspDiagnosisRepresentationDataResolution
Description	Data resolution of the alternative Diagnosis Representation for this scale.
Multiplicity	1..1
Value Range	0..Inf
Default Value	-
Type	ECUC-FLOAT-PARAM-DEF

2.3.1.115 Dcm/DcmConfigSet/DcmDsp/DcmDspRoutine/DcmDspStartRoutine/DcmDspStartRoutineOut/DcmDspStartRoutineOutSignal/DcmDspArgumentScaling/DcmDspAlternativeDataProps/DcmDspLinearScale/DcmDspDiagnosisRepresentationDataUpperRange

Short Name	DcmDspDiagnosisRepresentationDataUpperRange

Description	Upper Range for this scale of the data in the alternative Diagnosis Representation.
Multiplicity	1..1
Value Range	-Inf..Inf
Default Value	-
Type	ECUC-FLOAT-PARAM-DEF

2.3.1.116 Dcm/DcmConfigSet/DcmDsp/DcmDspRoutine/DcmDspStartRoutine/DcmDspStartRoutineOut/DcmDspStartRoutineOutSignal/DcmDspRoutineSignalLength

Short Name	DcmDspRoutineSignalLength
Description	Provide the length in bits of the signal in the RoutineControl request/response.
Multiplicity	0..1
Value Range	0..65535
Default Value	-
Type	ECUC-INTEGER-PARAM-DEF

2.3.1.117 Dcm/DcmConfigSet/DcmDsp/DcmDspRoutine/DcmDspStopRoutine/DcmDspStopRoutineIn/DcmDspStopRoutineInSignal/DcmDspArgumentScaling/DcmDspAlternativeDataProps

Short Name	DcmDspAlternativeDataProps
Description	This container contains the configuration (parameters) of an alternative Diagnosis Representation by the means of ECU configuration parameters.The physical unit of the alternative data representation is defined by the DataPrototype referenced by DcmDspExternal SRDataElementClass. Additionally the definition of a text table mapping can be a defined for DcmDspDataTypeCategory TEXTTABLE and SCALE_LINEAR_AND_TEXTTABLE
Introduction	The physical unit of the alternative data representation is defined by the DataPrototype referenced by DcmDspExternalSRDataElementClass. Additionally the definition of a text table mapping can be a defined for DcmDspDataTypeCategory TEXTTABLE and SCALE_LINEAR_AND_TEXTTABLE
Multiplicity	0..1
Default Value	-
Type	ECUC-PARAM-CONF-CONTAINER-DEF

2.3.1.118 Dcm/DcmConfigSet/DcmDsp/DcmDspRoutine/DcmDspStopRoutine/DcmDspStopRoutineIn/DcmDspStopRoutineInSignal/

DcmDspArgumentScaling/DcmDspAlternativeDataProps/DcmDspLinearScale

Short Name	DcmDspLinearScale
Description	This container contains the configuration (parameters) of an linear scale of the alternative Diagnosis Representation.
Multiplicity	1..1
Default Value	-
Type	ECUC-PARAM-CONF-CONTAINER-DEF

2.3.1.119 Dcm/DcmConfigSet/DcmDsp/DcmDspRoutine/DcmDspStopRoutine/DcmDspStopRoutineIn/DcmDspStopRoutineInSignal/DcmDspArgumentScaling/DcmDspAlternativeDataProps/DcmDspLinearScale/DcmDspDiagnosisRepresentationDataLowerRange

Short Name	DcmDspDiagnosisRepresentationDataLowerRange
Description	Lower Range for this scale of the data in the alternative Diagnosis Representation.
Multiplicity	1..1
Value Range	-Inf..Inf
Default Value	-
Type	ECUC-FLOAT-PARAM-DEF

2.3.1.120 Dcm/DcmConfigSet/DcmDsp/DcmDspRoutine/DcmDspStopRoutine/DcmDspStopRoutineIn/DcmDspStopRoutineInSignal/DcmDspArgumentScaling/DcmDspAlternativeDataProps/DcmDspLinearScale/DcmDspDiagnosisRepresentationDataOffset

Short Name	DcmDspDiagnosisRepresentationDataOffset
Description	Data offset of the alternative Diagnosis Representation for this scale.
Multiplicity	1..1
Value Range	0..Inf
Default Value	0
Type	ECUC-FLOAT-PARAM-DEF

2.3.1.121 Dcm/DcmConfigSet/DcmDsp/DcmDspRoutine/DcmDspStopRoutine/DcmDspStopRoutineIn/DcmDspStopRoutineInSignal/DcmDspArgumentScaling/DcmDspAlternativeDataProps/DcmDspLinearScale/DcmDspDiagnosisRepresentationDataResolution

Short Name	DcmDspDiagnosisRepresentationDataResolution
Description	Data resolution of the alternative Diagnosis Representation for this scale.
Multiplicity	1..1

Value Range	0..Inf
Default Value	-
Type	ECUC-FLOAT-PARAM-DEF

2.3.1.122 Dcm/DcmConfigSet/DcmDsp/DcmDspRoutine/DcmDspStopRoutine/DcmDspStopRoutineIn/DcmDspStopRoutineInSignal/DcmDspArgumentScaling/DcmDspAlternativeDataProps/DcmDspLinearScale/DcmDspDiagnosisRepresentationDataUpperRange

Short Name	DcmDspDiagnosisRepresentationDataUpperRange
Description	Upper Range for this scale of the data in the alternative Diagnosis Representation.
Multiplicity	1..1
Value Range	-Inf..Inf
Default Value	-
Type	ECUC-FLOAT-PARAM-DEF

2.3.1.123 Dcm/DcmConfigSet/DcmDsp/DcmDspRoutine/DcmDspStopRoutine/DcmDspStopRoutineIn/DcmDspStopRoutineInSignal/DcmDspRoutineSignalLength

Short Name	DcmDspRoutineSignalLength
Description	Provide the length in bits of the signal in the RoutineControl request/response
Multiplicity	0..1
Value Range	0..65535
Default Value	-
Type	ECUC-INTEGER-PARAM-DEF

2.3.1.124 Dcm/DcmConfigSet/DcmDsp/DcmDspRoutine/DcmDspStopRoutine/DcmDspStopRoutineOut/DcmDspStopRoutineOutSignal/DcmDspArgumentScaling/DcmDspAlternativeDataProps

Short Name	DcmDspAlternativeDataProps
Description	This container contains the configuration (parameters) of an alternative Diagnosis Representation by the means of ECU configuration parameters.The physical unit of the alternative data representation is defined by the DataPrototype referenced by DcmDspExternalSRDataElementClass. Additionally the definition of a text table mapping can be a defined for DcmDspDataTypeCategory TEXTTABLE and SCALE_LINEAR_AND_TEXTTABLE

Introduction	The physical unit of the alternative data representation is defined by the DataPrototype referenced by DcmDspExternalSRDataElementClass. Additionally the definition of a text table mapping can be a defined for DcmDspDataTypeCategory TEXTTABLE and SCALE_LINEAR_AND_TEXTTABLE
Multiplicity	0..1
Default Value	-
Type	ECUC-PARAM-CONF-CONTAINER-DEF

2.3.1.125 Dcm/DcmConfigSet/DcmDsp/DcmDspRoutine/DcmDspStopRoutine/DcmDspStopRoutineOut/DcmDspStopRoutineOutSignal/DcmDspArgumentScaling/DcmDspAlternativeDataProps/DcmDspLinearScale

Short Name	DcmDspLinearScale
Description	This container contains the configuration (parameters) of an linear scale of the alternative Diagnosis Representation.
Multiplicity	1..1
Default Value	-
Type	ECUC-PARAM-CONF-CONTAINER-DEF

2.3.1.126 Dcm/DcmConfigSet/DcmDsp/DcmDspRoutine/DcmDspStopRoutine/DcmDspStopRoutineOut/DcmDspStopRoutineOutSignal/DcmDspArgumentScaling/DcmDspAlternativeDataProps/DcmDspLinearScale/DcmDspDiagnosisRepresentationDataLowerRange

Short Name	DcmDspDiagnosisRepresentationDataLowerRange
Description	Lower Range for this scale of the data in the alternative Diagnosis Representation.
Multiplicity	1..1
Value Range	-Inf..Inf
Default Value	-
Type	ECUC-FLOAT-PARAM-DEF

2.3.1.127 Dcm/DcmConfigSet/DcmDsp/DcmDspRoutine/DcmDspStopRoutine/DcmDspStopRoutineOut/DcmDspStopRoutineOutSignal/DcmDspArgumentScaling/DcmDspAlternativeDataProps/DcmDspLinearScale/DcmDspDiagnosisRepresentationDataOffset

Short Name	DcmDspDiagnosisRepresentationDataOffset
Description	Data offset of the alternative Diagnosis Representation for this scale.
Multiplicity	1..1

Value Range	0..Inf
Default Value	0
Type	ECUC-FLOAT-PARAM-DEF

2.3.1.128 Dcm/DcmConfigSet/DcmDsp/DcmDspRoutine/DcmDspStopRoutine/DcmDspStopRoutineOut/DcmDspStopRoutineOutSignal/DcmDspArgumentScaling/DcmDspAlternativeDataProps/DcmDspLinearScale/DcmDspDiagnosisRepresentationDataResolution

Short Name	DcmDspDiagnosisRepresentationDataResolution
Description	Data resolution of the alternative Diagnosis Representation for this scale.
Multiplicity	1..1
Value Range	0..Inf
Default Value	-
Type	ECUC-FLOAT-PARAM-DEF

2.3.1.129 Dcm/DcmConfigSet/DcmDsp/DcmDspRoutine/DcmDspStopRoutine/DcmDspStopRoutineOut/DcmDspStopRoutineOutSignal/DcmDspArgumentScaling/DcmDspAlternativeDataProps/DcmDspLinearScale/DcmDspDiagnosisRepresentationDataUpperRange

Short Name	DcmDspDiagnosisRepresentationDataUpperRange
Description	Upper Range for this scale of the data in the alternative Diagnosis Representation.
Multiplicity	1..1
Value Range	-Inf..Inf
Default Value	-
Type	ECUC-FLOAT-PARAM-DEF

2.3.1.130 Dcm/DcmConfigSet/DcmDsp/DcmDspRoutine/DcmDspStopRoutine/DcmDspStopRoutineOut/DcmDspStopRoutineOutSignal/DcmDspRoutineSignalLength

Short Name	DcmDspRoutineSignalLength
Description	Provide the length in bits of the signal in the RoutineControl request/response
Multiplicity	0..1
Value Range	0..65535
Default Value	-
Type	ECUC-INTEGER-PARAM-DEF

2.3.1.131 Dcm/DcmConfigSet/DcmDsp/DcmDspRoutine/DcmRbDspIsRIDAvailableFnc

Short Name	DcmRbDspIsRIDAvailableFnc
Description	Function name to request the application to check if the routine identifier (RID) requested is supported in the variant.The user can leave the field empty if the check is not required for this RID
Introduction	Function name to request the application to check if the routine identifier (RID) requested is supported in the variant.The user can leave the field empty if the check is not required for this RID
Multiplicity	0..1
Default Value	-
Type	ECUC-FUNCTION-NAME-DEF

2.3.1.132 Dcm/DcmConfigSet/DcmDsp/DcmDspRoutine/DcmRbDspReqSequenceErrorEnabled

Short Name	DcmRbDspReqSequenceErrorEnabled
Introduction	True:For a Routine which supports both RequestResults/Stop and Start Subfunction,NRC 0x24 shall be triggered if Routine Control tester request with RequestResults subfunction(0x03/Stop subfunction(0x02) comes for a Routine which is not started (without a prior Routine Control request with Start subfunction(0x01)) False:For a Routine which supports both RequestResults/Stop and Start Subfunction,positive response shall be triggered if Routine Control tester request with RequestResults subfunction(0x03)/Stop subfunction(0x02) comes without a prior Routine Control request with Start subfunction(0x01)
Multiplicity	0..1
Default Value	true
Type	ECUC-BOOLEAN-PARAM-DEF

Description: Indicates Request Sequence Error (NRC 0x24) shall be triggered for a RequestResults/Stop subfunction with/without a prior Start subfunction request for Routines which supports both RequestResults and Start subfunctions/ Stop and Start Subfunctions. True :For a Routine which supports both RequestResults/Stop and Start Subfunction,NRC 0x24 shall be triggered if Routine Control tester request with RequestResults subfunction(0x03)/Stop subfunction(0x02) comes for a Routine which is not started (without a prior Routine Control request with Start subfunction(0x01)) False:For a Routine which supports both RequestResults/Stop and Start Subfunction,positive response

shall be triggered if valid Routine Control tester request with RequestResults subfunction(0x03)/Stop subfunction(0x02) comes without a prior Routine Control request with Start subfunction(0x01) Dependency: Applicable only if Start subfunction (DcmDspStartRoutineSupported configured as TRUE or left unconfigured) and RequestResults subfunction(DcmDspRequestResultsRoutineSupported configured as TRUE)/Stop subfunction(DcmDspStopRoutineSupported configured as TRUE) is supported for RID, otherwise not valid.If the parameter is left unconfigured , default value will be considered as TRUE and the request sequence error will be enabled.

2.3.1.133 Dcm/DcmConfigSet/DcmDsp/DcmDspRoutine/DcmRbDspRoutineUseAsynchronousServerCallPoint

Short Name	DcmRbDspRoutineUseAsynchronousServerCallPoint
Description	Only relevant in case DcmDspRoutineUsePort is set to TRUE.This parameter is used to select the type of Server Handling required.Set the value to TRUE for Asynchronous server call point handling else Synchronous server call point handling is carried out
Multiplicity	0..1
Default Value	false
Type	ECUC-BOOLEAN-PARAM-DEF

2.3.1.134 Dcm/DcmConfigSet/DcmDsp/DcmDspRoutine/DcmRbDspRoutineUserRoutineModeRuleFnc

Short Name	DcmRbDspRoutineUserRoutineModeRuleFnc
Description	Function name used by the application to perform user specific checks before accessing a Routine.In case the checks fails, NRC value set by the application can be returned to the tester. In case the function is not configured for the Routine, the DcmAppl API DcmAppl_UserRIDModeRuleService shall be called.
Introduction	Function name used by the application to perform user specific checks before accessing a Routine.In case the checks fails, NRC value set by the application can be returned to the tester. In case the function is not configured for the Routine, the DcmAppl API DcmAppl_UserRIDModeRuleService shall be called.
Multiplicity	0..1
Default Value	-
Type	ECUC-FUNCTION-NAME-DEF

2.3.1.135 Dcm/DcmConfigSet/DcmDsp/DcmDspRoutine/DcmRbDspSto-

pRoutineOnSessionChangeDcmRbDspStopRoutineOnSession-Change

Short Name	DcmRbDspStopRoutineOnSessionChange
Description	Indicates if the StopRoutine operation needs to be performed on any session change. True = Stop this Routine if it is active on transition to default session / Stop this routine if it is not supported in the new nondefault session. False = No action on session transitions Dependency: Applicable only if StopRoutine operation is supported for RID (DcmDspStopRoutine-Supported = true), otherwise not valid.If the parameter is left unconfigured FALSE will be considered as the default value.
Introduction	true = stopRoutine is supported on session change false = stopRoutine is not supported on session change
Multiplicity	0..1
Default Value	false
Type	ECUC-BOOLEAN-PARAM-DEF

2.3.1.136 Dcm/DcmConfigSet/DcmDsp/DcmDspSecurity/DcmDspSecurity-Row/DcmRbDspSecurityNumAttLock

Short Name	DcmRbDspSecurityNumAttLock
Description	Number of failed security accesses after which security access is locked. 0 indicates that the security access will be left untouched and the ECU never locks.
Multiplicity	1..1
Value Range	0..255
Default Value	-
Type	ECUC-INTEGER-PARAM-DEF

2.3.1.137 Dcm/DcmConfigSet/DcmDsp/DcmDspSecurity/DcmDspSecurity-Row/DcmRbDspSecurityRowUseAsynchronousServerCallPoint

Short Name	DcmRbDspSecurityRowUseAsynchronousServerCallPoint
Description	Only relevant in case DcmDspSecurityUsePort is set to USE_DATA_ASYNCH_CLIENT_SERVER.This parameter is used to select the type of Server Handling required.Set the value to TRUE for Asynchronous server call point handling else Synchronous server call point handling is carried out
Multiplicity	0..1

Default Value	false
Type	ECUC-BOOLEAN-PARAM-DEF

2.3.1.138 Dcm/DcmConfigSet/DcmDsp/DcmDspSecurity/DcmDspSecurity-Row/DcmRbDspSecurityRowUseFlexibleLength

Short Name	DcmRbDspSecurityRowUseFlexibleLength
Description	Parameter to enable/ disable use of flexible Seed Length and Key Length for all security levels If configured as TRUE, flexible length is used for Seed and Key handling
Multiplicity	0..1
Default Value	false
Type	ECUC-BOOLEAN-PARAM-DEF

2.3.1.139 Dcm/DcmConfigSet/DcmDsp/DcmDspSecurity/DcmRbDspSecu-rityApplCallbackOnDelayTimeExpiry

Short Name	DcmRbDspSecurityApplCallbackOnDelayTimeExpiry
Description	Parameter to enable/disable the application callback on security delay time expiry. If set to TRUE, then Dcm wil call application DcmAppl_UptFailedAtmCntOnDelay-TimeExpiry(). If set to FALSE or left unconfigured, then Dcm will use autosar defined behavior. DcmDspSecurity ResetAttemptCounterOnTimeout and DcmRbDspSecu-rityApplCallbackOnDelayTimeExpiry can not be config-ured at the time. Only one of these should be config-ured at the same time.
Multiplicity	0..1
Default Value	-
Type	ECUC-BOOLEAN-PARAM-DEF

2.3.1.140 Dcm/DcmConfigSet/DcmDsp/DcmDspSecurity/DcmRbDspSecuri-tyDelayTimerMonitor

Short Name	DcmRbDspSecurityDelayTimerMonitor

Description	According to ISO-14229-1 (2006-12-01) Security Access service shall return a NRC requiredTimeDelayNotExpired when requested action will not be taken because theclient?s latest attempt to gain security access was initiated before the server?s required timeout period had been elapsed. This provides an option to OEMs to either decide to send this NRC for either a REQUEST SEED sub-function or SEND KEY sub-function. It is possible to choose here for which of the two it is required to monitor the delay time and send this NRC with this vendor specific attribute.
Multiplicity	0..1
Default Value	REQUEST_SEED
Enumeration Literals	REQUEST_SEED, SEND_KEY
Type	ECUC-ENUMERATION-PARAM-DEF

2.3.1.141 Dcm/DcmConfigSet/DcmDsp/DcmDspSecurity/DcmRbDspSecurityNRCExceedNoOfAttempt

Short Name	DcmRbDspSecurityNRCExceedNoOfAttempt
Description	This parameter is to choose NRC return value in case of Exceeded Number Invalid Unlock Attempts. Literal “NumAttDelayAndNumAttLock” is in accordance to ISO 14229-1; For every invalid key sent, NRC 35 (INVALID_KEY) is returned till configured no of attempts (NumAttDelay) is reached. When Failed Attempts reaches or exceeds configured value (> = NummAttDelay) NRC 36 (EXCEEDED NUMBER OF ATTEMPTS) is returned. Or In case the failed attempts reaches or exceeds no of attempts to Lock (NumAttLock) ,NRC 36 will be returned. The Literal “NumAttLock” specifys that only on reaching or exceeding Number of attempts for Lock (NumAttLock) ,NRC 36 will be sent. Untill then NRC 35 is returned on invalid attempt.
Multiplicity	0..1
Default Value	NumAttDelayAndNumAttLock
Enumeration Literals	NumAttDelayAndNumAttLock, NumAttLock
Type	ECUC-ENUMERATION-PARAM-DEF

2.3.1.142 Dcm/DcmConfigSet/DcmDsp/DcmDspSecurity/DcmRbDspSecurityStoreSeed

Short Name	DcmRbDspSecurityStoreSeed
Description	DcmDspSecurity - DcmDspSecurityStoreSeed = TRUE, print DCM_CFG_ON

Multiplicity	1..1
Default Value	-
Type	ECUC-BOOLEAN-PARAM-DEF

2.3.1.143 Dcm/DcmConfigSet/DcmDsp/DcmDspSession/DcmRbDspKWPDefaultSesRef

Short Name	DcmRbDspKWPDefaultSesRef
Description	The DcmDspKWPDefaultSesRef is a reference container to DcmDspSessionRow. If the KWP protocol is using the Dcm Dsp session and security configurations (i.e. DcmDspSesSecUsedInProtocol is configured to KWP) then configure a Default Session for KWP protocol
Multiplicity	0..1
Default Value	-
Type	ECUC-REFERENCE-DEF

2.3.1.144 Dcm/DcmConfigSet/DcmDsp/DcmDspSession/DcmRbDspSessionParamRecordEnabled

Short Name	DcmRbDspSessionParamRecordEnabled
Description	If DcmDspSessionParamRecordEnabled = true, then the session parameter record will be part of session control service positive response. If DcmDspSession ParamRecordEnabled = false, then the session parameter record will not be part of session control service positive response. Session parameter record conatins the P2 Can Server and P2* Can Server times according to ISO15765-3 CAN requirements
Multiplicity	0..1
Default Value	true
Type	ECUC-BOOLEAN-PARAM-DEF

2.3.1.145 Dcm/DcmConfigSet/DcmDsp/DcmDspVehInfo/DcmRbDspVehInfoSupportFnc

Short Name	DcmRbDspVehInfoSupportFnc
Description	Function name to trigger the application to check if the requested infotype is supported or not.User can leave the field empty, if the check is not required for this Infotype.
Multiplicity	0..1
Default Value	-

Type	ECUC-FUNCTION-NAME-DEF

2.3.1.146 Dcm/DcmConfigSet/DcmDsp/DcmRbDspAutoConfigureDEMOBD-PIDs

Short Name	DcmRbDspAutoConfigureDEMOBDPIDs
Description	If the parameter is configured as TRUE, DCM will forward configuration of DEM relevant OBD Pid's [PID$01,PID$1C,PID$21,PID$30,PID$31,PID$41,PID$4D, PID$4E]. All these pid's are configured only for service$01.Ensure that static configuration doesn't exist for these pid's, otherwise it will result in build error because of multiple configuration.
Multiplicity	0..1
Default Value	false
Type	ECUC-BOOLEAN-PARAM-DEF

2.3.1.147 Dcm/DcmConfigSet/DcmDsp/DcmRbDspEcuResetRow

Short Name	DcmRbDspEcuResetRow
Description	Container to define the resetTypes and the type of reset(jump to bootloader or normal reset) for each subfunction of EcuReset Service. ResetTypes supported by the ECU but not configured under this container will be considered for normal reset.
Multiplicity	0..*
Default Value	-
Type	ECUC-PARAM-CONF-CONTAINER-DEF

2.3.1.148 Dcm/DcmConfigSet/DcmDsp/DcmRbDspEcuResetRow/DcmRbD-spEcuResetType

Short Name	DcmRbDspEcuResetType
Description	Value of the Ecu Reset Type (subfunction of EcuReset Service)
Introduction	Value of the Ecu Reset Type (subfunction of EcuReset Service)
Multiplicity	1..1
Value Range	0..127
Default Value	-
Type	ECUC-INTEGER-PARAM-DEF

2.3.1.149 Dcm/DcmConfigSet/DcmDsp/DcmRbDspEcuResetRow/DcmRbDspResetForBoot

Short Name	DcmRbDspResetForBoot
Description	This parameter defines whether this resetType allows to jump to Bootloader (OEM Bootloader/ System Supplier Bootloader/ Drive to Drive Jump). If the resetType is not allowed to jump to Boot/Drive, i.e., used for normal reset, the value DCM_NO_BOOT shall be chosen.ResetTypes supported by the ECU but not configured under the container DcmDspEcuResetRow will be considered as DCM_NO_BOOT (normal reset)
Multiplicity	1..1
Default Value	-
Enumeration Literals	DCM_NO_BOOT, DCM_OEM_BOOT, DCM_SYS_BOOT, DCM_DRIVE_TO_DRIVE
Type	ECUC-ENUMERATION-PARAM-DEF

2.3.1.150 Dcm/DcmConfigSet/DcmDsp/DcmRbDspReadDTC

Short Name	DcmRbDspReadDTC
Description	This container contains the configuration (parameters) of the service Read DTC Information - 0x19 (per sub-function type) This container contains Rows of DcmDspReadDTCRow
Multiplicity	0..1
Default Value	-
Type	ECUC-PARAM-CONF-CONTAINER-DEF

2.3.1.151 Dcm/DcmConfigSet/DcmDsp/DcmRbDspReadDTC/DcmRbDspReadDTCMaxNumDTCRead

Short Name	DcmRbDspReadDTCMaxNumDTCRead
Description	Maximum number of DTC that can be read in 1 service call
Multiplicity	1..1
Value Range	1..255
Default Value	-
Type	ECUC-INTEGER-PARAM-DEF

2.3.1.152 Dcm/DcmConfigSet/DcmDsp/DcmRbDspReadDTC/DcmRbDspReadDTCMaxNumRecordRead

Short Name	DcmRbDspReadDTCMaxNumRecordRead
Description	Max Number of Records that can be read in 1 service call
Multiplicity	1..1
Value Range	1..255
Default Value	-
Type	ECUC-INTEGER-PARAM-DEF

2.3.1.153 Dcm/DcmConfigSet/DcmDsp/DcmRbDspRoutineRange

Short Name	DcmRbDspRoutineRange
Description	This container defines the Routine Ranges
Multiplicity	0..*
Default Value	-
Type	ECUC-PARAM-CONF-CONTAINER-DEF

2.3.1.154 Dcm/DcmConfigSet/DcmDsp/DcmRbDspRoutineRange/DcmRbDspRoutineRangeHasGaps

Short Name	DcmRbDspRoutineRangeHasGaps
Description	Parameter specifying if there are gaps in the Routine range (parameter set to TRUE) or not (parameter set to FALSE)
Multiplicity	1..1
Default Value	-
Type	ECUC-BOOLEAN-PARAM-DEF

2.3.1.155 Dcm/DcmConfigSet/DcmDsp/DcmRbDspRoutineRange/DcmRbDspRoutineRangeIdentifierLowerLimit

Short Name	DcmRbDspRoutineRangeIdentifierLowerLimit
Description	Lower limit of Routine range.
Multiplicity	1..1
Value Range	0..65535
Default Value	-
Type	ECUC-INTEGER-PARAM-DEF

2.3.1.156 Dcm/DcmConfigSet/DcmDsp/DcmRbDspRoutineRange/DcmRbDspRoutineRangeIdentifierUpperLimit

Short Name	DcmRbDspRoutineRangeIdentifierUpperLimit
Description	Upper limit of Routine range.
Multiplicity	1..1
Value Range	0..65535
Default Value	-
Type	ECUC-INTEGER-PARAM-DEF

2.3.1.157 Dcm/DcmConfigSet/DcmDsp/DcmRbDspRoutineRange/DcmRbDspRoutineRangeInfoByte

Short Name	DcmRbDspRoutineRangeInfoByte
Description	Manufacturer specific value reported to the tester for the all record identifiers. For OBD RIDS routine info byte value is always 0 as per J1979-2. For other RIDS routine info byte value shall be same as configured value
Multiplicity	0..1
Value Range	0..255
Default Value	-
Type	ECUC-INTEGER-PARAM-DEF

2.3.1.158 Dcm/DcmConfigSet/DcmDsp/DcmRbDspRoutineRange/DcmRbDspRoutineRangeIsRidAvailableFnc

Short Name	DcmRbDspRoutineRangeIsRidAvailableFnc
Description	Function name to request from application if a specific Routine is available within the range or not.
Introduction	Only relevant if DcmRbDspRoutineRangeUsePort is set to false. This parameter is related to the interface Xxx_IsRidAvailable.
Multiplicity	0..1
Default Value	-
Type	ECUC-FUNCTION-NAME-DEF

2.3.1.159 Dcm/DcmConfigSet/DcmDsp/DcmRbDspRoutineRange/DcmRbDspRoutineRangeReqSequenceErrorEnabled

Short Name	DcmRbDspRoutineRangeReqSequenceErrorEnabled

Description	Indicates Request Sequence Error (NRC 0x24) shall be triggered for a RequestResults/Stop subfunction with/ without a prior Start subfunction request for Routines which supports both RequestResults and Start subfunctions/ Stop and Start Subfunctions.
Introduction	True:For a Routine which supports both RequestResults/Stop and Start Subfunction,NRC 0x24 shall be triggered if Routine Control tester request with RequestResults subfunction(0x03/Stop subfunction(0x02) comes for a Routine which is not started (without a prior Routine Control request with Start subfunction(0x01)) False:For a Routine which supports both RequestResults/Stop and Start Subfunction,positive response shall be triggered if Routine Control tester request with RequestResults subfunction(0x03)/Stop subfunction(0x02) comes without a prior Routine Control request with Start subfunction(0x01)
Multiplicity	0..1
Default Value	true
Type	ECUC-BOOLEAN-PARAM-DEF

2.3.1.160 Dcm/DcmConfigSet/DcmDsp/DcmRbDspRoutineRange/DcmRbDspRoutineRangeRequestRoutineResults

Short Name	DcmRbDspRoutineRangeRequestRoutineResults
Description	Provides the configuration of RequestResult subservice for a Range of Routines for RoutineControl service.
Introduction	Existence indicates that the RequestRoutineResults in the Range Routines in RoutineControl is supported.
Multiplicity	0..1
Default Value	-
Type	ECUC-PARAM-CONF-CONTAINER-DEF

2.3.1.161 Dcm/DcmConfigSet/DcmDsp/DcmRbDspRoutineRange/DcmRbDspRoutineRangeRequestRoutineResults/DcmRbDspRequestRoutineResultsIn

Short Name	DcmRbDspRequestRoutineResultsIn
Description	Provide description of input parameter of RequestResults subservice for RoutineControl service.
Multiplicity	0..1
Default Value	-
Type	ECUC-PARAM-CONF-CONTAINER-DEF

2.3.1.162 Dcm/DcmConfigSet/DcmDsp/DcmRbDspRoutineRange/DcmRbDspRoutineRangeRequestRoutineResults/DcmRbDspRequestRoutineResultsIn/DcmRbDspRoutineRangeRequestRoutineResultsInSignal

Short Name	DcmRbDspRoutineRangeRequestRoutineResultsInSignal
Description	Provide description of a routine signal used in RoutineControl service.
Multiplicity	1..*
Default Value	-
Type	ECUC-PARAM-CONF-CONTAINER-DEF

2.3.1.163 Dcm/DcmConfigSet/DcmDsp/DcmRbDspRoutineRange/DcmRbDspRoutineRangeRequestRoutineResults/DcmRbDspRequestRoutineResultsIn/DcmRbDspRoutineRangeRequestRoutineResultsInSignal/DcmRbDspArgumentScaling

Short Name	DcmRbDspArgumentScaling
Description	This container contains the configuration (parameters) of an alternative Diagnosis Representation. Out if this the scaling between Diagnosis and ECU internal representation and vice versa can be calculated.
Multiplicity	0..1
Default Value	-
Type	ECUC-CHOICE-CONTAINER-DEF

2.3.1.164 Dcm/DcmConfigSet/DcmDsp/DcmRbDspRoutineRange/DcmRbDspRoutineRangeRequestRoutineResults/DcmRbDspRequestRoutineResultsIn/DcmRbDspRoutineRangeRequestRoutineResultsInSignal/DcmRbDspArgumentScaling/DcmRbDspAlternativeArgumentData

Short Name	DcmRbDspAlternativeArgumentData
Description	This container contains the configuration (parameters) of an alternative Diagnosis Representation by the means of a ArgumentDataPrototype.
Multiplicity	0..0
Default Value	-
Type	ECUC-PARAM-CONF-CONTAINER-DEF

2.3.1.165 Dcm/DcmConfigSet/DcmDsp/DcmRbDspRoutineRange/DcmRbDspRoutineRangeRequestRoutineResults/DcmRbDspRequestRoutineResultsIn/DcmRbDspRoutineRangeRequestRoutineResultsInSignal/DcmRbDspArgumentScaling/DcmRbDspAlterna-

tiveArgumentData/DcmRbDataElement

Short Name	DcmRbDataElement
Description	Alternative Diagnosis Representation for the data defined by the means of a ArgumentDataPrototype
Multiplicity	0..0
Default Value	-
Type	ECUC-FOREIGN-REFERENCE-DEF

2.3.1.166 Dcm/DcmConfigSet/DcmDsp/DcmRbDspRoutineRange/DcmRbDspRoutineRangeRequestRoutineResults/DcmRbDspRequestRoutineResultsIn/DcmRbDspRoutineRangeRequestRoutineResultsInSignal/DcmRbDspArgumentScaling/DcmRbDspAlternativeDataProps

Short Name	DcmRbDspAlternativeDataProps
Description	This container contains the configuration (parameters) of an alternative Diagnosis Representation by the means of ECU configuration parameters.The physical unit of the alternative data representation is defined by the DataPrototype referenced by DcmDspExternal SRDataElementClass. Additionally the definition of a text table mapping can be a defined for DcmDspDataTypeCategory TEXTTABLE and SCALE_LINEAR_AND_TEXTTABLE
Introduction	The physical unit of the alternative data representation is defined by the DataPrototype referenced by DcmDspExternalSRDataElementClass. Additionally the definition of a text table mapping can be a defined for DcmDspDataTypeCategory TEXTTABLE and SCALE_LINEAR_AND_TEXTTABLE
Multiplicity	0..1
Default Value	-
Type	ECUC-PARAM-CONF-CONTAINER-DEF

2.3.1.167 Dcm/DcmConfigSet/DcmDsp/DcmRbDspRoutineRange/DcmRbDspRoutineRangeRequestRoutineResults/DcmRbDspRequestRoutineResultsIn/DcmRbDspRoutineRangeRequestRoutineResultsInSignal/DcmRbDspArgumentScaling/DcmRbDspAlternativeDataProps/DcmRbDspLinearScale

Short Name	DcmRbDspLinearScale
Description	This container contains the configuration (parameters) of an linear scale of the alternative Diagnosis Representation.
Multiplicity	1..1
Default Value	-

Type	ECUC-PARAM-CONF-CONTAINER-DEF

2.3.1.168 Dcm/DcmConfigSet/DcmDsp/DcmRbDspRoutineRange/DcmRbDspRoutineRangeRequestRoutineResults/DcmRbDspRequestRoutineResultsIn/DcmRbDspRoutineRangeRequestRoutineResultsInSignal/DcmRbDspArgumentScaling/DcmRbDspAlternativeDataProps/DcmRbDspLinearScale/DcmRbDspDiagnosisRepresentationDataLowerRange

Short Name	DcmRbDspDiagnosisRepresentationDataLowerRange
Description	Lower Range for this scale of the data in the alternative Diagnosis Representation.
Multiplicity	1..1
Value Range	-Inf..Inf
Default Value	-
Type	ECUC-FLOAT-PARAM-DEF

2.3.1.169 Dcm/DcmConfigSet/DcmDsp/DcmRbDspRoutineRange/DcmRbDspRoutineRangeRequestRoutineResults/DcmRbDspRequestRoutineResultsIn/DcmRbDspRoutineRangeRequestRoutineResultsInSignal/DcmRbDspArgumentScaling/DcmRbDspAlternativeDataProps/DcmRbDspLinearScale/DcmRbDspDiagnosisRepresentationDataOffset

Short Name	DcmRbDspDiagnosisRepresentationDataOffset
Description	Data offset of the alternative Diagnosis Representation for this scale.
Multiplicity	1..1
Value Range	0..Inf
Default Value	0
Type	ECUC-FLOAT-PARAM-DEF

2.3.1.170 Dcm/DcmConfigSet/DcmDsp/DcmRbDspRoutineRange/DcmRbDspRoutineRangeRequestRoutineResults/DcmRbDspRequestRoutineResultsIn/DcmRbDspRoutineRangeRequestRoutineResultsInSignal/DcmRbDspArgumentScaling/DcmRbDspAlternativeDataProps/DcmRbDspLinearScale/DcmRbDspDiagnosisRepresentationDataResolution

Short Name	DcmRbDspDiagnosisRepresentationDataResolution
Description	Data resolution of the alternative Diagnosis Representation for this scale.
Multiplicity	1..1
Value Range	0..Inf
Default Value	-
Type	ECUC-FLOAT-PARAM-DEF

2.3.1.171 Dcm/DcmConfigSet/DcmDsp/DcmRbDspRoutineRange/DcmRbDspRoutineRangeRequestRoutineResults/DcmRbDspRequestRoutineResultsIn/DcmRbDspRoutineRangeRequestRoutineResultsInSignal/DcmRbDspArgumentScaling/DcmRbDspAlternativeDataProps/DcmRbDspLinearScale/DcmRbDspDiagnosisRepresentationDataUpperRange

Short Name	DcmRbDspDiagnosisRepresentationDataUpperRange
Description	Upper Range for this scale of the data in the alternative Diagnosis Representation.
Multiplicity	1..1
Value Range	-Inf..Inf
Default Value	-
Type	ECUC-FLOAT-PARAM-DEF

2.3.1.172 Dcm/DcmConfigSet/DcmDsp/DcmRbDspRoutineRange/DcmRbDspRoutineRangeRequestRoutineResults/DcmRbDspRequestRoutineResultsIn/DcmRbDspRoutineRangeRequestRoutineResultsInSignal/DcmRbDspArgumentScaling/DcmRbDspAlternativeDataType

Short Name	DcmRbDspAlternativeDataType
Description	This container contains the configuration (parameters) of an alternative Diagnosis Representation by the means of an ApplicationDataType.
Introduction	Additionally the definition of a text table mapping can be a defined for ApplicationDataTypes of category TEXTTABLE and SCALE_LINEAR_AND_TEXTTABLE.
Multiplicity	0..1
Default Value	-
Type	ECUC-PARAM-CONF-CONTAINER-DEF

2.3.1.173 Dcm/DcmConfigSet/DcmDsp/DcmRbDspRoutineRange/DcmRbDspRoutineRangeRequestRoutineResults/DcmRbDspRequestRoutineResultsIn/DcmRbDspRoutineRangeRequestRoutineResultsInSignal/DcmRbDspArgumentScaling/DcmRbDspAlternativeDataType/DcmRbApplicationDataType

Short Name	DcmRbApplicationDataType
Description	Alternative Diagnosis Representation for the data defined by the means of a ApplicationPrimitiveDataType of category VALUE or BOOLEAN.
Multiplicity	1..1
Default Value	-
Type	ECUC-FOREIGN-REFERENCE-DEF

2.3.1.174 Dcm/DcmConfigSet/DcmDsp/DcmRbDspRoutineRange/DcmRbDspRoutineRangeRequestRoutineResults/DcmRbDspRequestRoutineResultsIn/DcmRbDspRoutineRangeRequestRoutineResultsInSignal/DcmRbDspArgumentScaling/DcmRbDspAlternativeDataType/DcmRbDspTextTableMapping

Short Name	DcmRbDspTextTableMapping
Description	This container contains the configuration (parameters) of the mapping a DataPrototype typed by AutosarDataType that refer to a CompuMethods of category TEXTTABLE or SCALE_LINEAR_AND_TEXTTABLE.
Introduction	Each DcmRbDspTextTableMapping defines a value pair which is used to map the ECU internal value (DcmRbDspInternalDataValue) to the vale used in the diagnosis representation (DcmRbDspDiagnosisRepresentationDataValue) and vice versa. The set of all DcmRbDspText TableMappings defines the whole mapping of an data.
Multiplicity	0..0
Default Value	-
Type	ECUC-PARAM-CONF-CONTAINER-DEF

2.3.1.175 Dcm/DcmConfigSet/DcmDsp/DcmRbDspRoutineRange/DcmRbDspRoutineRangeRequestRoutineResults/DcmRbDspRequestRoutineResultsIn/DcmRbDspRoutineRangeRequestRoutineResultsInSignal/DcmRbDspArgumentScaling/DcmRbDspAlternativeDataType/DcmRbDspTextTableMapping/DcmRbDspDiagnosisRepresentationDataValue

Short Name	DcmRbDspDiagnosisRepresentationDataValue
Description	The data value in the diagnosis representation.
Multiplicity	0..0
Value Range	0..18446744073709551615
Default Value	-
Type	ECUC-INTEGER-PARAM-DEF

2.3.1.176 Dcm/DcmConfigSet/DcmDsp/DcmRbDspRoutineRange/DcmRbDspRoutineRangeRequestRoutineResults/DcmRbDspRequestRoutineResultsIn/DcmRbDspRoutineRangeRequestRoutineResultsInSignal/DcmRbDspArgumentScaling/DcmRbDspAlternativeDataType/DcmRbDspTextTableMapping/DcmRbDspInternalDataValue

Short Name	DcmRbDspInternalDataValue
Description	The ECU internal data value.
Multiplicity	0..0

Value Range	0..18446744073709551615
Default Value	-
Type	ECUC-INTEGER-PARAM-DEF

2.3.1.177 Dcm/DcmConfigSet/DcmDsp/DcmRbDspRoutineRange/DcmRbDspRoutineRangeRequestRoutineResults/DcmRbDspRequestRoutineResultsIn/DcmRbDspRoutineRangeRequestRoutineResultsInSignal/DcmRbDspRoutineSignalEndianness

Short Name	DcmRbDspRoutineSignalEndianness
Description	Defines the endianness of the data belonging to a Request Routine Response In Signal.If projects configures OPAQUE no endianness conversion will be supported/applicable
Introduction	If no DcmRbDspRoutineSignalEndianness is defined the value of DcmRbDspDataDefaultEndianness is applicable.
Multiplicity	0..1
Default Value	-
Enumeration Literals	BIG_ENDIAN, LITTLE_ENDIAN, OPAQUE
Type	ECUC-ENUMERATION-PARAM-DEF

2.3.1.178 Dcm/DcmConfigSet/DcmDsp/DcmRbDspRoutineRange/DcmRbDspRoutineRangeRequestRoutineResults/DcmRbDspRequestRoutineResultsIn/DcmRbDspRoutineRangeRequestRoutineResultsInSignal/DcmRbDspRoutineSignalLength

Short Name	DcmRbDspRoutineSignalLength
Description	Provide the length of the Routine signal in bits.If DcmRbDspRoutineRequestResINSignal is configured as VARIABLE_LENGTH, then Dcm shall update the currentDataLength parameter of the configured DcmAppl API (DcmRbDspRequestResultsRoutineFnc) in BYTES before calling the API .CurrentDataLength parameter signifies the actual length of the variable length signal positioned at the end in request.
Multiplicity	0..1
Value Range	0..65535
Default Value	-
Type	ECUC-INTEGER-PARAM-DEF

2.3.1.179 Dcm/DcmConfigSet/DcmDsp/DcmRbDspRoutineRange/DcmRbDspRoutineRangeRequestRoutineResults/DcmRbDspRequestRoutineResultsIn/DcmRbDspRoutineRangeRequestRoutineRe-

sultsInSignal/DcmRbDspRoutineSignalPos

Short Name	DcmRbDspRoutineSignalPos
Description	Provide the position of the signal in the RoutineControl request/ response.The position is defined in bits.
Introduction	The position is defined in bits.
Multiplicity	1..1
Value Range	0..65535
Default Value	-
Type	ECUC-INTEGER-PARAM-DEF

2.3.1.180 Dcm/DcmConfigSet/DcmDsp/DcmRbDspRoutineRange/DcmRbDspRoutineRangeRequestRoutineResults/DcmRbDspRequestRoutineResultsIn/DcmRbDspRoutineRangeRequestRoutineResultsInSignal/DcmRbDspRoutineSignalType

Short Name	DcmRbDspRoutineSignalType
Description	Provide the type of the signal in the RoutineControl request/response.
Multiplicity	1..1
Default Value	-
Enumeration Literals	BOOLEAN, SINT16, SINT16_N, SINT32, SINT32_N, SINT8, SINT8_N, UINT16, UINT16_N, UINT32, UINT32_N, UINT8, UINT8_N, VARIABLE_LENGTH
Type	ECUC-ENUMERATION-PARAM-DEF

2.3.1.181 Dcm/DcmConfigSet/DcmDsp/DcmRbDspRoutineRange/DcmRbDspRoutineRangeRequestRoutineResults/DcmRbDspRequestRoutineResultsOut

Short Name	DcmRbDspRequestRoutineResultsOut
Description	Provide description of output parameter of RequestResult subservice for RoutineControl service.
Multiplicity	0..1
Default Value	-
Type	ECUC-PARAM-CONF-CONTAINER-DEF

2.3.1.182 Dcm/DcmConfigSet/DcmDsp/DcmRbDspRoutineRange/DcmRbDspRoutineRangeRequestRoutineResults/DcmRbDspRequestRoutineResultsOut/DcmRbDspRequestRoutineResultsOutSignal

Short Name	DcmRbDspRequestRoutineResultsOutSignal
Description	Provide description of a routine signal used in RoutineControl service.

Introduction	The ordering defined via the index attribute of the subcontainers in this list represents the order of the dataOutN elements in the XXX_RequestResult function call.
Multiplicity	1..*
Default Value	-
Type	ECUC-PARAM-CONF-CONTAINER-DEF

2.3.1.183 Dcm/DcmConfigSet/DcmDsp/DcmRbDspRoutineRange/DcmRbDspRoutineRangeRequestRoutineResults/DcmRbDspRequestRoutineResultsOut/DcmRbDspRequestRoutineResultsOutSignal/DcmRbDspArgumentScaling

Short Name	DcmRbDspArgumentScaling
Description	This container contains the configuration (parameters) of an alternative Diagnosis Representation. Out if this the scaling between Diagnosis and ECU internal representation and vice versa can be calculated.
Multiplicity	0..1
Default Value	-
Type	ECUC-CHOICE-CONTAINER-DEF

2.3.1.184 Dcm/DcmConfigSet/DcmDsp/DcmRbDspRoutineRange/DcmRbDspRoutineRangeRequestRoutineResults/DcmRbDspRequestRoutineResultsOut/DcmRbDspRequestRoutineResultsOutSignal/DcmRbDspArgumentScaling/DcmRbDspAlternativeArgumentData

Short Name	DcmRbDspAlternativeArgumentData
Description	This container contains the configuration (parameters) of an alternative Diagnosis Representation by the means of a ArgumentDataPrototype.
Multiplicity	0..0
Default Value	-
Type	ECUC-PARAM-CONF-CONTAINER-DEF

2.3.1.185 Dcm/DcmConfigSet/DcmDsp/DcmRbDspRoutineRange/DcmRbDspRoutineRangeRequestRoutineResults/DcmRbDspRequestRoutineResultsOut/DcmRbDspRequestRoutineResultsOutSignal/DcmRbDspArgumentScaling/DcmRbDspAlternativeArgumentData/DcmRbDataElement

Short Name	DcmRbDataElement

Description	Alternative Diagnosis Representation for the data defined by the means of a ArgumentDataPrototype
Multiplicity	0..0
Default Value	-
Type	ECUC-FOREIGN-REFERENCE-DEF

2.3.1.186 Dcm/DcmConfigSet/DcmDsp/DcmRbDspRoutineRange/DcmRbDspRoutineRangeRequestRoutineResults/DcmRbDspRequestRoutineResultsOut/DcmRbDspRequestRoutineResultsOutSignal/DcmRbDspArgumentScaling/DcmRbDspAlternativeDataProps

Short Name	DcmRbDspAlternativeDataProps
Description	This container contains the configuration (parameters) of an alternative Diagnosis Representation by the means of ECU configuration parameters.The physical unit of the alternative data representation is defined by the DataPrototype referenced by DcmDspExternal SRDataElementClass. Additionally the definition of a text table mapping can be a defined for DcmDspDataTypeCategory TEXTTABLE and SCALE_LINEAR_AND_TEXTTABLE
Introduction	The physical unit of the alternative data representation is defined by the DataPrototype referenced by DcmDspExternalSRDataElementClass. Additionally the definition of a text table mapping can be a defined for DcmDspDataTypeCategory TEXTTABLE and SCALE_LINEAR_AND_TEXTTABLE
Multiplicity	0..1
Default Value	-
Type	ECUC-PARAM-CONF-CONTAINER-DEF

2.3.1.187 Dcm/DcmConfigSet/DcmDsp/DcmRbDspRoutineRange/DcmRbDspRoutineRangeRequestRoutineResults/DcmRbDspRequestRoutineResultsOut/DcmRbDspRequestRoutineResultsOutSignal/DcmRbDspArgumentScaling/DcmRbDspAlternativeDataProps/DcmRbDspLinearScale

Short Name	DcmRbDspLinearScale
Description	This container contains the configuration (parameters) of an linear scale of the alternative Diagnosis Representation.
Multiplicity	1..1
Default Value	-
Type	ECUC-PARAM-CONF-CONTAINER-DEF

2.3.1.188 Dcm/DcmConfigSet/DcmDsp/DcmRbDspRoutineRange/DcmRbDspRoutineRangeRequestRoutineResults/DcmRbDspRequestRou-

tineResultsOut/DcmRbDspRequestRoutineResultsOutSignal/DcmRbDspArgumentScaling/DcmRbDspAlternativeDataProps/DcmRbDspLinearScale/DcmRbDspDiagnosisRepresentationDataLowerRange

Short Name	DcmRbDspDiagnosisRepresentationDataLowerRange
Description	Lower Range for this scale of the data in the alternative Diagnosis Representation.
Multiplicity	1..1
Value Range	-Inf..Inf
Default Value	-
Type	ECUC-FLOAT-PARAM-DEF

2.3.1.189 Dcm/DcmConfigSet/DcmDsp/DcmRbDspRoutineRange/DcmRbDspRoutineRangeRequestRoutineResults/DcmRbDspRequestRoutineResultsOut/DcmRbDspRequestRoutineResultsOutSignal/DcmRbDspArgumentScaling/DcmRbDspAlternativeDataProps/DcmRbDspLinearScale/DcmRbDspDiagnosisRepresentationDataOffset

Short Name	DcmRbDspDiagnosisRepresentationDataOffset
Description	Data offset of the alternative Diagnosis Representation for this scale.
Multiplicity	1..1
Value Range	0..Inf
Default Value	0
Type	ECUC-FLOAT-PARAM-DEF

2.3.1.190 Dcm/DcmConfigSet/DcmDsp/DcmRbDspRoutineRange/DcmRbDspRoutineRangeRequestRoutineResults/DcmRbDspRequestRoutineResultsOut/DcmRbDspRequestRoutineResultsOutSignal/DcmRbDspArgumentScaling/DcmRbDspAlternativeDataProps/DcmRbDspLinearScale/DcmRbDspDiagnosisRepresentationDataResolution

Short Name	DcmRbDspDiagnosisRepresentationDataResolution
Description	Data resolution of the alternative Diagnosis Representation for this scale.
Multiplicity	1..1
Value Range	0..Inf
Default Value	-
Type	ECUC-FLOAT-PARAM-DEF

2.3.1.191 Dcm/DcmConfigSet/DcmDsp/DcmRbDspRoutineRange/DcmRbDspRoutineRangeRequestRoutineResults/DcmRbDspRequestRoutineResultsOut/DcmRbDspRequestRoutineResultsOutSignal/DcmRbDspArgumentScaling/DcmRbDspAlternativeDataProps/

DcmRbDspLinearScale/DcmRbDspDiagnosisRepresentation-DataUpperRange

Short Name	DcmRbDspDiagnosisRepresentationDataUpperRange
Description	Upper Range for this scale of the data in the alternative Diagnosis Representation.
Multiplicity	1..1
Value Range	-Inf..Inf
Default Value	-
Type	ECUC-FLOAT-PARAM-DEF

2.3.1.192 Dcm/DcmConfigSet/DcmDsp/DcmRbDspRoutineRange/DcmRbDspRoutineRangeRequestRoutineResults/DcmRbDspRequestRoutineResultsOut/DcmRbDspRequestRoutineResultsOutSignal/DcmRbDspArgumentScaling/DcmRbDspAlternativeDataType

Short Name	DcmRbDspAlternativeDataType
Description	This container contains the configuration (parameters) of an alternative Diagnosis Representation by the means of an ApplicationDataType.
Introduction	Additionally the definition of a text table mapping can be a defined for ApplicationDataTypes of category TEXTTABLE and SCALE_LINEAR_AND_TEXTTABLE.
Multiplicity	0..1
Default Value	-
Type	ECUC-PARAM-CONF-CONTAINER-DEF

2.3.1.193 Dcm/DcmConfigSet/DcmDsp/DcmRbDspRoutineRange/DcmRbDspRoutineRangeRequestRoutineResults/DcmRbDspRequestRoutineResultsOut/DcmRbDspRequestRoutineResultsOutSignal/DcmRbDspArgumentScaling/DcmRbDspAlternativeDataType/DcmRbApplicationDataType

Short Name	DcmRbApplicationDataType
Description	Alternative Diagnosis Representation for the data defined by the means of a ApplicationPrimitiveDataType of category VALUE or BOOLEAN.
Multiplicity	1..1
Default Value	-
Type	ECUC-FOREIGN-REFERENCE-DEF

2.3.1.194 Dcm/DcmConfigSet/DcmDsp/DcmRbDspRoutineRange/DcmRbDspRoutineRangeRequestRoutineResults/DcmRbDspRequestRoutineResultsOut/DcmRbDspRequestRoutineResultsOutSignal/DcmRbDspArgumentScaling/DcmRbDspAlternativeDataType/

DcmRbDspTextTableMapping

Short Name	DcmRbDspTextTableMapping
Description	This container contains the configuration (parameters) of the mapping a DataPrototype typed by Autosar-DataType that refer to a CompuMethods of category TEXTTABLE or SCALE_LINEAR_AND_TEXTTABLE.
Introduction	Each DcmRbDspTextTableMapping defines a value pair which is used to map the ECU internal value (DcmRbD-spInternalDataValue) to the vale used in the diagnosis representation (DcmRbDspDiagnosisRepresentation-DataValue) and vice versa. The set of all DcmRbDspText TableMappings defines the whole mapping of an data.
Multiplicity	0..0
Default Value	-
Type	ECUC-PARAM-CONF-CONTAINER-DEF

2.3.1.195 Dcm/DcmConfigSet/DcmDsp/DcmRbDspRoutineRange/DcmRbD-spRoutineRangeRequestRoutineResults/DcmRbDspRequestRou-tineResultsOut/DcmRbDspRequestRoutineResultsOutSignal/ DcmRbDspArgumentScaling/DcmRbDspAlternativeDataType/ DcmRbDspTextTableMapping/DcmRbDspDiagnosisRepresenta-tionDataValue

Short Name	DcmRbDspDiagnosisRepresentationDataValue
Description	The data value in the diagnosis representation.
Multiplicity	0..0
Value Range	0..18446744073709551615
Default Value	-
Type	ECUC-INTEGER-PARAM-DEF

2.3.1.196 Dcm/DcmConfigSet/DcmDsp/DcmRbDspRoutineRange/DcmRbD-spRoutineRangeRequestRoutineResults/DcmRbDspRequestRou-tineResultsOut/DcmRbDspRequestRoutineResultsOutSignal/ DcmRbDspArgumentScaling/DcmRbDspAlternativeDataType/ DcmRbDspTextTableMapping/DcmRbDspInternalDataValue

Short Name	DcmRbDspInternalDataValue
Description	The ECU internal data value.
Multiplicity	0..0
Value Range	0..18446744073709551615
Default Value	-
Type	ECUC-INTEGER-PARAM-DEF

2.3.1.197 Dcm/DcmConfigSet/DcmDsp/DcmRbDspRoutineRange/DcmRbDspRoutineRangeRequestRoutineResults/DcmRbDspRequestRoutineResultsOut/DcmRbDspRequestRoutineResultsOutSignal/DcmRbDspRoutineSignalEndianness

Short Name	DcmRbDspRoutineSignalEndianness
Description	Defines the endianness of the data belonging to a Request Routine Response Out Signal.
Introduction	If no DcmDspRoutineSignalEndianness is defined the value of DcmDspDataDefaultEndianness is applicable.
Multiplicity	0..1
Default Value	-
Enumeration Literals	BIG_ENDIAN, LITTLE_ENDIAN, OPAQUE
Type	ECUC-ENUMERATION-PARAM-DEF

2.3.1.198 Dcm/DcmConfigSet/DcmDsp/DcmRbDspRoutineRange/DcmRbDspRoutineRangeRequestRoutineResults/DcmRbDspRequestRoutineResultsOut/DcmRbDspRequestRoutineResultsOutSignal/DcmRbDspRoutineSignalLength

Short Name	DcmRbDspRoutineSignalLength
Description	Provide the length in bits of the signal in the RoutineControl request/response
Multiplicity	0..1
Value Range	0..65535
Default Value	-
Type	ECUC-INTEGER-PARAM-DEF

2.3.1.199 Dcm/DcmConfigSet/DcmDsp/DcmRbDspRoutineRange/DcmRbDspRoutineRangeRequestRoutineResults/DcmRbDspRequestRoutineResultsOut/DcmRbDspRequestRoutineResultsOutSignal/DcmRbDspRoutineSignalPos

Short Name	DcmRbDspRoutineSignalPos
Description	Provide the position of the signal in the RoutineControl request/response.
Introduction	The position is defined in bits.
Multiplicity	1..1
Value Range	0..65535
Default Value	-
Type	ECUC-INTEGER-PARAM-DEF

2.3.1.200 Dcm/DcmConfigSet/DcmDsp/DcmRbDspRoutineRange/DcmRbDspRoutineRangeRequestRoutineResults/DcmRbDspRequestRoutineResultsOut/DcmRbDspRequestRoutineResultsOutSignal/DcmRbDspRoutineSignalType

Short Name	DcmRbDspRoutineSignalType
Description	Provide the type of the signal in the RoutineControl request/response.
Multiplicity	1..1
Default Value	-
Enumeration Literals	BOOLEAN, SINT16, SINT16_N, SINT32, SINT32_N, SINT8, SINT8_N, UINT16, UINT16_N, UINT32, UINT32_N, UINT8, UINT8_N, VARIABLE_LENGTH
Type	ECUC-ENUMERATION-PARAM-DEF

2.3.1.201 Dcm/DcmConfigSet/DcmDsp/DcmRbDspRoutineRange/DcmRbDspRoutineRangeRequestRoutineResults/DcmRbDspRoutineRangeInterfaceArgumentIntegrity

Short Name	DcmRbDspRoutineRangeInterfaceArgumentIntegrity
Description	Defines the value of ClientServerOperation.diagArgIntegrity for the created C/S interface of this routine
Multiplicity	0..1
Default Value	true
Type	ECUC-BOOLEAN-PARAM-DEF

2.3.1.202 Dcm/DcmConfigSet/DcmDsp/DcmRbDspRoutineRange/DcmRbDspRoutineRangeRequestRoutineResults/DcmRbDspRoutineRangeRequestRoutineResultsCommonAuthorizationRef

Short Name	DcmRbDspRoutineRangeRequestRoutineResults CommonAuthorizationRef
Description	Reference to DcmDspCommonAuthorization
Introduction	Common authorization configuration taken from the referenced DcmRbDspRoutineRangeRequestRoutineResultsCommonAuthorizationRef. If there is no reference, no check on the commonly defined authorization conditions shall be done to get the routine result.
Multiplicity	0..1
Default Value	-
Type	ECUC-REFERENCE-DEF

2.3.1.203 Dcm/DcmConfigSet/DcmDsp/DcmRbDspRoutineRange/DcmRbDspRoutineRangeRequestRoutineResults/DcmRbDspRoutineRan-

geRequestRoutineResultsFncDcmRbDspRoutineRangeRequestRoutineResultsFnc

Short Name	DcmRbDspRoutineRangeRequestRoutineResultsFnc
Description	Function name for request to application the results of a routine in a Range. (RoutineRange_RequestResults-function)
Introduction	This parameter is related to the interface Xxx_RequestResults.
Multiplicity	0..1
Default Value	-
Type	ECUC-FUNCTION-NAME-DEF

2.3.1.204 Dcm/DcmConfigSet/DcmDsp/DcmRbDspRoutineRange/DcmRbDspRoutineRangeRequestRoutineResults/DcmRbDspRoutineRangeRequestRoutineResultsRoleRef

Short Name	DcmRbDspRoutineRangeRequestRoutineResultsRoleRef
Description	Reference to DcmDspAuthenticationRow that defines a role in that this routine can be started.
Multiplicity	0..32
Default Value	-
Type	ECUC-REFERENCE-DEF

2.3.1.205 Dcm/DcmConfigSet/DcmDsp/DcmRbDspRoutineRange/DcmRbDspRoutineRangeStartRoutine

Short Name	DcmRbDspRoutineRangeStartRoutine
Description	Provides the configuration of Start subservice for RoutineControl service.
Multiplicity	0..1
Default Value	-
Type	ECUC-PARAM-CONF-CONTAINER-DEF

2.3.1.206 Dcm/DcmConfigSet/DcmDsp/DcmRbDspRoutineRange/DcmRbDspRoutineRangeStartRoutine/DcmRbDspRoutineRangeInterfaceArgumentIntegrity

Short Name	DcmRbDspRoutineRangeInterfaceArgumentIntegrity
Description	Defines the value of ClientServerOperation.diagArgIntegrity for the created C/S interface of this routine
Multiplicity	0..1
Default Value	true
Type	ECUC-BOOLEAN-PARAM-DEF

2.3.1.207 Dcm/DcmConfigSet/DcmDsp/DcmRbDspRoutineRange/DcmRbD-

spRoutineRangeStartRoutine/DcmRbDspRoutineRangeStartRoutineCommonAuthorizationRef

Short Name	DcmRbDspRoutineRangeStartRoutineCommonAuthorizationRef
Description	Reference to DcmDspCommonAuthorization
Introduction	Common authorization configuration taken from the referenced DcmRbDspStartRoutineCommonAuthorizationRef. If there is no reference, no check on the commonly defined authorization conditions shall be done to start the routine.
Multiplicity	0..1
Default Value	-
Type	ECUC-REFERENCE-DEF

2.3.1.208 Dcm/DcmConfigSet/DcmDsp/DcmRbDspRoutineRange/DcmRbDspRoutineRangeStartRoutine/DcmRbDspRoutineRangeStartRoutineFnc

Short Name	DcmRbDspRoutineRangeStartRoutineFnc
Description	Function name for request to application to start a routine. (Routine_ Start-function)
Introduction	This parameter is related to the interface Xxx_Start.
Multiplicity	0..1
Default Value	-
Type	ECUC-FUNCTION-NAME-DEF

2.3.1.209 Dcm/DcmConfigSet/DcmDsp/DcmRbDspRoutineRange/DcmRbDspRoutineRangeStartRoutine/DcmRbDspRoutineRangeStartRoutineRoleRef

Short Name	DcmRbDspRoutineRangeStartRoutineRoleRef
Description	Reference to DcmDspAuthenticationRow that defines a role in that this routine can be started.
Multiplicity	0..32
Default Value	-
Type	ECUC-REFERENCE-DEF

2.3.1.210 Dcm/DcmConfigSet/DcmDsp/DcmRbDspRoutineRange/DcmRbDspRoutineRangeStartRoutine/DcmRbDspStartRoutineIn

Short Name	DcmRbDspStartRoutineIn

Description	Provide description of input parameter of Start subservice for RoutineControl service
Multiplicity	0..1
Default Value	-
Type	ECUC-PARAM-CONF-CONTAINER-DEF

2.3.1.211 Dcm/DcmConfigSet/DcmDsp/DcmRbDspRoutineRange/DcmRbDspRoutineRangeStartRoutine/DcmRbDspStartRoutineIn/DcmRbDspStartRoutineInSignal

Short Name	DcmRbDspStartRoutineInSignal
Description	Provide description of a routine signal used in RoutineControl service.
Introduction	The ordering defined via the index attribute of the subcontainers in this list represents the order of the dataInN elements in the XXX_Start function call.
Multiplicity	1..*
Default Value	-
Type	ECUC-PARAM-CONF-CONTAINER-DEF

2.3.1.212 Dcm/DcmConfigSet/DcmDsp/DcmRbDspRoutineRange/DcmRbDspRoutineRangeStartRoutine/DcmRbDspStartRoutineIn/DcmRbDspStartRoutineInSignal/DcmRbDspArgumentScaling

Short Name	DcmRbDspArgumentScaling
Description	This container contains the configuration (parameters) of an alternative Diagnosis Representation. Out if this the scaling between Diagnosis and ECU internal representation and vice versa can be calculated.
Multiplicity	0..1
Default Value	-
Type	ECUC-CHOICE-CONTAINER-DEF

2.3.1.213 Dcm/DcmConfigSet/DcmDsp/DcmRbDspRoutineRange/DcmRbDspRoutineRangeStartRoutine/DcmRbDspStartRoutineIn/DcmRbDspStartRoutineInSignal/DcmRbDspArgumentScaling/DcmRbDspAlternativeDataProps

Short Name	DcmRbDspAlternativeDataProps

Description	This container contains the configuration (parameters) of an alternative Diagnosis Representation by the means of ECU configuration parameters.The physical unit of the alternative data representation is defined by the DataPrototype referenced by DcmDspExternal SRDataElementClass. Additionally the definition of a text table mapping can be a defined for DcmDspDataTypeCategory TEXTTABLE and SCALE_LINEAR_AND_TEXTTABLE
Introduction	The physical unit of the alternative data representation is defined by the DataPrototype referenced by DcmDspExternalSRDataElementClass. Additionally the definition of a text table mapping can be a defined for DcmDspDataTypeCategory TEXTTABLE and SCALE_LINEAR_AND_TEXTTABLE
Multiplicity	0..1
Default Value	-
Type	ECUC-PARAM-CONF-CONTAINER-DEF

2.3.1.214 Dcm/DcmConfigSet/DcmDsp/DcmRbDspRoutineRange/DcmRbDspRoutineRangeStartRoutine/DcmRbDspStartRoutineIn/DcmRbDspStartRoutineInSignal/DcmRbDspArgumentScaling/DcmRbDspAlternativeDataProps/DcmRbDspLinearScale

Short Name	DcmRbDspLinearScale
Description	This container contains the configuration (parameters) of an linear scale of the alternative Diagnosis Representation.
Multiplicity	1..1
Default Value	-
Type	ECUC-PARAM-CONF-CONTAINER-DEF

2.3.1.215 Dcm/DcmConfigSet/DcmDsp/DcmRbDspRoutineRange/DcmRbDspRoutineRangeStartRoutine/DcmRbDspStartRoutineIn/DcmRbDspStartRoutineInSignal/DcmRbDspArgumentScaling/DcmRbDspAlternativeDataProps/DcmRbDspLinearScale/DcmRbDspDiagnosisRepresentationDataLowerRange

Short Name	DcmRbDspDiagnosisRepresentationDataLowerRange
Description	Lower Range for this scale of the data in the alternative Diagnosis Representation.
Multiplicity	1..1
Value Range	-Inf..Inf
Default Value	-
Type	ECUC-FLOAT-PARAM-DEF

2.3.1.216 Dcm/DcmConfigSet/DcmDsp/DcmRbDspRoutineRange/DcmRbDspRoutineRangeStartRoutine/DcmRbDspStartRoutineIn/DcmRbDspStartRoutineInSignal/DcmRbDspArgumentScaling/DcmRbDspAlternativeDataProps/DcmRbDspLinearScale/DcmRbDspDiagnosisRepresentationDataOffset

Short Name	DcmRbDspDiagnosisRepresentationDataOffset
Description	Data offset of the alternative Diagnosis Representation for this scale.
Multiplicity	1..1
Value Range	0..Inf
Default Value	0
Type	ECUC-FLOAT-PARAM-DEF

2.3.1.217 Dcm/DcmConfigSet/DcmDsp/DcmRbDspRoutineRange/DcmRbDspRoutineRangeStartRoutine/DcmRbDspStartRoutineIn/DcmRbDspStartRoutineInSignal/DcmRbDspArgumentScaling/DcmRbDspAlternativeDataProps/DcmRbDspLinearScale/DcmRbDspDiagnosisRepresentationDataResolution

Short Name	DcmRbDspDiagnosisRepresentationDataResolution
Description	Data resolution of the alternative Diagnosis Representation for this scale.
Multiplicity	1..1
Value Range	0..Inf
Default Value	-
Type	ECUC-FLOAT-PARAM-DEF

2.3.1.218 Dcm/DcmConfigSet/DcmDsp/DcmRbDspRoutineRange/DcmRbDspRoutineRangeStartRoutine/DcmRbDspStartRoutineIn/DcmRbDspStartRoutineInSignal/DcmRbDspArgumentScaling/DcmRbDspAlternativeDataProps/DcmRbDspLinearScale/DcmRbDspDiagnosisRepresentationDataUpperRange

Short Name	DcmRbDspDiagnosisRepresentationDataUpperRange
Description	Upper Range for this scale of the data in the alternative Diagnosis Representation.
Multiplicity	1..1
Value Range	-Inf..Inf
Default Value	-
Type	ECUC-FLOAT-PARAM-DEF

2.3.1.219 Dcm/DcmConfigSet/DcmDsp/DcmRbDspRoutineRange/DcmRbDspRoutineRangeStartRoutine/DcmRbDspStartRoutineIn/DcmRbDspStartRoutineInSignal/DcmRbDspArgumentScaling/DcmRbDspAlternativeDataType

Short Name	DcmRbDspAlternativeDataType
Description	This container contains the configuration (parameters) of an alternative Diagnosis Representation by the means of an ApplicationDataType.
Introduction	Additionally the definition of a text table mapping can be a defined for ApplicationDataTypes of category TEXTTABLE and SCALE_LINEAR_AND_TEXTTABLE.
Multiplicity	0..1
Default Value	-
Type	ECUC-PARAM-CONF-CONTAINER-DEF

2.3.1.220 Dcm/DcmConfigSet/DcmDsp/DcmRbDspRoutineRange/DcmRbDspRoutineRangeStartRoutine/DcmRbDspStartRoutineIn/DcmRbDspStartRoutineInSignal/DcmRbDspArgumentScaling/DcmRbDspAlternativeDataType/DcmRbApplicationDataType

Short Name	DcmRbApplicationDataType
Description	Alternative Diagnosis Representation for the data defined by the means of a ApplicationPrimitiveDataType of category VALUE or BOOLEAN.
Multiplicity	1..1
Default Value	-
Type	ECUC-FOREIGN-REFERENCE-DEF

2.3.1.221 Dcm/DcmConfigSet/DcmDsp/DcmRbDspRoutineRange/DcmRbDspRoutineRangeStartRoutine/DcmRbDspStartRoutineIn/DcmRbDspStartRoutineInSignal/DcmRbDspArgumentScaling/DcmRbDspAlternativeDataType/DcmRbDspTextTableMapping

Short Name	DcmRbDspTextTableMapping
Description	This container contains the configuration (parameters) of the mapping a DataPrototype typed by AutosarDataType that refer to a CompuMethods of category TEXTTABLE or SCALE_LINEAR_AND_TEXTTABLE.
Introduction	Each DcmRbDspTextTableMapping defines a value pair which is used to map the ECU internal value (DcmRbDspInternalDataValue) to the vale used in the diagnosis representation (DcmRbDspDiagnosisRepresentationDataValue) and vice versa. The set of all DcmRbDspText TableMappings defines the whole mapping of an data.

Multiplicity	0..0
Default Value	-
Type	ECUC-PARAM-CONF-CONTAINER-DEF

2.3.1.222 Dcm/DcmConfigSet/DcmDsp/DcmRbDspRoutineRange/DcmRbDspRoutineRangeStartRoutine/DcmRbDspStartRoutineIn/DcmRbDspStartRoutineInSignal/DcmRbDspArgumentScaling/DcmRbDspAlternativeDataType/DcmRbDspTextTableMapping/DcmRbDspDiagnosisRepresentationDataValue

Short Name	DcmRbDspDiagnosisRepresentationDataValue
Description	The data value in the diagnosis representation.
Multiplicity	0..0
Value Range	0..18446744073709551615
Default Value	-
Type	ECUC-INTEGER-PARAM-DEF

2.3.1.223 Dcm/DcmConfigSet/DcmDsp/DcmRbDspRoutineRange/DcmRbDspRoutineRangeStartRoutine/DcmRbDspStartRoutineIn/DcmRbDspStartRoutineInSignal/DcmRbDspArgumentScaling/DcmRbDspAlternativeDataType/DcmRbDspTextTableMapping/DcmRbDspInternalDataValue

Short Name	DcmRbDspInternalDataValue
Description	The ECU internal data value.
Multiplicity	0..0
Value Range	0..18446744073709551615
Default Value	-
Type	ECUC-INTEGER-PARAM-DEF

2.3.1.224 Dcm/DcmConfigSet/DcmDsp/DcmRbDspRoutineRange/DcmRbDspRoutineRangeStartRoutine/DcmRbDspStartRoutineIn/DcmRbDspStartRoutineInSignal/DcmRbDspRoutineSignalEndianness

Short Name	DcmRbDspRoutineSignalEndianness
Description	Defines the endianness of the data belonging to a Start Routine In Signal.
Introduction	If no DcmRbDspRoutineSignalEndianness is defined the value of DcmDspDataDefaultEndianness is applicable.
Multiplicity	0..1
Default Value	-
Enumeration Literals	BIG_ENDIAN, LITTLE_ENDIAN, OPAQUE

Type	ECUC-ENUMERATION-PARAM-DEF

2.3.1.225 Dcm/DcmConfigSet/DcmDsp/DcmRbDspRoutineRange/DcmRbDspRoutineRangeStartRoutine/DcmRbDspStartRoutineIn/DcmRbDspStartRoutineInSignal/DcmRbDspRoutineSignalLength

Short Name	DcmRbDspRoutineSignalLength
Description	Provide the length in bits of the signal in the RoutineControl request/response
Multiplicity	0..1
Value Range	0..65535
Default Value	-
Type	ECUC-INTEGER-PARAM-DEF

2.3.1.226 Dcm/DcmConfigSet/DcmDsp/DcmRbDspRoutineRange/DcmRbDspRoutineRangeStartRoutine/DcmRbDspStartRoutineIn/DcmRbDspStartRoutineInSignal/DcmRbDspRoutineSignalPos

Short Name	DcmRbDspRoutineSignalPos
Description	Provide the position of the signal in the RoutineControl request/response.
Introduction	The position is defined in bits.
Multiplicity	1..1
Value Range	0..65535
Default Value	-
Type	ECUC-INTEGER-PARAM-DEF

2.3.1.227 Dcm/DcmConfigSet/DcmDsp/DcmRbDspRoutineRange/DcmRbDspRoutineRangeStartRoutine/DcmRbDspStartRoutineIn/DcmRbDspStartRoutineInSignal/DcmRbDspRoutineSignalType

Short Name	DcmRbDspRoutineSignalType
Description	Provide the type of the signal in the RoutineControl request/response.
Multiplicity	1..1
Default Value	-
Enumeration Literals	BOOLEAN, SINT16, SINT16_N, SINT32, SINT32_N, SINT8, SINT8_N, UINT16, UINT16_N, UINT32, UINT32_N, UINT8, UINT8_N, VARIABLE_LENGTH
Type	ECUC-ENUMERATION-PARAM-DEF

2.3.1.228 Dcm/DcmConfigSet/DcmDsp/DcmRbDspRoutineRange/DcmRbDspRoutineRangeStartRoutine/DcmRbDspStartRoutineOut

Short Name	DcmRbDspStartRoutineOut

Description	Provide description of output parameter of Start subservice for RoutineControl service.
Multiplicity	0..1
Default Value	-
Type	ECUC-PARAM-CONF-CONTAINER-DEF

2.3.1.229 Dcm/DcmConfigSet/DcmDsp/DcmRbDspRoutineRange/DcmRbDspRoutineRangeStartRoutine/DcmRbDspStartRoutineOut/DcmRbDspStartRoutineOutSignal

Short Name	DcmRbDspStartRoutineOutSignal
Description	Provide description of a routine signal used in RoutineControl service.
Introduction	The ordering defined via the index attribute of the subcontainers in this list represents the order of the dataOutN elements in the XXX_Start function call.
Multiplicity	1..*
Default Value	-
Type	ECUC-PARAM-CONF-CONTAINER-DEF

2.3.1.230 Dcm/DcmConfigSet/DcmDsp/DcmRbDspRoutineRange/DcmRbDspRoutineRangeStartRoutine/DcmRbDspStartRoutineOut/DcmRbDspStartRoutineOutSignal/DcmRbDspArgumentScaling

Short Name	DcmRbDspArgumentScaling
Description	This container contains the configuration (parameters) of an alternative Diagnosis Representation. Out if this the scaling between Diagnosis and ECU internal representation and vice versa can be calculated.
Multiplicity	0..1
Default Value	-
Type	ECUC-CHOICE-CONTAINER-DEF

2.3.1.231 Dcm/DcmConfigSet/DcmDsp/DcmRbDspRoutineRange/DcmRbDspRoutineRangeStartRoutine/DcmRbDspStartRoutineOut/DcmRbDspStartRoutineOutSignal/DcmRbDspArgumentScaling/DcmRbDspAlternativeDataProps

Short Name	DcmRbDspAlternativeDataProps

Description	This container contains the configuration (parameters) of an alternative Diagnosis Representation by the means of ECU configuration parameters.The physical unit of the alternative data representation is defined by the DataPrototype referenced by DcmDspExternal SRDataElementClass. Additionally the definition of a text table mapping can be a defined for DcmDspDataTypeCategory TEXTTABLE and SCALE_LINEAR_AND_TEXTTABLE
Introduction	The physical unit of the alternative data representation is defined by the DataPrototype referenced by DcmDspExternalSRDataElementClass. Additionally the definition of a text table mapping can be a defined for DcmDspDataTypeCategory TEXTTABLE and SCALE_LINEAR_AND_TEXTTABLE
Multiplicity	0..1
Default Value	-
Type	ECUC-PARAM-CONF-CONTAINER-DEF

2.3.1.232 Dcm/DcmConfigSet/DcmDsp/DcmRbDspRoutineRange/DcmRbDspRoutineRangeStartRoutine/DcmRbDspStartRoutineOut/DcmRbDspStartRoutineOutSignal/DcmRbDspArgumentScaling/DcmRbDspAlternativeDataProps/DcmRbDspLinearScale

Short Name	DcmRbDspLinearScale
Description	This container contains the configuration (parameters) of an linear scale of the alternative Diagnosis Representation.
Multiplicity	1..1
Default Value	-
Type	ECUC-PARAM-CONF-CONTAINER-DEF

2.3.1.233 Dcm/DcmConfigSet/DcmDsp/DcmRbDspRoutineRange/DcmRbDspRoutineRangeStartRoutine/DcmRbDspStartRoutineOut/DcmRbDspStartRoutineOutSignal/DcmRbDspArgumentScaling/DcmRbDspAlternativeDataProps/DcmRbDspLinearScale/DcmRbDspDiagnosisRepresentationDataLowerRange

Short Name	DcmRbDspDiagnosisRepresentationDataLowerRange
Description	Lower Range for this scale of the data in the alternative Diagnosis Representation.
Multiplicity	1..1
Value Range	-Inf..Inf
Default Value	-
Type	ECUC-FLOAT-PARAM-DEF

2.3.1.234 Dcm/DcmConfigSet/DcmDsp/DcmRbDspRoutineRange/DcmRbDspRoutineRangeStartRoutine/DcmRbDspStartRoutineOut/DcmRbDspStartRoutineOutSignal/DcmRbDspArgumentScaling/DcmRbDspAlternativeDataProps/DcmRbDspLinearScale/DcmRbDspDiagnosisRepresentationDataOffset

Short Name	DcmRbDspDiagnosisRepresentationDataOffset
Description	Data offset of the alternative Diagnosis Representation for this scale.
Multiplicity	1..1
Value Range	0..Inf
Default Value	0
Type	ECUC-FLOAT-PARAM-DEF

2.3.1.235 Dcm/DcmConfigSet/DcmDsp/DcmRbDspRoutineRange/DcmRbDspRoutineRangeStartRoutine/DcmRbDspStartRoutineOut/DcmRbDspStartRoutineOutSignal/DcmRbDspArgumentScaling/DcmRbDspAlternativeDataProps/DcmRbDspLinearScale/DcmRbDspDiagnosisRepresentationDataResolution

Short Name	DcmRbDspDiagnosisRepresentationDataResolution
Description	Data resolution of the alternative Diagnosis Representation for this scale.
Multiplicity	1..1
Value Range	0..Inf
Default Value	-
Type	ECUC-FLOAT-PARAM-DEF

2.3.1.236 Dcm/DcmConfigSet/DcmDsp/DcmRbDspRoutineRange/DcmRbDspRoutineRangeStartRoutine/DcmRbDspStartRoutineOut/DcmRbDspStartRoutineOutSignal/DcmRbDspArgumentScaling/DcmRbDspAlternativeDataProps/DcmRbDspLinearScale/DcmRbDspDiagnosisRepresentationDataUpperRange

Short Name	DcmRbDspDiagnosisRepresentationDataUpperRange
Description	Upper Range for this scale of the data in the alternative Diagnosis Representation.
Multiplicity	1..1
Value Range	-Inf..Inf
Default Value	-
Type	ECUC-FLOAT-PARAM-DEF

2.3.1.237 Dcm/DcmConfigSet/DcmDsp/DcmRbDspRoutineRange/DcmRbDspRoutineRangeStartRoutine/DcmRbDspStartRoutineOut/DcmRbDspStartRoutineOutSignal/DcmRbDspArgumentScaling/DcmRbDspAlternativeDataType

Short Name	DcmRbDspAlternativeDataType
Description	This container contains the configuration (parameters) of an alternative Diagnosis Representation by the means of an ApplicationDataType.
Introduction	Additionally the definition of a text table mapping can be a defined for ApplicationDataTypes of category TEXTTABLE and SCALE_LINEAR_AND_TEXTTABLE.
Multiplicity	0..1
Default Value	-
Type	ECUC-PARAM-CONF-CONTAINER-DEF

2.3.1.238 Dcm/DcmConfigSet/DcmDsp/DcmRbDspRoutineRange/DcmRbDspRoutineRangeStartRoutine/DcmRbDspStartRoutineOut/DcmRbDspStartRoutineOutSignal/DcmRbDspArgumentScaling/DcmRbDspAlternativeDataType/DcmRbApplicationDataType

Short Name	DcmRbApplicationDataType
Description	Alternative Diagnosis Representation for the data defined by the means of a ApplicationPrimitiveDataType of category VALUE or BOOLEAN.
Multiplicity	1..1
Default Value	-
Type	ECUC-FOREIGN-REFERENCE-DEF

2.3.1.239 Dcm/DcmConfigSet/DcmDsp/DcmRbDspRoutineRange/DcmRbDspRoutineRangeStartRoutine/DcmRbDspStartRoutineOut/DcmRbDspStartRoutineOutSignal/DcmRbDspArgumentScaling/DcmRbDspAlternativeDataType/DcmRbDspTextTableMapping

Short Name	DcmRbDspTextTableMapping
Description	This container contains the configuration (parameters) of the mapping a DataPrototype typed by AutosarDataType that refer to a CompuMethods of category TEXTTABLE or SCALE_LINEAR_AND_TEXTTABLE.
Introduction	Each DcmRbDspTextTableMapping defines a value pair which is used to map the ECU internal value (DcmRbDspInternalDataValue) to the vale used in the diagnosis representation (DcmRbDspDiagnosisRepresentationDataValue) and vice versa. The set of all DcmRbDspText TableMappings defines the whole mapping of an data.

Multiplicity	0..0
Default Value	-
Type	ECUC-PARAM-CONF-CONTAINER-DEF

2.3.1.240 Dcm/DcmConfigSet/DcmDsp/DcmRbDspRoutineRange/DcmRbDspRoutineRangeStartRoutine/DcmRbDspStartRoutineOut/DcmRbDspStartRoutineOutSignal/DcmRbDspArgumentScaling/DcmRbDspAlternativeDataType/DcmRbDspTextTableMapping/DcmRbDspDiagnosisRepresentationDataValue

Short Name	DcmRbDspDiagnosisRepresentationDataValue
Description	The data value in the diagnosis representation.
Multiplicity	0..0
Value Range	0..18446744073709551615
Default Value	-
Type	ECUC-INTEGER-PARAM-DEF

2.3.1.241 Dcm/DcmConfigSet/DcmDsp/DcmRbDspRoutineRange/DcmRbDspRoutineRangeStartRoutine/DcmRbDspStartRoutineOut/DcmRbDspStartRoutineOutSignal/DcmRbDspArgumentScaling/DcmRbDspAlternativeDataType/DcmRbDspTextTableMapping/DcmRbDspInternalDataValue

Short Name	DcmRbDspInternalDataValue
Description	The ECU internal data value.
Multiplicity	0..0
Value Range	0..18446744073709551615
Default Value	-
Type	ECUC-INTEGER-PARAM-DEF

2.3.1.242 Dcm/DcmConfigSet/DcmDsp/DcmRbDspRoutineRange/DcmRbDspRoutineRangeStartRoutine/DcmRbDspStartRoutineOut/DcmRbDspStartRoutineOutSignal/DcmRbDspRoutineSignalEndianness

Short Name	DcmRbDspRoutineSignalEndianness
Description	Defines the endianness of the data belonging to a Start Routine Out Signal.
Introduction	If no DcmRbDspRoutineSignalEndianness is defined the value of DcmDspDataDefaultEndianness is applicable.
Multiplicity	0..1
Default Value	-

Enumeration Literals	BIG_ENDIAN, LITTLE_ENDIAN, OPAQUE
Type	ECUC-ENUMERATION-PARAM-DEF

2.3.1.243 Dcm/DcmConfigSet/DcmDsp/DcmRbDspRoutineRange/DcmRbDspRoutineRangeStartRoutine/DcmRbDspStartRoutineOut/DcmRbDspStartRoutineOutSignal/DcmRbDspRoutineSignalLength

Short Name	DcmRbDspRoutineSignalLength
Description	Provide the length in bits of the signal in the RoutineControl request/response.
Multiplicity	0..1
Value Range	0..65535
Default Value	-
Type	ECUC-INTEGER-PARAM-DEF

2.3.1.244 Dcm/DcmConfigSet/DcmDsp/DcmRbDspRoutineRange/DcmRbDspRoutineRangeStartRoutine/DcmRbDspStartRoutineOut/DcmRbDspStartRoutineOutSignal/DcmRbDspRoutineSignalPos

Short Name	DcmRbDspRoutineSignalPos
Description	Provide the position of the signal in the RoutineControl request/response.
Introduction	The position is defined in bits.
Multiplicity	0..1
Value Range	0..65535
Default Value	-
Type	ECUC-INTEGER-PARAM-DEF

2.3.1.245 Dcm/DcmConfigSet/DcmDsp/DcmRbDspRoutineRange/DcmRbDspRoutineRangeStartRoutine/DcmRbDspStartRoutineOut/DcmRbDspStartRoutineOutSignal/DcmRbDspRoutineSignalType

Short Name	DcmRbDspRoutineSignalType
Description	Provide the type of the signal in the RoutineControl request/response.
Multiplicity	1..1
Default Value	-
Enumeration Literals	BOOLEAN, SINT16, SINT16_N, SINT32, SINT32_N, SINT8, SINT8_N, UINT16, UINT16_N, UINT32, UINT32_N, UINT8, UINT8_N, VARIABLE_LENGTH
Type	ECUC-ENUMERATION-PARAM-DEF

2.3.1.246 Dcm/DcmConfigSet/DcmDsp/DcmRbDspRoutineRange/DcmRbDspRoutineRangeStopRoutine

Short Name	DcmRbDspRoutineRangeStopRoutine
Description	Provides the configuration of Stop subservice for RoutineControl service.
Introduction	Existence indicates that the StopRoutine in the RoutineControl is supported.
Multiplicity	0..1
Default Value	-
Type	ECUC-PARAM-CONF-CONTAINER-DEF

2.3.1.247 Dcm/DcmConfigSet/DcmDsp/DcmRbDspRoutineRange/DcmRbDspRoutineRangeStopRoutine/DcmRbDspRoutineRangeInterfaceArgumentIntegrity

Short Name	DcmRbDspRoutineRangeInterfaceArgumentIntegrity
Description	Defines the value of ClientServerOperation.diagArgIntegrity for the created C/S interface of this routine
Multiplicity	0..1
Default Value	true
Type	ECUC-BOOLEAN-PARAM-DEF

2.3.1.248 Dcm/DcmConfigSet/DcmDsp/DcmRbDspRoutineRange/DcmRbDspRoutineRangeStopRoutine/DcmRbDspRoutineRangeStopRoutineCommonAuthorizationRef

Short Name	DcmRbDspRoutineRangeStopRoutineCommonAuthorizationRef
Description	Reference to DcmDspCommonAuthorization
Introduction	Common authorization configuration taken from the referenced DcmRbDspStopRoutineCommonAuthorizationRef. If there is no reference, no check on the commonly defined authorization conditions shall be done to stop the routine.
Multiplicity	0..1
Default Value	-
Type	ECUC-REFERENCE-DEF

2.3.1.249 Dcm/DcmConfigSet/DcmDsp/DcmRbDspRoutineRange/DcmRbDspRoutineRangeStopRoutine/DcmRbDspRoutineRangeStopRoutineFnc

Short Name	DcmRbDspRoutineRangeStopRoutineFnc

Description	Function name for request to application to stop a routine. (Routine_ Stop-function)
Introduction	This parameter is related to the interface Xxx_Stop.
Multiplicity	0..1
Default Value	-
Type	ECUC-FUNCTION-NAME-DEF

2.3.1.250 Dcm/DcmConfigSet/DcmDsp/DcmRbDspRoutineRange/DcmRbDspRoutineRangeStopRoutine/DcmRbDspRoutineRangeStopRoutineRoleRef

Short Name	DcmRbDspRoutineRangeStopRoutineRoleRef
Description	Reference to DcmDspAuthenticationRow that defines a role in that this routine can be started.
Multiplicity	0..32
Default Value	-
Type	ECUC-REFERENCE-DEF

2.3.1.251 Dcm/DcmConfigSet/DcmDsp/DcmRbDspRoutineRange/DcmRbDspRoutineRangeStopRoutine/DcmRbDspStopRoutineIn

Short Name	DcmRbDspStopRoutineIn
Description	Provide description of input parameter of Stop subservice for RoutineControl service.
Multiplicity	0..1
Default Value	-
Type	ECUC-PARAM-CONF-CONTAINER-DEF

2.3.1.252 Dcm/DcmConfigSet/DcmDsp/DcmRbDspRoutineRange/DcmRbDspRoutineRangeStopRoutine/DcmRbDspStopRoutineIn/DcmRbDspStopRoutineInSignal

Short Name	DcmRbDspStopRoutineInSignal
Description	Provide description of a routine signal used in RoutineControl service.
Introduction	The ordering defined via the index attribute of the subcontainers in this list represents the order of the dataInN elements in the XXX_Stop function call.
Multiplicity	1..*
Default Value	-
Type	ECUC-PARAM-CONF-CONTAINER-DEF

2.3.1.253 Dcm/DcmConfigSet/DcmDsp/DcmRbDspRoutineRange/DcmRbDspRoutineRangeStopRoutine/DcmRbDspStopRoutineIn/DcmRbD-

spStopRoutineInSignal/DcmRbDspArgumentScaling

Short Name	DcmRbDspArgumentScaling
Description	This container contains the configuration (parameters) of an alternative Diagnosis Representation. Out if this the scaling between Diagnosis and ECU internal representation and vice versa can be calculated.
Multiplicity	0..1
Default Value	-
Type	ECUC-CHOICE-CONTAINER-DEF

2.3.1.254 Dcm/DcmConfigSet/DcmDsp/DcmRbDspRoutineRange/DcmRbDspRoutineRangeStopRoutine/DcmRbDspStopRoutineIn/DcmRbDspStopRoutineInSignal/DcmRbDspArgumentScaling/DcmRbDspAlternativeDataProps

Short Name	DcmRbDspAlternativeDataProps
Description	This container contains the configuration (parameters) of an alternative Diagnosis Representation by the means of ECU configuration parameters.The physical unit of the alternative data representation is defined by the DataPrototype referenced by DcmDspExternal SRDataElementClass. Additionally the definition of a text table mapping can be a defined for DcmDspDataTypeCategory TEXTTABLE and SCALE_LINEAR_AND_TEXTTABLE
Introduction	The physical unit of the alternative data representation is defined by the DataPrototype referenced by DcmDspExternalSRDataElementClass. Additionally the definition of a text table mapping can be a defined for DcmDspDataTypeCategory TEXTTABLE and SCALE_LINEAR_AND_TEXTTABLE
Multiplicity	0..1
Default Value	-
Type	ECUC-PARAM-CONF-CONTAINER-DEF

2.3.1.255 Dcm/DcmConfigSet/DcmDsp/DcmRbDspRoutineRange/DcmRbDspRoutineRangeStopRoutine/DcmRbDspStopRoutineIn/DcmRbDspStopRoutineInSignal/DcmRbDspArgumentScaling/DcmRbDspAlternativeDataProps/DcmRbDspLinearScale

Short Name	DcmRbDspLinearScale
Description	This container contains the configuration (parameters) of an linear scale of the alternative Diagnosis Representation.
Multiplicity	1..1

Default Value	-
Type	ECUC-PARAM-CONF-CONTAINER-DEF

2.3.1.256 Dcm/DcmConfigSet/DcmDsp/DcmRbDspRoutineRange/DcmRbDspRoutineRangeStopRoutine/DcmRbDspStopRoutineIn/DcmRbDspStopRoutineInSignal/DcmRbDspArgumentScaling/DcmRbDspAlternativeDataProps/DcmRbDspLinearScale/DcmRbDspDiagnosisRepresentationDataLowerRange

Short Name	DcmRbDspDiagnosisRepresentationDataLowerRange
Description	Lower Range for this scale of the data in the alternative Diagnosis Representation.
Multiplicity	1..1
Value Range	-Inf..Inf
Default Value	-
Type	ECUC-FLOAT-PARAM-DEF

2.3.1.257 Dcm/DcmConfigSet/DcmDsp/DcmRbDspRoutineRange/DcmRbDspRoutineRangeStopRoutine/DcmRbDspStopRoutineIn/DcmRbDspStopRoutineInSignal/DcmRbDspArgumentScaling/DcmRbDspAlternativeDataProps/DcmRbDspLinearScale/DcmRbDspDiagnosisRepresentationDataOffset

Short Name	DcmRbDspDiagnosisRepresentationDataOffset
Description	Data offset of the alternative Diagnosis Representation for this scale.
Multiplicity	1..1
Value Range	0..Inf
Default Value	0
Type	ECUC-FLOAT-PARAM-DEF

2.3.1.258 Dcm/DcmConfigSet/DcmDsp/DcmRbDspRoutineRange/DcmRbDspRoutineRangeStopRoutine/DcmRbDspStopRoutineIn/DcmRbDspStopRoutineInSignal/DcmRbDspArgumentScaling/DcmRbDspAlternativeDataProps/DcmRbDspLinearScale/DcmRbDspDiagnosisRepresentationDataResolution

Short Name	DcmRbDspDiagnosisRepresentationDataResolution
Description	Data resolution of the alternative Diagnosis Representation for this scale.
Multiplicity	1..1
Value Range	0..Inf
Default Value	-
Type	ECUC-FLOAT-PARAM-DEF

2.3.1.259 Dcm/DcmConfigSet/DcmDsp/DcmRbDspRoutineRange/DcmRbD-

spRoutineRangeStopRoutine/DcmRbDspStopRoutineIn/DcmRbDspStopRoutineInSignal/DcmRbDspArgumentScaling/DcmRbDspAlternativeDataProps/DcmRbDspLinearScale/DcmRbDspDiagnosisRepresentationDataUpperRange

Short Name	DcmRbDspDiagnosisRepresentationDataUpperRange
Description	Upper Range for this scale of the data in the alternative Diagnosis Representation.
Multiplicity	1..1
Value Range	-Inf..Inf
Default Value	-
Type	ECUC-FLOAT-PARAM-DEF

2.3.1.260 Dcm/DcmConfigSet/DcmDsp/DcmRbDspRoutineRange/DcmRbDspRoutineRangeStopRoutine/DcmRbDspStopRoutineIn/DcmRbDspStopRoutineInSignal/DcmRbDspArgumentScaling/DcmRbDspAlternativeDataType

Short Name	DcmRbDspAlternativeDataType
Description	This container contains the configuration (parameters) of an alternative Diagnosis Representation by the means of an ApplicationDataType.
Introduction	Additionally the definition of a text table mapping can be a defined for ApplicationDataTypes of category TEXTTABLE and SCALE_LINEAR_AND_TEXTTABLE.
Multiplicity	0..1
Default Value	-
Type	ECUC-PARAM-CONF-CONTAINER-DEF

2.3.1.261 Dcm/DcmConfigSet/DcmDsp/DcmRbDspRoutineRange/DcmRbDspRoutineRangeStopRoutine/DcmRbDspStopRoutineIn/DcmRbDspStopRoutineInSignal/DcmRbDspArgumentScaling/DcmRbDspAlternativeDataType/DcmRbApplicationDataType

Short Name	DcmRbApplicationDataType
Description	Alternative Diagnosis Representation for the data defined by the means of a ApplicationPrimitiveDataType of category VALUE or BOOLEAN.
Multiplicity	1..1
Default Value	-
Type	ECUC-FOREIGN-REFERENCE-DEF

2.3.1.262 Dcm/DcmConfigSet/DcmDsp/DcmRbDspRoutineRange/DcmRbDspRoutineRangeStopRoutine/DcmRbDspStopRoutineIn/DcmRbDspStopRoutineInSignal/DcmRbDspArgumentScaling/DcmRbDspAl-

ternativeDataType/DcmRbDspTextTableMapping

Short Name	DcmRbDspTextTableMapping
Description	This container contains the configuration (parameters) of the mapping a DataPrototype typed by Autosar-DataType that refer to a CompuMethods of category TEXTTABLE or SCALE_LINEAR_AND_TEXTTABLE.
Introduction	Each DcmRbDspTextTableMapping defines a value pair which is used to map the ECU internal value (DcmRbD-spInternalDataValue) to the vale used in the diagnosis representation (DcmRbDspDiagnosisRepresentation-DataValue) and vice versa. The set of all DcmRbDspText TableMappings defines the whole mapping of an data.
Multiplicity	0..0
Default Value	-
Type	ECUC-PARAM-CONF-CONTAINER-DEF

2.3.1.263 Dcm/DcmConfigSet/DcmDsp/DcmRbDspRoutineRange/DcmRbD-spRoutineRangeStopRoutine/DcmRbDspStopRoutineIn/DcmRbD-spStopRoutineInSignal/DcmRbDspArgumentScaling/DcmRbDspAl-ternativeDataType/DcmRbDspTextTableMapping/DcmRbDspDiag-nosisRepresentationDataValue

Short Name	DcmRbDspDiagnosisRepresentationDataValue
Description	The data value in the diagnosis representation.
Multiplicity	0..0
Value Range	0..18446744073709551615
Default Value	-
Type	ECUC-INTEGER-PARAM-DEF

2.3.1.264 Dcm/DcmConfigSet/DcmDsp/DcmRbDspRoutineRange/DcmRbD-spRoutineRangeStopRoutine/DcmRbDspStopRoutineIn/DcmRbD-spStopRoutineInSignal/DcmRbDspArgumentScaling/DcmRbDspAl-ternativeDataType/DcmRbDspTextTableMapping/DcmRbDspIn-ternalDataValue

Short Name	DcmRbDspInternalDataValue
Description	The ECU internal data value.
Multiplicity	0..0
Value Range	0..18446744073709551615
Default Value	-
Type	ECUC-INTEGER-PARAM-DEF

2.3.1.265 Dcm/DcmConfigSet/DcmDsp/DcmRbDspRoutineRange/DcmRbD-

spRoutineRangeStopRoutine/DcmRbDspStopRoutineIn/DcmRbDspStopRoutineInSignal/DcmRbDspRoutineSignalEndianness

Short Name	DcmRbDspRoutineSignalEndianness
Description	Defines the endianness of the data belonging to a Stop Routine In Signal.
Introduction	If no DcmRbDspRoutineSignalEndianness is defined the value of DcmDspDataDefaultEndianness is applicable.
Multiplicity	0..1
Default Value	-
Enumeration Literals	BIG_ENDIAN, LITTLE_ENDIAN, OPAQUE
Type	ECUC-ENUMERATION-PARAM-DEF

2.3.1.266 Dcm/DcmConfigSet/DcmDsp/DcmRbDspRoutineRange/DcmRbDspRoutineRangeStopRoutine/DcmRbDspStopRoutineIn/DcmRbDspStopRoutineInSignal/DcmRbDspRoutineSignalLength

Short Name	DcmRbDspRoutineSignalLength
Description	Provide the length in bits of the signal in the RoutineControl request/response
Multiplicity	0..1
Value Range	0..65535
Default Value	-
Type	ECUC-INTEGER-PARAM-DEF

2.3.1.267 Dcm/DcmConfigSet/DcmDsp/DcmRbDspRoutineRange/DcmRbDspRoutineRangeStopRoutine/DcmRbDspStopRoutineIn/DcmRbDspStopRoutineInSignal/DcmRbDspRoutineSignalPos

Short Name	DcmRbDspRoutineSignalPos
Description	Provide the position of the signal in the RoutineControl request/response.
Introduction	The position is defined in bits.
Multiplicity	1..1
Value Range	0..65535
Default Value	-
Type	ECUC-INTEGER-PARAM-DEF

2.3.1.268 Dcm/DcmConfigSet/DcmDsp/DcmRbDspRoutineRange/DcmRbDspRoutineRangeStopRoutine/DcmRbDspStopRoutineIn/DcmRbDspStopRoutineInSignal/DcmRbDspRoutineSignalType

Short Name	DcmRbDspRoutineSignalType

Description	Provide the type of the signal in the RoutineControl request/response.
Multiplicity	1..1
Default Value	-
Enumeration Literals	BOOLEAN, SINT16, SINT16_N, SINT32, SINT32_N, SINT8, SINT8_N, UINT16, UINT16_N, UINT32, UINT32_N, UINT8, UINT8_N, VARIABLE_LENGTH
Type	ECUC-ENUMERATION-PARAM-DEF

2.3.1.269 Dcm/DcmConfigSet/DcmDsp/DcmRbDspRoutineRange/DcmRbDspRoutineRangeStopRoutine/DcmRbDspStopRoutineOut

Short Name	DcmRbDspStopRoutineOut
Description	Provide description of output parameter of Stop subservice for RoutineControl service.
Multiplicity	0..1
Default Value	-
Type	ECUC-PARAM-CONF-CONTAINER-DEF

2.3.1.270 Dcm/DcmConfigSet/DcmDsp/DcmRbDspRoutineRange/DcmRbDspRoutineRangeStopRoutine/DcmRbDspStopRoutineOut/DcmRbDspStopRoutineOutSignal

Short Name	DcmRbDspStopRoutineOutSignal
Description	Provide description of a routine signal used in RoutineControl service.
Introduction	The ordering defined via the index attribute of the subcontainers in this list represents the order of the dataOutN elements in the XXX_Stop function call.
Multiplicity	1..*
Default Value	-
Type	ECUC-PARAM-CONF-CONTAINER-DEF

2.3.1.271 Dcm/DcmConfigSet/DcmDsp/DcmRbDspRoutineRange/DcmRbDspRoutineRangeStopRoutine/DcmRbDspStopRoutineOut/DcmRbDspStopRoutineOutSignal/DcmRbDspArgumentScaling

Short Name	DcmRbDspArgumentScaling
Description	This container contains the configuration (parameters) of an alternative Diagnosis Representation. Out if this the scaling between Diagnosis and ECU internal representation and vice versa can be calculated.
Multiplicity	0..1
Default Value	-

Type	ECUC-CHOICE-CONTAINER-DEF

2.3.1.272 Dcm/DcmConfigSet/DcmDsp/DcmRbDspRoutineRange/DcmRbDspRoutineRangeStopRoutine/DcmRbDspStopRoutineOut/DcmRbDspStopRoutineOutSignal/DcmRbDspArgumentScaling/DcmRbDspAlternativeDataProps

Short Name	DcmRbDspAlternativeDataProps
Description	This container contains the configuration (parameters) of an alternative Diagnosis Representation by the means of ECU configuration parameters.The physical unit of the alternative data representation is defined by the DataPrototype referenced by DcmDspExternal SRDataElementClass. Additionally the definition of a text table mapping can be a defined for DcmDspDataTypeCategory TEXTTABLE and SCALE_LINEAR_AND_TEXTTABLE
Introduction	The physical unit of the alternative data representation is defined by the DataPrototype referenced by DcmDspExternalSRDataElementClass. Additionally the definition of a text table mapping can be a defined for DcmDspDataTypeCategory TEXTTABLE and SCALE_LINEAR_AND_TEXTTABLE
Multiplicity	0..1
Default Value	-
Type	ECUC-PARAM-CONF-CONTAINER-DEF

2.3.1.273 Dcm/DcmConfigSet/DcmDsp/DcmRbDspRoutineRange/DcmRbDspRoutineRangeStopRoutine/DcmRbDspStopRoutineOut/DcmRbDspStopRoutineOutSignal/DcmRbDspArgumentScaling/DcmRbDspAlternativeDataProps/DcmRbDspLinearScale

Short Name	DcmRbDspLinearScale
Description	This container contains the configuration (parameters) of an linear scale of the alternative Diagnosis Representation.
Multiplicity	1..1
Default Value	-
Type	ECUC-PARAM-CONF-CONTAINER-DEF

2.3.1.274 Dcm/DcmConfigSet/DcmDsp/DcmRbDspRoutineRange/DcmRbDspRoutineRangeStopRoutine/DcmRbDspStopRoutineOut/DcmRbDspStopRoutineOutSignal/DcmRbDspArgumentScaling/DcmRbDspAlternativeDataProps/DcmRbDspLinearScale/DcmRbDspDiagnosisRepresentationDataLowerRange

Short Name	DcmRbDspDiagnosisRepresentationDataLowerRange

Description	Lower Range for this scale of the data in the alternative Diagnosis Representation.
Multiplicity	1..1
Value Range	-Inf..Inf
Default Value	-
Type	ECUC-FLOAT-PARAM-DEF

2.3.1.275 Dcm/DcmConfigSet/DcmDsp/DcmRbDspRoutineRange/DcmRbDspRoutineRangeStopRoutine/DcmRbDspStopRoutineOut/DcmRbDspStopRoutineOutSignal/DcmRbDspArgumentScaling/DcmRbDspAlternativeDataProps/DcmRbDspLinearScale/DcmRbDspDiagnosisRepresentationDataOffset

Short Name	DcmRbDspDiagnosisRepresentationDataOffset
Description	Data offset of the alternative Diagnosis Representation for this scale.
Multiplicity	1..1
Value Range	0..Inf
Default Value	0
Type	ECUC-FLOAT-PARAM-DEF

2.3.1.276 Dcm/DcmConfigSet/DcmDsp/DcmRbDspRoutineRange/DcmRbDspRoutineRangeStopRoutine/DcmRbDspStopRoutineOut/DcmRbDspStopRoutineOutSignal/DcmRbDspArgumentScaling/DcmRbDspAlternativeDataProps/DcmRbDspLinearScale/DcmRbDspDiagnosisRepresentationDataResolution

Short Name	DcmRbDspDiagnosisRepresentationDataResolution
Description	Data resolution of the alternative Diagnosis Representation for this scale.
Multiplicity	1..1
Value Range	0..Inf
Default Value	-
Type	ECUC-FLOAT-PARAM-DEF

2.3.1.277 Dcm/DcmConfigSet/DcmDsp/DcmRbDspRoutineRange/DcmRbDspRoutineRangeStopRoutine/DcmRbDspStopRoutineOut/DcmRbDspStopRoutineOutSignal/DcmRbDspArgumentScaling/DcmRbDspAlternativeDataProps/DcmRbDspLinearScale/DcmRbDspDiagnosisRepresentationDataUpperRange

Short Name	DcmRbDspDiagnosisRepresentationDataUpperRange
Description	Upper Range for this scale of the data in the alternative Diagnosis Representation.
Multiplicity	1..1

Value Range	-Inf..Inf
Default Value	-
Type	ECUC-FLOAT-PARAM-DEF

2.3.1.278 Dcm/DcmConfigSet/DcmDsp/DcmRbDspRoutineRange/DcmRbDspRoutineRangeStopRoutine/DcmRbDspStopRoutineOut/DcmRbDspStopRoutineOutSignal/DcmRbDspArgumentScaling/DcmRbDspAlternativeDataType

Short Name	DcmRbDspAlternativeDataType
Description	This container contains the configuration (parameters) of an alternative Diagnosis Representation by the means of an ApplicationDataType.
Introduction	Additionally the definition of a text table mapping can be a defined for ApplicationDataTypes of category TEXTTABLE and SCALE_LINEAR_AND_TEXTTABLE.
Multiplicity	0..1
Default Value	-
Type	ECUC-PARAM-CONF-CONTAINER-DEF

2.3.1.279 Dcm/DcmConfigSet/DcmDsp/DcmRbDspRoutineRange/DcmRbDspRoutineRangeStopRoutine/DcmRbDspStopRoutineOut/DcmRbDspStopRoutineOutSignal/DcmRbDspArgumentScaling/DcmRbDspAlternativeDataType/DcmRbApplicationDataType

Short Name	DcmRbApplicationDataType
Description	Alternative Diagnosis Representation for the data defined by the means of a ApplicationPrimitiveDataType of category VALUE or BOOLEAN.
Multiplicity	1..1
Default Value	-
Type	ECUC-FOREIGN-REFERENCE-DEF

2.3.1.280 Dcm/DcmConfigSet/DcmDsp/DcmRbDspRoutineRange/DcmRbDspRoutineRangeStopRoutine/DcmRbDspStopRoutineOut/DcmRbDspStopRoutineOutSignal/DcmRbDspArgumentScaling/DcmRbDspAlternativeDataType/DcmRbDspTextTableMapping

Short Name	DcmRbDspTextTableMapping
Description	This container contains the configuration (parameters) of the mapping a DataPrototype typed by AutosarDataType that refer to a CompuMethods of category TEXTTABLE or SCALE_LINEAR_AND_TEXTTABLE.

Introduction	Each DcmRbDspTextTableMapping defines a value pair which is used to map the ECU internal value (DcmRbDspInternalDataValue) to the vale used in the diagnosis representation (DcmRbDspDiagnosisRepresentationDataValue) and vice versa. The set of all DcmRbDspTextTableMappings defines the whole mapping of an data.
Multiplicity	0..0
Default Value	-
Type	ECUC-PARAM-CONF-CONTAINER-DEF

2.3.1.281 Dcm/DcmConfigSet/DcmDsp/DcmRbDspRoutineRange/DcmRbDspRoutineRangeStopRoutine/DcmRbDspStopRoutineOut/DcmRbDspStopRoutineOutSignal/DcmRbDspArgumentScaling/DcmRbDspAlternativeDataType/DcmRbDspTextTableMapping/DcmRbDspDiagnosisRepresentationDataValue

Short Name	DcmRbDspDiagnosisRepresentationDataValue
Description	The data value in the diagnosis representation.
Multiplicity	0..0
Value Range	0..18446744073709551615
Default Value	-
Type	ECUC-INTEGER-PARAM-DEF

2.3.1.282 Dcm/DcmConfigSet/DcmDsp/DcmRbDspRoutineRange/DcmRbDspRoutineRangeStopRoutine/DcmRbDspStopRoutineOut/DcmRbDspStopRoutineOutSignal/DcmRbDspArgumentScaling/DcmRbDspAlternativeDataType/DcmRbDspTextTableMapping/DcmRbDspInternalDataValue

Short Name	DcmRbDspInternalDataValue
Description	The ECU internal data value.
Multiplicity	0..0
Value Range	0..18446744073709551615
Default Value	-
Type	ECUC-INTEGER-PARAM-DEF

2.3.1.283 Dcm/DcmConfigSet/DcmDsp/DcmRbDspRoutineRange/DcmRbDspRoutineRangeStopRoutine/DcmRbDspStopRoutineOut/DcmRbDspStopRoutineOutSignal/DcmRbDspRoutineSignalEndianness

Short Name	DcmRbDspRoutineSignalEndianness
Description	Defines the endianness of the data belonging to a Stop Routine Out Signal.

Introduction	If no DcmRbDspRoutineSignalEndianness is defined the value of DcmDspDataDefaultEndianness is applicable.
Multiplicity	0..1
Default Value	-
Enumeration Literals	BIG_ENDIAN, LITTLE_ENDIAN, OPAQUE
Type	ECUC-ENUMERATION-PARAM-DEF

2.3.1.284 Dcm/DcmConfigSet/DcmDsp/DcmRbDspRoutineRange/DcmRbDspRoutineRangeStopRoutine/DcmRbDspStopRoutineOut/DcmRbDspStopRoutineOutSignal/DcmRbDspRoutineSignalLength

Short Name	DcmRbDspRoutineSignalLength
Description	Provide the length in bits of the signal in the RoutineControl request/response
Multiplicity	0..1
Value Range	0..65535
Default Value	-
Type	ECUC-INTEGER-PARAM-DEF

2.3.1.285 Dcm/DcmConfigSet/DcmDsp/DcmRbDspRoutineRange/DcmRbDspRoutineRangeStopRoutine/DcmRbDspStopRoutineOut/DcmRbDspStopRoutineOutSignal/DcmRbDspRoutineSignalPos

Short Name	DcmRbDspRoutineSignalPos
Description	Provide the position of the signal in the RoutineControl request/response.
Introduction	The position is defined in bits.
Multiplicity	1..1
Value Range	0..65535
Default Value	-
Type	ECUC-INTEGER-PARAM-DEF

2.3.1.286 Dcm/DcmConfigSet/DcmDsp/DcmRbDspRoutineRange/DcmRbDspRoutineRangeStopRoutine/DcmRbDspStopRoutineOut/DcmRbDspStopRoutineOutSignal/DcmRbDspRoutineSignalType

Short Name	DcmRbDspRoutineSignalType
Description	Provide the type of the signal in the RoutineControl request/response.
Multiplicity	1..1
Default Value	-
Enumeration Literals	BOOLEAN, SINT16, SINT16_N, SINT32, SINT32_N, SINT8, SINT8_N, UINT16, UINT16_N, UINT32, UINT32_N, UINT8, UINT8_N, VARIABLE_LENGTH

Type	ECUC-ENUMERATION-PARAM-DEF

2.3.1.287 Dcm/DcmConfigSet/DcmDsp/DcmRbDspRoutineRange/DcmRbDspRoutineRangeStopRoutineOnSessionChange

Short Name	DcmRbDspRoutineRangeStopRoutineOnSessionChange
Description	Indicates if the StopRoutine operation needs to be performed on any session change. True = Stop this Routine if it is active on transition to default session / Stop this routine if it is not supported in the new nondefault session. False = No action on session transitions Dependency: Applicable only if StopRoutine operation is supported for RID (DcmDspStopRoutineSupported = true), otherwise not valid.If the parameter is left unconfigured FALSE will be considered as the default value.
Introduction	true = stopRoutine is supported on session change false = stopRoutine is not supported on session change
Multiplicity	0..1
Default Value	false
Type	ECUC-BOOLEAN-PARAM-DEF

2.3.1.288 Dcm/DcmConfigSet/DcmDsp/DcmRbDspRoutineRange/DcmRbDspRoutineRangeUseAsynchronousServerCallPoint

Short Name	DcmRbDspRoutineRangeUseAsynchronousServerCallPoint
Description	Only relevant in case DcmRbDspRoutineRangeUsePort is set to TRUE. This parameter is used to select the type of Server Handling required. Set the value to TRUE for Asynchronous server call point handling else Synchronous server call point handling is carried out
Multiplicity	0..1
Default Value	false
Type	ECUC-BOOLEAN-PARAM-DEF

2.3.1.289 Dcm/DcmConfigSet/DcmDsp/DcmRbDspRoutineRange/DcmRbDspRoutineRangeUsePort

Short Name	DcmRbDspRoutineRangeUsePort

Description	If this parameter is set to true, the DCM uses an R-Port requiring a PortInteface RoutineRange_{RangeRoutineName} where {RangeRoutineName} is the name of the container DcmDspRoutineRange. In that case, DcmRbDspRoutineRangeIsRidAvailableFnc, DcmRbDspRoutineRangeStartRoutineFnc, DcmRbDspRoutineRangeStopRoutineFnc and DcmRbDspRoutineRangeRequestRoutineResultsFnc will be ignored and the RTE APIs are used. When the parameter DcmRbDspRoutineRangeUsePort is false, the DCM calls the functions defined in DcmRbDspRoutineRangeIsRidAvailableFnc, DcmRbDspRoutineRangeStartRoutineFnc, DcmRbDspRoutineRangeStopRoutineFnc and DcmRbDspRoutineRangeRequestRoutineResultsFnc.
Multiplicity	1..1
Default Value	-
Type	ECUC-BOOLEAN-PARAM-DEF

2.3.1.290 Dcm/DcmConfigSet/DcmDsp/DcmRbDspRoutineRange/DcmRbDspRoutineRangeUserRoutineModeRuleFnc

Short Name	DcmRbDspRoutineRangeUserRoutineModeRuleFnc
Description	Function name used by the application to perform user specific checks before accessing a Routine.
Introduction	In case the checks fails, NRC value set by the application can be returned to the tester. In case the function is not configured for the Routine, the DcmAppl API DcmAppl_UserRIDModeRuleService shall be called.
Multiplicity	0..1
Default Value	-
Type	ECUC-FUNCTION-NAME-DEF

2.3.1.291 Dcm/DcmConfigSet/DcmDsp/DcmRbDspSequenceCheckRequestDownloadDisabled

Short Name	DcmRbDspSequenceCheckRequestDownloadDisabled

Description	Parameter which decides whether sequence check is enabled or disabled for RequestDownload.If the parameter is set to FALSE , a sequence check shall be done according to the ISO 14229-1 NRC sequence if an RequestDownload request is received if Upload/Download is already active (A previous RequestUpload which is not yet completed with a successful RequestTransferExit request. A negative response with NRC set to 0x22 (ConditionsNotCorrect) shall be send in this case. If the parameter is set to TRUE, no sequence check will be done if a RequestDownload request is received when upload/download is already active.If the parameter is left unconfigured ,then FALSE will be considered as the default value and therefore the sequence check will be enabled..
Multiplicity	0..1
Default Value	false
Type	ECUC-BOOLEAN-PARAM-DEF

2.3.1.292 Dcm/DcmConfigSet/DcmDsp/DcmRbDspSequenceCheckRequestUploadDisabled

Short Name	DcmRbDspSequenceCheckRequestUploadDisabled
Description	Parameter whihc decides whether sequence check is enabled or disabled for RequestUpload.If the parameter is set to FALSE , a sequence check shall be done according to the ISO 14229-1 NRC sequence if an RequestUpload request is received if Upload is already active (A previous RequestUpload which is not yet completed with a successful RequestTransferExit request. A negative response with NRC set to 0x22 (ConditionsNotCorrect) shall be send in this case. If the parameter is set to TRUE, no sequence check will be done if a RequestUpload request is received when upload is already active.If the parameter is left unconfigured ,then FALSE will be considered as the default value and therefore the sequence check will be enabled.
Multiplicity	0..1
Default Value	false
Type	ECUC-BOOLEAN-PARAM-DEF

2.3.1.293 Dcm/DcmConfigSet/DcmDsp/DcmRbDspSesSecUsedInProtocol

Short Name	DcmRbDspSesSecUsedInProtocol

Description	The DcmRbDspSesSecUsedInProtocol is a reference container to DcmDslProtocolRow which uses the Dsp Session and Security Configuration.Configure the protocol that will be using the DcmDspSession and DcmDspSecurity Services. If configured as UDS: UDS protocol shall use the DcmDsp session and security configuration. If configured as KWP: KWP protocol shall use the DcmDsp session and security configuration.
Multiplicity	0..1
Default Value	UDS
Enumeration Literals	UDS, KWP
Type	ECUC-ENUMERATION-PARAM-DEF

2.3.1.294 Dcm/DcmConfigSet/DcmDsp/DcmRbDspTransferDataMaxLenChkRequired

Short Name	DcmRbDspTransferDataMaxLenChkRequired
Description	Parameter which decides whether the length for TransferData requests/ response following a RequestDownload / RequestUpload needs to be checked against the maxNumberOfBlockLength passed in the RequestUpload/RequestDownload response.If the parameter is configured as TRUE or left unconfigured , Dcm will check if the length for TransferData requests/ response following a RequestDownload / RequestUpload is equal to the parameter maxNumberOfBlockLength passed in response of RequestDownload/RequestUpload service.If the parameter is configured as FALSE, Dcm will not check the length against maxNumberOfBlock-Length passed in the RequestUpload/RequestDown-load response
Multiplicity	0..1
Default Value	true
Type	ECUC-BOOLEAN-PARAM-DEF

2.3.1.295 Dcm/DcmConfigSet/DcmProcessingConditions/DcmModeCondition/DcmRbSwcSRDataElementValue

Short Name	DcmRbSwcSRDataElementValue
Description	This Parameter is used as constant value of type numerical.This value is used for comparison agsinst DcmSRDataElementRef.Enter the value in decimal eg:5,11.
Multiplicity	0..1

Default Value	-
Type	ECUC-INTEGER-PARAM-DEF

2.3.1.296 Dcm/DcmGeneral/DcmRbGeneral

Short Name	DcmRbGeneral
Description	This container contains the configuration (parameters) for Component wide parameters
Multiplicity	1..1
Default Value	-
Type	ECUC-PARAM-CONF-CONTAINER-DEF

2.3.1.297 Dcm/DcmGeneral/DcmRbGeneral/DcmRbApplCallRequiredOnReqReception

Short Name	DcmRbApplCallRequiredOnReqReception
Description	Parameter to enable/disable the invocation of the application function from Dcm StartofReception/ CopyRxData/RxIndication when a request is received at the Dcm. Autosar stack already ensures with Autosar 4.0 that every N-PDU is copied immediately to Application buffer (DSL/DSD). This parameter enables the feature which will provide an option for the Customer Teams to start working the receiving data (DSP) even before the reception is completed. This is achieved through the call-outs DcmAppl_ StartOfReception, DcmAppl_CopyRxData and DcmAppl_TpRxIndication from Dcm which can provide the buffer pointer, total length, currently received length etc.. If set to TRUE, then DcmAppl_ StartOfReception, DcmAppl_CopyRxData and DcmAppl_TpRxIndication will be called from Dcm. If set to FALSE or left unconfigured, then the mentioned functions will not be called from Dcm
Multiplicity	0..1
Default Value	false
Type	ECUC-BOOLEAN-PARAM-DEF

2.3.1.298 Dcm/DcmGeneral/DcmRbGeneral/DcmRbApplTxConfirmationRequiredAlways

Short Name	DcmRbApplTxConfirmationRequiredAlways

Description	Parameter to enable the Tx confirmation call for application. If the parameter is set to TRUE, then confirmation to the application after transmission of response will be provided even if service is located inside DSP. Also confirmation to the application will be sent in case of suppression of responses in case ApplTxConfirmationRequiredAlways is TRUE. If ApplTxConfirmationRequiredAlways is TRUE, Dcm Appl API DcmAppl_DcmConfirmation will be called from Dcm.
Multiplicity	0..1
Default Value	true
Type	ECUC-BOOLEAN-PARAM-DEF

2.3.1.299 Dcm/DcmGeneral/DcmRbGeneral/DcmRbDcmClientSIDTable

Short Name	DcmRbDcmClientSIDTable
Description	Reference to a DcmDsdServiceTable that is used for diagnostic request processing for this protocol via DcmClient. A reference to a DcmDsdServiceTable containing Dcm internal services has to be configured here. This is needed for the DcmClient feature where the actual DcmDsdServiceTable referenced from the DcmDslProtocolRow→DcmDslProtocolSIDTable contains external services, which will be called by Dcm when a request is received on thet protocol. The external service implementation will use the callback Dcm_DcmClientServiceHandler() to call the Dcm internal services configured in DcmRbDcmClientSIDTable to collect the data from the Host ECU and then will merge that with the data from the Client ECU and send as a final response via the external service handler.
Multiplicity	0..1
Default Value	-
Type	ECUC-REFERENCE-DEF

2.3.1.300 Dcm/DcmGeneral/DcmRbGeneral/DcmRbDemSREnabled

Short Name	DcmRbDemSREnabled
Description	Enable / Disable Generation of Sender-Receiver Interfaces for Dem
Multiplicity	0..1
Default Value	false
Type	ECUC-BOOLEAN-PARAM-DEF

2.3.1.301 Dcm/DcmGeneral/DcmRbGeneral/DcmRbDiagAdaptSupport

Short Name	DcmRbDiagAdaptSupport
Description	Preprocessor switch to enable or disable rba_DiagAdapt Support in DCM.This is used to support vendor specific MEDC17 services along with DCM.
Multiplicity	0..1
Default Value	false
Type	ECUC-BOOLEAN-PARAM-DEF

2.3.1.302 Dcm/DcmGeneral/DcmRbGeneral/DcmRbDspSecurityStoreDelay-CountAndTimerOnJump

Short Name	DcmRbDspSecurityStoreDelayCountAndTimerOnJump
Description	Parameter to allow the user to decide on whether Delay-Count and DelayTimer values needs to be stored during Boot Loader interactions. If set to TRUE, DelayCount and DelayTimer values shall be stored during Boot Loader interactions. If set to FALSE, DelayCount and DelayTimer values shall not be stored during Boot Loader interac-tions. If the parameter is left unconfigured, DelayCount and DelayTimer values shall not be stored during Boot Loader interactions.
Multiplicity	0..1
Default Value	false
Type	ECUC-BOOLEAN-PARAM-DEF

2.3.1.303 Dcm/DcmGeneral/DcmRbGeneral/DcmRbEcuResetTime

Short Name	DcmRbEcuResetTime
Description	The parameter provides the time taken for reset of the ECU and it is relevant only in case of bootloader jump. If the parameter is configured and during jump from boot using Dsp supported EcuReset service(0x11) or DiagnosticSessionControl service(0x10), if the time configured is greater than the remaining P2/P2* time ,Dcm shall again trigger a wait pend response and shall wait for a positive confirmation before triggering function to perform the reset. The parameter if configured should be in multiples of DcmTaskTime and shoule be less than the P2*Max configured for the protocol used for jump to bootloader.

Introduction	The AUTOSAR configuration standard is to use SI units, so this parameter is defined as float value in seconds. DCM configuration tools must convert this float value to the appropriate value format for the use in the software implementation of DCM. A negative value is not allowed
Multiplicity	0..1
Value Range	0..100
Default Value	-
Type	ECUC-FLOAT-PARAM-DEF

2.3.1.304 Dcm/DcmGeneral/DcmRbGeneral/DcmRbMaxRequestsInParallel

Short Name	DcmRbMaxRequestsInParallel
Description	Maximum number of requests allowed to be processed in parallel in Default session. If the configured value is equal to 1 then parallel protocol processing is not enabled as only one protocol can be active at a time and protocols are preempted based on protocol priority. If configured value is greater than 1 then parallel protocol processing is enabled, and separate resources are allocated for each protocol and requests can be processed in parallel without preemption. By default the value is 1. Currently the maximum value is restricted to 2 and can be extended in the future. Features like Parallel Obd Uds, Queuing and Page buffer won't be supported when this parameter is configured to 2.
Multiplicity	0..1
Value Range	1..2
Default Value	1
Type	ECUC-INTEGER-PARAM-DEF

2.3.1.305 Dcm/DcmGeneral/DcmRbGeneral/DcmRbOSCounterId

Short Name	DcmRbOSCounterId
Description	The counterID is used by OStimer.The user shall configure the parameter as a string, which would be the name of the counter in the AUTOSAR OS
Multiplicity	0..1
Default Value	-
Type	ECUC-STRING-PARAM-DEF

2.3.1.306 Dcm/DcmGeneral/DcmRbGeneral/DcmRbOSCounterMacro

Short Name	DcmRbOSCounterMacro
Description	Macro used to calculate the current system time (Ticks)
Multiplicity	0..1
Default Value	-
Type	ECUC-STRING-PARAM-DEF

2.3.1.307 Dcm/DcmGeneral/DcmRbGeneral/DcmRbOSTimerUse

Short Name	DcmRbOSTimerUse
Multiplicity	1..1
Default Value	-
Enumeration Literals	ProjectSpecific, AutosarSpecific, CyclicCount
Type	ECUC-ENUMERATION-PARAM-DEF

Description: Choose between Project Specific , AUTOSAR Specific and Cyclic Count Project Specific : CPU timer will be used for timer monitoring. Corresponding Project Specific OScounterMacro and OScounterID has to be configured with this option Project can make necessary changes in file os_bsw.h file which will be included when project specific is chosen AUTOSAR Specific : CPU timer will be used for timer monitoring. Corresponding AUTOSAR Specific OScounterMacro and OScounterID has to be configured with this option Autosar specific needs to be selected if the project wants to use mcu counter. Project can make necessary changes in file Dcm_Cfg_Timer.h file which will be included when autosar specific is chosen Cyclic Count: The Dcm scheduled cycle time will be used as reference for calculating the time. Cyclic Count E.g : If Dcm_mainfunction is scheduled in 5ms rastor & is called twice; The total time elapsed is calculated as Dcm task time X 2 Cyclic Count : With this option , there is no need to configure OScounterID & OScounterMAcro

2.3.1.308 Dcm/DcmGeneral/DcmRbGeneral/DcmRbParallelProcessingAllowed

Short Name	DcmRbParallelProcessingAllowed
Description	If Configured - TRUE , then separate resources are allocated for OBD and non-OBD execution and the requests from these two protocols are handled separately without Pre-Emption. If set to true, pduRCancelTransmit and pduRCancelReceive should be enabled. If Configure - FALSE . Then only one Protocol is active at a time and Protocols are Pre-Empted based on Protocol Priority

Multiplicity	0..1
Default Value	false
Type	ECUC-BOOLEAN-PARAM-DEF

2.3.1.309 Dcm/DcmGeneral/DcmRbGeneral/DcmRbQueuingOfRequestEnabled

Short Name	DcmRbQueuingOfRequestEnabled
Description	Parameter to enable/disable the queuing of request reception, which is received in parallel to a running request processing. If set to TRUE, Queuing of request shall be enabled in Dcm. The queuing shall be done with a depth of 1 queued buffer returning FC_CTS to the tester. If the third request arrives, BUFREQ_E_NOT_OK shall be returned to the lower layer and the request shall be dropped. If set to FALSE or left unconfigured, then Queuing of request shall be disabled in Dcm.
Multiplicity	0..1
Default Value	-
Type	ECUC-BOOLEAN-PARAM-DEF

2.3.1.310 Dcm/DcmGeneral/DcmRbGeneral/DcmRbRTEsupportsPRportFunctionality

Short Name	DcmRbRTEsupportsPRportFunctionality
Description	Enables generation of PR port (incase conditions of generation of said Port are met) provided the RTE component used supports full PR port functionality.
Multiplicity	1..1
Default Value	false
Type	ECUC-BOOLEAN-PARAM-DEF

2.3.1.311 Dcm/DcmGeneral/DcmRbGeneral/DcmRbRespondToRequestAfterEcuReset

Short Name	DcmRbRespondToRequestAfterEcuReset

Description	The parameter is used to enable/disable the processing of new tester requests after the successful processing of Dsp EcuReset Service(0x11) request and before the actual reset happens. TRUE - DCM shall respond to further requests after sending the positive response of EcuReset (call of Dcm_ TpTxConfirmation) during reset- processing FALSE - DCM shall ignore all further requests during reset-processing. If the parameter is left unconfig-ured ,the default value (TRUE)will be taken.
Multiplicity	0..1
Default Value	true
Type	ECUC-BOOLEAN-PARAM-DEF

2.3.1.312 Dcm/DcmGeneral/DcmRbGeneral/DcmRbRestoringEnabled

Short Name	DcmRbRestoringEnabled
Description	Enable / Disable restoring of Diagnostic information (Jump from BootLoader)
Multiplicity	0..1
Default Value	false
Type	ECUC-BOOLEAN-PARAM-DEF

2.3.1.313 Dcm/DcmGeneral/DcmRbGeneral/DcmRbSignalDefaultValue

Short Name	DcmRbSignalDefaultValue
Description	Default value to be configured for Signal Data with gaps.The value config-ured shall be used to fill the gaps if existing between signal data of a DID.The default value of this parameter is 0.
Multiplicity	0..1
Value Range	0..255
Default Value	0
Type	ECUC-INTEGER-PARAM-DEF

2.3.1.314 Dcm/DcmGeneral/DcmRbGeneral/DcmRbSingleBufferConceptDis-abled

Short Name	DcmRbSingleBufferConceptDisabled
Multiplicity	0..1
Default Value	false
Type	ECUC-BOOLEAN-PARAM-DEF

Description: Parameter to enable/disable the concept of single buffer sharing for

reception and transmission. If set to TRUE, single buffer concept shall be disabled and a separate reception and transmission buffer will be assigned to the protocol. If set to FALSE,single buffer concept shall be enabled and a single buffer will be shared for reception and transmission of protocols. If the parameter is left unconfigured, the single buffer concept shall be used.

In case Protocol Preemption is enabled (through the configuration of multiple protocols with different DcmDslProtocolPriority) , it is recommended to configure separate Buffers for Reception and Transmission to avoid data corruption.

Possible use case : Low Prio request is in progress, a high priority request is received as an interrupt. The request buffer is over written by the response of the low prio protocol. This scenario can occur when the control is in the service and the response is yet to be filled/ being filled and a SOR for a high prio request is received. After the RxIndication is processed and the control is given back to the service, with a single buffer concept, the request can potentially be over written by the response which will inturn lead to an incorrect final response for the high priority request.

2.3.1.315 Dcm/DcmGeneral/DcmRbGeneral/DcmRbStoringEnabled

Short Name	DcmRbStoringEnabled
Description	Enable / Disable storing of Diagnostic information (Jump to BootLoader)
Multiplicity	0..1
Default Value	false
Type	ECUC-BOOLEAN-PARAM-DEF

2.3.1.316 Dcm/DcmGeneral/DcmRbGeneral/DcmRbSwApplComponent-NameGeneration

Short Name	DcmRbSwApplComponentNameGeneration
Description	Enable / Disable generation of application Sw Component names in Flatview File(or Assembly connection file).This is mandatory when dcmDspExternal SRDataElementClass is configured.
Multiplicity	0..1
Default Value	-
Type	ECUC-BOOLEAN-PARAM-DEF

2.3.1.317 Dcm/DcmGeneral/DcmRbGeneral/DcmRbSwCompositionPath

Short Name	DcmRbSwCompositionPath

Description	This Parameter is used to provide path to SwComposition in Flatview File.This is mandatory when dcmDspExternalSRDataElementClass is configured. e.g RB/PT/PCT_XXX/SwComponentTypes/XXX. Also the last name in the string(in given example "XXX") will be considered to generate the name of the flatview file e.g Dcm_Cfg_SWCD_XXX_Connectors.arxml
Multiplicity	0..1
Default Value	-
Type	ECUC-STRING-PARAM-DEF

2.3.1.318 Dcm/DcmGeneral/DcmRbGeneral/DcmRbSwDcmComponentName

Short Name	DcmRbSwDcmComponentName
Description	This Parameter is used to provide name of Dcm software component in flatview file(or Assembly connection file).TThis is mandatory when dcmDspExternalSRDataElementClass is configured.
Multiplicity	0..1
Default Value	-
Type	ECUC-STRING-PARAM-DEF

2.3.1.319 Dcm/DcmGeneral/DcmRbGeneral/DcmRbSwDcmComponentNameGeneration

Short Name	DcmRbSwDcmComponentNameGeneration
Description	Enable / Disable generation of Dcm Component name in Flatview File(or Assembly connection file).This is mandatory when dcmDspExternal SRDataElementClass is configured.
Multiplicity	0..1
Default Value	-
Type	ECUC-BOOLEAN-PARAM-DEF

2.3.1.320 Dcm/DcmGeneral/DcmRbGeneral/DcmRbVINMaxNumRespPend

Short Name	DcmRbVINMaxNumRespPend

Description	This parameter is introduced to fetch the Vehicle Identification Number(VIN) from VIN DID, When this parameter is configured, DCM will try to fetch the VIN data until VIN wait pend count reaches the configured range. If this parameter is not configured, then DcmDslDiagRespMaxNumRespPend parameter value will be considered as default value.
Introduction	DCM will wait to fetch VIN DID data till the configured time elapses.
Multiplicity	0..1
Value Range	0..255
Default Value	-
Type	ECUC-INTEGER-PARAM-DEF

2.3.1.321 Dcm/DcmGeneral/DcmRbOBDonUDSEnabled

Short Name	DcmRbOBDonUDSEnabled
Description	If this parameter is set to true, OBDonUDS protocol is enabled. If this parameter is not configured or set to false, OBDonUDS protocol is disabled.
Multiplicity	0..1
Default Value	false
Type	ECUC-BOOLEAN-PARAM-DEF

2.3.1.322 Dcm/DcmGeneral/DcmSecurityEventRefs/DIAG_SEV_AUTHENTICATION_SUCCESSFUL

Short Name	DIAG_SEV_AUTHENTICATION_SUCCESSFUL
Description	Successfully authenticated (via Authentication $29) Tags: atp.Status=draft
Multiplicity	0..1
Default Value	-
Type	ECUC-REFERENCE-DEF

2.3.1.323 Dcm/DcmGeneral/DcmSecurityEventRefs/DIAG_SEV_CERTIFICATE_FAILURE

Short Name	DIAG_SEV_CERTIFICATE_FAILURE
Description	The tester tried an authentication with an invalid certificate. Tags: atp.Status=draft
Multiplicity	0..1
Default Value	-

Type	ECUC-REFERENCE-DEF

2.3.1.324 Dcm/DcmGeneral/DcmSecurityEventRefs/DIAG_SEV_-CLEAR_DTC_SUCCESSFUL

Short Name	DIAG_SEV_CLEAR_DTC_SUCCESSFUL
Description	DTC information has been cleared by SID 0x14 ClearDTCInformation Tags: atp.Status=draft
Multiplicity	0..1
Default Value	-
Type	ECUC-REFERENCE-DEF

2.3.1.325 Dcm/DcmGeneral/DcmSecurityEventRefs/DIAG_SEV_COMMUNI-CATION_CONTROL_SWITCHED_OFF

Short Name	DIAG_SEV_COMMUNICATION_CONTROL_SWITCHED_OFF
Description	ECU communication was switched off with service CommunicationControl. Tags: atp.Status=draft
Multiplicity	0..1
Default Value	-
Type	ECUC-REFERENCE-DEF

2.3.1.326 Dcm/DcmGeneral/DcmSecurityEventRefs/DIAG_SEV_DTC_SET-TING_SWITCHED_OFF

Short Name	DIAG_SEV_DTC_SETTING_SWITCHED_OFF
Description	DTC setting switched off with service 0x85 (ControlDTCSetting). Tags: atp.Status=draft
Multiplicity	0..1
Default Value	-
Type	ECUC-REFERENCE-DEF

2.3.1.327 Dcm/DcmGeneral/DcmSecurityEventRefs/DIAG_SEV_ECU_RESET

Short Name	DIAG_SEV_ECU_RESET
Description	ECU had been reset by SID 0x11 ResetECU. Tags: atp.Status=draft
Multiplicity	0..1
Default Value	-
Type	ECUC-REFERENCE-DEF

2.3.1.328 Dcm/DcmGeneral/DcmSecurityEventRefs/DIAG_SEV_ECU_UNLOCK_SUCCESSFUL

Short Name	DIAG_SEV_ECU_UNLOCK_SUCCESSFUL
Description	Successfully unlocked the ECU (via Security Access $27), per Level Tags: atp.Status=draft
Multiplicity	0..1
Default Value	-
Type	ECUC-REFERENCE-DEF

2.3.1.329 Dcm/DcmGeneral/DcmSecurityEventRefs/DIAG_SEV_INCORRECT_MESSAGE_LENGTH_OR_FORMAT

Short Name	DIAG_SEV_INCORRECT_MESSAGE_LENGTH_OR_FORMAT
Description	Received request message does not match the specified length or the format of the parameters is different from the specified ones for that service. NRC 0x13 (incorrectMessageLengthOrInvalidFormat) was returned. Tags: atp.Status=draft
Multiplicity	0..1
Default Value	-
Type	ECUC-REFERENCE-DEF

2.3.1.330 Dcm/DcmGeneral/DcmSecurityEventRefs/DIAG_SEV_NUMBER_OF_FAILED_AUTHENTICATION_ATTEMPTS_EXCEEDED

Short Name	DIAG_SEV_NUMBER_OF_FAILED_AUTHENTICATION_ATTEMPTS_EXCEEDED
Description	Number of failed authentication attempts - OEM specific Tags: atp.Status=draft
Multiplicity	0..0
Default Value	-
Type	ECUC-REFERENCE-DEF

2.3.1.331 Dcm/DcmGeneral/DcmSecurityEventRefs/DIAG_SEV_REQUESTED_ACTIONS_REQUIRES_AUTHENTICATION

Short Name	DIAG_SEV_REQUESTED_ACTIONS_REQUIRES_AUTHENTICATION

Description	A diagnostic request was received while the required authentication to execute this service is not given. NRC 0x34 (authenticationRequired) was returned. Tags: atp.Status=draft
Multiplicity	0..1
Default Value	-
Type	ECUC-REFERENCE-DEF

2.3.1.332 Dcm/DcmGeneral/DcmSecurityEventRefs/DIAG_SEV_REQUEST_-DOWNLOAD

Short Name	DIAG_SEV_REQUEST_DOWNLOAD
Description	A diagnostic data download sequence was requested with SID 0x34. Tags: atp.Status=draft
Multiplicity	0..1
Default Value	-
Type	ECUC-REFERENCE-DEF

2.3.1.333 Dcm/DcmGeneral/DcmSecurityEventRefs/DIAG_SEV_RE-QUEST_OUT_OF_RANGE

Short Name	DIAG_SEV_REQUEST_OUT_OF_RANGE
Description	A diagnostic request with a parameter that is out of range of the allowed bounds was received. A NRC 0x31 (requestOutOfRange) was returned. Tags: atp.Status=draft
Multiplicity	0..1
Default Value	-
Type	ECUC-REFERENCE-DEF

2.3.1.334 Dcm/DcmGeneral/DcmSecurityEventRefs/DIAG_SEV_RE-QUEST_SEQUENCE_ERROR

Short Name	DIAG_SEV_REQUEST_SEQUENCE_ERROR
Description	A diagnostic service was received in the wrong sequence order. NRC 0x24 (requestSequenceError) was returned. Tags: atp.Status=draft
Multiplicity	0..1
Default Value	-
Type	ECUC-REFERENCE-DEF

2.3.1.335 Dcm/DcmGeneral/DcmSecurityEventRefs/DIAG_SEV_SECURI-

TY_ACCESS_DENIEDDIAG_SEV_SECURITY_ACCESS_DENIED

Short Name	DIAG_SEV_SECURITY_ACCESS_DENIED
Description	Tester has sent an diagnostic request without meeting the servers security level requirements for that service. NRC 0x33 (securityAccessDenied) was returned. Tags: atp.Status=draft
Multiplicity	0..1
Default Value	-
Type	ECUC-REFERENCE-DEF

2.3.1.336 Dcm/DcmGeneral/DcmSecurityEventRefs/DIAG_SEV_SECURITY_ACCESS_INVALID_KEY

Short Name	DIAG_SEV_SECURITY_ACCESS_INVALID_KEY
Description	Security Access failed because the tester has sent an invalid key to the server. An NRC 0x35 (invalidKey) was returned. Tags: atp.Status=draft
Multiplicity	0..1
Default Value	-
Type	ECUC-REFERENCE-DEF

2.3.1.337 Dcm/DcmGeneral/DcmSecurityEventRefs/DIAG_SEV_SECURITY_ACCESS_NUMBER_OF_ATTEMPTS_EXCEEDED

Short Name	DIAG_SEV_SECURITY_ACCESS_NUMBER_OF_ATTEMPTS_EXCEEDED
Description	Security Access failed and the number of invalid attempts has been exceeded. NRC 0x36 (exceedNumberOfAttempts) was returned. Tags: atp.Status=draft
Multiplicity	0..1
Default Value	-
Type	ECUC-REFERENCE-DEF

2.3.1.338 Dcm/DcmGeneral/DcmSecurityEventRefs/DIAG_SEV_SECURITY_ACCESS_REQUIRED_TIME_DELAY_NOT_EXPIRED

Short Name	DIAG_SEV_SECURITY_ACCESS_REQUIRED_TIME_DELAY_NOT_EXPIRED
Description	A further attempt to unlock the ECU was taken before the delay time had expired. NRC 0x37 (requiredTimeDelayNotExpired) was sent. Tags: atp.Status=draft
Multiplicity	0..1

Default Value	-
Type	ECUC-REFERENCE-DEF

2.3.1.339 Dcm/DcmGeneral/DcmSecurityEventRefs/DIAG_SEV_SERVICE_NOT_SUPPORTED

Short Name	DIAG_SEV_SERVICE_NOT_SUPPORTED
Description	A diagnostic service was requested that is not supported or not supported in the active session. NRC 0x11 (serviceNotSupported) or NRC 0x7F (serviceNotSupportedInActiveSession) was returned. Tags: atp.Status=draft
Multiplicity	0..1
Default Value	-
Type	ECUC-REFERENCE-DEF

2.3.1.340 Dcm/DcmGeneral/DcmSecurityEventRefs/DIAG_SEV_SUBFUNCTION_NOT_SUPPORTED

Short Name	DIAG_SEV_SUBFUNCTION_NOT_SUPPORTED
Description	A subfunction of a diagnostic service was requested that is not supported for the requested service. Tags: atp.Status=draft
Multiplicity	0..1
Default Value	-
Type	ECUC-REFERENCE-DEF

2.3.1.341 Dcm/DcmGeneral/DcmSecurityEventRefs/DIAG_SEV_WRITE_DATA

Short Name	DIAG_SEV_WRITE_DATA
Description	Diagnostic data identifier has been written by SID 0x2E. Tags: atp.Status=draft
Multiplicity	0..1
Default Value	-
Type	ECUC-REFERENCE-DEF

2.3.1.342 Dcm/DcmGeneral/DcmSecurityEventRefs/DIAG_SEV_WRITE_INV_DATA

Short Name	DIAG_SEV_WRITE_INV_DATA

Description	Tester writes invalid data with service WriteDataByIdentifier. Tags: atp.Status=draft
Multiplicity	0..1
Default Value	-
Type	ECUC-REFERENCE-DEF

2.3.2 Additional Services

The following API are available in version v12.7.0 in addition to those defined in AR v22-11

Syntax	`Dcm_Dsp_RestoreDelayCount`
Description	Restores the Delay count values for each security level configured in the Security Access service.

Syntax	`Dcm_Dsp_SecaPowerOnDelayIni`
Description	Initialisation of PowerOn delay timer for all configured security levels.

Syntax	`Dcm_Dsp_SecaSessIni`
Description	Re-Initialisaton of all security parameters due to session timeout or change in session.

Syntax	`Dcm_GetActiveDid`
Parameter Out	`dataDid_u16` Pointer to a variable for updating of the DID under processing. The DID value returned is valid only if return value is E_OK.
Description	This API will update the normal DID currently being processed by Dsp-supported RDBI/WDBI/IOCBI services. The API can be polled from Read/ConditionCheckRead/ReadDataLength functions during RDBI request processing, from Write function during WDBI request processing and from FreezeCurrentState/ResetToDefault/ShortTermAdjustment/ReturnCon-trolToEcu/Read function during IOCBI request processing. The API can be used to identify the appropriate data requested by the application. If the API is called from anywhere outside the mentioned application APIs while processing the requested services, the DID value may not be correct. If the DID currently under process is a normal DID, E_OK is returned and a valid value is filled in the parameter passed. If the DID currently under process is a range DID, E_NOT_OK is returned.

Syntax	`Dcm_GetActiveProtocolRxBufferSize`
Parameter Out	`rxBufferLength` Address of global variable in which the buffer size should be written.

Description	API to get the Active protocol RX buffer size.

Syntax	`Dcm_GetActiveRid`
Parameter In	`dataRid_u16` Buffer to get the current active RID.
Description	API to get the service table id. This API has to be called from the application callbacks of Start, Stop and Req. Results routine configured for the project. Example scenarios: a) This API has to be called by the application: - a.1) during Start, Stop and Request Results operations for the normal tester request. - a.2) during cancelling the pending operations either Start or Stop or Request Results due to CANCEL operation triggered by Dcm by setting the Opstatus to DCM_CANCEL in the RC Ini function. - a.3) during the stop of the routines due to session transitions (either due to protocol start/stop or session time out/change). By "Active RID", it means the active RID index under processing and not the RIDs of already-started routines.
Return Values	E_OK - E_OK - The parameter dataRid_u16 contains valid RID value. E_NOT_OK - The parameter dataRid_u16 contains invalid data if none of the Routine is under processing.

Syntax	`Dcm_GetActiveServiceTable`
Parameter Out	`ActiveServiceTable` Address of global variable passed by the application.
Description	API to get the service table id.

Syntax	`Dcm_GetActiveSourceDataId`
Parameter In	`dataSrcDid_u16` Parameter for updating of the Source DID under processing. The value returned is valid only if return value is E_OK.
Parameter In	`posnSrcDataRec_u8` Parameter to update the position in the source Data Record for DID under process.
Parameter In	`adrMemSize_u8` Memory size/length of the source DID under progress.
Description	This API will update the Source Data Identifiers sent during a 0x2c request. The API can be polled from Read/ConditionCheckRead/ReadDataLength functions during RDBI request processing. If a DDDI is defined with one or more source DIDs using 0x2C request, when read, this API can be used to determine the current source DID being processed.

Return Values	E_OK - The source DID under processing is a normal DID, and the parameter dataSrcDid_u16 contains valid Source DID value. E_NOT_OK - The DID under processing is a range DID, and the parameter dataDid_u16 contains invalid data.

Syntax	`Dcm_GetControlDTCSetting_EnableStatus`
Description	Function to get the status of control dtc setting. Can be used by other SWCs and BSW modules in case RTE is not supported.

Syntax	`Dcm_GetDIDData`
Parameter InOut	`idxDidIndexType_st` index in Dcm_DIDConfig or Dcm_DIDRangeConfig based on dataRange_b status.
Parameter Out	`targetBuffer` Pointer to the buffer where the data is to be written.
Description	Reads the DID data of the requested DID by calling the corresponding configured function for reading the data. The DID value is not written in the buffer.

Syntax	`Dcm_GetDIDRangeStatus`
Parameter In	`did` The DID for which the length is requested.
Parameter Out	`idxDidIndexType_st` Index of the requested DID in Dcm_DIDRangeConfig structure if it is a range DID.
Description	Calculations to check if the DID is in the DID Range limits.

Syntax	`Dcm_GetIndexOfDID`
Parameter In	`did` The DID for which the index is requested.
Parameter Out	`idxDidIndexType_st` Structure with index in Dcm_DID-Config or Dcm_DIDRangeConfig based on the status of dataRange_b.
Description	Determines the index of DID / Range DID (specified by the IN parameter). Writes the index to the idxIn-dex_u16 element in structure idxDidIndexType_st (the OUT parameter). For a normal DID, the value of index is in Dcm_DIDConfig structure array, and for a Range DID is in Dcm_DIDRangeConfig_cast struc-ture array. The “dataRange_b” element of structure idxDidIndexType_st is set to FALSE if DID is found in the DCM_DIDConfig structure array.

Syntax	`Dcm_GetKwpTimingValues`
Parameter In	`TimerMode` Mode of timing.

Parameter Out	`TimerServerCurrent` Pointer to structure where timings are written by DCM.
Description	API to get the different set of timings.

Syntax	`Dcm_GetLengthOfDIDIndex`
Parameter In	`idxDidIndexType_st` index in Dcm_DIDConfig or Dcm_DIDRangeConfig based on the dataRange_b parameter.
Parameter In	`did` DID in the request.
Parameter Out	`length` Length calculated if E_OK is returned.
Description	Calculates the length of the ID at position index in Dcm_DIDConfig.

Syntax	`Dcm_GetMediumOfActiveConnection`
Parameter Out	`ActiveMediumId` Variable passed by the rba_DiagAdapt to get active Medium information of ComMChannel for the active connection.
Description	API to get Medium information (ie. Can, Flexray, etc.) available in the ComMChannel for the active connection. This is explicitly provided only for rba_DiagAdapt. This should be called by only rba_DiagAdapt when a new protocol request is triggered from the Tester.

Syntax	`Dcm_GetP2Timings`
Parameter In	`dSessionId_u8` Session Id for which timings have to be retrieved.
Parameter Out	`dP2Timing_pu32` Pointer to P2max timings of the session requested.
Parameter Out	`dP2StarTiming_pu32` Pointer to P2Starmax timings of the session requested.
Description	This API is used to get the timings of a particular session. This API is used in the jump to boot feature implementation to retrieve the same session timings.

Syntax	`Dcm_GetSupportOfIndex`
Parameter In	`index` Index in Dcm_DIDConfig.
Parameter In	`direction` Check for read or write support (DCM_SUPPORT_READ,DCM_SUPPORT_WRITE).
Parameter In	`rangestatus` Status to indicate if the DID is configured as a range DID.
Parameter Out	`NegRespCode` Pointer to a Byte in which to store a negative Response code in the event of a detection of an error in the request.

Description	Calculate if the ID at position index in the Dcm_DIDConfig is supported at this point in time or not. Returns the following Values: DCM_SUPPORT_OK - ID is supported. DCM_SUPPORT_SESSION_VIOLATED - ID is not supported in the current session; negative response code is set to NegRespCode. DCM_SUPPORT_SECURITY_VIOLATED - ID is not supported in the current security level; negative response code is set to NegRespCode. DCM_SUPPORT_CONDITION_VIOLATED - ID is not supported as per configured callback function; negative response code is set to NegRespCode. DCM_SUPPORT_CONDITION_PENDING - Checking needs more time. Call again.

Syntax	`Dcm_PrepareKwpTimingValues`
Parameter In	`TimerServerNew` Pointer to structure which timings are to be validated.
Description	API to validate the given set of timings.

Syntax	`Dcm_ProcessingDone`
Parameter In	`pMsgContext` Message-related information for one diagnostic protocol identifier.
Description	When this function is called, a positive response will be sent based on the data contained in pMsgContext. The application does not have to care about any timing requirement to process a request.

Syntax	`Dcm_OBDProcessingDone(const Dcm_MsgContextType* pMsgContext)`
Parameter In	`pMsgContext` Message-related information for one diagnostic protocol identifier.
Description	Initializes the transmission structure before response transmission for OBD.

Syntax	`Dcm_ProcessPage`
Parameter In	`FilledPageLen` Filled data length in current page.
Description	Application requests transmission of filled page.

Syntax	`Dcm_RestartP3timer`
Description	Called by Kline TP to restart P3 timer after getting the first byte of request.

Syntax	`Dcm_RoeSendFinalResponse`
Parameter In	`Dcm_MsgType MsgPtr` Pointer to buffer containing the response.
Parameter In	`Dcm_MsgLenType MsgLen` Length of the response.
Parameter In	`PduIdType DcmRxPduId` Rx pduid on which ROE request is received.

Description	This API will be called by the application to transmit the final response when the window timer expires in the application.

Syntax	`Dcm_SetKwpDefaultTimingValues`
Description	API to set the default timings.

Syntax	`Dcm_SetKwpTimingValues`
Description	API to set the given set of timings.

Syntax	`Dcm_SetNegResponse`
Parameter In	`pMsgContext` Message-related information for one diagnostic protocol identifier.
Parameter In	`ErrorCode` Error code to be set. This error code will be taken as negative response code (NRC).
Description	If this function is called, a negative response will be sent instead of a positive response. The negative response code is taken from the first call. Duplicate calls are allowed but the errorCode transmitted is ignored. The transmission of the negative response will be performed after calling Dcm_ProcessingDone.

Syntax	`Dcm_SetP3MaxMonitoring`
Parameter In	`active` TRUE = P3 monitoring required, FALSE = P3 monitoring not required.
Description	Function called by service to enable the P3 monitoring.

Syntax	`Dcm_StartPagedProcessing`
Parameter In	`pMsgContext` Message context table given by the service.
Description	With this API, the application gives the complete response length to DCM and starts PagedBuffer handling. Complete response length information (in bytes) is given in pMsgContext→ resDataLen. Callback functions are used to provide paged buffer handling in DSP and RTE.

2.4 Limitations

2.4.1 DcmDspSupportedAddressAndLengthFormatIdentifier

The lower and high nibbles of the `DcmDspSupportedAddressAndLengthFormatIdentifier` container must be comprehended between 1 and 4.

2.4.2 DcmDspAddressAndLengthFormatIdentifier

Since the `DcmDspAddressAndLengthFormatIdentifier` is one byte in size, it cannot support more than 256 values.

2.4.3 Communication Control Service (0x28)

Sub-functions 0x04 and 0x05 of the Communication Control Service (0x28) are currently not supported, so Dcm will not expect any Node Identification Number in the request message.

2.4.4 Authentication Service (0x29)

Service 0x29 implements authentication via PKI certificated exchange. Authentication with challenge-response (ACR), and secure diagnostic communication with Diffie-Hellmann key exchange is not supported.

The "Verify target identification" feature is not supported. The same functionality is available in KeyM, so this feature will be removed from Dcm in a future release of Autosar. As a consequence, the related DcmDspAuthenticationTargetIdentificationModeRuleRef parameter is not supported.

The paged buffer functionality is not currently supported.

The port Authentication_<Connection> is available as a ProvidedPort in the RTA-BSW implementation of Dcm. Autosar 19-11 incorrectly specifies it as a RequiredPort, and this will be corrected in a future release of Autosar.

The following configuration parameters specified in Autosar 19-11 have no use case with respect to the requirements. They are not supported in the RTA-BSW implementation of Dcm, and they will be removed in a future release of Autosar.

- DcmDspAuthenticationCertificatePublicKeyStoreJobRef
- DcmDspAuthenticationECUCertificateKeyElementRef
- DcmDspAuthenticationPublicKeyElementRef

2.4.5 PostBuild

Currently only PostBuild selectable is supported in Dcm and is limited to postbuild parameters DcmDslProtocolRowUsed, DcmDsdServiceUsed and DcmDspPidUsed.

If these parameters are not configured then by default these will be considered as supported in all postbuild variants. If any of these parameters are not supported or not needed in any postbuild variant then that parameter should be configured as FALSE for that variant.

NOTE: Functionality associated with DCM_CFG_POSTBUILD_SUPPORT is not supported anymore and will be removed.

If postbuild is enabled, the postbuild variant should be passed to Dcm_Init as the parameter. For the RDPI/ROE protocol, the PostBuild variants must match the referenced protocol. Currently, configuring different PostBuild variants for the

RDPI protocol, as opposed to the referenced protocol, is not supported.

If postbuild is disabled, NULL_PTR should be passed to Dcm_Init as the parameter.

2.4.6 Co-Operative Scheduling

Dcm does not provide the option to use Co-Operative scheduling. Services can only be scheduled with the Dcm Main function scheduler.

2.4.7 ControlDTCSetting

ControlOptionRecords other than 0xFFFFFF are no longer supported, so only all DTC events can be enabled/disabled.

2.4.8 Asynchronous Server Point Handling

Dcm supports Asynchronous Server Point handling for DidData, SecurityAccess and Routine Control Interfaces, but this feature does not include Range DIDs.

2.4.9 Atomic NV Data Interface port

The functionality for DcmDspDidUsePort USE_ATOMIC_NV_DATA_INTERFACE is not yet fully supported.

2.4.10 DcmDspExternalSRDataElementClass for Atomic DIDs

DcmDspExternalSRDataElementClass is not supported for Atomic DIDs (i.e. DIDs having DcmDspDidUsePort as USE_ATOMIC_SENDER_RECEIVER_INTERFACE or USE_ATOMIC_SENDER_RECEIVER_INTERFACE_AS_SERVICE or USE_ATOMIC_NV_DATA_INTERFACE).

An individual reference is needed for each DcmDspExternalSRDataElementClass enabled Data element, and so atomic behaviour (similar behaviour for all sub-containers) at DID level cannot be implemented.

2.4.11 DcmDspDiagnosisScaling for Atomic DIDs

DcmDspDiagnosisScaling is not supported for Atomic DIDs (i.e. DIDs having DcmDspDidUsePort as USE_ATOMIC_SENDER_RECEIVER_INTERFACE or USE_ATOMIC_SENDER_RECEIVER_INTERFACE_AS_SERVICE or USE_ATOMIC_NV_DATA_INTERFACE).

For Atomic DIDs, the generated Structure is an Implementation DataType, but a reference to ApplicationDataType is necessary for DiagnosisScaling-enabled DataElements, and this is not possible for ImplementationDataTypes.

2.4.12 Generic Connections

Dcm supports Generic Connections to reduce the number of DcmDslConnections per protocol. Generic Connections are currently not supported for ROE or RDPI.

2.4.13 Parallel Diagnostic Processing

Parallel Obd Uds, Queuing, Page buffer, SharedPdu, KWP protocol and Periodic Transmission are not currently supported when the parallel protocol processing is enabled (DcmRbMaxRequestsInParallel is 2).

2.5 Exclusions

The following features from AR v22-11 are not supported in version v12.7.0.

API functions not currently implemented:

- `Dcm_BndMWriteBlockFinish`

The following configuration parameters from AR v22-11 are not supported in version v12.7.0.

- Dcm/DcmConfigSet/DcmDsd/DcmDsdServiceTable/DcmDsdService/DcmDsdServiceRoleRef
- Dcm/DcmConfigSet/DcmDsd/DcmDsdServiceTable/DcmDsdService/DcmDsdSidTabSecurityLevelRef
- Dcm/DcmConfigSet/DcmDsd/DcmDsdServiceTable/DcmDsdService/DcmDsdSidTabSessionLevelRef
- Dcm/DcmConfigSet/DcmDsd/DcmDsdServiceTable/DcmDsdService/DcmDsdSubService/DcmDsdSubServiceRoleRef
- Dcm/DcmConfigSet/DcmDsd/DcmDsdServiceTable/DcmDsdService/DcmDsdSubService/DcmDsdSubServiceSecurityLevelRef
- Dcm/DcmConfigSet/DcmDsd/DcmDsdServiceTable/DcmDsdService/DcmDsdSubService/DcmDsdSubServiceSessionLevelRef
- Dcm/DcmConfigSet/DcmDsd/DcmDsdServiceTable/DcmDsdService/DcmDsdSubService/DcmDsdSubServiceUsed
- Dcm/DcmConfigSet/DcmDsp/DcmDspAuthentication/DcmDspAuthenticationConnection/DcmDspAuthenticationCertificatePublicKeyStoreJobRef
- Dcm/DcmConfigSet/DcmDsp/DcmDspAuthentication/DcmDspAuthenticationConnection/DcmDspAuthenticationECUCertificateKeyElementRef
- Dcm/DcmConfigSet/DcmDsp/DcmDspAuthentication/DcmDspAuthenticationConnection/DcmDspAuthenticationPublicKeyElementRef
- Dcm/DcmConfigSet/DcmDsp/DcmDspAuthentication/DcmDspAuthenticationDeauthenticatedRoleRef
- Dcm/DcmConfigSet/DcmDsp/DcmDspAuthentication/DcmDspAuthenticationTransmitCertificate/DcmDspAuthenticationTransmitCertificateFunction
- Dcm/DcmConfigSet/DcmDsp/DcmDspComControl/DcmDspComControlAllChannel/DcmDspComControlAllChannelUsed
- Dcm/DcmConfigSet/DcmDsp/DcmDspComControl/DcmDspComControlSpecificChannel/DcmDspComControlSpecificChannelUsed

- Dcm/DcmConfigSet/DcmDsp/DcmDspCommonAuthorization/DcmDspCommonAuthorizationSecurityLevelRef
- Dcm/DcmConfigSet/DcmDsp/DcmDspCommonAuthorization/DcmDspCommonAuthorizationSessionRef
- Dcm/DcmConfigSet/DcmDsp/DcmDspData/DcmDspDataEcuSignal
- Dcm/DcmConfigSet/DcmDsp/DcmDspData/DcmDspDataGetScalingInfoFnc
- Dcm/DcmConfigSet/DcmDsp/DcmDspData/DcmDspDataInfoRef
- Dcm/DcmConfigSet/DcmDsp/DcmDspData/DcmDspDataReadEcuSignal
- Dcm/DcmConfigSet/DcmDsp/DcmDspData/DcmDspDiagnosisScaling/DcmDspAlternativeDataInterface
- Dcm/DcmConfigSet/DcmDsp/DcmDspData/DcmDspDiagnosisScaling/DcmDspAlternativeDataInterface/DcmDataElement
- Dcm/DcmConfigSet/DcmDsp/DcmDspData/DcmDspDiagnosisScaling/DcmDspAlternativeDataInterface/DcmNestedDataElement
- Dcm/DcmConfigSet/DcmDsp/DcmDspData/DcmDspDiagnosisScaling/DcmDspAlternativeDataInterface/DcmPortInterfaceMapping
- Dcm/DcmConfigSet/DcmDsp/DcmDspData/DcmDspDiagnosisScaling/DcmDspAlternativeDataProps/DcmDspTextTableMapping
- Dcm/DcmConfigSet/DcmDsp/DcmDspData/DcmDspDiagnosisScaling/DcmDspAlternativeDataProps/DcmDspTextTableMapping/DcmDspDiagnosisRepresentationDataValue
- Dcm/DcmConfigSet/DcmDsp/DcmDspData/DcmDspDiagnosisScaling/DcmDspAlternativeDataProps/DcmDspTextTableMapping/DcmDspInternalDataValue
- Dcm/DcmConfigSet/DcmDsp/DcmDspData/DcmDspDiagnosisScaling/DcmDspAlternativeDataType/DcmDspTextTableMapping
- Dcm/DcmConfigSet/DcmDsp/DcmDspData/DcmDspDiagnosisScaling/DcmDspAlternativeDataType/DcmDspTextTableMapping/DcmDspDiagnosisRepresentationDataValue
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- Dcm/DcmConfigSet/DcmDsp/DcmDspData/DcmDspDiagnosisScaling/DcmDspAlternativeDiagnosticDataElement
- Dcm/DcmConfigSet/DcmDsp/DcmDspData/DcmDspDiagnosisScaling/DcmDspAlternativeDiagnosticDataElement/DcmDspDiagnosticDataElementRef
- Dcm/DcmConfigSet/DcmDsp/DcmDspData/DcmDspOdxDataDescription
- Dcm/DcmConfigSet/DcmDsp/DcmDspDataInfo
- Dcm/DcmConfigSet/DcmDsp/DcmDspDataInfo/DcmDspDataScalingInfoSize
- Dcm/DcmConfigSet/DcmDsp/DcmDspDid/DcmDspDidBndMBlockIdRef

- Dcm/DcmConfigSet/DcmDsp/DcmDspDid/DcmDspDidRef
- Dcm/DcmConfigSet/DcmDsp/DcmDspDid/DcmDspDidSignal/DcmDsp-DataByteSize
- Dcm/DcmConfigSet/DcmDsp/DcmDspDid/DcmDspDidSignal/DcmDspDidSig-nalCompositeRef
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- Dcm/DcmConfigSet/DcmDsp/DcmDspDid/DcmDspDidSignalCompositePool/DcmDspDidSignal
- Dcm/DcmConfigSet/DcmDsp/DcmDspDid/DcmDspDidSignalCompositePool/DcmDspDidSignal/DcmDspDataByteSize
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- Dcm/DcmConfigSet/DcmDsp/DcmDspDid/DcmDspDidSignalCompositePool/DcmDspDidSignal/DcmDspDidSignalCompositeRef
- Dcm/DcmConfigSet/DcmDsp/DcmDspDid/DcmDspDidUsed
- Dcm/DcmConfigSet/DcmDsp/DcmDspDidInfo/DcmDspDidControl/DcmDsp-DidControlRoleRef
- Dcm/DcmConfigSet/DcmDsp/DcmDspDidInfo/DcmDspDidControl/DcmDsp-DidControlSecurityLevelRef
- Dcm/DcmConfigSet/DcmDsp/DcmDspDidInfo/DcmDspDidControl/DcmDsp-DidControlSessionRef
- Dcm/DcmConfigSet/DcmDsp/DcmDspDidInfo/DcmDspDidRead/DcmDsp-DidReadRoleRef
- Dcm/DcmConfigSet/DcmDsp/DcmDspDidInfo/DcmDspDidRead/DcmDsp-DidReadSecurityLevelRef
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- Dcm/DcmConfigSet/DcmDsp/DcmDspEcuReset/DcmDspEcuResetRow
- Dcm/DcmConfigSet/DcmDsp/DcmDspEcuReset/DcmDspEcuResetRow/DcmDspEcuResetId
- Dcm/DcmConfigSet/DcmDsp/DcmDspEcuReset/DcmDspEcuResetRow/DcmResponseToEcuReset

- Dcm/DcmConfigSet/DcmDsp/DcmDspMemory/DcmDspAddressAndLength-FormatIdentifier/DcmDspSupportedAddressAndLengthFormatIdentifier
- Dcm/DcmConfigSet/DcmDsp/DcmDspMemory/DcmDspMemoryIdInfo/DcmDspReadMemoryRangeByLabelInfo/DcmDspReadMemoryRangeSecurityLevelRef
- Dcm/DcmConfigSet/DcmDsp/DcmDspMemory/DcmDspMemoryIdInfo/DcmDspReadMemoryRangeByLabelInfo/DcmDspReadMemoryRangeSessionLevelRef
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- Dcm/DcmConfigSet/DcmDsp/DcmDspMemoryTransfer/DcmDspMemory-TransferIdInfo
- Dcm/DcmConfigSet/DcmDsp/DcmDspMemoryTransfer/DcmDspMemory-TransferUsePort
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- Dcm/DcmConfigSet/DcmDsp/DcmDspPid/DcmDspPidData/DcmDspPidService01/DcmDspDiagnosisScaling
- Dcm/DcmConfigSet/DcmDsp/DcmDspPid/DcmDspPidData/DcmDspPidService01/DcmDspDiagnosisScaling/DcmDspAlternativeDataInterface
- Dcm/DcmConfigSet/DcmDsp/DcmDspPid/DcmDspPidData/DcmDspPidService01/DcmDspDiagnosisScaling/DcmDspAlternativeDataInterface/DcmDataElement

- Dcm/DcmConfigSet/DcmDsp/DcmDspPid/DcmDspPidData/DcmDspPidService01/DcmDspDiagnosisScaling/DcmDspAlternativeDataInterface/DcmNestedDataElement
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- Dcm/DcmConfigSet/DcmDsp/DcmDspPid/DcmDspPidData/DcmDspPidService01/DcmDspDiagnosisScaling/DcmDspAlternativeDataType/DcmApplicationDataType
- Dcm/DcmConfigSet/DcmDsp/DcmDspPid/DcmDspPidData/DcmDspPidService01/DcmDspDiagnosisScaling/DcmDspAlternativeDataType/DcmDspTextTableMapping
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- Dcm/DcmConfigSet/DcmDsp/DcmDspRequestFileTransfer
- Dcm/DcmConfigSet/DcmDsp/DcmDspRequestFileTransfer/DcmRequestFileTransferFileSizeOrDirInfoParameterLength
- Dcm/DcmConfigSet/DcmDsp/DcmDspRequestFileTransfer/DcmRequestFileTransferLengthFormatIdentifier
- Dcm/DcmConfigSet/DcmDsp/DcmDspRequestFileTransfer/DcmRequestFileTransferMaxFileAndDirName
- Dcm/DcmConfigSet/DcmDsp/DcmDspRequestFileTransfer/DcmRequestFileTransferUsePort
- Dcm/DcmConfigSet/DcmDsp/DcmDspRoe/DcmDspRoeMaxNumChangeOf-

DataIdentfierEventsDcmDspRoeMaxNumChangeOfDataIdentfierEvents

- Dcm/DcmConfigSet/DcmDsp/DcmDspRoe/DcmDspRoeMaxNumComparisionOfValueEvents
- Dcm/DcmConfigSet/DcmDsp/DcmDspRoe/DcmDspRoeMaxNumberOfStoredDTCStatusChangedEvents
- Dcm/DcmConfigSet/DcmDsp/DcmDspRoe/DcmDspRoeMaxSupportedDIDLength
- Dcm/DcmConfigSet/DcmDsp/DcmDspRoe/DcmDspRoeSchedulerRate
- Dcm/DcmConfigSet/DcmDsp/DcmDspRoe/DcmDspRoeStoreEventSupport
- Dcm/DcmConfigSet/DcmDsp/DcmDspRoutine/DcmDspRequestRoutineResults/DcmDspRequestRoutineResultsConfirmationEnabled
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- Dcm/DcmConfigSet/DcmDsp/DcmDspRoutine/DcmDspRequestRoutineResults/DcmDspRequestRoutineResultsIn/DcmDspRequestRoutineResultsInSignal/DcmDspArgumentScaling/DcmDspAlternativeDataType/DcmDspTextTableMapping/DcmDspInternalDataValue
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- Dcm/DcmConfigSet/DcmDsp/DcmDspRoutine/DcmDspRequestRoutineResults/DcmDspRequestRoutineResultsIn/DcmDspRequestRoutineResultsInSignal/DcmDspArgumentScaling/DcmDspAlternativeDiagnosticDataElement/DcmDspDiagnosticDataElementRef
- Dcm/DcmConfigSet/DcmDsp/DcmDspRoutine/DcmDspRequestRoutineResults/DcmDspRequestRoutineResultsIn/DcmDspRequestRoutineResultsInSignal/DcmDspRequestRoutineResultsInSignalCompositeSignalRef

- Dcm/DcmConfigSet/DcmDsp/DcmDspRoutine/DcmDspRequestRoutineRe-sults/DcmDspRequestRoutineResultsIn/DcmDspRequestRoutineRe-sultsInSignal/DcmDspRoutineParameterSize
- Dcm/DcmConfigSet/DcmDsp/DcmDspRoutine/DcmDspRequestRoutineRe-sults/DcmDspRequestRoutineResultsIn/DcmDspRequestRoutineRe-sultsInSignalCompositePool
- Dcm/DcmConfigSet/DcmDsp/DcmDspRoutine/DcmDspRequestRoutineRe-sults/DcmDspRequestRoutineResultsIn/DcmDspRequestRoutineRe-sultsInSignalCompositePool/DcmDspRequestRoutineResultsInSignal
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- Dcm/DcmConfigSet/DcmDsp/DcmDspRoutine/DcmDspRequestRoutineRe-sults/DcmDspRequestRoutineResultsIn/DcmDspRequestRoutineRe-sultsInSignalCompositePool/DcmDspRequestRoutineResultsInSignal/DcmDspArgumentScaling/DcmDspAlternativeArgumentData
- Dcm/DcmConfigSet/DcmDsp/DcmDspRoutine/DcmDspRequestRoutineRe-sults/DcmDspRequestRoutineResultsIn/DcmDspRequestRoutineRe-sultsInSignalCompositePool/DcmDspRequestRoutineResultsInSignal/DcmDspArgumentScaling/DcmDspAlternativeArgumentData/DcmDataEle-ment
- Dcm/DcmConfigSet/DcmDsp/DcmDspRoutine/DcmDspRequestRoutineRe-sults/DcmDspRequestRoutineResultsIn/DcmDspRequestRoutineRe-sultsInSignalCompositePool/DcmDspRequestRoutineResultsInSignal/DcmDspArgumentScaling/DcmDspAlternativeDataType
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- Dcm/DcmConfigSet/DcmDsp/DcmDspRoutine/DcmDspRequestRoutineRe-sults/DcmDspRequestRoutineResultsIn/DcmDspRequestRoutineRe-sultsInSignalCompositePool/DcmDspRequestRoutineResultsInSignal/DcmDspArgumentScaling/DcmDspAlternativeDataType/DcmDspText-TableMapping
- Dcm/DcmConfigSet/DcmDsp/DcmDspRoutine/DcmDspRequestRoutineRe-sults/DcmDspRequestRoutineResultsIn/DcmDspRequestRoutineRe-sultsInSignalCompositePool/DcmDspRequestRoutineResultsInSignal/DcmDspArgumentScaling/DcmDspAlternativeDataType/DcmDspText-TableMapping/DcmDspDiagnosisRepresentationDataValue
- Dcm/DcmConfigSet/DcmDsp/DcmDspRoutine/DcmDspRequestRoutineRe-sults/DcmDspRequestRoutineResultsIn/DcmDspRequestRoutineRe-sultsInSignalCompositePool/DcmDspRequestRoutineResultsInSignal/DcmDspArgumentScaling/DcmDspAlternativeDataType/DcmDspText-

TableMapping/DcmDspInternalDataValue

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- Dcm/DcmConfigSet/DcmDsp/DcmDspRoutine/DcmDspRequestRoutineResults/DcmDspRequestRoutineResultsIn/DcmDspRequestRoutineResultsInSignalCompositePool/DcmDspRequestRoutineResultsInSignal/DcmDspRequestRoutineResultsInSignalCompositeSignalRef
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- Dcm/DcmConfigSet/DcmDsp/DcmDspRoutine/DcmDspRequestRoutineResults/DcmDspRequestRoutineResultsOut/DcmDspRequestRoutineResultsOutSignalCompositePool/DcmDspRequestRoutineResultsOutSignal
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- Dcm/DcmConfigSet/DcmDsp/DcmDspRoutine/DcmDspRequestRoutineResults/DcmDspRequestRoutineResultsOut/DcmDspRequestRoutineResultsOutSignalCompositePool/DcmDspRequestRoutineResultsOutSignal/DcmDspArgumentScaling/DcmDspAlternativeArgumentData/DcmDataElement
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DataTypeDcmApplicationDataType

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- Dcm/DcmConfigSet/DcmDsp/DcmDspRoutine/DcmDspRequestRoutineResults/DcmDspRequestRoutineResultsOut/DcmDspRequestRoutineResultsOutSignalCompositePool/DcmDspRequestRoutineResultsOutSignal/DcmDspArgumentScaling/DcmDspAlternativeDataType/DcmDspTextTableMapping/DcmDspDiagnosisRepresentationDataValue
- Dcm/DcmConfigSet/DcmDsp/DcmDspRoutine/DcmDspRequestRoutineResults/DcmDspRequestRoutineResultsOut/DcmDspRequestRoutineResultsOutSignalCompositePool/DcmDspRequestRoutineResultsOutSignal/DcmDspArgumentScaling/DcmDspAlternativeDataType/DcmDspTextTableMapping/DcmDspInternalDataValue
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- Dcm/DcmConfigSet/DcmDsp/DcmDspRoutine/DcmDspRequestRoutineResults/DcmDspRequestRoutineResultsOut/DcmDspRequestRoutineResultsOutSignalCompositePool/DcmDspRequestRoutineResultsOutSignal/DcmDspRequestRoutineResultsOutSignalCompositeSignalRef
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- Dcm/DcmConfigSet/DcmDsp/DcmDspRoutine/DcmDspRequestRoutineResults/DcmDspRequestRoutineResultsRoleRef
- Dcm/DcmConfigSet/DcmDsp/DcmDspRoutine/DcmDspRoutineUsed
- Dcm/DcmConfigSet/DcmDsp/DcmDspRoutine/DcmDspStartRoutine/DcmDspStartRoutineConfirmationEnabled
- Dcm/DcmConfigSet/DcmDsp/DcmDspRoutine/DcmDspStartRoutine/DcmDspStartRoutineConfirmationFnc
- Dcm/DcmConfigSet/DcmDsp/DcmDspRoutine/DcmDspStartRoutine/DcmDspStartRoutineIn/DcmDspStartRoutineInSignal/DcmDspArgumentScaling/DcmDspAlternativeArgumentData
- Dcm/DcmConfigSet/DcmDsp/DcmDspRoutine/DcmDspStartRoutine/DcmDspStartRoutineIn/DcmDspStartRoutineInSignal/DcmDspArgumentScaling/DcmDspAlternativeArgumentData/DcmDataElement
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- Dcm/DcmConfigSet/DcmDsp/DcmDspRoutine/DcmDspStartRoutine/DcmDspStartRoutineIn/DcmDspStartRoutineInSignal/DcmDspArgumentScaling/DcmDspAlternativeDataType/DcmDspTextTableMapping/DcmDspDiagnosisRepresentationDataValue
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- Dcm/DcmConfigSet/DcmDsp/DcmDspRoutine/DcmDspStartRoutine/DcmDspStartRoutineIn/DcmDspStartRoutineInSignal/DcmDspArgumentScaling/DcmDspAlternativeDiagnosticDataElement
- Dcm/DcmConfigSet/DcmDsp/DcmDspRoutine/DcmDspStartRoutine/DcmDspStartRoutineIn/DcmDspStartRoutineInSignal/DcmDspArgumentScaling/DcmDspAlternativeDiagnosticDataElement/DcmDspDiagnosticDataElementRef
- Dcm/DcmConfigSet/DcmDsp/DcmDspRoutine/DcmDspStartRoutine/DcmDspStartRoutineIn/DcmDspStartRoutineInSignal/DcmDspRoutineParameterSize
- Dcm/DcmConfigSet/DcmDsp/DcmDspRoutine/DcmDspStartRoutine/DcmDspStartRoutineIn/DcmDspStartRoutineInSignal/DcmDspStartRoutineInSignalCompositeSignalRef
- Dcm/DcmConfigSet/DcmDsp/DcmDspRoutine/DcmDspStartRoutine/DcmDspStartRoutineIn/DcmDspStartRoutineInSignalCompositePool
- Dcm/DcmConfigSet/DcmDsp/DcmDspRoutine/DcmDspStartRoutine/DcmDspStartRoutineIn/DcmDspStartRoutineInSignalCompositePool/DcmDspStartRoutineInSignal
- Dcm/DcmConfigSet/DcmDsp/DcmDspRoutine/DcmDspStartRoutine/DcmD-

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- Dcm/DcmConfigSet/DcmDsp/DcmDspRoutine/DcmDspStartRoutine/DcmDspStartRoutineOut/DcmDspStartRoutineOutSignalCompositePool/DcmDspStartRoutineOutSignal/DcmDspArgumentScaling/DcmDspAlternativeArgumentData/DcmDataElement
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StartRoutineOutSignal/DcmDspRoutineSignalEndianness

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- Dcm/DcmConfigSet/DcmDsp/DcmDspRoutine/DcmDspStartRoutine/DcmDspStartRoutineOut/DcmDspStartRoutineOutSignalCompositePool/DcmDspStartRoutineOutSignal/DcmDspStartRoutineOutSignalCompositeSignalRef
- Dcm/DcmConfigSet/DcmDsp/DcmDspRoutine/DcmDspStartRoutine/DcmDspStartRoutineRoleRef
- Dcm/DcmConfigSet/DcmDsp/DcmDspRoutine/DcmDspStopRoutine/DcmDspStopRoutineConfirmationEnabled
- Dcm/DcmConfigSet/DcmDsp/DcmDspRoutine/DcmDspStopRoutine/DcmDspStopRoutineConfirmationFnc
- Dcm/DcmConfigSet/DcmDsp/DcmDspRoutine/DcmDspStopRoutine/DcmDspStopRoutineIn/DcmDspStopRoutineInSignal/DcmDspArgumentScaling/DcmDspAlternativeArgumentData
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- Dcm/DcmConfigSet/DcmDsp/DcmDspRoutine/DcmDspStopRoutine/DcmDspStopRoutineIn/DcmDspStopRoutineInSignal/DcmDspArgumentScaling/DcmDspAlternativeDiagnosticDataElement
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- Dcm/DcmConfigSet/DcmDsp/DcmDspRoutine/DcmDspStopRoutine/DcmDspStopRoutineIn/DcmDspStopRoutineInSignal/DcmDspRoutineParameterSize

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- Dcm/DcmConfigSet/DcmDsp/DcmDspRoutine/DcmDspStopRoutine/DcmDspStopRoutineIn/DcmDspStopRoutineInSignalCompositePool
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- Dcm/DcmConfigSet/DcmDsp/DcmDspRoutine/DcmDspStopRoutine/DcmDspStopRoutineIn/DcmDspStopRoutineInSignalCompositePool/DcmDspStopRoutineInSignal/DcmDspArgumentScaling/DcmDspAlternativeDataType/DcmDspTextTableMapping/DcmDspInternalDataValue
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pRoutineInSignal/DcmDspArgumentScaling/DcmDspAlternativeDiagnostic-DataElement/DcmDspDiagnosticDataElementRef

- Dcm/DcmConfigSet/DcmDsp/DcmDspRoutine/DcmDspStopRoutine/DcmD-spStopRoutineIn/DcmDspStopRoutineInSignalCompositePool/DcmDspSto-pRoutineInSignal/DcmDspRoutineParameterSize
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- Dcm/DcmConfigSet/DcmDsp/DcmDspRoutine/DcmDspStopRoutine/DcmDspStopRoutineRoleRef
- Dcm/DcmConfigSet/DcmProcessingConditions/DcmModeCondition/DcmModeConditionCertificateCompareElementRef
- Dcm/DcmConfigSet/DcmProcessingConditions/DcmModeCondition/DcmSwcDataElementValue
- Dcm/DcmConfigSet/DcmProcessingConditions/DcmModeCondition/DcmSwcDataElementValue/DcmSwcDataElementArray
- Dcm/DcmConfigSet/DcmProcessingConditions/DcmModeCondition/DcmSwcDataElementValue/DcmSwcDataElementArray/DcmSwcDataElementArrayElement
- Dcm/DcmConfigSet/DcmProcessingConditions/DcmModeCondition/DcmSwcDataElementValue/DcmSwcDataElementArray/DcmSwcDataElementArrayElement/DcmSwcDataElementArrayElementIndex
- Dcm/DcmConfigSet/DcmProcessingConditions/DcmModeCondition/DcmSwcDataElementValue/DcmSwcDataElementArray/DcmSwcDataElementArrayElement/DcmSwcDataElementArrayElementValue
- Dcm/DcmConfigSet/DcmProcessingConditions/DcmModeCondition/DcmSwcDataElementValue/DcmSwcDataElementPrimitive
- Dcm/DcmConfigSet/DcmProcessingConditions/DcmModeCondition/DcmSwcDataElementValue/DcmSwcDataElementPrimitive/DcmSwcDataElementPrimitiveValue
- Dcm/DcmConfigSet/DcmProcessingConditions/DcmModeRule/DcmArgumentRef
- Dcm/DcmGeneral/DcmHeaderFileInclusion
- Dcm/DcmGeneral/DcmSecureCoding
- Dcm/DcmGeneral/DcmSecureCoding/DcmSecureCodingAuthenticationJobRef
- Dcm/DcmGeneral/DcmSecureCoding/DcmSecureCodingDataBufferSize
- Dcm/DcmGeneral/DcmSecureCoding/DcmSecureCodingDataIdentifier
- Dcm/DcmGeneral/DcmSecureCoding/DcmSecureCodingValidationRoutine
- Dcm/DcmGeneral/DcmSecurityEventRefs/SEV_UDS_AUTHENTICATION_SUCCESSFUL
- Dcm/DcmGeneral/DcmSecurityEventRefs/SEV_UDS_CERTIFICATE_FAILURE
- Dcm/DcmGeneral/DcmSecurityEventRefs/SEV_UDS_CLEAR_DTC_SUC-

CESSFULSEV_UDS_CLEAR_DTC_SUCCESSFUL

- Dcm/DcmGeneral/DcmSecurityEventRefs/SEV_UDS_COMMUNICATION_-CONTROL_SWITCHED_OFF
- Dcm/DcmGeneral/DcmSecurityEventRefs/SEV_UDS_DTC_SET-TING_SWITCHED_OFF
- Dcm/DcmGeneral/DcmSecurityEventRefs/SEV_UDS_ECU_RESET
- Dcm/DcmGeneral/DcmSecurityEventRefs/SEV_UDS_ECU_UNLOCK_SUC-CESSFUL
- Dcm/DcmGeneral/DcmSecurityEventRefs/SEV_UDS_INCORRECT_MES-SAGE_LENGTH_OR_FORMAT
- Dcm/DcmGeneral/DcmSecurityEventRefs/SEV_UDS_NUMBER_OF_-FAILED_AUTHENTICATION_ATTEMPTS_EXCEEDED
- Dcm/DcmGeneral/DcmSecurityEventRefs/SEV_UDS_REQUESTED_ACTION-S_REQUIRES_AUTHENTICATION
- Dcm/DcmGeneral/DcmSecurityEventRefs/SEV_UDS_REQUEST_DOWNLOAD
- Dcm/DcmGeneral/DcmSecurityEventRefs/SEV_UDS_RE-QUEST_OUT_OF_RANGE
- Dcm/DcmGeneral/DcmSecurityEventRefs/SEV_UDS_REQUEST_SE-QUENCE_ERROR
- Dcm/DcmGeneral/DcmSecurityEventRefs/SEV_UDS_SECURITY_ACCESS_-DENIED
- Dcm/DcmGeneral/DcmSecurityEventRefs/SEV_UDS_SECURITY_AC-CESS_INVALID_KEY
- Dcm/DcmGeneral/DcmSecurityEventRefs/SEV_UDS_SECURITY_AC-CESS_NUMBER_OF_ATTEMPTS_EXCEEDED
- Dcm/DcmGeneral/DcmSecurityEventRefs/SEV_UDS_SECURITY_AC-CESS_REQUIRED_TIME_DELAY_NOT_EXPIRED
- Dcm/DcmGeneral/DcmSecurityEventRefs/SEV_UDS_SERVICE_NOT_SUP-PORTED
- Dcm/DcmGeneral/DcmSecurityEventRefs/SEV_UDS_SUBFUNCTION_NOT_-SUPPORTED
- Dcm/DcmGeneral/DcmSecurityEventRefs/SEV_UDS_WRITE_DATA
- Dcm/DcmGeneral/DcmSecurityEventRefs/SEV_UDS_WRITE_INVALID_DATA

2.6 Deviations

2.6.1 Configuration Deviations

Configuration parameters which differ from the AR v22-11 specification are listed below.

2.6.1.1 Dcm/DcmConfigSet/DcmDsd/DcmDsdServiceRequestManufactur-

erNotificationDcmDsdServiceRequestManufacturerNotification

[RTA-BSW] Introduction	The R-Port is named ServiceRequestManufacturerNotification_{Name} where {Name} is the name of the container DcmDsdServiceRequestManufacturerNotification. The lowerMultiplicity is 0: If DcmRbDsdRequestManufacturerNotificationEnabled = false the Indication API is not available.
[AUTOSAR] Introduction	The R-Port is named ServiceRequestManufacturerNotification_{Name} where {Name} is the name of the container DcmDsdServiceRequestManufacturerNotification. The lowerMultiplicity is 0: If container DcmDsdService RequestManufacturerNotification does not exist the Indication API is not available.

2.6.1.2 Dcm/DcmConfigSet/DcmDsd/DcmDsdServiceRequestSupplier-Notification

[RTA-BSW] Introduction	The R-Port is named ServiceRequestSupplierNotification_<SWC> where <SWC> is the name of the container DcmDsdServiceRequestSupplierNotification. The lowerMultiplicity is 0: If DcmDsdRequestSupplierNotification = false the Indication API is not available.
[AUTOSAR] Introduction	The R-Port is named ServiceRequestSupplierNotification_<SWC> where <SWC> is the name of the container DcmDsdServiceRequestSupplierNotification. The lowerMultiplicity is 0: If the container DcmDsdRequestSupplierNotification does not exist the Indication API is not available.

2.6.1.3 Dcm/DcmConfigSet/DcmDsd/DcmDsdServiceTable/DcmDsdService/DcmDsdServiceUsed

[RTA-BSW] Lower Multiplicity	0
[AUTOSAR] Lower Multiplicity	1

2.6.1.4 Dcm/DcmConfigSet/DcmDsd/DcmDsdServiceTable/DcmDsdService/DcmDsdSubService

[RTA-BSW] Description	This container contains the configuration (DSD parameters) for a subservice of a service. Only those services may have subservices, which have the DcmDsdSidTabSubfuncAvail configured as TRUE. It is not advisable to configure Sub-Functions for RoutineControl Service(0x31).Any DcmDsdSubService configuration for RoutineControl Service(0x31) service will be ignored.
[AUTOSAR] Description	This container contains the configuration (DSD parameters) for a subservice of a service. Only those services may have subservices, which have the DcmDsdSidTabSubfuncAvail configured as TRUE.

2.6.1.5 Dcm/DcmConfigSet/DcmDsd/DcmDsdServiceTable/DcmDsdService/DcmDsdSubService/DcmDsdSubServiceFnc

[RTA-BSW] Description	Callback function of the ECU Supplier specific component for the particular Subservice. The function's prototype is as described for <Module>_<DiagnosticService>_<SubService>. Hint: Dcm_OpStatusType is used instead of Dcm_ExtendedOpStatusType. At the moment this parameter is supported only for the subservice 0x04-TransmitCertificate in Service 0x29-Authentication and Service 0x19-RDTC.
[AUTOSAR] Description	Callback function of the ECU Supplier specific component for the particular service. The function's prototype is as described for <Module>_<DiagnosticService>_<SubService>.

2.6.1.6 Dcm/DcmConfigSet/DcmDsl/DcmDslProtocol/DcmDslProtocolRow/DcmDemClientRef

[RTA-BSW] Lower Multiplicity	0
[AUTOSAR] Lower Multiplicity	1

2.6.1.7 Dcm/DcmConfigSet/DcmDsl/DcmDslProtocol/DcmDslProtocolRow/DcmDslConnection/DcmDslMainConnection/DcmDslProtocolRx

[RTA-BSW] Description	This container contains the configuration parameters of a reception channel in a diagnostic connection. The following parameters needs to be configured per protocol. Please keep in mind, that the parameter DcmDslProtocolRxPduId can be given several times per protocol and that the combination from DcmDslProtocolRxPduId and DcmDslProtocolRxAddrType is unique.
[AUTOSAR] Description	This container contains the configuration parameters of a reception channel in a diagnostic connection.

2.6.1.8 Dcm/DcmConfigSet/DcmDsl/DcmDslProtocol/DcmDslProtocolRow/DcmDslConnection/DcmDslMainConnection/DcmDslProtocolRx/DcmDslProtocolRxPduId

[RTA-BSW] Description	Identifier of the PDU that is used for this reception channel. The user need not update this field. Instead the user may invoke generateId for Dcm inorder to automatically configure the same.
[AUTOSAR] Description	Identifier of the PDU that is used for this reception channel.

2.6.1.9 Dcm/DcmConfigSet/DcmDsl/DcmDslProtocol/DcmDslProtocolRow/DcmDslConnection/DcmDslMainConnection/DcmDslProtocolRxConnectionId

[RTA-BSW] Description	Unique identifier of the tester which uses this connection for diagnostic communication. The user not need update this field. Instead the user may invoke DcmForwarder inorder to automatically configure the same.
[AUTOSAR] Description	Unique identifier of the tester which uses this connection for diagnostic communication.

2.6.1.10 Dcm/DcmConfigSet/DcmDsl/DcmDslProtocol/DcmDslProtocolRow/DcmDslConnection/DcmDslMainConnection/DcmDslProtocolTx

[RTA-BSW] Lower Multiplicity	1
[AUTOSAR] Lower Multiplicity	0

2.6.1.11 Dcm/DcmConfigSet/DcmDsl/DcmDslProtocol/DcmDslProtocolRow/DcmDslConnection/DcmDslMainConnection/DcmDslProtocolTx/DcmDslTxConfirmationPduId

[RTA-BSW] Description	Identifier of the PDU that is used by the lower level module for transmission confirmation of responses on this channel. The user need not update this field. Instead, the user may invoke generateId for Dcm inorder to automatically configure the same.
[AUTOSAR] Description	Identifier of the PDU that is used by the lower level module for transmission confirmation of responses on this channel.

2.6.1.12 Dcm/DcmConfigSet/DcmDsl/DcmDslProtocol/DcmDslProtocolRow/DcmDslConnection/DcmDslPeriodicTransmission/DcmDslPeriodicConnection/DcmDslPeriodicTxConfirmationPduId

[RTA-BSW] Description	Identifier of the PDU that is used by the lower level module for transmission confirmation of responses on this channel. The user need not update this field. Instead, the user may invoke generateId for Dcm inorder to automatically configure the same.
[AUTOSAR] Description	Identifier of the PDU that is used by the lower level module for transmission confirmation of responses on this channel.

2.6.1.13 Dcm/DcmConfigSet/DcmDsl/DcmDslProtocol/DcmDslProtocolRow/DcmDslConnection/DcmDslPeriodicTransmission/DcmDslPeriodicConnection/DcmDslPeriodicTxPduRef

[RTA-BSW] Description	Reference to a Pdu in EcuC that is used for this periodic transmission channel. It is only needed for Type2 Periodic Transmissions configurations.One or several DcmDslPeriodicTxPduRef can be defined if Periodic Transmission is enabled and TYPE2 Periodic Transmissions is configured.DcmDslPeriodicTxPduRef does not need to be defined if Periodic Transmission is not enabled or TYPE1 Periodic Transmissions is configured.
[AUTOSAR] Description	Reference to a Pdu in EcuC that is used for this periodic transmission channel.

2.6.1.14 Dcm/DcmConfigSet/DcmDsl/DcmDslProtocol/DcmDslProtocolRow/DcmDslProtocolMaximumResponseSize

[RTA-BSW] Description: This parameter is mandatory and defines the maximum length of the response message in case DcmPagedBufferEnabled == TRUE In case the user leaves the parameter unconfigured, Dcm shall fill the value during generate Id as 4095 for CAN(If CanTpFlexibleDataRateSupport is not configured/configured as FALSE in CanTp) and as 65535 for other protocols (Flexray, IP,CAN (if CanTpFlexibleDataRateSupport ios configured as TRUE)). Values configured by the user wil be overwritten by Dcm in this case.If paged buffer is enabled and the generated Response for a Request is longer than DcmDslProtocolMaximumResponseSize configured for the active protocol, the Dcm service shall respond with NRC 0x14 (DCM_E_RESPONSETOOLONG).Vendor specific services using paged buffer should also do a check based on this parameter. If paged buffer is disabled and the generated Response for a Request is longer than Dcm_MsgContextType structure element resMaxDataLen, the Dcm shall respond with NRC 0x14 (DCM_E_RESPONSETOOLONG).

[AUTOSAR] Description: This parameter is mandatory and defines the maximum length of the response message in case DcmPagedBufferEnabled == TRUE

2.6.1.15 Dcm/DcmConfigSet/DcmDsl/DcmDslProtocol/DcmDslProtocolRow/DcmDslProtocolPriority

[RTA-BSW] Description	Protocol priority used during protocol preemption. A higher priority protocol may preempt a lower priority protocol. If Protocol Premption is required then pduRCancelTransmit and pduRCancelReceive should be enabled
[AUTOSAR] Description	Protocol priority used during protocol preemption. A higher priority protocol may preempt a lower priority protocol.

2.6.1.16 Dcm/DcmConfigSet/DcmDsl/DcmDslProtocol/DcmDslProtocolRow/DcmDslProtocolRowUsed

[RTA-BSW] Lower Multiplicity	0
[AUTOSAR] Lower Multiplicity	1

2.6.1.17 Dcm/DcmConfigSet/DcmDsl/DcmDslProtocol/DcmDslProtocolRow/DcmDslProtocolRxBufferRef

[RTA-BSW] Lower Multiplicity	0
[AUTOSAR] Lower Multiplicity	1

2.6.1.18 Dcm/DcmConfigSet/DcmDsl/DcmDslProtocol/DcmDslProtocolRow/DcmDslProtocolSIDTable

[RTA-BSW] Lower Multiplicity	0
[AUTOSAR] Lower Multiplicity	1

2.6.1.19 Dcm/DcmConfigSet/DcmDsl/DcmDslProtocol/DcmDslProtocolRow/DcmDslProtocolTransType

[RTA-BSW] Description	This parameter is used only if the protocol is of type DCM_ROE_ON_xxx or DCM_PERIODICTRANS_ON_xxx. It selects the transmission type of the protocol. TYPE1: Messages for ROE uses the DcmTxPduId used for normal diagnostic responses. The outgoing messages must be synchronized with normal outgoing messages, which have a higher priority. TYPE2: Messages for ROE or Periodic Transmission are sent on a separate DcmTxPduId.
[AUTOSAR] Description	This parameter is used only if the protocol is of type DCM_ROE_ON_xxx. It selects the transmission type of the protocol.

2.6.1.20 Dcm/DcmConfigSet/DcmDsl/DcmDslProtocol/DcmDslProtocolRow/DcmDslProtocolTxBufferRef

[RTA-BSW] Lower Multiplicity	0
[AUTOSAR] Lower Multiplicity	1

2.6.1.21 Dcm/DcmConfigSet/DcmDsl/DcmDslProtocol/DcmDslProtocolRow/DcmDslProtocolType

[RTA-BSW] Enumeration Literals: DCM_OBD_ON_CAN, DCM_OBD_ON_FLEXRAY, DCM_OBD_ON_IP, DCM_OBD_ON_KLINE, DCM_PERIODICTRANS_ON_CAN, DCM_PERIODICTRANS_ON_FLEXRAY, DCM_PERIODICTRANS_ON_IP, DCM_ROE_ON_CAN, DCM_ROE_ON_FLEXRAY, DCM_ROE_ON_IP, DCM_SUPPLIER_1, DCM_SUPPLIER_10, DCM_SUPPLIER_11, DCM_SUPPLIER_12, DCM_SUPPLIER_13, DCM_SUPPLIER_14, DCM_SUPPLIER_15, DCM_SUPPLIER_2,

DCM_SUPPLIER_3, DCM_SUPPLIER_4, DCM_SUPPLIER_5, DCM_SUPPLIER_6, DCM_SUPPLIER_7, DCM_SUPPLIER_8, DCM_SUPPLIER_9, DCM_UDS_ON_CAN, DCM_UDS_ON_FLEXRAY, DCM_UDS_ON_IP, DCM_ACEA_ON_CAN, DCM_ACEA_ON_FLEXRAY, DCM_KWP_ON_KLINE, DCM_KWP_ON_CAN, DCM_CARB_ON_KLINE

[AUTOSAR] Enumeration Literals: DCM_OBD_ON_CAN, DCM_OBD_ON_FLEXRAY, DCM_OBD_ON_IP, DCM_PERIODICTRANS_ON_CAN, DCM_PERIODICTRANS_ON_FLEXRAY, DCM_PERIODICTRANS_ON_IP, DCM_ROE_ON_CAN, DCM_ROE_ON_FLEXRAY, DCM_ROE_ON_IP, DCM_SUPPLIER_1, DCM_SUPPLIER_10, DCM_SUPPLIER_11, DCM_SUPPLIER_12, DCM_SUPPLIER_13, DCM_SUPPLIER_14, DCM_SUPPLIER_15, DCM_SUPPLIER_2, DCM_SUPPLIER_3, DCM_SUPPLIER_4, DCM_SUPPLIER_5, DCM_SUPPLIER_6, DCM_SUPPLIER_7, DCM_SUPPLIER_8, DCM_SUPPLIER_9, DCM_UDS_ON_CAN, DCM_UDS_ON_FLEXRAY, DCM_UDS_ON_IP, DCM_UDS_ON_LIN

2.6.1.22 Dcm/DcmConfigSet/DcmDsp/DcmDspAuthentication/DcmDspAuthenticationConnection/DcmDspAuthenticationClientChallengeSignJobRef

[RTA-BSW] Description	Reference to a job used to sign the client challenge. Mandatory only if sub-service 0x02 - verifyCertificateBiDirectional is configured.
[AUTOSAR] Description	Reference to a job used to sign the client challenge.
[RTA-BSW] Lower Multiplicity	0
[AUTOSAR] Lower Multiplicity	1

2.6.1.23 Dcm/DcmConfigSet/DcmDsp/DcmDspAuthentication/DcmDspAuthenticationConnection/DcmDspAuthenticationConnectionCertificateRef

[RTA-BSW] Description	Reference to a KeyMCertificate used to store the certificate within the KeyM. Mandatory only if DcmRbDspAuthenticationConcept != DCM_ACR_SYMMETRIC.
[AUTOSAR] Description	Reference to a KeyMCertificate used to store the certificate within the KeyM.
[RTA-BSW] Lower Multiplicity	0
[AUTOSAR] Lower Multiplicity	1

2.6.1.24 Dcm/DcmConfigSet/DcmDsp/DcmDspAuthentication/DcmDspAuthenticationConnection/DcmDspAuthenticationECUCertificateRef

[RTA-BSW] Description	Reference to a KeyMCertificate with the server certificate for bi-directional authentication. Mandatory only if DcmRbDspAuthenticationConcept != DCM_ACR_SYMMETRIC and sub-service 0x02 - verifyCertificateBiDirectional or DcmRbDspAuthenticationConcept == DCM_ACR_SYMMETRIC and sub-service 0x07 - verifyProofOfOwnershipBidirectional is configured. .
[AUTOSAR] Description	Reference to a KeyMCertificate with the server certificate for bi-directional authentication.

2.6.1.25 Dcm/DcmConfigSet/DcmDsp/DcmDspAuthentication/DcmDspAuthenticationConnection/DcmDspAuthenticationRoleElementRef

[RTA-BSW] Description	Reference to a certificate data element with the role in the certificate. Mandatory only if DcmRbDspAuthenticationConcept != DCM_ACR_SYMMETRIC.
[AUTOSAR] Description	Reference to a certificate data element with the role in the certificate.
[RTA-BSW] Lower Multiplicity	0
[AUTOSAR] Lower Multiplicity	1

2.6.1.26 Dcm/DcmConfigSet/DcmDsp/DcmDspAuthentication/DcmDspAuthenticationConnection/DcmDspAuthenticationWhiteListServicesElementRef

[RTA-BSW] Lower Multiplicity	0
[AUTOSAR] Lower Multiplicity	1

2.6.1.27 Dcm/DcmConfigSet/DcmDsp/DcmDspAuthentication/DcmDspAuthenticationDefaultSessionTimeOut

[RTA-BSW] Maximum Value	4294967295
[AUTOSAR] Maximum Value	INF

2.6.1.28 Dcm/DcmConfigSet/DcmDsp/DcmDspAuthentication/DcmDspAuthenticationRow

[RTA-BSW] Upper Multiplicity	32

[AUTOSAR] Upper Multiplicity	31

2.6.1.29 Dcm/DcmConfigSet/DcmDsp/DcmDspComControl/DcmDspComControlSetting/DcmDspComControlCommunicationReEnableModeRuleRef

[RTA-BSW] Description	Reference to DcmModeRule Mode rule which controls re-enabling of communication by DCM. The DCM module shall call BswM_Dcm_CommunicationMode_CurrentState(DCM_ENABLE_RX_TX_NORM_NM)) in case that the referenced mode rule is not fulfilled anymore.
[AUTOSAR] Description	Reference to DcmModeRule Mode rule which controls re-enabling

2.6.1.30 Dcm/DcmConfigSet/DcmDsp/DcmDspData/DcmDspDataByteSize

[RTA-BSW] Description	Defines the array length in bytes or the the maximum array length for variable datalengths. If DcmDspDataByteSize is configured, then DcmDspDidByteOffset shall be used. If the the parameter DcmDspDataUsePort is configured as USE_DATA_SENDER_RECEIVER/USE_BLOCK_ID, the user need not update this field. Instead the user may invoke generateId for Dcm inorder to automatically configure the same depending on parameter DcmDspDataType/DcmDspDataBlockIdRef→NvMNvBlockLength.respectively
[AUTOSAR] Description	Defines the array length in bytes or the maximum array length for variable datalengths.

2.6.1.31 Dcm/DcmConfigSet/DcmDsp/DcmDspData/DcmDspDataType

[RTA-BSW] Enumeration Literals	BOOLEAN, SINT16, SINT16_N, SINT32, SINT32_N, SINT8, SINT8_N, UINT16, UINT16_N, UINT32, UINT32_N, UINT8, UINT8_DYN, UINT8_N
[AUTOSAR] Enumeration Literals	BOOLEAN, FLOAT, FLOAT_N, SINT16, SINT16_N, SINT32, SINT32_N, SINT8, SINT8_N, UINT16, UINT16_N, UINT32, UINT32_N, UINT8, UINT8_DYN, UINT8_N

2.6.1.32 Dcm/DcmConfigSet/DcmDsp/DcmDspData/DcmDspDataUsePort

[RTA-BSW] Enumeration Literals	USE_BLOCK_ID, USE_DATA_ASYNCH_CLIENT_SERVER, USE_DATA_ASYNCH_FNC, USE_DATA_ASYNCH_FNC_PROXY, USE_DATA_SENDER_RECEIVER, USE_DATA_SENDER_RECEIVER_AS_SERVICE, USE_DATA_SYNCH_CLIENT_SERVER, USE_DATA_SYNCH_FNC, USE_DATA_SYNCH_FNC_PROXY, USE_DATA_RDBI_PAGED_FNC, USE_DATA_ASYNCH_FNC_ERROR

[AUTOSAR] Enumeration Literals	USE_DATA_ASYNCH_CLIENT_SERVER, USE_DATA_ASYNCH_CLIENT_SERVER_ERROR, USE_DATA_ASYNCH_FNC, USE_DATA_ASYNCH_FNC_ERROR, USE_DATA_ASYNCH_FNC_PROXY, USE_DATA_SENDER_RECEIVER, USE_DATA_SENDER_RECEIVER_AS_SERVICE, USE_DATA_SYNCH_CLIENT_SERVER, USE_DATA_SYNCH_FNC, USE_DATA_SYNCH_FNC_PROXY, USE_ECU_SIGNAL

2.6.1.33 Dcm/DcmConfigSet/DcmDsp/DcmDspData/DcmDspExternalSR-DataElementClass/DcmSubElementInDataElementInstance/DcmSubElementInDataElementInstanceRef

[RTA-BSW] Introduction	Supported are VariableDataPrototypes in SenderReceiverInterfaces and NvDataInterfaces and ParameterDataPrototypes in ParameterInterfaces (read only). This reference is applicable if the AutosarDataPrototype is typed with a ApplicationCompositeDataType. Note: One or more class attributes are too long to display. Please refer to diagram INTERNAL_DcmDspSRData for the class attributes.
[AUTOSAR] Introduction	Supported are VariableDataPrototypes in SenderReceiverInterfaces and NvDataInterfaces and ParameterDataPrototypes in ParameterInterfaces (read only). This reference is applicable if the AutosarDataPrototype is typed with a ApplicationCompositeDataType.

2.6.1.34 Dcm/DcmConfigSet/DcmDsp/DcmDspData/DcmDspExternalSR-DataElementClass/DcmSubElementInImplDataElementInstance/DcmSubElementInImplDataElementInstanceRef

[RTA-BSW] Introduction	Supported are VariableDataPrototypes in SenderReceiverInterfaces and NvDataInterfaces and ParameterDataPrototypes in ParameterInterfaces (read only). This reference is applicable if the AutosarDataPrototype is typed with a ImplementationDataType of category STRUCTURE or ARRAY. Please note that in case of ARRAY the index attribute in the target reference has to be set to select a single array element. Note: One or more class attributes are too long to display. Please refer to diagram INTERNAL_DcmDspSRData for the class attributes.
[AUTOSAR] Introduction	Supported are VariableDataPrototypes in SenderReceiverInterfaces and NvDataInterfaces and ParameterDataPrototypes in ParameterInterfaces (read only). This reference is applicable if the AutosarDataPrototype is typed with a ImplementationDataType of category STRUCTURE or ARRAY. Please note that in case of ARRAY the index attribute in the target reference has to be set to select a single array element.

2.6.1.35 Dcm/DcmConfigSet/DcmDsp/DcmDspDid/DcmDspDidSignal/DcmDspDidByteOffset

[RTA-BSW] Lower Multiplicity	0
[AUTOSAR] Lower Multiplicity	1

2.6.1.36 Dcm/DcmConfigSet/DcmDsp/DcmDspDid/DcmDspDidSignal/DcmDspDidDataRef

[RTA-BSW] Lower Multiplicity	1
[AUTOSAR] Lower Multiplicity	0

2.6.1.37 Dcm/DcmConfigSet/DcmDsp/DcmDspDid/DcmDspDidSupportInfo/DcmDspDidSupportInfoPos

[RTA-BSW] Description	Length of the support information in bytes.
[AUTOSAR] Description	Position of the support information in bytes.

2.6.1.38 Dcm/DcmConfigSet/DcmDsp/DcmDspDid/DcmDspDidUsePort

[RTA-BSW] Enumeration Literals	USE_ATOMIC_NV_DATA_INTERFACE, USE_ATOMIC_SENDER_RECEIVER_INTERFACE, USE_ATOMIC_SENDER_RECEIVER_INTERFACE_AS_SERVICE, USE_DATA_ELEMENT_SPECIFIC_INTERFACES
[AUTOSAR] Enumeration Literals	USE_ATOMIC_BNDM, USE_ATOMIC_NV_DATA_INTERFACE, USE_ATOMIC_SENDER_RECEIVER_INTERFACE, USE_ATOMIC_SENDER_RECEIVER_INTERFACE_AS_SERVICE, USE_DATA_ELEMENT_SPECIFIC_INTERFACES

2.6.1.39 Dcm/DcmConfigSet/DcmDsp/DcmDspMaxDidToRead

[RTA-BSW] Description	Indicates the maximum allowed DIDs in a single “ReadDataByIdentifier” request. If not configured then no limitation is applied,i.e,any number of DIDs in the request is supported.
[AUTOSAR] Description	Indicates the maximum allowed DIDs in a single “ReadDataByIdentifier” request.

2.6.1.40 Dcm/DcmConfigSet/DcmDsp/DcmDspMemoryTransfer/DcmDspAddressAndLengthFormatIdentifier

[RTA-BSW] Upper Multiplicity	0
[AUTOSAR] Upper Multiplicity	1

2.6.1.41 Dcm/DcmConfigSet/DcmDsp/DcmDspMemoryTransfer/DcmDsp-

MemoryTransferIdInfo/DcmDspMemoryIdValue

[RTA-BSW] Upper Multiplicity	0
[AUTOSAR] Upper Multiplicity	1

2.6.1.42 Dcm/DcmConfigSet/DcmDsp/DcmDspPeriodicTransmission/DcmDspPeriodicTransmissionMaxPeriodicFastTransmissions

[RTA-BSW] Description	Defines the maximum number of periodic connections used for fast transmission. Not configuring the parameter or configuring value as 0 would be considered as no limitation on the maximum number of periodic connections used for fast transmission.
[AUTOSAR] Description	Defines the maximum number of periodic connections used for fast transmission.

2.6.1.43 Dcm/DcmConfigSet/DcmDsp/DcmDspPeriodicTransmission/DcmDspPeriodicTransmissionMaxPeriodicMediumTransmissions

[RTA-BSW] Description	Defines the maximum number of periodic connections used for medium transmission. Not configuring the parameter or configuring value as 0 would be considered as no limitation on the maximum number of periodic connections used for medium transmission.
[AUTOSAR] Description	Defines the maximum number of periodic connections used for medium transmission.

2.6.1.44 Dcm/DcmConfigSet/DcmDsp/DcmDspPeriodicTransmission/DcmDspPeriodicTransmissionMaxPeriodicSlowTransmissions

[RTA-BSW] Description	Defines the maximum number of periodic connections used for slow transmission. Not configuring the parameter or configuring value as 0 would be considered as no limitation on the maximum number of periodic connections used for fast transmission.
[AUTOSAR] Description	Defines the maximum number of periodic connections used for slow transmission.

2.6.1.45 Dcm/DcmConfigSet/DcmDsp/DcmDspPid/DcmDspPidData/DcmDspPidDataByteSize

[RTA-BSW] Description	Defines the array length in bytes or the the maximum array length for variable datalengths.
[AUTOSAR] Description	Defines the array length in bytes or the maximum array length for variable datalengths.
[RTA-BSW] Upper Multiplicity	0
[AUTOSAR] Upper Multiplicity	1

2.6.1.46 Dcm/DcmConfigSet/DcmDsp/DcmDspPid/DcmDspPidData/DcmDspPidService01/DcmDspPidDataUsePort

[RTA-BSW] Introduction	If this parameter is set to USE_DATA_SYNCH_CLIENT_SERVER, the Dcm will have an R-Port requiring the interface DataServices_{Data}. If this parameter is set to USE_DATA_SENDER_RECEIVER, the DCM will have an R-Port requiring a SenderReceiverInterface The R-Port is named DataServices_{Data} where {Data} is the name of the container DcmDspPidData.
[AUTOSAR] Introduction	If this parameter is set to USE_DATA_SYNCH_CLIENT_SERVER, the Dcm will have an R-Port requiring the interface DataServices_{Data}. If this parameter is set to USE_DATA_SENDER_RECEIVER or USE_DATA_SENDER_RECEIVER_AS_SERVICE, the DCM will have an R-Port requiring a SenderReceiverInterface. The R-Port is named DataServices_{Data} where {Data} is the name of the container DcmDspPidData.
[RTA-BSW] Enumeration Literals	USE_DATA_SENDER_RECEIVER, USE_DATA_SYNCH_CLIENT_SERVER, USE_DATA_SYNCH_FNC
[AUTOSAR] Enumeration Literals	USE_DATA_SENDER_RECEIVER, USE_DATA_SENDER_RECEIVER_AS_SERVICE, USE_DATA_SYNCH_CLIENT_SERVER, USE_DATA_SYNCH_FNC

2.6.1.47 Dcm/DcmConfigSet/DcmDsp/DcmDspPid/DcmDspPidData/DcmDspPidService01/DcmDspPidService01ExternalSRDataElementClass/DcmDataElementInstance

[RTA-BSW] Upper Multiplicity	0
[AUTOSAR] Upper Multiplicity	1

2.6.1.48 Dcm/DcmConfigSet/DcmDsp/DcmDspPid/DcmDspPidData/DcmDspPidService01/DcmDspPidService01ExternalSRDataElementClass/DcmDataElementInstance/DcmDataElementInstanceRef

[RTA-BSW] Multiplicity	0..0
[AUTOSAR] Multiplicity	1..1

2.6.1.49 Dcm/DcmConfigSet/DcmDsp/DcmDspPid/DcmDspPidData/DcmDspPidService01/DcmDspPidService01ExternalSRDataElementClass/DcmSubElementInDataElementInstance

[RTA-BSW] Upper Multiplicity	0
[AUTOSAR] Upper Multiplicity	1

2.6.1.50 Dcm/DcmConfigSet/DcmDsp/DcmDspPid/DcmDspPidData/DcmDspPidService01/DcmDspPidService01ExternalSRDataElementClass/DcmSubElementInDataElementInstance/DcmSubElementIn-

DataElementInstanceRefDcmSubElementInDataElementInstanceRef

[RTA-BSW] Introduction	Supported are VariableDataPrototypes in SenderReceiverInterfaces and NvDataInterfaces and ParameterDataPrototypes in ParameterInterfaces (read only). This reference is applicable if the AutosarDataPrototype is typed with a ApplicationCompositeDataType. Note: One or more class attributes are too long to display. Please refer to diagram INTERNAL_DcmDspSRData for the class attributes.
[AUTOSAR] Introduction	Supported are VariableDataPrototypes in SenderReceiverInterfaces and NvDataInterfaces and ParameterDataPrototypes in ParameterInterfaces (read only). This reference is applicable if the AutosarDataPrototype is typed with a ApplicationCompositeDataType.
[RTA-BSW] Multiplicity	0..0
[AUTOSAR] Multiplicity	1..1

2.6.1.51 Dcm/DcmConfigSet/DcmDsp/DcmDspPid/DcmDspPidData/DcmDspPidService01/DcmDspPidService01ExternalSRDataElementClass/DcmSubElementInImplDataElementInstance

[RTA-BSW] Upper Multiplicity	0
[AUTOSAR] Upper Multiplicity	1

2.6.1.52 Dcm/DcmConfigSet/DcmDsp/DcmDspPid/DcmDspPidData/DcmDspPidService01/DcmDspPidService01ExternalSRDataElementClass/DcmSubElementInImplDataElementInstance/DcmSubElementInImplDataElementInstanceRef

[RTA-BSW] Introduction	Supported are VariableDataPrototypes in SenderReceiverInterfaces and NvDataInterfaces and ParameterDataPrototypes in ParameterInterfaces (read only). This reference is applicable if the AutosarDataPrototype is typed with a ImplementationDataType of category STRUCTURE or ARRAY. Please note that in case of ARRAY the index attribute in the target reference has to be set to select a single array element. Note: One or more class attributes are too long to display. Please refer to diagram INTERNAL_DcmDspSRData for the class attributes.

[AUTOSAR] Introduction	Supported are VariableDataPrototypes in SenderReceiverInterfaces and NvDataInterfaces and ParameterDataPrototypes in ParameterInterfaces (read only). This reference is applicable if the AutosarDataPrototype is typed with a ImplementationDataType of category STRUCTURE or ARRAY. Please note that in case of ARRAY the index attribute in the target reference has to be set to select a single array element.
[RTA-BSW] Multiplicity	0..0
[AUTOSAR] Multiplicity	1..1

2.6.1.53 Dcm/DcmConfigSet/DcmDsp/DcmDspPid/DcmDspPidUsed

[RTA-BSW] Description	Allow to activate or deactivate the usage of a PID, for multi purpose ECUs. For Multi Variant configuration: The user must ensure that this parameter is set to either true or false for all configured variants.
[AUTOSAR] Description	Allow to activate or deactivate the usage of a PID, for multi purpose ECUs
[RTA-BSW] Introduction	true = PID is available false = PID is not available.
[AUTOSAR] Introduction	true = PID is available false = PID is not available
[RTA-BSW] Lower Multiplicity	0
[AUTOSAR] Lower Multiplicity	1

2.6.1.54 Dcm/DcmConfigSet/DcmDsp/DcmDspPowerDownTime

[RTA-BSW] Description	This parameter indicates to the client the minimum time of the stand-by sequence the server will remain in the power-down sequence. The resolution of this parameter is one second per count.Values 0x00-0xFE are valid representing 0-254s.The resolution of this parameter is one second per count. The following values are valid: 00 - FE hex: 0 - 254 s powerDownTime; FF hex: indicates a failure or time not available. This value needs to be defined by the integrator according to the ECU capabilities. This parameter has to be available if the service EcuReset, sub-service enableRapid PowerShutDown is configured.

[AUTOSAR] Description	This parameter indicates to the client the minimum time of the stand-by sequence the server will remain in the power-down sequence.

2.6.1.55 Dcm/DcmConfigSet/DcmDsp/DcmDspReadDTCInformation

[RTA-BSW] Description	This container contains the configuration for the UDS service ReadDTCInformation (0x19).
[AUTOSAR] Description	Provides further configuration for the UDS service ReadDTCInformation processing.

2.6.1.56 Dcm/DcmConfigSet/DcmDsp/DcmDspRoutine/DcmDspRequestRoutineResults/DcmDspRequestRoutineResultsIn

[RTA-BSW] Upper Multiplicity	0
[AUTOSAR] Upper Multiplicity	1

2.6.1.57 Dcm/DcmConfigSet/DcmDsp/DcmDspRoutine/DcmDspRequestRoutineResults/DcmDspRequestRoutineResultsIn/DcmDspRequestRoutineResultsInSignal/DcmDspRoutineSignalEndianness

[RTA-BSW] Upper Multiplicity	0
[AUTOSAR] Upper Multiplicity	1

2.6.1.58 Dcm/DcmConfigSet/DcmDsp/DcmDspRoutine/DcmDspRequestRoutineResults/DcmDspRequestRoutineResultsIn/DcmDspRequestRoutineResultsInSignal/DcmDspRoutineSignalPos

[RTA-BSW] Multiplicity	0..0
[AUTOSAR] Multiplicity	1..1

2.6.1.59 Dcm/DcmConfigSet/DcmDsp/DcmDspRoutine/DcmDspRequestRoutineResults/DcmDspRequestRoutineResultsIn/DcmDspRequestRoutineResultsInSignal/DcmDspRoutineSignalType

[RTA-BSW] Upper Multiplicity	0
[AUTOSAR] Upper Multiplicity	1
[RTA-BSW] Enumeration Literals	BOOLEAN, SINT16, SINT16_N, SINT32, SINT32_N, SINT8, SINT8_N, UINT16, UINT16_N, UINT32, UINT32_N, UINT8, UINT8_N, VARIABLE_LENGTH
[AUTOSAR] Enumeration Literals	BOOLEAN, FLOAT, FLOAT_N, SINT16, SINT16_N, SINT32, SINT32_N, SINT8, SINT8_N, UINT16, UINT16_N, UINT32, UINT32_N, UINT8, UINT8_N, VARIABLE_LENGTH

2.6.1.60 Dcm/DcmConfigSet/DcmDsp/DcmDspRoutine/DcmDspRequestRoutineResults/DcmDspRequestRoutineResultsOut/DcmDspRequestRoutineResultsOutSignal/DcmDspRoutineSignalType

[RTA-BSW] Lower Multiplicity	1
[AUTOSAR] Lower Multiplicity	0
[RTA-BSW] Enumeration Literals	BOOLEAN, SINT16, SINT16_N, SINT32, SINT32_N, SINT8, SINT8_N, UINT16, UINT16_N, UINT32, UINT32_N, UINT8, UINT8_N, VARIABLE_LENGTH
[AUTOSAR] Enumeration Literals	BOOLEAN, FLOAT, FLOAT_N, SINT16, SINT16_N, SINT32, SINT32_N, SINT8, SINT8_N, UINT16, UINT16_N, UINT32, UINT32_N, UINT8, UINT8_N, VARIABLE_LENGTH

2.6.1.61 Dcm/DcmConfigSet/DcmDsp/DcmDspRoutine/DcmDspRequestRoutineResults/DcmDspRoutineInterfaceArgumentIntegrity

[REMOVED] Default Value	false

2.6.1.62 Dcm/DcmConfigSet/DcmDsp/DcmDspRoutine/DcmDspRoutineInfoByte

[RTA-BSW] Description	Manufacturer specific value reported to the tester for the all record identifiers. For OBD RIDS routine info byte value is always 0 as per J1979-2. For other RIDS routine info byte value shall be same as configured value
[AUTOSAR] Description	Manufacturer specific value reported to the tester for the record identifiers 0xE000 to OxE1FF. (OBD use cases)

2.6.1.63 Dcm/DcmConfigSet/DcmDsp/DcmDspRoutine/DcmDspStartRoutine/DcmDspRoutineInterfaceArgumentIntegrity

[REMOVED] Default Value	false

2.6.1.64 Dcm/DcmConfigSet/DcmDsp/DcmDspRoutine/DcmDspStartRoutine/DcmDspStartRoutineIn/DcmDspStartRoutineInSignal/DcmDspRoutineSignalType

[RTA-BSW] Lower Multiplicity	1
[AUTOSAR] Lower Multiplicity	0

[RTA-BSW] Enumeration Literals	BOOLEAN, SINT16, SINT16_N, SINT32, SINT32_N, SINT8, SINT8_N, UINT16, UINT16_N, UINT32, UINT32_N, UINT8, UINT8_N, VARIABLE_LENGTH
[AUTOSAR] Enumeration Literals	BOOLEAN, FLOAT, FLOAT_N, SINT16, SINT16_N, SINT32, SINT32_N, SINT8, SINT8_N, UINT16, UINT16_N, UINT32, UINT32_N, UINT8, UINT8_N, VARIABLE_LENGTH

2.6.1.65 Dcm/DcmConfigSet/DcmDsp/DcmDspRoutine/DcmDspStartRoutine/DcmDspStartRoutineOut/DcmDspStartRoutineOutSignal/DcmDspRoutineSignalType

[RTA-BSW] Lower Multiplicity	1
[AUTOSAR] Lower Multiplicity	0
[RTA-BSW] Enumeration Literals	BOOLEAN, SINT16, SINT16_N, SINT32, SINT32_N, SINT8, SINT8_N, UINT16, UINT16_N, UINT32, UINT32_N, UINT8, UINT8_N, VARIABLE_LENGTH
[AUTOSAR] Enumeration Literals	BOOLEAN, FLOAT, FLOAT_N, SINT16, SINT16_N, SINT32, SINT32_N, SINT8, SINT8_N, UINT16, UINT16_N, UINT32, UINT32_N, UINT8, UINT8_N, VARIABLE_LENGTH

2.6.1.66 Dcm/DcmConfigSet/DcmDsp/DcmDspRoutine/DcmDspStopRoutine/DcmDspRoutineInterfaceArgumentIntegrity

[REMOVED] Default Value	false

2.6.1.67 Dcm/DcmConfigSet/DcmDsp/DcmDspRoutine/DcmDspStopRoutine/DcmDspStopRoutineIn/DcmDspStopRoutineInSignal/DcmDspRoutineSignalType

[RTA-BSW] Lower Multiplicity	1
[AUTOSAR] Lower Multiplicity	0
[RTA-BSW] Enumeration Literals	BOOLEAN, SINT16, SINT16_N, SINT32, SINT32_N, SINT8, SINT8_N, UINT16, UINT16_N, UINT32, UINT32_N, UINT8, UINT8_N, VARIABLE_LENGTH
[AUTOSAR] Enumeration Literals	BOOLEAN, FLOAT, FLOAT_N, SINT16, SINT16_N, SINT32, SINT32_N, SINT8, SINT8_N, UINT16, UINT16_N, UINT32, UINT32_N, UINT8, UINT8_N, VARIABLE_LENGTH

2.6.1.68 Dcm/DcmConfigSet/DcmDsp/DcmDspRoutine/DcmDspStopRoutine/DcmDspStopRoutineOut/DcmDspStopRoutineOutSignal/DcmDspRoutineSignalType

[RTA-BSW] Lower Multiplicity	1
[AUTOSAR] Lower Multiplicity	0
[RTA-BSW] Enumeration Literals	BOOLEAN, SINT16, SINT16_N, SINT32, SINT32_N, SINT8, SINT8_N, UINT16, UINT16_N, UINT32, UINT32_N, UINT8, UINT8_N, VARIABLE_LENGTH
[AUTOSAR] Enumeration Literals	BOOLEAN, FLOAT, FLOAT_N, SINT16, SINT16_N, SINT32, SINT32_N, SINT8, SINT8_N, UINT16, UINT16_N, UINT32, UINT32_N, UINT8, UINT8_N, VARIABLE_LENGTH

2.6.1.69 Dcm/DcmConfigSet/DcmDsp/DcmDspSecurity/DcmDspSecurityMaxAttemptCounterReadoutTime

[RTA-BSW] Lower Multiplicity	0
[AUTOSAR] Lower Multiplicity	1

2.6.1.70 Dcm/DcmConfigSet/DcmDsp/DcmDspSecurity/DcmDspSecurityResetAttemptCounterOnTimeout

[RTA-BSW] Description	This configuration parameter shall control the resetting behavior of SecurityAccess AttemptCounter upon Delay Timer expiration. The configuration parameters DcmDspSecurityResetAttemptCounterOnTimeout and DcmRbDspSecurityApplCallbackOnDelayTimeExpiry shall not be configured as TRUE at the same time, either DcmDspSecurityResetAttemptCounterOnTimeout or DcmRbDspSecurityApplCallbackOnDelayTimeExpiry shall be configured as TRUE.
[AUTOSAR] Description	This configuration parameter shall control the resetting behavior of SecurityAccess AttemptCounter upon Delay Timer expiration.

2.6.1.71 Dcm/DcmConfigSet/DcmDsp/DcmDspSecurity/DcmDspSecurityRow/DcmDspSecurityADRSize

[RTA-BSW] Maximum Value	4294967285
[AUTOSAR] Maximum Value	4294967295

2.6.1.72 Dcm/DcmConfigSet/DcmDsp/DcmDspSecurity/DcmDspSecurity-Row/DcmDspSecurityKeySize

[RTA-BSW] Maximum Value	4294967285
[AUTOSAR] Maximum Value	4294967295

2.6.1.73 Dcm/DcmConfigSet/DcmDsp/DcmDspSecurity/DcmDspSecurity-Row/DcmDspSecuritySeedSize

[RTA-BSW] Maximum Value	4294967285
[AUTOSAR] Maximum Value	4294967295

2.6.1.74 Dcm/DcmConfigSet/DcmDsp/DcmDspSession/DcmDspSession-Row/DcmDspSessionForBoot

[REMOVED] Default Value	DCM_NO_BOOT

2.6.1.75 Dcm/DcmConfigSet/DcmPageBufferCfg/DcmPagedBufferEnabled

[RTA-BSW] Introduction	true = Paged buffer handling enabled false = Paged Buffer handling disabled If set to true, pduRCancelTransmit and pduRCancelReceive should be enabled.
[AUTOSAR] Introduction	true = Paged buffer handling enabled false = Paged Buffer handling disabled

2.6.1.76 Dcm/DcmGeneral/DcmTaskTime

[RTA-BSW] Value Range	1E-4..0.1
[AUTOSAR] Value Range	0..INF

2.7 API Definition

Table 2.446.Dcm Supported API functions

`Dcm_Init`	Initialises the Dcm module.
`Dcm_MainFunction`	Processes the tasks of the main loop.
`Dcm_GetVersionInfo`	Returns the version information of the Dcm module.
`Dcm_DemTriggerOnDTCStatus`	Triggers on changes of the UDS DTC status byte.
`Dcm_GetVin`	Returns the VIN (as defined in SAE J1979-DA).
`Dcm_SetDeauthenticatedRole`	Sets a new role used in deauthenticated state for that connection.

`Dcm_GetSecurityLevel`	Returns the active security level value.
`Dcm_GetSesCtrlType`	Returns the active session control type.
`Dcm_GetActiveProtocol`	Returns the active protocol name.
`Dcm_ResetToDefaultSession`	Allows the application to reset the current session to Default session.
`Dcm_TriggerOnEvent`	A call to this function allows the triggering of an event linked to a ResponseOnEvent request.
`Dcm_SetActiveDiagnostic`	Activates and deactivates the call of ComM_DCM_ActiveDiagnostic() function.
`Dcm_StartOfReception`	Called at the start of receiving an N-SDU.
`Dcm_CopyRxData`	Called to provide the received data of an I-PDU segment (N-PDU) to the upper layer.
`Dcm_TpRxIndication`	Called after an I-PDU has been received via the TP API and the result indicates whether the transmission was successful or not.
`Dcm_CopyTxData`	Called to acquire the transmit data of an I-PDU segment (N-PDU).
`Dcm_TpTxConfirmation`	Called after the I-PDU has been transmitted on its network and the result indicates whether the transmission was successful or not.
`Dcm_TxConfirmation`	The lower layer communication interface module confirms the transmission of an IPDU.
`Dcm_ComM_NoComModeEntered`	Informs the Dcm module about a ComM mode change to COMM_NO_COMMUNICATION.
`Dcm_ComM_SilentComModeEntered`	Informs the Dcm module about a ComM mode change to COMM_SILENT_COMMUNICATION.
`Dcm_ComM_FullComModeEntered`	Informs the Dcm module about a ComM mode change to COMM_FULL_COMMUNICATION.
`Dcm_CsmAsyncJobFinished`	Called from Csm upon finishing an asynchronous job processing.
`Dcm_KeyMAsyncCertificateVerifyFinished`	Called from KeyM upon finishing an asynchronous certificate verification.

For further information on these functions, please refer to AUTOSAR_SWS_DiagnosticCommunicationManager.pdf

2.8 Configuration Advice

> **NOTE**
>
> Before configuring this module, please note that some of the default values used by ConfGen to produce the default ECU Configuration can be changed, and this is done by either adding a Conf-Gen Enhancer Mapping (CEM) or by adding a rule to an *algo.properties* file. For more information, see the "Configuration Generation" section of the RTA-CAR User Guide.

2.8.1 Multicore support

The following should be noted with regard to Dcm's support for Multicore.

- When the diagnostic services are configured in the application scheduler (e.g rba_DiagAdapt scheduler) a specific order of calls is expected. The application scheduler will close the diagnostic service processing if the service init is called.
- During Init, no protection to interrupts/spinlocks are added as it is expected that interrupts and OS on all cores would be enabled only after the Init sequence is completed (although there's no harm in having a lock call during init).
- Dcm_DslDsd_ResetROEScheduler should normally only be called within the Dcm Main Function context, but if the project requires that it is called outside the Main Function context, this is only allowed inside DcmAppl_DcmSesCtrlChangeIndication().
- The S/R communication for IOCBI requires the existence of ControlMask in the request, and also a DID signal structure with more than one signal on the application side.

The scaling feature is introduced for the routine control service and is supported only for client server interfaces of routine.

2.8.2 Interrupt Protection Locks

Dcm uses Exclusive-Areas (implemented by Interrupt-Locks) to support a multi-task system architecture. The following Interrupt Protection locks exist within Dcm:

- SchM_Enter_Dcm_Global()
- SchM_Exit_Dcm_Global()

The following locks are mainly introduced to maintain data consistency between P2Max, P2StrMax and P3Max:

- SchM_Enter_Dcm_DslTimer()
- SchM_Exit_Dcm_DslTimer()

The Exclusive-Area APIs in Dcm_Cfg_SchM.h should be reviewed

2.8.3 Asynchronous Server Point Handling

Dcm supports Asynchronous Server Point handling for DidData, SecurityAccess and Routine Control Interfaces, which can be enabled by setting the following parameters to TRUE:

- DcmRbDspDataUseAsynchronousServerCallPoint (For DidData)
- DcmRbDspSecurityRowUseAsynchronousServerCallPoint (For SECA)
- DcmRbDspRoutineUseAsynchronousServerCallPoint (For Routine Control)

In all cases, dcmDspDataUsePort should be set to USE_DATA_ASYNCH_CLIENT_SERVER.

This feature does not include Range DIDs.

2.8.4 Generic Connection Handling

All configured PDUs (DcmDslProtocolRxPduRef and DcmDslProtocolTxPduRef) under a Generic Connection should be consistent (have the same MetaData Reference).

The Source and Target Address received during Start Of Reception on a Generic Connection will be interpreted in Big Endian format with the Source Address occupying the first two bytes of the MetaDataPtr and Target Address occupying the next two bytes. The expected Target Address per protocol can be configured using the DcmDspProtocolEcuAddr parameter. If this parameter is not configured, the Target Address will be considered to be zero.

2.8.5 Parallel Processing

Parallel Processing allows OBD and another non-OBD protocol to be handled at the same time. This feature is enabled via the DcmRbParallelProcessingAllowed parameter.

Enabling Parallel Processing will involve the generation of separate resources, which will result in an increase in RAM consumption.

As transmissions can occur in parallel, separate channels should be configured for these protocols.

DcmAppl_SupplierNotification is common interface for both protocols, so the application should ensure its re-entrancy for OBD and non-OBD requests (i.e. can be done with RxPduId, Source Address etc).

The DcmRbQueuingOfRequestEnabled and DcmRbSingleBufferConceptDisabled extension parameters apply only for non-OBD protocol handling.

If DcmRbMaxRequestsInParallel is configured to 2 then the requests can be processed in Parallel. When the parameter is set to 1 or not configured then the requests won't be processed in Parallel, and the processing of requests will be based on protocol priority.

Default value and Minimum value for DcmRbMaxRequestsInParallel is 1 and

currently the Maximum value for this parameter is restricted to 2.

If same service is requested through different protocols, the second request will be delayed until the first request is completed as per [SWS_Dcm_01604]. The service interpreter is considered as a resource and since both the requests needs to call the same interpreter in order to process the service request, second request will be delayed until first request is processed. During this, if P2 timer is expired NRC0x78 response will be sent to second request.

2.8.6 Memory mapping

A template is delivered for the MemMap, which should be modified based on project requirements and included as a header file.

2.8.7 Enable Primary ECU

Configuring to DEM_OBD_MASTER_ECU will throw an error. There is no difference whether DemOBDSupport is configured as DEM_OBD_MASTER_ECU or DEM_OBD_PRIMARY_ECU. Configuring to DEM_OBD_MASTER_ECU may mislead someone to believe that the Dem supports all features that are legally required for a master ECU, which is not guaranteed currently. So, it should be configured as DEM_OBD_PRIMARY_ECU. Set `Dem/DemGeneral/DemOBDSupport` to DEM_OBD_PRIMARY_ECU for OBD Primary ECU.

2.8.8 DEM-SR Interfaces

Dcm supports the generation of Sender-Receiver elements for Dem. This feature is enabled/disabled by the DcmRbDemSREnabled extension parameter.

Dem forwards the contents to Dcm in the DcmDspData container with the short name starting with FOR_DEM. If DcmRbDemSREnabled is set to TRUE, ensure that the short name of the Dcm-related DcmDspData container element does not start with FOR_DEM as this will result in it being treated as a Dem-forwarded element and not as Dcm's element.

2.8.9 UDS Services

This subsection provides some specific advice on Dcm's support for the indicated UDS Services.

2.8.9.1 ClearDiagnostic service (0x14)

To integrate service 0x14 (ClearDiagnosticInformation) in Dcm, the service has to be configured inside DcmDsdServiceTable container (/AUTOSAR/EcucDefs/Dcm/DcmConfigSet/DcmDsd/DcmDsdServiceTable).

For Dcm specific CDI, there is no need to configure an INI function, and it should be set to "NULL_PTR". This is because cancellation of a Clear operation is not supported by DEM.

2.8.9.2 ReadDTCInformation service (0x19)

This service allows a client to read the status of server-resident Diagnostic Trouble Code (DTC) information from any server, or group of servers within a

vehicle. Unless otherwise required by the particular sub-function, the server returns all DTC information. When the RDTC page buffer is enabled, the RDTC service must be configured in any of the service tables with at least one of the following sub-functions: 0x02, 0x0A, 0x15, 0x08, 0x13, 0x14, 0x17.

OBD service 0x02 should be configured first to use RTA-BSW specific RDTC sub-function 0x05 (Dcm_Dsp_GetFreezeFrameDataByRecord). This sub-function should be used only to read freeze frame data for Record number 0.

2.8.9.3 SecurityAccessInformation service (0x27)

Dcm allows DcmDspSecurityADRSize, DcmDspSecuritySeedSize and DcmDspSecurityKeySize to be configured as more than 255, so the DataType of arguments (ADR size, SEED size and KEY size) in the application functions Xxx_GetSeed() and Xxx_CompareKey() must be modified from UINT8 to UINT32.

2.8.9.4 Authentication (0x29)

The 0x29 service is configured in the DcmDsdServiceTables container (/Dcm/DcmConfigSet/DcmDsd/DcmDsdServiceTables). It interacts with the KeyM and Csm components, so these need to be integrated if service 0x29 is configured.

The Dcm_ReadAccessRights, Dcm_WriteAccessRights and Dcm_GetPersistedStatusOfAuthenticationConnections callouts are required to store the authentication related information in non-volatile memory, and so need to be implemented if the authentication status of a tester needs to be persisted.

2.8.9.5 ReadDataByPeriodicIdentifier service (0x2A)

For RDPI Type2 requests, the response will be transmitted through the TxPdu configured for Periodic Transmission to the connection from which the request is received. The periodic transmission rate should be configured considering DCM raster, TXPdu and the number of PDIDs to be transmitted.

Scheduler Type #1: When the PeriodicRateCounter reaches zero then the transmission of all the scheduled PDIDs will commence, starting with the first in the list of scheduled PDIDs.

The NumPeriodicAddr PDIDs (the number of available protocol specific periodic data response message address information IDs allocated per scheduler call) will be transmitted at each scheduler call, using the addresses that are available. This continues until all NumPDID scheduled PDIDs have been transmitted, and will restart the next time PeriodicRateCounter reaches zero.

2.8.9.6 ControlDTCSetting (0x85)

The ControlDTCSetting service allows you to enable or disable DTC events. This service is implemented as specified in Autosar 4.3.1, so if ControlOptionRecord is enabled, values other than 0xFFFFFF are not supported.

The ControlDTCSetting service only works with AR 4.3.1 DEM interfaces for enabling/disabling DTC events.

2.8.9.7 ResponseOnEvent (0x86)

The following configuration items should be noted.

- The configuration parameter "DcmDslProtocolTransType" should specify TYPE1 or TYPE2.
- For the DcmDsdSidTabServiceId (0x19) parameter the sub function should only be "DcmDsdSubServiceId (0x0E).
- DcmDsdSubServiceFnc = Dcm_Dsp_ReportFailedDTC for reporting the DTC via RDTC service.

2.8.10 DcmDspDidUsePort

The IOCBI service supports only Atomic SR interface, so DcmDspDidUsePort should be configured either with USE_ATOMIC_SENDER_RECEIVER_INTERFACE or USE_ATOMIC_SENDER_RECEIVER_INTERFACE_AS_SERVICE. The DcmDspDidControlEnableMasks and DcmDspDidControlEnableMask parameters must be configured if DcmDspDidUsePort is configured as Atomic SR, and the DcmRbAtomicSenderReceiverCommunication parameter must be configured as TRUE.

2.8.11 DcmDspDidControlMaskSize

Configuring a DcmDspDidControlMaskSize larger than 4 is only allowed if DcmDspDataUsePort is set to USE_DATA_ASYNCH_CLIENT_SERVER or USE_DATA_SYNCH_CLIENT_SERVER. In this case, DcmDspDidControlMask should be equal to DCM_CONTROLMASK_EXTERNAL. (Note: Configuring DcmDspDidControlMask with DCM_CONTROLMASK_EXTERNAL is not supported if the DcmDspDataUsePort of the signal is configured as USE_DATA_ASYNCH_FNC or USE_DATA_SYNCH_FNC).

2.8.12 DcmDspEnableObdMirror

ReadDataByIdentifier for OBD DID range (0xF400 - 0xF8FF) is supported when DcmDspEnableObdMirror is set to TRUE. Configure Service 0x01 in OBD to support ReadDataByIdentifier for OBD DID range of 0xF400 to 0xF4FF, Service 0x06 in OBD to support ReadDataByIdentifier for OBD DID range of 0xF600 to 0xF6FF and Service 0x09 in OBD to support ReadDataByIdentifier for OBD DID range of 0xF800 to 0xF8FF.

Routine Control for OBD RID (0xE000 - 0xE003) is supported when DcmDspEnableObdMirror is set to TRUE. Project will configure OBD Service $08 and TID the same as classic OBD. DCM will communicate to tester in J1979-2 format. When DcmDspEnableObdMirror is set to FALSE, project will not configure availability RIDs.

If DcmDspEnableObdMirror is TRUE, then project shall configure service 0x01 in DcmDsdService mandatorily. If DcmDspEnableObdMirror is FALSE, then project shall not configure availability DID in the range (0xF400-0xF4FF, 0xF600-0xF6FF and 0xF800-0xF8FF).

2.8.13 Full Communication Mode indication

Before initiating the transmission of the diagnostic answer, Dcm will wait the Full Communication mode indication, which is passed from ComM via a call to Dcm_-ComM_FullComModeEntered. The time for Dcm to wait should be no longer than the P2ServerMax calculated from the moment the request was received.

2.8.14 Support Single Did in a DidRange

Dcm allows DcmDspDidIdentifier to be configured between DcmDspDidRangeIdentifierLowerLimit and DcmDspDidRangeIdentifierUpperLimit, while DcmDspDidRangeHasGaps for that DcmDspDidRange is set to True. The project needs to ensure that such a configuration is intended.

2.8.15 DcmRbEcuResetTime

The extension parameter DcmRbEcuResetTime provides the time taken for the reset of the ECU and it is relevant only in case of jump from boot using the Dsp-supported EcuReset (0x11) or DiagnosticSessionControl (0x10) services.

If configured, DcmRbEcuResetTime should be a multiple of DcmTaskTime and less than the P2*Max(DcmDspSessionP2StarServerMax) configured for the protocol used for jump to bootloader. As P2*Max is different for different sessions, DcmRbEcuResetTime should be configured to a value less than the minimum value of P2*Max among all the sessions.

2.8.16 DcmRbRTEsupport

This extension parameter has now been removed. You should instead use the EcuCRbRteInUse parameter (in EcuCRbRteInUseEcuC/EcuCRbSettings), which should be set to TRUE.

2.8.17 DcmRbRTEsupportsPRportFunctionality

This extension parameter should be configured at your discretion. If the RTE component used by the Project supports complete PR port functionality, then this parameter should be set as TRUE, otherwise it should be set to FALSE.

For DIDs having both Read and Write functionality, a PR port will be generated if the parameter has been set as TRUE, otherwise separate P and R ports will be generated.

2.8.18 DcmDslProtocolRxConnectionId

For every DcmDslMainConnection in the project, the parameter DcmDslProtocolRxConnectionId has to be mandatorily configured.

2.8.19 DcmRbDsdSidTabSidInitFunction

This extension parameter should not be configured for a Dcm Internal service. For an external service, this parameter is optional, and should be configured with a valid external Init function only if the service needs Init handling. The external init function has to be implemented by the application that is implementing the

external service.

2.8.20 Function Protoype changes

The function prototype for all Dcm Services has been modified to include the parameters OpStatus, pMsgcontext, and a NegativeResponseCode and it supports a return of type Std_ReturnType.

2.8.21 FrTpMaxBufferSize

Dcm has previously referred to the local FrTpMaxBufferSize parameter of FrTp to validate that the configured buffer size was less than or equal to the maximum buffer size that FrTp can transmit. As Dcm is independent of protocol, this validation check has now been removed (and the FrTpMaxBufferSize parameter has been removed from FrTp).

2.8.22 CanTp configuration for OBD

When the OBD protocol is configured on CAN, and the RxPduId is shared between UDS and OBD, then ISO_14229-3 dictates that the CAN frame shall be 8 bytes and any unused bytes shall be padded. You must ensure that the canTpRxPaddingActivation parameter in the CanTp configuration is turned ON for the corresponding Nsdu.

2.8.23 ServiceRequestNotification

The Datasize parameter of ClientServerInterface→ServiceRequestNotification has been changed to uint32 (from earlier implementation of uint16) so that there is no loss of data if PduLengthTypeEnum has been configured to uint32.

2.8.24 Xxx_Indication

Autosar states that the Datasize argument for Xxx_Indication can only be of type uint16, but note that the pointer used to store Rx buffer size is of PduLengthType, which itself can be configured to uint8, uint16 or uint32.

2.8.25 Cancellation of Dcm Services

Previously, when cancellation of a Dcm service was requested, this was handled via a direct call of the Ini of the respective service requiring cancellation. Now, the service to be cancelled is called with the OpStatus parameter set to DCM_-CANCEL (because the service, upon invocation with OpStatus set to DCM_-CANCEL, will call its Ini function to cancel).

2.8.26 Reset to Default Session

The Dcm_ResetToDefaultSession API should always return E_OK and accept the request, even when Dcm is busy (i.e. processing Tester request / ROE / RDPI request). The request will be taken when the API is called, and then processed in the next Dcm MainFunction cycle.

2.8.27 Queuing in Dcm

Dcm will not queue a second request until the first request is fully received (i.e.

until positive Dcm_TpRxIndication for the first request is received).

2.8.27.1 Dcm Routing path for Tp modules

The default setting for the Dcm Routing path for Tp modules is TRUE, but this can be overridden by configuring IsDcmRoutingForTpOnly in algo.properties. The valid options for IsDcmRoutingForTpOnly are as follows:

- **manprop_IsDcmRoutingForTpOnly=true:** This is the default in order to retain compatibility with existing user configurations. If there is an improper system description configuration for Tp path, ConfGen will still support the generation of Dcm Routing path for Tp modules.
- **manprop_IsDcmRoutingForTpOnly=false:** You need to follow the AR system template for Tp configuration.

2.8.28 Manual Configuration

The following parameters must be manually configured in the ISOLAR-AB application after running the Confgen tool.

- Dcm/DcmConfigSet/DcmDsl/DcmDslProtocol/DcmDslProtocolRow/DcmDslConnection/DcmDslMainConnection/DcmDslProtocolComMChannelRef
- Dcm/DcmConfigSet/DcmDsl/DcmDslProtocol/DcmDslProtocolRow/DcmDslConnection/DcmDslMainConnection/DcmDslProtocolRx
- Dcm/DcmConfigSet/DcmDsl/DcmDslProtocol/DcmDslProtocolRow/DcmDslConnection/DcmDslMainConnection/DcmDslProtocolTx
- Dcm/DcmConfigSet/DcmDsl/DcmDslProtocol/DcmDslProtocolRow/DcmDslConnection/DcmDslPeriodicTransmission
- Dcm/DcmConfigSet/DcmDsl/DcmDslProtocol/DcmDslProtocolRow/DcmDslConnection/DcmDslResponseOnEvent

2.8.29 Configuring Type 2 (UUDT) Periodic Messages

[SWS_Dcm_01101] states that all periodic responses (scheduled responses, not the initial response) will use dedicated IF-PDUs, and that transmission will be done through PduR. Only Unacknowledged Unsegmented Data Transfer (UUDT) messages are supported.

Thus, when configuring Dcm, the configuration for periodic messages of Type 2 (UUDT) should be done via Dcm > PduR > CanIf. Tx messages should not be routed via CanTp.

The DcmDslPeriodicTxPduRef configuration parameter must be referred to the PDU that will be subsequently routed by PduR directly to the CanIf module, otherwise a Dcm 0x2A periodic positive response is sent, but the periodic messages are not transmitted afterwards.

2.8.30 ProtocolPreemptionHandlingConfiguration for DcmRbSingleBufferConceptDisabled

If Protocol Preemption is enabled/required (via the configuration of multiple protocols with different DcmDslProtocolPriority), the recommendation is to configure separate Buffers for Reception and Transmission (using the configuration parameter DcmRbSingleBufferConceptDisabled) to avoid data corruption.

A possible use case where corruption may occur: a low-priority request is in progress when a high-priority request is received as an interrupt. The request buffer is overwritten by the response of the low-priority protocol. This scenario can occur when the control is in the service and the response has yet to be filled when a StartofReception for a high-priority request is received. After the RxIndication is processed and control is given back to the service (with a single buffer concept) the request can be overwritten by the response, which will, in turn, lead to an incorrect final response for the high-priority request.

2.9 Integration Advice

2.9.1 Init Functions

Dcm_Init has to be called during the startup phase of the ECU.

Dcm_Dsp_SecaPowerOnDelayIni initialises the "Power On Delay", which is required by the Security Access service. It must be scheduled separately to avoid it being called for all "initialisation", and ensure that it is only called for the "Power On Init".

2.9.2 DeInit Functions

No specific DeInit function is supported by Dcm. Calling Dcm_Init again would cause the reinitialisation of the Dcm, which would lead to termination of existing activity without any confirmation or indication.

2.9.3 Scheduled Functions

The cyclic function Dcm_MainFunction must be configured and scheduled, and Dcm_Init needs to be configured in the initialisation sequence.

Dcm_Init should be called along with Dem_Init only after NvM_ReadAll is completed. This is to ensure that NvM data is available for successful restoration in bootblock and during jump from boot to application.

2.9.4 Protocol Pre-emption

During protocol preemption, the Dem_DcmCancelOperation API needs to be called from DcmAppl_StartProtocol in order to cancel the pending operations in Dem that were triggered by the earlier service, if the earlier service is vendor specific.

2.9.5 Handling of DID APIs

The following 3 APIs are available to get DID data:

- Dcm_GetIndexOfDID
- Dcm_GetLengthOfDIDIndex
- Dcm_GetDIDData

The following sequence should be followed:

1. Dcm_GetIndexOfDID should be called first to reset the DID structure to default values (e.g. integer types to zero, boolean types to false and OpStatus to DCM_INITIAL) and setup DID index data.
2. Then Dcm_GetLengthOfDIDIndex should be called to setup DID length. Dcm_GetIndexOfDID must be finished successfully before calling Dcm_GetLengthOfDIDIndex.
3. The Dcm_GetDIDData is called to get DID data. To ensure that the DID structure is properly updated, both Dcm_GetIndexOfDID and Dcm_GetLengthOfDIDIndex should be finished successfully before calling Dcm_GetDIDData. Configured ports for data (in DCM) will be used to get data from application.

2.9.6 DID Data Size and DID Signal

The following points should be noted during the configuration for DID Data Size and DID Signal:

- All the DIDs within a project shall use `DataByteSize`.
- For position, `DcmDspDidByteOffset` shall be configured.
- For Data Size, `DcmDspDataByteSize` shall be configured. If the parameter `DcmDspDataUsePort` is configured as *USE_DATA_SENDER_RECEIVER/USE_BLOCK_ID*, the User need not update this field. Instead they may invoke `generateId` for Dcm, in order to automatically configure the same depending on the parameters `DcmDspDataType/DcmDspDataBlockIdRef` → `NvMNvBlockLength` respectively.

2.9.7 PduR_DcmCancelReceive

When low priority protocol reception is in progress and a high priority protocol request is received, Dcm will always call PduR_DcmCancelReceive to cancel low priority protocol reception, so the PduRCancelReceive parameter in the PduR configuration should be set to TRUE.

2.9.8 Interrupt Protection Locks

The following Interrupt Protection locks exist within Dcm:

- SchM_Enter_Dcm_Global()
- SchM_Exit_Dcm_Global()

The following locks are mainly introduced to maintain data consistency between P2Max, P2StrMax and P3Max:

- SchM_Enter_Dcm_DsldTimer()
- SchM_Exit_Dcm_DsldTimer()

Locks can be implemented as No Nested or Nested by integrator.

2.9.9 DcmAppl API/Functions

The following API/Functions are included with the Dcm module. For full details of the In/Out Parameters and Return Vales for each API/Function, see the commented documentation in the relevant <name>.c file.

Table 2.447.DcmAppl Supported API functions

`DcmAppl_CheckEncryptionCompressionMethod`	This API provides the compression and encryption method from the tester request to the application.
`DcmAppl_ConfirmationRespPend`	This API confirms the successful transmission of a waitpend response.
`DcmAppl_CopyRxData`	This API is called to inform the application about the data received in each call.
`DcmAppl_Dcm_CheckPermissionForTransferData`	Called by Dcm to allow the application to do checks before calling DcmAppl_Dcm_WriteMemory.
`DcmAppl_Dcm_ProcessRequestDownload`	This function provides the compression and encryption method from the tester request to the application.
`DcmAppl_Dcm_ProcessRequestTransferExit`	This API terminates a download/upload process (for the implementation of UDS service RequestTransferExit).
`DcmAppl_Dcm_ProcessRequestUpload`	This function provides the compression and encryption method from the tester request to the application.
`DcmAppl_Dcm_ReadMemory`	This api implements Memory read functionality.
`Dcm_WriteMemory`	This API implements Write to memory functionality.
`DcmAppl_DcmCancelPagedBufferProcessing`	This API indicates the service to stop paged buffer transmission.
`DcmAppl_DcmCheckCDTCRecord`	This api checks the validity of the controlOptionRecord passed in the CDTC request.
`DcmAppl_DcmCheckControlMaskAndState`	This api is invoked from IOCBI service if the control mask or control status bits are present in the request.

DcmAppl_DcmCheckRdbiResponseLength	Application function called to validate total response length if there's at least one valid DID in the multi-DID RDBI request.
DcmAppl_DcmCheckRoutineControlOptionRecord	Called by Dcm_DcmRoutineControl if a received request contains a control option record.
DcmAppl_DcmCheckSeed	Request to application for validation of passed seed value.
DcmAppl_DcmCheckWDBIReqLen	Used by the application to perform exact length check for the request if variable length signals are configured for the requested DID.
DcmAppl_DcmClearRoeEventInformation	This API is used to reset the ROE scheduler and then clear the ROE transmission queue.
DcmAppl_DcmCommControlConditionCheck	This api is called after the request length check for the UDS service Communication control (0x28) request.
DcmAppl_DcmComModeError	Called by DCM to inform the application that DCM is not in FULL COMMUNICATION or is unable to move to FULL COMMUNICATION MODE for the channel which is having the passed NetworkId.
DcmAppl_DcmConfirmation	This function confirms the successful transmission or a transmission error of a diagnostic service.
DcmAppl_DcmConfirmation_DcmNegResp	This API is invoked as an indication called by DSD after sending the DSD-generated Neg response and after sending the General rejectresponse by DSL.
DcmAppl_DcmConfirmation_GeneralReject	This API is invoked as an indication called by DSD after sending the General Reject response due to max number of configured NRC 0x78 being reached.
DcmAppl_DcmConfirmationRDPI	This API confirms the successful/unsuccessful transmission of periodic response.
DcmAppl_DcmControlDtcSettingEnableStatus	This API is called when the control dtc setting status is changed.
DcmAppl_DcmEcuResetPreparation	This API prepares for reset. Code should be added for the actions to be performed before Ecu Reset.

DcmAppl_DcmGetNRCForMinLengthCheck	This API is used by the application to get the NRC to be returned as minimum length check has failed for vendor-specific service supporting sub-function.
DcmAppl_DcmGetPermissionForMemoryAccess_u8	This API is used to check the memory read and write access permission.
DcmAppl_DcmGetPermTxWarmResp	This function is called to get the permission for transmission of the Warm response/Warm request stored in Boot.
DcmAppl_DcmGetRemainingResponseLength	This API is used only for the KWP2000 services and when the splitting responses feature is enabled in the configuration (by enabling the container DcmDslKWPSplitResponseFeatureRB).
DcmAppl_DcmGetRxPermission	This API is called by Dcm_StartOfReception before processing the new request.
DcmAppl_DcmGetSesChgPermission	Request to the application to check if the conditions allow the requested session control to be started.
DcmAppl_DcmGetStoreType	This API is used to retrieve the Type of Storage required by the User for Jump to Boot.
DcmAppl_DcmGetUpdatedDelayForPowerOn	This API is used for providing the means to the application to change the power On delay for security access service on COLD start.
DcmAppl_DcmGetUpdatedDelayTime	Request to application to provide the updated delay time in milliseconds.
DcmAppl_DcmIndicationFuncTpr	This API is called by DCM if functional tester present request is received with the suppress positive bit set in the request.
DcmAppl_DcmIsComModeDefault	This api is called to check whether the ComM channel identified by the passed Network ID is in default ComMode (DCM_ENABLE_RX_TX_NORM_NM).
DcmAppl_DcmMemSizeCheckForTransferData	Callout function called by DCM to allow the application to do checks before calling DcmAppl_Dcm_ReadMemory/DcmAppl_Dcm_WriteMemory to read-/write the data in the TransferData service.
DcmAppl_DcmModifyResponse	This function allows the application to conditionally modify the responses from all the services.

DcmAppl_DcmReadDataNRC	If the ReadData function has returned E_NOT_OK this function is called to obtain the NRC from the application.
DcmAppl_DcmReadDddiFromNvM	This API is called for each DDDID that is configured.
DcmAppl_DcmRespWarmInit	This API is called for every Jump To Boot transition.
DcmAppl_DcmSecaInvalidKey	If the Key sent by the tester is invalid for a particular security level then this API is called with the index of the security table.
DcmAppl_DcmSecaValidKey	If the Key sent by the tester is valid this API is called with the index of the security table which contains the security level for the key was sent.
DcmAppl_DcmSecurityLevelLocked	For each security level the number of unsuccessful attempts after which the security level is locked will be configured (Lock Count).
DcmAppl_DcmSesCtrlChangeIndication	This API is invoked as an indication to the application that value of active session control is changed.
DcmAppl_DcmStartProtocol	The application can use this callback function to check whether the conditions for further processing of the protocol identified by the passed ProtocolID.
DcmAppl_DcmStopProtocol	Invoked if a running diagnostic request is preempted by a higher-priority request - the application is requested to abort further processing of the running request and finish with Dcm_ProcessingDone().
DcmAppl_DcmStoreRespForJump	This API is invoked for Storing of Response information when the user-specific service is used for Jumping to Boot.
DcmAppl_DcmSwitchCommunicationControl	This api is called after sending the confirmation for the UDS service Communication control (0x28).
DcmAppl_DcmUpdateRxTable	Called by Dcm_Init during Power On so the application can switch the rx table based on the ECU type (master/slave).

DcmAppl_DcmWriteOrClearDddiInNvM	Called by DDDID definition from Dcm when a DDDID is defined/cleared so the application can store/clear the DDDID definition in NvM.
DcmAppl_DDDI_Read_Memory_Condition	Check if it is permitted to read from the memory of the ECU as per the specification in a DynamicalDefinedID record.
DcmAppl_DisableDTCSetting	This api is called to get permission to disable DTC Setting.
DcmAppl_EnableDTCSetting	Invoked after calling "Dem_EnableDTC-Setting" the return value of which is passed to DcmAppl_EnableDTCSetting as DtcEnableStatus.
DcmAppl_ManufacturerNotification	Indicates to the manufacturer application the successful reception of a new request.
DcmAppl_OBD_Mode04	This API is used for check conditions for OBD Service 0x04.
DcmAppl_P3TimeoutIndication	This API is used to indicate to the application that P3 timeout in DSL has occurred.
DcmAppl_StartOfReception	Inform the application regarding the the reception of a new request.
DcmAppl_SupplierNotification	This API indicates to the application the successful reception of a new request and it is called immediately before processing the diagnostic service XXX_Dcm‹DiagnosticService›.
DcmAppl_Switch_DcmBootLoaderReset	Notify the application regarding Jump to BootLoader.
DcmAppl_Switch_DcmDiagnosticSessionControl	This API is used to notify the application regarding session transition.
DcmAppl_Switch_DcmDriveToDriveReset	This API is used to notify the application regarding jump to drive to drive.
DcmAppl_Switch_DcmEcuReset	This api is called to notify the application regarding the ECU reset type in the tester request.
DcmAppl_Switch_DcmExecuteDscReset	Informs the application to execute reset where the reset is triggered due to DSC service when used for jump to boot.
DcmAppl_Switch_DcmExecuteEcuReset	Informs the application to execute reset where the reset is triggered due to EcuReset service when used for jump to boot/drive.

DcmAppl_Switch_DcmExecuteReset	This API is invoked to notify the application that Dcm shall trigger Ecu Reset soon after this API is called.
DcmAppl_Switch_DcmSysSupplierReset	This API is used to notify the application regarding jump to Sys SupplierBootLoader.
DcmAppl_TpRxIndication	This API is called to inform the application regarding the completion of copying of the request into the Dcm buffer.
DcmAppl_UserCommCtrlReEnableModeRuleService	Checks if communication control state needs to be re-enabled back to the default state DCM_ENABLE_RX_TX_NORM_NM.
DcmAppl_UserDIDModeRuleService	This API is invoked to do specific checks on the DID read/write/control operation before processing the service.
DcmAppl_UserDTCSettingEnableModeRuleService	This API is invoked to check for conditions to decide if DTC Setting state needs to be enabled.
DcmAppl_UserMemoryRangeModeRuleService	This API is used to do specific checks on the DID read/write/control operation before processing the service.
DcmAppl_UserRIDModeRuleService	Function to do specific checks on the RID in the routine control requests before processing the service.
DcmAppl_UserServiceModeRuleService	This API is used to do specific checks on the SID before processing the service.
DcmAppl_UserSubServiceModeRuleService	This API is invoked to do specific checks on the subfunction parameter before processing the service.
Dcm_GetCounterValue	This API is used to get the OS ticks for the autosar specific/Mcu OS counter.
Dcm_GetProgConditions	This API is used for restoring the Programming Information by the application.
Dcm_SetProgConditions	This API is used for setting the Programming Information in the application.

2.9.10 OS Resources

The number of the active OS resources, and their usage, depends on the user configuration of the AUTOSAR System and Basic Software. It is the user's responsibility to ensure that the `ENTER` and `EXIT` resource API are implemented to disable/enable interrupts as required. Alternatively they must be configured in the OS with the correct task/ISR allocation.

Dcm uses the following OS resources

- `Dcm_Global`
- `Dcm_DsldTimer`

2.10 Hardware Requirements

Table 2.448.Dcm Hardware Requirements

Support	Usage
FLASH	62462 bytes
RAM	6603 bytes

Usage calculated based on a sample configuration.

3 Diagnostic Event Manager

Module name	Dem
Package	RTA-DIAG
AUTOSAR Module ID	54
AUTOSAR Version	23-11

3.1 Introduction

The Diagnostic Event Manager (Dem) is located in the system services abstraction layer, forming part of the diagnostic (RTA-DIAG) stack.

Dem is responsible for processing, storing and retrieving information relating to diagnostic events and their associated data.

Dem offers interfaces that make this stored data available to the application layer and to other BSW modules. In particular, Dem is accessed by Dcm to respond to tester requests by reading data from fault memory (e.g. during development/testing and for post-sale fault analysis).

Dem also provides the change status of fault information to Fim.

3.1.1 Interaction/Dependencies

The diagram in the RTA-DIAG Introduction section shows the position of Dem (and the other RTA-DIAG modules) in the Autosar layered architecture. The interaction/dependencies of Dem with other modules is summarised here.

Diagnostic Communication Manager (Dcm)

- Dcm forwards the requests to Dem from an external or onboard diagnostic/test system, and is also responsible for the assembly of response messages (e.g. DTC, status information, etc.) back to the requestor.

Diagnostic Log and Trace (Dlt)

- Dlt provides Dem with a generic Logging and Tracing functionality. Dem informs and updates Dlt on changes to an event status and provides access on the current event-related data in order to log and trace this information.

Function Inhibition Manager (Fim)

- Dem informs and updates FiM on changes to an event status in order to enable/disable functionality (according to assigned dependencies).

NVRAM Manager (NvM)

- The Dem module retrieves information related to fault memory on Dcm's behalf.

Diagnostic Communication Manager for J1939 (J1939Dcm)

- J1939Dcm controls the SAE J1939 diagnostics communication protocol,

which is used by Dem.

rba_DiagLib

- The rba_Diaglib provides common functionality and utils for all RTA-DIAG modules, including Dem.

In addition to the above, SWCs (over RTE) will access/retrieve current event status from Dem, provide event-related data to Dem, and retrieve data from Dem.

Module Configuration

This module can be generated by the RTA-BSW Configuration Generator.

For any unsupported automatic configuration, the module can be manually configured by the user.

3.2 Supported Features

- Diagnostic Events
- Diagnostic Trouble Codes (DTCs) and DTC Groups
- Monitored Components
- Event Status Management
- Event Memory Management
- Debouncing of Diagnostic Events
- Store “conspicuous events” in NvM for system analysis support
- Fault Confirmation
- Event Combination
- Enable and Storage Conditions of Diagnostic Events
- Event Related Data
- Operation Cycle Management
- Operation Cycle Counters
- Aging of Diagnostic Events
- Healing of Diagnostic Events
- OBD-Specific Functionality
- J1939-Specific Functionality
- DemRatio for NON-OBD DTCs

3.2.1 DTC and MIL Calculation for Zero Emissions Vehicles

For zero-emission vehicles (ZEVs), a Malfunction Indicator Lamp (MIL) may not be triggered in the conventional sense. Therefore for the purposes of PID $21 and PID $4D, MIL activated shall be treated as a confirmed ZEV propulsion related DTC (OBD DTC) stored in a given ECU. This parameter ID would provide a user with a window into how far the operator has driven the vehicle with a confirmed ZEV

propulsion DTC.

The configuration switch 'DemRbOBDCalculatePID21AndPID4DOnConfirmedDTC' defines the calculation of PID $21 and PID $4D based on MIL status or confirmed DTC. If this configuration switch is enabled then calculation of PID $21 and PID $4D based on confirmed DTC. If this configuration switch is disabled then calculation of PID $21 and PID $4D based on MIL Status.

3.3 Extensions

3.3.1 Configuration Extensions

The following configuration parameters are available in version v12.7.0 in addition to those defined in AR v22-11

3.3.1.1 Dem/DemConfigSet/DemComponent/DemRbComponentAllowedRecoveries

Short Name	DemRbComponentAllowedRecoveries
Description	Number of allowed recoveries at a component during one ignition cycle. To limit system toggeling for unrobust monitorings.
Multiplicity	0..1
Value Range	0..255
Default Value	255
Type	ECUC-INTEGER-PARAM-DEF

3.3.1.2 Dem/DemConfigSet/DemComponent/DemRbResetPerformedRecoveriesValue

Short Name	DemRbResetPerformedRecoveriesValue
Description	Performed recoveries counter of a DemComponent will be reset on the operation cycle change. Hint: "DemRbComponentAllowedRecoveries" controls whether a component might be recovered (event reports PASSED). Related config parameter "DemRbSupportRecoveryLimitation" must be turned on.
Multiplicity	0..1
Default Value	false
Type	ECUC-BOOLEAN-PARAM-DEF

3.3.1.3 Dem/DemConfigSet/DemDTC/DemRbAlternativeDTC

Short Name	DemRbAlternativeDTC
Description	Alternative Diagnostic Trouble Code value
Introduction	(Range: 0x000000 and 0xFFFFFF are reserved for DTC groups by ISO 14229-1)

Multiplicity	0..1
Value Range	1..16777214
Default Value	-
Type	ECUC-INTEGER-PARAM-DEF

3.3.1.4 Dem/DemConfigSet/DemDTCAttributes/DemAgingAllowed

Short Name	DemAgingAllowed
Description	Defines if a DTC can be aged.
Introduction	TRUE: aging allowed FALSE: aging not allowed
Multiplicity	1..1
Default Value	false
Type	ECUC-BOOLEAN-PARAM-DEF

3.3.1.5 Dem/DemConfigSet/DemDTCAttributes/DemEventMemoryEntryFdcThresholdStorageValue

Short Name	DemEventMemoryEntryFdcThresholdStorageValue
Description	Threshold to allocate an event memory entry and to capture the Freeze Frame.
Multiplicity	0..1
Value Range	1..126
Default Value	-
Type	ECUC-INTEGER-PARAM-DEF

3.3.1.6 Dem/DemConfigSet/DemDebounceCounterBasedClass/DemRbDebounceCounterDecrementStepSizeAlternative

Short Name	DemRbDebounceCounterDecrementStepSizeAlternative
Description	Defines alternative step size for decrementation of the internal fault detection counter (PREPASSED) (e.g. for plant mode software). If the parameter is not configured, the value of “DemDebounceCounterDecrementStepSize” shall be considered.
Multiplicity	0..1
Value Range	1..32768
Default Value	-
Type	ECUC-INTEGER-PARAM-DEF

3.3.1.7 Dem/DemConfigSet/DemDebounceCounterBasedClass/DemRbDebounceCounterFdcThresholdStorageValue

Short Name	DemRbDebounceCounterFdcThresholdStorageValue
Description	Threshold to allocate an event memory entry and to capture the Freeze-Frame.
Multiplicity	0..1
Value Range	1..32767
Default Value	-
Type	ECUC-INTEGER-PARAM-DEF

3.3.1.8 Dem/DemConfigSet/DemDebounceCounterBasedClass/DemRbDebounceCounterIncrementStepSizeAlternative

Short Name	DemRbDebounceCounterIncrementStepSizeAlternative
Description	Defines alternative step size for incrementation of the internal fault detection counter (PREFAILED) (e.g. for plant mode software). If the parameter is not configured, the value of “DemDebounceCounterIncrementStepSize” shall be considered.
Multiplicity	0..1
Value Range	1..32767
Default Value	-
Type	ECUC-INTEGER-PARAM-DEF

3.3.1.9 Dem/DemConfigSet/DemDebounceCounterBasedClass/DemRbDebounceCounterJumpDownAlternative

Short Name	DemRbDebounceCounterJumpDownAlternative
Description	Switch for the activation of Jump-Down. true: Jump-Down activated false: Jump-Down deactivated If the parameter is not configured, the value of DemDebounceCounterJumpDown shall be considered.
Multiplicity	0..1
Default Value	-
Type	ECUC-BOOLEAN-PARAM-DEF

3.3.1.10 Dem/DemConfigSet/DemDebounceCounterBasedClass/DemRb-DebounceCounterJumpDownValueAlternative

Short Name	DemRbDebounceCounterJumpDownValueAlternative
Description	Defines Alternative Jump down value of the internal fault detection counter. If the parameter is not configured, the value of “DemDebounceCounterJump-DownValue” shall be considered.
Multiplicity	0..1
Value Range	-32768..32767
Default Value	-
Type	ECUC-INTEGER-PARAM-DEF

3.3.1.11 Dem/DemConfigSet/DemDebounceCounterBasedClass/DemRb-DebounceCounterJumpUpAlternative

Short Name	DemRbDebounceCounterJumpUpAlternative
Description	Switch for the activation of Jump-Up. true: Jump-Up activated false: Jump-Up deactivated If the parameter is not configured, the value of “DemDebounceCounterJumpUp” shall be considered
Multiplicity	0..1
Default Value	-
Type	ECUC-BOOLEAN-PARAM-DEF

3.3.1.12 Dem/DemConfigSet/DemDebounceCounterBasedClass/DemRb-DebounceCounterJumpUpValueAlternative

Short Name	DemRbDebounceCounterJumpUpValueAlternative
Description	Defines Alternative Jump Up value of the internal fault detection counter. If the parameter is not configured, the value of DemDebounceCounter-JumpUpValue shall be considered.
Multiplicity	0..1
Value Range	-32768..32767
Default Value	-
Type	ECUC-INTEGER-PARAM-DEF

3.3.1.13 Dem/DemConfigSet/DemDebounceCounterBasedClass/DemRbDebounceCounterSuspiciousThreshold

Short Name	DemRbDebounceCounterSuspiciousThreshold
Description	Threshold to indicate the event is suspicious.
Multiplicity	0..1
Value Range	1..32767
Default Value	-
Type	ECUC-INTEGER-PARAM-DEF

3.3.1.14 Dem/DemConfigSet/DemDebounceCounterBasedClass/DemRbShiftView

Short Name	DemRbShiftView
Description	This parameter is added for DGS Dem. It specifies where to arrange calibration parameters of the ratiold in asap file (shiftview). DGS DSM creates diagnostic calibration parameters for ratios. If your module has own calibration parameters then give your assigned function name in a2l-file and DGS-DSM will shift the diagnostic calibration parameters to this a2l function. Otherwise define a function name for the diagnostic calibration, only.
Multiplicity	0..1
Default Value	-
Type	ECUC-STRING-PARAM-DEF

3.3.1.15 Dem/DemConfigSet/DemDebounceTimeBaseClass

Short Name	DemDebounceTimeBaseClass
Description	This container contains the configuration of Debounce Time Based Class.
Multiplicity	0..65535
Default Value	-
Type	ECUC-PARAM-CONF-CONTAINER-DEF

3.3.1.16 Dem/DemConfigSet/DemDebounceTimeBaseClass/DemRbDebounceTimeSuspiciousThreshold

Short Name	DemRbDebounceTimeSuspiciousThreshold
Description	Threshold to indicate the event is suspicious.
Multiplicity	0..1
Value Range	0.001..3600

Default Value	-
Type	ECUC-FLOAT-PARAM-DEF

3.3.1.17 Dem/DemConfigSet/DemDebounceTimeBaseClass/DemRb-ShiftView

Short Name	DemRbShiftView
Description	This parameter is added for DGS Dem. It specifies where to arrange calibration parameters of the ratiold in asap file (shiftview). DGS DSM creates diagnostic calibration parameters for ratios. If your module has own calibration parameters then give your assigned function name in a2l-file and DGS-DSM will shift the diagnostic calibration parameters to this a2l function. Otherwise define a function name for the diagnostic calibration, only.
Multiplicity	0..1
Default Value	-
Type	ECUC-STRING-PARAM-DEF

3.3.1.18 Dem/DemConfigSet/DemEventParameter/DemDebounceAlgorithmClass/DemDebounceMonitorInternal/DemCallbackGetFDC/DemRbGeneralCallbackGetFDCFncUsed

Short Name	DemRbGeneralCallbackGetFDCFncUsed
Description	If configured to TRUE, the global callback has to be configured under DemGeneral/demRbGeneral/demRb-GlobalCallbackGetFaultDetectionCounter. General function name used for “GetFaultDetectionCounter”.
Multiplicity	0..1
Default Value	false
Type	ECUC-BOOLEAN-PARAM-DEF

3.3.1.19 Dem/DemConfigSet/DemEventParameter/DemDebounceAlgorithmClass/DemDebounceTimeBase

Short Name	DemDebounceTimeBase
Description	This container contains the configuration (parameters) for time based debouncing.
Multiplicity	0..1
Default Value	-
Type	ECUC-PARAM-CONF-CONTAINER-DEF

3.3.1.20 Dem/DemConfigSet/DemEventParameter/DemDebounceAlgorithmClass/DemDebounceTimeBase/DemDebounceTimeBaseRef

Short Name	DemDebounceTimeBaseRef
Description	This reference selects the DemDebounceTimeBaseClass applied for the debouncing of the DemEventParameter.
Multiplicity	1..1
Default Value	-
Type	ECUC-REFERENCE-DEF

3.3.1.21 Dem/DemConfigSet/DemEventParameter/DemIndicatorAttribute/DemRbIndicatorApiControl

Short Name	DemRbIndicatorApiControl
Description	This configuration switch defines whether the indicator state is controlled using the API Dem_SetIndicatorStatusForEvent, according to DemIndicator-Behaviour
Multiplicity	0..1
Default Value	false
Type	ECUC-BOOLEAN-PARAM-DEF

3.3.1.22 Dem/DemConfigSet/DemEventParameter/DemRbDoc

Short Name	DemRbDoc
Long Name	Documentation
Description	This container contains the documentation for the Event.
Multiplicity	0..1
Default Value	-
Type	ECUC-PARAM-CONF-CONTAINER-DEF

3.3.1.23 Dem/DemConfigSet/DemEventParameter/DemRbDoc/DemRbDocFailedCriteria

Short Name	DemRbDocFailedCriteria
Long Name	Failure detection criteria
Description	Verbal description of the recognition method for the FAILED/PREFAILED status, i.e. how it is detected that a failure is present. If applicable, describe what limit values or thresholds are used. If several alternative strategies are implemented, all of them shall be described.
Multiplicity	1..1

Default Value	-
Type	ECUC-ADD-INFO-PARAM-DEF

3.3.1.24 Dem/DemConfigSet/DemEventParameter/DemRbDoc/DemRbDoc-FailedDebouncing

Short Name	DemRbDocFailedDebouncing
Long Name	Monitoring-internal failure debouncing
Description	This shall describe how long or how often the failure detection criteria has to be detected before the FAILED/PREFAILED status is reported to the Dem. Note: Describe monitoring-internal debouncing only. If no monitoring-internal debouncing is implemented, then enter an empty paragraph.
Multiplicity	1..1
Default Value	-
Type	ECUC-ADD-INFO-PARAM-DEF

3.3.1.25 Dem/DemConfigSet/DemEventParameter/DemRbDoc/DemRbDoc-FaultReaction

Short Name	DemRbDocFaultReaction
Long Name	Monitoring-internal fault reaction
Description	Describe what further actions are triggered in case of a failure besides the reporting to the Dem. Note: Do not describe FiM-related failure reactions. If no failure reactions are triggered, enter an empty paragraph.
Multiplicity	1..1
Default Value	-
Type	ECUC-ADD-INFO-PARAM-DEF

3.3.1.26 Dem/DemConfigSet/DemEventParameter/DemRbDoc/DemRbDoc-PassedCriteria

Short Name	DemRbDocPassedCriteria
Long Name	Passed detection criteria
Description	Verbal description of the recognition method for the PASSED/PREPASSED status, i.e. how it is detected that no failure is present. If applicable, describe what limit values or thresholds are used. If several alternative strategies are implemented, all of them shall be described.
Multiplicity	1..1
Default Value	-

Type	ECUC-ADD-INFO-PARAM-DEF

3.3.1.27 Dem/DemConfigSet/DemEventParameter/DemRbDoc/DemRbDoc-PassedDebouncing

Short Name	DemRbDocPassedDebouncing
Long Name	Monitoring-internal passed debouncing
Description	This shall describe how long or how often the passed detection criteria has to be detected before the PASSED/PREPASSED status is reported to the Dem. Note: Describe monitoring-internal debouncing only. If no monitoring-internal debouncing is implemented, then enter an empty paragraph.
Multiplicity	1..1
Default Value	-
Type	ECUC-ADD-INFO-PARAM-DEF

3.3.1.28 Dem/DemConfigSet/DemEventParameter/DemRbDoc/DemRb-DocTestPrecondition

Short Name	DemRbDocTestPrecondition
Long Name	Monitoring-internal enable condition
Description	Under which preconditions does a test take place? Note: Describe monitoring-internal preconditions only (not Dem-configured enable conditions). If no monitoring-internal precondition is implemented, then enter an empty paragraph.
Multiplicity	1..1
Default Value	-
Type	ECUC-ADD-INFO-PARAM-DEF

3.3.1.29 Dem/DemConfigSet/DemEventParameter/DemRbDoc/DemRbFDoc

Short Name	DemRbFDoc
Long Name	Formalized description
Description	This container contains the formalized documentation of the monitoring reporting this Dem event. If a (PRE)-PASSED/(PRE)FAILED status can be detected in multiple ways, create one container instance for each way.
Multiplicity	0..*
Default Value	-
Type	ECUC-PARAM-CONF-CONTAINER-DEF

3.3.1.30 Dem/DemConfigSet/DemEventParameter/DemRbDoc/DemRbF-

Doc/DemRbFDocApplObdLaw

Short Name	DemRbFDocApplObdLaw
Long Name	OBD law reference
Description	Filling out this element is needed for legally required events. If the event is not legally required, please select “none”. For further information contact your local legislation expert.
Multiplicity	1..1
Default Value	-
Enumeration Literals	none, e13_CCR_P_1968_2, e13_CCR_P_1971_1, e70_220_EEC_ECE_R83, EU_595_2009_EC_ECE_R49, other
Type	ECUC-ENUMERATION-PARAM-DEF

3.3.1.31 Dem/DemConfigSet/DemEventParameter/DemRbDoc/DemRbF-Doc/DemRbFDocApplObdLawOther

Short Name	DemRbFDocApplObdLawOther
Long Name	OBD law reference other
Description	This element has to be filled out, if ‘other’ is selected for OBD law reference. For further information contact your local legislation expert.
Multiplicity	0..1
Default Value	-
Type	ECUC-STRING-PARAM-DEF

3.3.1.32 Dem/DemConfigSet/DemEventParameter/DemRbDoc/DemRbF-Doc/DemRbFDocApplObdLawSection

Short Name	DemRbFDocApplObdLawSection
Long Name	OBD law section
Description	Filling out this element is needed for legally required events. For further information contact your local legislation expert.
Multiplicity	0..1
Default Value	-
Type	ECUC-STRING-PARAM-DEF

3.3.1.33 Dem/DemConfigSet/DemEventParameter/DemRbDoc/DemRbF-Doc/DemRbFDocFailedCriterion

Short Name	DemRbFDocFailedCriterion
Long Name	Failure detection criterion

Description	Formal description of the recognition method for the FAILED/PREFAILED status, i.e. how it is detected that a failure is present. This shall be formulated as an expression based on measurement variables and calibration parameters. If applicable measurement variables and calibration parameters are not available and there is exactly one instance of the formalized description container, then enter an empty paragraph. If applicable measurement variables and calibration parameters are not available and there is more than one instance of the formalized description container, then enter an textual description.
Multiplicity	1..1
Default Value	-
Type	ECUC-ADD-INFO-PARAM-DEF

3.3.1.34 Dem/DemConfigSet/DemEventParameter/DemRbDoc/DemRbFDoc/DemRbFDocFailedDebouncing

Short Name	DemRbFDocFailedDebouncing
Long Name	Monitoring-internal failure debouncing
Description	This shall describe how long or how often the failure detection criteria has to be detected before the FAILED/PREFAILED status is reported to the Dem. This shall be formulated as an expression based on measurement variables and calibration parameters. Note: Describe monitoring-internal debouncing only. If no monitoring-internal debouncing is implemented, then enter an empty paragraph.
Multiplicity	1..1
Default Value	-
Type	ECUC-ADD-INFO-PARAM-DEF

3.3.1.35 Dem/DemConfigSet/DemEventParameter/DemRbDoc/DemRbFDoc/DemRbFDocMonitoredComponent

Short Name	DemRbFDocMonitoredComponent
Long Name	Monitored component
Description	What is monitored? Select the monitored component or monitored system from the list.
Multiplicity	1..1
Default Value	-
Type	ECUC-REFERENCE-DEF

3.3.1.36 Dem/DemConfigSet/DemEventParameter/DemRbDoc/DemRbF-Doc/DemRbFDocMonitoredComponentOther

Short Name	DemRbFDocMonitoredComponentOther
Long Name	Monitored component other
Description	Specification of the monitored component or monitored system, if not included in the list (DemFDocMonitoredComponent).
Multiplicity	0..1
Default Value	-
Type	ECUC-STRING-PARAM-DEF

3.3.1.37 Dem/DemConfigSet/DemEventParameter/DemRbDoc/DemRbF-Doc/DemRbFDocMonitoringDeactivation

Short Name	DemRbFDocMonitoringDeactivation
Long Name	Monitoring deactivation
Description	Typically a simple condition which deactivates the monitoring (at least report to DEM will be suppressed/ deactivated). This might e.g. be a monitor-internal, calibratable parameter to deactivate the calculation of the fault detection criterion. The condition does not represent a physical necessity. Also it does not change during ECU runtime. Therefore it is not an enable condition. If no deactivation condition is implemented, then enter an empty paragraph.
Multiplicity	1..1
Default Value	-
Type	ECUC-ADD-INFO-PARAM-DEF

3.3.1.38 Dem/DemConfigSet/DemEventParameter/DemRbDoc/DemRbF-Doc/DemRbFDocMonitoringFrequency

Short Name	DemRbFDocMonitoringFrequency
Long Name	Monitoring frequency
Description	Describes how often the monitoring algorithm is calculated. “continuous”: does not have a precondition or may only depend on the following preconditions: battery voltage, ignition, altitude, ambient temperature, barometric pressure, can initialization “multiple”: uses additional preconditions (besides the ones listed above) but runs always if preconditions are fulfilled. “once per driving cycle”: only runs once per driving cycle “once per driving cycle keep alive”: only runs during postrun “once per lifetime”: only runs once per ecu lifetime

Multiplicity	1..1
Default Value	-
Enumeration Literals	continuous, multiple, once_per_driving_cycle, once_per_driving_cycle_keep_alive, once_per_lifetime
Type	ECUC-ENUMERATION-PARAM-DEF

3.3.1.39 Dem/DemConfigSet/DemEventParameter/DemRbDoc/DemRbFDoc/DemRbFDocMonitoringStrategy

Short Name	DemRbFDocMonitoringStrategy
Long Name	Monitoring strategy
Description	Brief description of the basic idea for the fault detection criterion of the monitoring. Select appropriate value of the monitoring strategy from the list. If none of the values fits, choose “other” and enter description in “Other Monitoring strategy”.
Multiplicity	1..1
Default Value	-
Type	ECUC-REFERENCE-DEF

3.3.1.40 Dem/DemConfigSet/DemEventParameter/DemRbDoc/DemRbFDoc/DemRbFDocMonitoringStrategyComprehensiveComp

Short Name	DemRbFDocMonitoringStrategyComprehensiveComp
Long Name	General monitoring strategy for comprehensive component
Description	General classification of the monitoring strategy for Comprehensive Components according to the definitions of the California Air Ressources Board CARB. In case a “Comprehensive Component” monitoring strategy is selected above please enter here the general monitoring strategy according to CARBs definition. For monitored components other than comprehensive components select “none”.
Multiplicity	1..1
Default Value	-
Enumeration Literals	none, Input_Open_Circuit, Input_Out_of_Range_Low, Input_Out_of_Range_High, Input_Rationality_Low, Input_Rationality_High, Input_Other_Rationality, Output_Open_Circuit, Output_Shorted_Low, Output_Shorted_High, Output_Functional
Type	ECUC-ENUMERATION-PARAM-DEF

3.3.1.41 Dem/DemConfigSet/DemEventParameter/DemRbDoc/DemRbF-Doc/DemRbFDocMonitoringStrategyOther

Short Name	DemRbFDocMonitoringStrategyOther
Long Name	Other Monitoring strategy
Description	Brief description of the basic idea for the fault detection criterion of the monitoring. Only needed if “Monitoring strategy” “other” is selected.
Multiplicity	0..1
Default Value	-
Type	ECUC-STRING-PARAM-DEF

3.3.1.42 Dem/DemConfigSet/DemEventParameter/DemRbDoc/DemRbF-Doc/DemRbFDocPassedCriterion

Short Name	DemRbFDocPassedCriterion
Long Name	Passed detection criterion
Description	Formal description of the recognition method for the PASSED/PREPASSED status, i.e. how it is detected that no failure is present. This shall be formulated as an expression based on measurement variables and calibration parameters. If applicable measurement variables and calibration parameters are not available and there is exactly one instance of the formalized description container, then enter an empty paragraph. If applicable measurement variables and calibration parameters are not available and there is more than one instance of the formalized description container, then enter an textual description.
Multiplicity	1..1
Default Value	-
Type	ECUC-ADD-INFO-PARAM-DEF

3.3.1.43 Dem/DemConfigSet/DemEventParameter/DemRbDoc/DemRbF-Doc/DemRbFDocPassedDebouncing

Short Name	DemRbFDocPassedDebouncing
Long Name	Monitoring-internal passed debouncing

Description	This shall describe how long or how often the passed detection criteria has to be detected before the PASSED/PREPASSED status is reported to the Dem. This shall be formulated as an expression based on measurement variables and calibration parameters. Note: Describe monitoring-internal debouncing only. If no monitoring-internal debouncing is implemented, then enter an empty paragraph.
Multiplicity	1..1
Default Value	-
Type	ECUC-ADD-INFO-PARAM-DEF

3.3.1.44 Dem/DemConfigSet/DemEventParameter/DemRbDoc/DemRbFDoc/DemRbFDocTestPrecondition

Short Name	DemRbFDocTestPrecondition
Long Name	Test precondition
Description	Describe all atomic conditions which must be fulfilled so that the fault detection criterion can be calculated with a meaningful result. Atomic condition means a single condition such as “battery voltage above minimum”. Describe in formal syntax if feasible - otherwise use text. Note: The fault detection criterion shall not be repeated here, neither partly nor completely. Note: Do not repeat a fault detection inactive condition “Monitoring deactivation”.
Multiplicity	1..1
Default Value	-
Type	ECUC-ADD-INFO-PARAM-DEF

3.3.1.45 Dem/DemConfigSet/DemEventParameter/DemRbEventClass

Short Name	DemRbEventClass
Description	This container contains the configuaration (parameters) for EventClass
Multiplicity	1..1
Default Value	-
Type	ECUC-PARAM-CONF-CONTAINER-DEF

3.3.1.46 Dem/DemConfigSet/DemEventParameter/DemRbEventClass/DemRbDgsDfcName

Short Name	DemRbDgsDfcName

Description	This parameter is added for DGS Dem. It specifies the legacy DFC (Diagnostic Fault Code) name used in DGS systems.
Multiplicity	0..1
Default Value	-
Type	ECUC-STRING-PARAM-DEF

3.3.1.47 Dem/DemConfigSet/DemEventParameter/DemRbEventClass/DemRbEventCategoryRef

Short Name	DemRbEventCategoryRef
Description	Reference to category of the event [NON-AUTOSAR EXTENSION]
Multiplicity	0..1
Default Value	-
Type	ECUC-REFERENCE-DEF

3.3.1.48 Dem/DemConfigSet/DemEventParameter/DemRbEventClass/DemRbEventCustAttribute

Short Name	DemRbEventCustAttribute
Description	This container contains the configuration(parameters) for event customer attribute values and reference.
Multiplicity	0..*
Default Value	-
Type	ECUC-PARAM-CONF-CONTAINER-DEF

3.3.1.49 Dem/DemConfigSet/DemEventParameter/DemRbEventClass/DemRbEventCustAttribute/DemRbCustAttributeRef

Short Name	DemRbCustAttributeRef
Description	The configuration of the event attribute done by creating a reference
Multiplicity	1..1
Default Value	-
Type	ECUC-REFERENCE-DEF

3.3.1.50 Dem/DemConfigSet/DemEventParameter/DemRbEventClass/DemRbEventCustAttribute/DemRbCustAttributeValue

Short Name	DemRbCustAttributeValue
Description	The configuration of the event attributes done by assigning a value
Multiplicity	1..1
Default Value	-

Type	ECUC-STRING-PARAM-DEF

3.3.1.51 Dem/DemConfigSet/DemEventParameter/DemRbEventClass/DemRbEventStatusBitStorageTestFailed

Short Name	DemRbEventStatusBitStorageTestFailed
Description	This parameter specifies if the failure of this event (UDS status bit 0 of DTC mapped to the event) is stored to the non-volatile memory over power cycles or not. This setting is only relevant if the global setting DemStatusBitStorageTestFailed is set to FALSE. [NON-AUTOSAR EXTENSION]
Multiplicity	0..1
Default Value	false
Type	ECUC-BOOLEAN-PARAM-DEF

3.3.1.52 Dem/DemConfigSet/DemEventParameter/DemRbEventClass/DemRbEventUniqueId

Short Name	DemRbEventUniqueId
Description	This contains a free definable unique id. It is used with the BFM.
Multiplicity	0..1
Value Range	0..0xFFFFFFFF
Default Value	-
Type	ECUC-INTEGER-PARAM-DEF

3.3.1.53 Dem/DemConfigSet/DemEventParameter/DemRbEventClass/DemRbShiftView

Short Name	DemRbShiftView
Long Name	Mapping for documentation
Description	Reference to the name of the SW-component calculating the DemEvent. This parameter is added for DGS Dem. It is used in further processing to assign calibration parameters of the considered event, generated during software build in asap file to the calculating SW-component("shiftview"). DGS DSM creates diagnostic calibration parameters for events(= DGS DFCs). If your module has own calibration parameters then give your assigned function name in a2l-file and DGS-DSM will shift the diagnostic calibration parameters to this a2l function. Otherwise define a function name for the diagnostic calibration, only.
Multiplicity	0..1

Default Value	-
Type	ECUC-STRING-PARAM-DEF

3.3.1.54 Dem/DemConfigSet/DemEventParameter/DemRbEventDescription

Short Name	DemRbEventDescription
Description	The purpose of the specified event shall be described.
Multiplicity	0..1
Default Value	-
Type	ECUC-STRING-PARAM-DEF

3.3.1.55 Dem/DemConfigSet/DemEventParameter/DemRbEventEcucPartitionRef

Short Name	DemRbEventEcucPartitionRef
Description	Maps this Dem Event to one Ecuc partition.
Multiplicity	0..1
Default Value	-
Type	ECUC-SYMBOLIC-NAME-REFERENCE-DEF

3.3.1.56 Dem/DemConfigSet/DemEventParameter/DemRbIgnorePFCCycleQualificationForPDTCRemoval

Short Name	DemRbIgnorePFCCycleQualificationForPDTCRemoval
Description	This parameter defines whether the deletion of the permanent DTC subsequent to a code clear requires the PFC cycle qualification to be satisfied as well (see DemOBDPFCCycleRef). This parameter is relevant only for OBD relevant events and only when Permanent DTC feature is enabled.(see DemMaxNumberEventEntryPermanent).
Introduction	true: The PFC cycle qualification is not necessary for the PDTC removal following a code clear. false: The PFC cycle qualification is required for the PDTC removal following a code clear.
Multiplicity	0..1
Default Value	false
Type	ECUC-BOOLEAN-PARAM-DEF

3.3.1.57 Dem/DemGeneral/DemClient/DemRbClientPriority

Short Name	DemRbClientPriority

Description	This parameter defines a unique priority value that a DemClient can have. Lower the value higher the priority.
Multiplicity	1..1
Value Range	1..255
Default Value	-
Type	ECUC-INTEGER-PARAM-DEF

3.3.1.58 Dem/DemGeneral/DemDataElementClass/DemInternalDataElementClass/DemRbDataElementReadFnc

Short Name	DemRbDataElementReadFnc
Description	Defines the name of the project specific C function "ReadDataElement" which will provide the desired internal data. Please refer to Dem_PrjEnvDataElement.c_tpl. This parameter must be configured if DemInternalDataElement is configured as DEM_PRJ_SPECIFIC_DATA_ON_RETRIEVE or DEM_PRJ_SPECIFIC_DATA_ON_REPORT. Otherwise it must not be configured.
Multiplicity	0..1
Default Value	-
Type	ECUC-FUNCTION-NAME-DEF

3.3.1.59 Dem/DemGeneral/DemEventMemorySet/DemMirrorMemory

Short Name	DemMirrorMemory
Description	This container contains the mirror event memory specific parameters of the Dem module.
Multiplicity	0..1
Default Value	-
Type	ECUC-PARAM-CONF-CONTAINER-DEF

3.3.1.60 Dem/DemGeneral/DemEventMemorySet/DemMirrorMemory/DemMaxNumberEventEntryMirror

Short Name	DemMaxNumberEventEntryMirror
Description	Maximum number of events which can be stored in the mirror memory
Multiplicity	1..1
Value Range	1..255
Default Value	-
Type	ECUC-INTEGER-PARAM-DEF

3.3.1.61 Dem/DemGeneral/DemEventMemorySet/DemPrimaryMemory/DemEnvironmentDataCapture

Short Name	DemEnvironmentDataCapture
Description	DemEnvironmentDataCapture defines the point in time, when the data actually is captured.
Multiplicity	0..1
Default Value	-
Enumeration Literals	DEM_CAPTURE_ASYNCHRONOUS_TO_REPORTING, DEM_CAPTURE_SYNCHRONOUS_TO_REPORTING
Type	ECUC-ENUMERATION-PARAM-DEF

3.3.1.62 Dem/DemGeneral/DemEventMemorySet/DemPrimaryMemory/DemRbImmediateNvStorageLimit

Short Name	DemRbImmediateNvStorageLimit
Description	This parameter defines the maximum number of occurrences, a specific event memory entry is allowed, to be stored in NVRAM immediately (refer to DemNvStorageStrategy).
Multiplicity	0..1
Value Range	2..255
Default Value	-
Type	ECUC-INTEGER-PARAM-DEF

3.3.1.63 Dem/DemGeneral/DemEventMemorySet/DemUserDefinedMemory/DemEnvironmentDataCapture

Short Name	DemEnvironmentDataCapture
Description	DemEnvironmentDataCapture defines the point in time, when the data actually is captured.
Multiplicity	0..1
Default Value	-
Enumeration Literals	DEM_CAPTURE_ASYNCHRONOUS_TO_REPORTING, DEM_CAPTURE_SYNCHRONOUS_TO_REPORTING
Type	ECUC-ENUMERATION-PARAM-DEF

3.3.1.64 Dem/DemGeneral/DemEventMemorySet/DemUserDefinedMemory/DemRbImmediateNvStorageLimit

Short Name	DemRbImmediateNvStorageLimit

Description	This parameter defines the maximum number of occurrences, a specific event memory entry is allowed, to be stored in NVRAM immediately (refer to DemNvStorageStrategy).
Multiplicity	0..1
Value Range	2..255
Default Value	-
Type	ECUC-INTEGER-PARAM-DEF

3.3.1.65 Dem/DemGeneral/DemGeneralOBD/DemRbGeneralOBD

Short Name	DemRbGeneralOBD
Description	This container contains the configuration (parameters) of the BSW Dem. Delete container if ObdBasic is not required.
Multiplicity	0..1
Default Value	-
Type	ECUC-PARAM-CONF-CONTAINER-DEF

3.3.1.66 Dem/DemGeneral/DemGeneralOBD/DemRbGeneralOBD/DemRbEnableApiSignalEndOfOBDIgnitionCycle

Short Name	DemRbEnableApiSignalEndOfOBDIgnitionCycle
Description	This configuration switch enables or disables the functionality of the API "Dem_SignalEndOfOBDIgnitionCycle". This API can be used to explicitly signal the end of the ignition cycle. Basically this API controls the storing and visibility of Permanent DTCs (see DemRbOBDPermanentDTCVisibilityTiming). TRUE: The storing and visibility of permanent DTCs is updated when the API Dem_SignalEndOfOBDIgnitionCycle gets called. FALSE: The storing and visibility of permanent DTCs is updated when the ignition cycle is restarted (see DemRbOBDIgnitionCycleRef).
Multiplicity	0..1
Default Value	false
Type	ECUC-BOOLEAN-PARAM-DEF

3.3.1.67 Dem/DemGeneral/DemGeneralOBD/DemRbGeneralOBD/DemRbOBDCalculatePID21

Short Name	DemRbOBDCalculatePID21
Description	[NON-AUTOSAR] This configuration switch enables or disables the calculation of PID21
Multiplicity	0..1

Default Value	false
Type	ECUC-BOOLEAN-PARAM-DEF

3.3.1.68 Dem/DemGeneral/DemGeneralOBD/DemRbGeneralOBD/DemRbOBDCalculatePID21AndPID4DOnConfirmedDTC

Short Name	DemRbOBDCalculatePID21AndPID4DOnConfirmedDTC
Description	[NON-AUTOSAR] Switch to enable calculation of PID $21 and PID $4D based on MIL Status or Confirmed DTC. Parameter is relevant only when DemRbOBDCalculatePID21 or DemRbOBDCalculatePID4D is true, and centralized handling feature is disabled. True: Calculation of PIDs based on Confirmed-DTC(s) False: Calculation of PIDs based on MIL Status
Multiplicity	0..1
Default Value	false
Type	ECUC-BOOLEAN-PARAM-DEF

3.3.1.69 Dem/DemGeneral/DemGeneralOBD/DemRbGeneralOBD/DemRbOBDCalculatePID31

Short Name	DemRbOBDCalculatePID31
Description	[NON-AUTOSAR] This configuration switch enables or disables the calculation of PID31
Multiplicity	0..1
Default Value	false
Type	ECUC-BOOLEAN-PARAM-DEF

3.3.1.70 Dem/DemGeneral/DemGeneralOBD/DemRbGeneralOBD/DemRbOBDCalculatePID4D

Short Name	DemRbOBDCalculatePID4D
Description	[NON-AUTOSAR] This configuration switch enables or disables the calculation of PID4D
Multiplicity	0..1
Default Value	false
Type	ECUC-BOOLEAN-PARAM-DEF

3.3.1.71 Dem/DemGeneral/DemGeneralOBD/DemRbGeneralOBD/DemRbOBDCalculatePID4E

Short Name	DemRbOBDCalculatePID4E

Description	[NON-AUTOSAR] This configuration switch enables or disables the calculation of PID4E
Multiplicity	0..1
Default Value	false
Type	ECUC-BOOLEAN-PARAM-DEF

3.3.1.72 Dem/DemGeneral/DemGeneralOBD/DemRbGeneralOBD/DemRbOBDCentralizedPID4DHandling

Short Name	DemRbOBDCentralizedPID4DHandling
Description	Switch to enable the centralized handling of PID $4D. Parameter is relevant only when DemRbOBDCalculatePID4D is true.
Introduction	true: centralized handling of PID $4D is enabled. Application shall inform the centrally calculated PID value to Dem via the API Dem_SetDataOfPID4D(). false: centralized handling of PID $4D is disabled. Dem calculates the value of the PID $4D on its own, which the application can read via the API Dem_GetDataOfPID4D().
Multiplicity	0..1
Default Value	false
Type	ECUC-BOOLEAN-PARAM-DEF

3.3.1.73 Dem/DemGeneral/DemGeneralOBD/DemRbGeneralOBD/DemRbOBDCentralizedPID4EHandling

Short Name	DemRbOBDCentralizedPID4EHandling
Description	Switch to enable the centralized handling of PID $4E. Parameter is relevant only when DemRbOBDCalculatePID4E is true.
Introduction	true: centralized handling of PID $4E is enabled. Application shall inform the centrally calculated PID value to Dem via the API Dem_SetDataOfPID4E(). false: centralized handling of PID $4E is disabled. Dem calculates the value of the PID $4D on its own, which the application can read via the API Dem_GetDataOfPID4E().
Multiplicity	0..1
Default Value	false
Type	ECUC-BOOLEAN-PARAM-DEF

3.3.1.74 Dem/DemGeneral/DemGeneralOBD/DemRbGeneralOBD/DemRbOBDDtcCodesConfigurable

Short Name	DemRbOBDDtcCodesConfigurable
Description	This configuration switch defines whether the OBD Dtc codes are configuratble or not. If configurable, the DemDtcCode has to be entered in the DemDtcClass. If not configurable, the 2 higher bytes of the DemDtc is the OBD Dtc code.
Multiplicity	0..1
Default Value	true
Type	ECUC-BOOLEAN-PARAM-DEF

3.3.1.75 Dem/DemGeneral/DemGeneralOBD/DemRbGeneralOBD/DemRbOBDFreezeFrameCapture

Short Name	DemRbOBDFreezeFrameCapture
Description	[NON-AUTOSAR] This parameter defines the point in time, when the OBD freeze frame data collection is done for the event memory entry. [Hint] The customer should ensure that the configuration is done correctly, according to their interpretation of the applicable regulations.
Multiplicity	0..1
Default Value	DEM_TRIGGER_PENDINGDTC
Enumeration Literals	DEM_TRIGGER_CONFIRMEDDTC, DEM_TRIGGER_PENDINGDTC, DEM_TRIGGER_PENDINGDTC_UPDATE_ON_CONFIRMEDDTC
Type	ECUC-ENUMERATION-PARAM-DEF

3.3.1.76 Dem/DemGeneral/DemGeneralOBD/DemRbGeneralOBD/DemRbOBDFreezeFrameVisibility

Short Name	DemRbOBDFreezeFrameVisibility
Description	[NON-AUTOSAR] This parameter defines the point of time, when the OBD freeze frame data is visible to scan tool tester.
Multiplicity	0..1
Default Value	DEM_OBDFF_CONFIRMED_DTCS
Enumeration Literals	DEM_OBDFF_PENDING_AND_CONFIRMED_DTCS, DEM_OBDFF_CONFIRMED_DTCS
Type	ECUC-ENUMERATION-PARAM-DEF

3.3.1.77 Dem/DemGeneral/DemGeneralOBD/DemRbGeneralOBD/DemR-

bOBDIgnitionCycleRefDemRbOBDIgnitionCycleRef

Short Name	DemRbOBDIgnitionCycleRef
Description	References the operationCycle which denotes the OBD ignition cycle. The OBD Ignition Cycle is required to compute the ignition cycle counter. It is also used to control the visibility of Permanent DTCs (see DemRbEnableApiSignalEndOfOBDIgnitionCycle)
Multiplicity	0..1
Default Value	-
Type	ECUC-REFERENCE-DEF

3.3.1.78 Dem/DemGeneral/DemGeneralOBD/DemRbGeneralOBD/DemRbOBDPermanentDTCVisibilityTiming

Short Name	DemRbOBDPermanentDTCVisibilityTiming
Description	[NON-AUTOSAR] This configuration switch defines whether the permanent DTCs are visible immediately, or only after the next init of the Dem or at the end of Ignition Cycle.
Multiplicity	0..1
Default Value	IMMEDIATELY
Enumeration Literals	END_OF_CYCLE, IMMEDIATELY
Type	ECUC-ENUMERATION-PARAM-DEF

3.3.1.79 Dem/DemGeneral/DemGeneralOBD/DemRbGeneralOBD/DemRbOBDPid01MisfireMonitorForcedReady

Short Name	DemRbOBDPid01MisfireMonitorForcedReady
Description	This parameter controls whether the misfire monitoring group within PID $01 is always indicated as “ready”. This parameter is relevant only for compression ignition vehicles. For spark ignition vehicles, the misfire group is automatically indicated as “ready” in PID$01, if it is supported. True: Misfire group status is forced as “ready” False: Misfire group status is not forced as “ready”
Multiplicity	0..1
Default Value	false
Type	ECUC-BOOLEAN-PARAM-DEF

3.3.1.80 Dem/DemGeneral/DemGeneralOBD/DemRbGeneralOBD/DemRbOBDProtocol

Short Name	DemRbOBDProtocol
Description	This parameter defines which OBD protocol shall be used (1) DEM_RB_J1979 (OBD Standard) (2) DEM_RB_J1979_2 (OBDonUDS) (3) DEM_RB_BOTH_PROTOCOL (switch between OBD standard and OBDonUDS at runtime)
Multiplicity	0..1
Default Value	DEM_RB_J1979
Enumeration Literals	DEM_RB_J1979, DEM_RB_J1979_2, DEM_RB_BOTH_PROTOCOL
Type	ECUC-ENUMERATION-PARAM-DEF

3.3.1.81 Dem/DemGeneral/DemGeneralOBD/DemRbGeneralOBD/DemRbOBDProtocolFnc

Short Name	DemRbOBDProtocolFnc
Description	Specifies the name of the function which returns the OBD protocol (refer to data type DemRb_ObdProtocolType for implementation) to be used. This parameter is optional, but becomes mandatory if DemRbOBDProtocol is set to DEM_RB_BOTH_PROTOCOL.
Multiplicity	0..1
Default Value	-
Type	ECUC-FUNCTION-NAME-DEF

3.3.1.82 Dem/DemGeneral/DemGeneralOBD/DemRbGeneralOBD/DemRbOBDSecondIgnitionCycleRef

Short Name	DemRbOBDSecondIgnitionCycleRef
Description	References the operationCycle which can be used in hybrid vehicles for the fueled engine operation ignition cycle.
Multiplicity	0..1
Default Value	-
Type	ECUC-REFERENCE-DEF

3.3.1.83 Dem/DemGeneral/DemGeneralOBD/DemRbGeneralOBD/DemRbOBDSupportBlockingPermanentDTC

Short Name	DemRbOBDSupportBlockingPermanentDTC

Description	This configuration switch enables the feature to block storing of OBD DTCs to Permanent memory.-CAUTION: THIS FEATURE MUST USED ONLY DURING PRODUCTION.!!! OTHERWISE OBD CONFORMITY IS VIOLATED.See document description of the callback API rba_DemObdBasic_IsPermanentDTCStorageDisabled
Multiplicity	0..1
Default Value	false
Type	ECUC-BOOLEAN-PARAM-DEF

3.3.1.84 Dem/DemGeneral/DemGeneralOBD/DemRbGeneralOBD/DemRbOBDSupportObdVariantCoding

Short Name	DemRbOBDSupportObdVariantCoding
Description	This configuration switch enables the feature to deactivate the OBD functionality. This feature could be used for variant coding to dynamically switch an OBD project to behave like a non OBD variant. This feature only controls the MIL status. Project specific implementation is necessary. Refer the interface documentation of the API rba_DemObdBasic_IsObdDeactivatedByVariant provided as a template file.
Multiplicity	0..1
Default Value	false
Type	ECUC-BOOLEAN-PARAM-DEF

3.3.1.85 Dem/DemGeneral/DemGeneralOBD/DemRbGeneralOBD/DemRbOBDonUDSFreezeFrameClassRef

Short Name	DemRbOBDonUDSFreezeFrameClassRef
Description	This reference defines the link to the freeze frame for emission related diagnostics over UDS.
Multiplicity	0..1
Default Value	-
Type	ECUC-REFERENCE-DEF

3.3.1.86 Dem/DemGeneral/DemGeneralOBD/DemRbGeneralOBD/DemRbObdRGIDZeroMappedReadinessGroup

Short Name	DemRbObdRGIDZeroMappedReadinessGroup
Description	[NON-AUTOSAR] This parameter specifies DTCs assigned to which readiness group (see DemEventOBDReadinessGroup) are considered by the API Dem_SetDTCFilterByReadinessGroup() when called with RGID 0x00.

Introduction	This parameter can be configured only when DemRbObdReadinessGroupIdentifierZeroValidity is configured as TRUE.
Multiplicity	0..1
Default Value	-
Enumeration Literals	DEM_OBD_RDY_NONE
Type	ECUC-ENUMERATION-PARAM-DEF

3.3.1.87 Dem/DemGeneral/DemGeneralOBD/DemRbGeneralOBD/DemRbObdReadinessGroupIdentifierZeroValidity

Short Name	DemRbObdReadinessGroupIdentifierZeroValidity
Description	[NON-AUTOSAR] Switch to decide whether the API Dem_SetDTCFilterByReadinessGroup() considers 0x00 as a valid Readiness Group Identifier (RGID) or not. See also DemRbObdRGIDZeroMappedReadinessGroup.
Introduction	true: Dem_SetDTCFilterByReadinessGroup() considers 0x00 as a valid RGID, resulting in positive response to service $19 56. false: Dem_SetDTCFilterByReadinessGroup() considers 0x00 as an invalid RGID, resulting in negative response to service $19 56. Note: If there are no DTCs that match the filter criteria (e.g., due to all relevant events being unavailable or no DTCs configured to belong to the relevant readiness group), then Dem_SetDTCFilterByReadinessGroup() considers 0x00 as an invalid RGID, even if DemRbObdReadinessGroupIdentifierZeroValidity is true, when DemRbObdRGIDZeroMappedReadinessGroup is configured but there are no DTCs that belong to that group.
Multiplicity	0..1
Default Value	false
Type	ECUC-BOOLEAN-PARAM-DEF

3.3.1.88 Dem/DemGeneral/DemGeneralOBD/DemRbGeneralOBD/DemRbProvidesOBDSnapshotDataAsPartOfUDSRequestForRecordFF

Short Name	DemRbProvidesOBDSnapshotDataAsPartOfUDSRequestForRecordFF

Description	Enable for project specific implementation to retrieve OBD snapshot data as part of UDS 0x19 0x04 “DTC” 0xFF request. true : - OBDonUDS: Records 0x00 and 0xF0, if available, shall also be part of the response. - WWH-OBD: Record 0x00, if available, shall also be part of the response. false : - OBDonUDS: Records 0x00 and 0xF0 shall not be part of the response. - WWH-OBD: Record 0x00 shall not be part of the response.
Multiplicity	0..1
Default Value	false
Type	ECUC-BOOLEAN-PARAM-DEF

3.3.1.89 Dem/DemGeneral/DemMaxNumberEventEntryEventBuffer

Short Name	DemMaxNumberEventEntryEventBuffer
Description	Specifies the size of the buffer for storing environmental data (freeze-frames and extended data) until they are processed and stored to the event memory.
Multiplicity	0..1
Value Range	1..250
Default Value	-
Type	ECUC-INTEGER-PARAM-DEF

3.3.1.90 Dem/DemGeneral/DemNvRamBlockId/DemRbNvRamImmediateStoragePriority

Short Name	DemRbNvRamImmediateStoragePriority
Description	If an immediate storage is triggered (e.g. via Dem_TriggerStorageToNvm()) then the blocks are stored to NvM in the order of decreasing priority. Some blocks must be configured to the highest priority (see documentation). The order between blocks which have the same priority is undefined.
Multiplicity	0..1
Value Range	-10000..10000
Default Value	0
Type	ECUC-INTEGER-PARAM-DEF

3.3.1.91 Dem/DemGeneral/DemOperationCycle/DemRbIsStartViaApiAllowed

Short Name	DemRbIsStartViaApiAllowed
Long Name	DemRbIsAllowedToBeStartedDirectly
Description	Is dependent cycle allowed to be started with Dem_RestartOperationCycle as well. (Only relevant if DemLeadingCycleRef is set)
Multiplicity	0..1
Default Value	false
Type	ECUC-BOOLEAN-PARAM-DEF

3.3.1.92 Dem/DemGeneral/DemRatio/DemRbSPNNumber

Short Name	DemRbSPNNumber
Description	This parameter defines customer specific SPN number for individual Ratio group.
Multiplicity	0..1
Value Range	0..524287
Default Value	-
Type	ECUC-INTEGER-PARAM-DEF

3.3.1.93 Dem/DemGeneral/DemRbGDoc

Short Name	DemRbGDoc
Description	This container contains the configuration (parameters) of the BSW Dem
Multiplicity	0..1
Default Value	-
Type	ECUC-PARAM-CONF-CONTAINER-DEF

3.3.1.94 Dem/DemGeneral/DemRbGDoc/DemRbGDocMonitoredComponent

Short Name	DemRbGDocMonitoredComponent
Description	Specify a monitored component or system.
Multiplicity	1..*
Default Value	-
Type	ECUC-PARAM-CONF-CONTAINER-DEF

3.3.1.95 Dem/DemGeneral/DemRbGDoc/DemRbGDocMonitoringStrategy

Short Name	DemRbGDocMonitoringStrategy

Description	Specify a monitoring strategy.
Multiplicity	1..*
Default Value	-
Type	ECUC-PARAM-CONF-CONTAINER-DEF

3.3.1.96 Dem/DemGeneral/DemRbGeneral

Short Name	DemRbGeneral
Description	This container contains the configuration (parameters) of the BSW Dem.
Multiplicity	1..1
Default Value	-
Type	ECUC-PARAM-CONF-CONTAINER-DEF

3.3.1.97 Dem/DemGeneral/DemRbGeneral/DemRbAllowDisableDTCSetting-Callback

Short Name	DemRbAllowDisableDTCSettingCallback
Description	Enter the name of a function which indicates whether disabling the DTCSet-ting for that particular event is allowed or not.If the function returns TRUE the event shall be ignored and if it returns FALSE the function shall be processed.
Multiplicity	0..1
Default Value	-
Type	ECUC-FUNCTION-NAME-DEF

3.3.1.98 Dem/DemGeneral/DemRbGeneral/DemRbBfm

Short Name	DemRbBfm
Multiplicity	0..1
Default Value	-
Type	ECUC-PARAM-CONF-CONTAINER-DEF

3.3.1.99 Dem/DemGeneral/DemRbGeneral/DemRbCalibParameters

Short Name	DemRbCalibParameters
Long Name	DemRbCalibtParameters
Description	This container contains the configuration (parameters) to calibrate data into separate structure. This is an experi-mental feature only for a specific pilot customer and may be discontinued in future without further notice
Multiplicity	0..1

Default Value	-
Type	ECUC-PARAM-CONF-CONTAINER-DEF

3.3.1.100 Dem/DemGeneral/DemRbGeneral/DemRbCalibParameters/DemRbCalibAgingAllowed

Short Name	DemRbCalibAgingAllowed
Description	This parameter calibrates aging allowed value. True: Calibrate data into separate structure. False: no effect.
Multiplicity	0..1
Default Value	false
Type	ECUC-BOOLEAN-PARAM-DEF

3.3.1.101 Dem/DemGeneral/DemRbGeneral/DemRbCalibParameters/DemRbCalibAgingCycleCounterThreshold

Short Name	DemRbCalibAgingCycleCounterThreshold
Description	This parameter calibrates Aging Threshold. True: Calibrate data into separate structure. False: no effect.
Multiplicity	0..1
Default Value	false
Type	ECUC-BOOLEAN-PARAM-DEF

3.3.1.102 Dem/DemGeneral/DemRbGeneral/DemRbCalibParameters/DemRbCalibDTCKind

Short Name	DemRbCalibDTCKind
Description	This parameter calibrates DTC kind value. True: Calibrate data into separate structure. False: no effect.
Multiplicity	0..1
Default Value	false
Type	ECUC-BOOLEAN-PARAM-DEF

3.3.1.103 Dem/DemGeneral/DemRbGeneral/DemRbCalibParameters/DemRbCalibEventAvailability

Short Name	DemRbCalibEventAvailability
Description	This parameter calibrates event availability value. True: Calibrate data into separate structure. False: no effect.
Multiplicity	0..1
Default Value	false
Type	ECUC-BOOLEAN-PARAM-DEF

3.3.1.104 Dem/DemGeneral/DemRbGeneral/DemRbCalibParameters/DemRbCalibEventConfirmationThreshold

Short Name	DemRbCalibEventConfirmationThreshold
Description	This parameter calibrates failure confirmation Threshold. True: Calibrate data into separate structure. False: no effect.
Multiplicity	0..1
Default Value	false
Type	ECUC-BOOLEAN-PARAM-DEF

3.3.1.105 Dem/DemGeneral/DemRbGeneral/DemRbCalibParameters/DemRbCalibEventPriority

Short Name	DemRbCalibEventPriority
Description	This parameter calibrates Event Priority. True: Calibrate data into separate structure. False: no effect.
Multiplicity	0..1
Default Value	false
Type	ECUC-BOOLEAN-PARAM-DEF

3.3.1.106 Dem/DemGeneral/DemRbGeneral/DemRbCalibParameters/DemRbCalibMaxNumberFreezeFrameRecords

Short Name	DemRbCalibMaxNumberFreezeFrameRecords
Description	This parameter calibrates Maximum freez frame records value. True: Calibrate data into separate structure. False: no effect.
Multiplicity	0..1
Default Value	false
Type	ECUC-BOOLEAN-PARAM-DEF

3.3.1.107 Dem/DemGeneral/DemRbGeneral/DemRbCalibParameters/DemRbCalibSeverity

Short Name	DemRbCalibSeverity
Description	This parameter moves the attributes DemDTCSeverity and DemWWHOBDDTCClass to separate data structures. The two attributes together contribute to the DTCSeverityMask of a DTC
Multiplicity	0..1
Default Value	false
Type	ECUC-BOOLEAN-PARAM-DEF

3.3.1.108 Dem/DemGeneral/DemRbGeneral/DemRbCalibParameters/DemRbCalibSignificance

Short Name	DemRbCalibSignificance
Description	This parameter calibrates significance value. True: Calibrate data into separate structure. False: no effect.
Multiplicity	0..1
Default Value	false
Type	ECUC-BOOLEAN-PARAM-DEF

3.3.1.109 Dem/DemGeneral/DemRbGeneral/DemRbCheckApiConsistency

Short Name	DemRbCheckApiConsistency
Description	This parameter controls the check done for the consistency of DEM C and RTE APIs [Redundant declaration warnings are expected if the APIs are consistent]. Configured as True → Consistency of APIs are checked Configured as False → Consistency of APIs are not checked
Multiplicity	0..1
Default Value	false
Type	ECUC-BOOLEAN-PARAM-DEF

3.3.1.110 Dem/DemGeneral/DemRbGeneral/DemRbClearDTCHandleEventMemoryOverflow

Short Name	DemRbClearDTCHandleEventMemoryOverflow
Description	[NON-AUTOSAR] This parameter shall be activated for handling the buffer overflow situation for cyclic events reported during clear. If TRUE,Re-Reporting Cyclic events shall make an entry into event memory after clearDTC is complete,if there is a buffer overflow. If FALSE, Re-Reporting Cyclic events shall not make an entry into event memory after clearDTC is complete,if there is a buffer overflow.
Multiplicity	0..1
Default Value	true
Type	ECUC-BOOLEAN-PARAM-DEF

3.3.1.111 Dem/DemGeneral/DemRbGeneral/DemRbClearDTCUpdateRelevantFlagNumberOfEventsPerCycle

Short Name	DemRbClearDTCUpdateRelevantFlagNumberOfEventsPerCycle
Description	Configures the number of events to be processed per DemMainCycle for updating the next report relevant for Memories flag. →If this parameter is not configured, By default 400 events shall be selected for updating the next report relevant for Memories flag per main function call. This parameter is considered only if DemRbClearDTCHandleEventMemoryOverflow = TRUE
Multiplicity	0..1
Value Range	1..65535
Default Value	400
Type	ECUC-INTEGER-PARAM-DEF

3.3.1.112 Dem/DemGeneral/DemRbGeneral/DemRbClearDtcClearsAllBits

Short Name	DemRbClearDtcClearsAllBits
Description	Control whether clear DTC will clear all the bits of the isobyte. If true is selected behavior is like AR4.0.*, AR4.1.*, AR4.2.1). In case of false the test-failed flag and warning indicator flag won't be cleared.
Multiplicity	0..1
Default Value	true
Type	ECUC-BOOLEAN-PARAM-DEF

3.3.1.113 Dem/DemGeneral/DemRbGeneral/DemRbClearProjectExtensionIsClearPendingNotification

Short Name	DemRbClearProjectExtensionIsClearPendingNotification
Description	This configuration switch defines, project specific callback to get the feedback from project if the clear triggered via this API is pending or not, before finish the ClearDTC action
Multiplicity	0..1
Default Value	false
Type	ECUC-BOOLEAN-PARAM-DEF

3.3.1.114 Dem/DemGeneral/DemRbGeneral/DemRbCustomizableDTCStatusByte

Short Name	DemRbCustomizableDTCStatusByte
Description	Activate separate calculation of a dtc statusbyte. This allows project specific extensions of the behavior of the dtc statusbyte, without influencing the “failsafe” behavior (e.g. testfailed flag of an event). Costs an additional statusbyte per event. If enabled use template file Dem_CustomStatusByte.h_tpl for projct specific implemenation.
Multiplicity	0..1
Default Value	false
Type	ECUC-BOOLEAN-PARAM-DEF

3.3.1.115 Dem/DemGeneral/DemRbGeneral/DemRbDTCSettingBlocksReporting

Short Name	DemRbDTCSettingBlocksReporting
Description	This parameter specifies the impact of ControlDTCSetting (UDS service 0x85) on the processing of event status reports. If DemRbDTCSettingBlocksReporting == TRUE , ControlDTCSetting will work similar to enable conditions. The event status update as well as the storage to event memory is blocked in this case. If DemRbDTCSettingBlocksReporting == FALSE , ControlDTCSetting will block the storage to event memory. The event status update itself is not blocked, allowing for system behavior.
Multiplicity	0..1
Default Value	true
Type	ECUC-BOOLEAN-PARAM-DEF

3.3.1.116 Dem/DemGeneral/DemRbGeneral/DemRbDataElementLastFailedSinceLastFailedExcludingTNCIncludesFailedCycle

Short Name	DemRbDataElementLastFailedSinceLastFailedExcludingTNCIncludesFailedCycle
Description	Enables/Disables if the counter for the internal data element class DEM_CYCLES_SINCE_LAST_FAILED_EXCLUDING_TNC includes the cycle in which the failed occurred.
Multiplicity	0..1
Default Value	true

Type	ECUC-BOOLEAN-PARAM-DEF

3.3.1.117 Dem/DemGeneral/DemRbGeneral/DemRbDebLimitRelevantFor-Jumping

Short Name	DemRbDebLimitRelevantForJumping
Description	If configured to false, the debouncer jump behavior is performed on change of counter direction (as defined by Autosar). If set to true, the debouncer jump behavior is only done when the counter additionally equals the passed/failed limit.
Multiplicity	0..1
Default Value	false
Type	ECUC-BOOLEAN-PARAM-DEF

3.3.1.118 Dem/DemGeneral/DemRbGeneral/DemRbDebounceTimeBased-TaskTime

Short Name	DemRbDebounceTimeBasedTaskTime
Description	This parameter allows to configure a different cyclic task time as the base for Timebase Debounce method.If this is configured then Dem_TimeBased-DebounceMainFunction must be scheduled in appropriate cycle task. Please note: This configuration value shall be equal to the value in the Basic Software Scheduler configuration of the RTE module. [RB-Comment] The timebase for this parameter is seconds.
Introduction	The AUTOSAR configuration standard is to use SI units, so this parameter is defined as float value in seconds. Dem configuration tools must convert this float value to the appropriate value format for the use in the software implementation of Dem. min: A negative value is not allowed. max: After event status was reported, processing shall be completed within 100ms in order to have the fault entry status information updated as soon as possible (e.g. for PID $01). upperMultiplicity: Exactly one TaskTime must be specified per configuration. lowerMultiplicity: Exactly one TaskTime must be specified per configuration.
Multiplicity	0..1
Value Range	0..0.1

Default Value	-
Type	ECUC-FLOAT-PARAM-DEF

3.3.1.119 Dem/DemGeneral/DemRbGeneral/DemRbDefaultEventCustAttribute

Short Name	DemRbDefaultEventCustAttribute
Description	Definition of available Customer Specific Attributes, which may be set per Event. Handling to be done in template file Dem_EventAttribute.h_tpl.
Multiplicity	0..*
Default Value	-
Type	ECUC-PARAM-CONF-CONTAINER-DEF

3.3.1.120 Dem/DemGeneral/DemRbGeneral/DemRbDefaultEventCustAttribute/DemRbCustAttributeDefaultValue

Short Name	DemRbCustAttributeDefaultValue
Description	DefaultValue if this parameter is not referred in an Event
Multiplicity	1..1
Default Value	-
Type	ECUC-STRING-PARAM-DEF

3.3.1.121 Dem/DemGeneral/DemRbGeneral/DemRbDisableAutomaticEventIds

Short Name	DemRbDisableAutomaticEventIds
Description	Enables/ Disables the automatic assignement of EventIds (see DemEventId). Hint: If automatic eventId assignement is disabled, the user has to manually assign the IDs and has to ensure a consecutive numbering from 1..numberofEventIds. Automatic number assignement is recommended.
Multiplicity	0..1
Default Value	false
Type	ECUC-BOOLEAN-PARAM-DEF

3.3.1.122 Dem/DemGeneral/DemRbGeneral/DemRbDisturbanceMemoryCallbackOnDisturbance

Short Name	DemRbDisturbanceMemoryCallbackOnDisturbance

Description	Enables/ Disables project specific callback at the time of disturbance for Disturbance memory
Multiplicity	0..1
Default Value	false
Type	ECUC-BOOLEAN-PARAM-DEF

3.3.1.123 Dem/DemGeneral/DemRbGeneral/DemRbDisturbanceMemoryExtendedDataClassRef

Short Name	DemRbDisturbanceMemoryExtendedDataClassRef
Description	This reference defines the link to an extended data class sampler.
Multiplicity	0..1
Default Value	-
Type	ECUC-REFERENCE-DEF

3.3.1.124 Dem/DemGeneral/DemRbGeneral/DemRbDisturbanceMemorySize

Short Name	DemRbDisturbanceMemorySize
Description	number of elements stored in the dirsturbance memory.
Multiplicity	0..1
Value Range	1..15
Default Value	15
Type	ECUC-INTEGER-PARAM-DEF

3.3.1.125 Dem/DemGeneral/DemRbGeneral/DemRbDisturbanceMemorySupported

Short Name	DemRbDisturbanceMemorySupported
Description	Disturbance memory is supported or not
Multiplicity	0..1
Default Value	false
Type	ECUC-BOOLEAN-PARAM-DEF

3.3.1.126 Dem/DemGeneral/DemRbGeneral/DemRbDocumentation

Short Name	DemRbDocumentation
Description	This container contains the configuration (parameters) of the BSW Dem
Multiplicity	0..1
Default Value	-
Type	ECUC-PARAM-CONF-CONTAINER-DEF

3.3.1.127 Dem/DemGeneral/DemRbGeneral/DemRbDocumentation/DemRb-DDL_CSV

Short Name	DemRbDDL_CSV
Description	Specify a DDL parameters
Multiplicity	0..1
Default Value	-
Type	ECUC-PARAM-CONF-CONTAINER-DEF

3.3.1.128 Dem/DemGeneral/DemRbGeneral/DemRbDocumentation/DemRb-DDL_CSV/DemRbIncludeAgingCycleInDDLExport

Short Name	DemRbIncludeAgingCycleInDDLExport
Description	Configurable DDL parameters
Introduction	true: Provide AgingCycle value in Dem_DDL.csv file false: Not Provide Aging-Cycle value in Dem_DDL.csv file
Multiplicity	0..1
Default Value	true
Type	ECUC-BOOLEAN-PARAM-DEF

3.3.1.129 Dem/DemGeneral/DemRbGeneral/DemRbDocumentation/DemRb-DDL_CSV/DemRbIncludeCallbackClearEventAllowedInDDLExport

Short Name	DemRbIncludeCallbackClearEventAllowedInDDLExport
Description	Configurable DDL parameters
Introduction	true: Provide CallbackClearEventAllowed value in Dem_DDL.csv file false: Not Provide CallbackClearEventAllowed value in Dem_DDL.csv file
Multiplicity	0..1
Default Value	true
Type	ECUC-BOOLEAN-PARAM-DEF

3.3.1.130 Dem/DemGeneral/DemRbGeneral/DemRbDocumentation/DemRb-DDL_CSV/DemRbIncludeCallbackEventDataChangedInDDLExport

Short Name	DemRbIncludeCallbackEventDataChangedInDDLExport
Description	Configurable DDL parameters
Introduction	true: Provide CallbackEventDataChanged value in Dem_DDL.csv file false: Not Provide CallbackEventDataChanged value in Dem_DDL.csv file
Multiplicity	0..1
Default Value	true
Type	ECUC-BOOLEAN-PARAM-DEF

3.3.1.131 Dem/DemGeneral/DemRbGeneral/DemRbDocumentation/DemRbDDL_CSV/DemRbIncludeCallbackEventStatusChangedInDDLExport

Short Name	DemRbIncludeCallbackEventStatusChangedInDDLExport
Description	Configurable DDL parameters
Introduction	true: Provide CallbackEventStatusChanged value in Dem_DDL.csv file false: Not Provide CallbackEventStatusChanged value in Dem_DDL.csv file
Multiplicity	0..1
Default Value	true
Type	ECUC-BOOLEAN-PARAM-DEF

3.3.1.132 Dem/DemGeneral/DemRbGeneral/DemRbDocumentation/DemRbDDL_CSV/DemRbIncludeCallbackInitMForEInDDLExport

Short Name	DemRbIncludeCallbackInitMForEInDDLExport
Description	Configurable DDL parameters
Introduction	true: Provide CallbackInitMForE value in Dem_DDL.csv file false: Not Provide CallbackInitMForE value in Dem_DDL.csv file
Multiplicity	0..1
Default Value	true
Type	ECUC-BOOLEAN-PARAM-DEF

3.3.1.133 Dem/DemGeneral/DemRbGeneral/DemRbDocumentation/DemRbDDL_CSV/DemRbIncludeCausalityDelayTimeInDDLExport

Short Name	DemRbIncludeCausalityDelayTimeInDDLExport
Description	Configurable DDL parameters
Introduction	true: Provide EvBufferTime value in Dem_DDL.csv file false: Not Provide EvBufferTime value in Dem_DDL.csv file
Multiplicity	0..1
Default Value	true
Type	ECUC-BOOLEAN-PARAM-DEF

3.3.1.134 Dem/DemGeneral/DemRbGeneral/DemRbDocumentation/DemRbDDL_CSV/DemRbIncludeDTCClassRefInDDLExport

Short Name	DemRbIncludeDTCClassRefInDDLExport
Description	Configurable DDL parameters
Introduction	true: Provide DTCClassRef value in Dem_DDL.csv file false: Not Provide DTCClassRef value in Dem_DDL.csv file

Multiplicity	0..1
Default Value	true
Type	ECUC-BOOLEAN-PARAM-DEF

3.3.1.135 Dem/DemGeneral/DemRbGeneral/DemRbDocumentation/DemRb-DDL_CSV/DemRbIncludeDTCValueInDDLExport

Short Name	DemRbIncludeDTCValueInDDLExport
Description	Configurable DDL parameters
Introduction	true: Provide DTC Value in Dem_DDL.csv file false: Not Provide DTC Value in Dem_DDL.csv file
Multiplicity	0..1
Default Value	false
Type	ECUC-BOOLEAN-PARAM-DEF

3.3.1.136 Dem/DemGeneral/DemRbGeneral/DemRbDocumentation/DemRb-DDL_CSV/DemRbIncludeDebounceAlgorithmClassInDDLExport

Short Name	DemRbIncludeDebounceAlgorithmClassInDDLExport
Description	Configurable DDL parameters
Introduction	true: Provide DebounceAlgorithmClass value in Dem_DDL.csv file false: Not Provide DebounceAlgorithmClass value in Dem_DDL.csv file
Multiplicity	0..1
Default Value	true
Type	ECUC-BOOLEAN-PARAM-DEF

3.3.1.137 Dem/DemGeneral/DemRbGeneral/DemRbDocumentation/DemRb-DDL_CSV/DemRbIncludeDgsDfcNameInDDLExport

Short Name	DemRbIncludeDgsDfcNameInDDLExport
Description	Configurable DDL parameters
Introduction	true: Provide DemRbDgsDfcName value in Dem_DDL.csv file false: Not Provide DemRbDgsDfcName value in Dem_DDL.csv file
Multiplicity	0..1
Default Value	true
Type	ECUC-BOOLEAN-PARAM-DEF

3.3.1.138 Dem/DemGeneral/DemRbGeneral/DemRbDocumentation/DemRb-DDL_CSV/DemRbIncludeEnableConditionsInDDLExport

Short Name	DemRbIncludeEnableConditionsInDDLExport
Description	Configurable DDL parameters

Introduction	true: Provide EnableConditions value in Dem_DDL.csv file false: Not Provide EnableConditions value in Dem_DDL.csv file
Multiplicity	0..1
Default Value	true
Type	ECUC-BOOLEAN-PARAM-DEF

3.3.1.139 Dem/DemGeneral/DemRbGeneral/DemRbDocumentation/DemRb-DDL_CSV/DemRbIncludeEventConfirmationThresholdInDDLExport

Short Name	DemRbIncludeEventConfirmationThresholdInDDLExport
Description	Configurable DDL parameters
Introduction	true: Provide EventConfirmationThreshold value in Dem_DDL.csv file false: Not Provide EventConfirmationThreshold value in Dem_DDL.csv file
Multiplicity	0..1
Default Value	true
Type	ECUC-BOOLEAN-PARAM-DEF

3.3.1.140 Dem/DemGeneral/DemRbGeneral/DemRbDocumentation/DemRb-DDL_CSV/DemRbIncludeEventIdInDDLExport

Short Name	DemRbIncludeEventIdInDDLExport
Description	Configurable DDL parameters
Introduction	true: Provide EventId value in Dem_DDL.csv file false: Not Provide EventId value in Dem_DDL.csv file
Multiplicity	0..1
Default Value	true
Type	ECUC-BOOLEAN-PARAM-DEF

3.3.1.141 Dem/DemGeneral/DemRbGeneral/DemRbDocumentation/DemRb-DDL_CSV/DemRbIncludeEventKindInDDLExport

Short Name	DemRbIncludeEventKindInDDLExport
Description	Configurable DDL parameters
Introduction	true: Provide EventKind value in Dem_DDL.csv file false: Not Provide EventKind Parameter value in Dem_DDL.csv file
Multiplicity	0..1
Default Value	true
Type	ECUC-BOOLEAN-PARAM-DEF

3.3.1.142 Dem/DemGeneral/DemRbGeneral/DemRbDocumentation/DemRb-DDL_CSV/DemRbIncludeEventParameterInDDLExport

Short Name	DemRbIncludeEventParameterInDDLExport
Description	Configurable DDL parameters
Introduction	true: Provide EventParameter value in Dem_DDL.csv file false: Not Provide EventParameter value in Dem_DDL.csv file
Multiplicity	0..1
Default Value	true
Type	ECUC-BOOLEAN-PARAM-DEF

3.3.1.143 Dem/DemGeneral/DemRbGeneral/DemRbDocumentation/DemRb-DDL_CSV/DemRbIncludeExtendedDataClassRefInDDLExport

Short Name	DemRbIncludeExtendedDataClassRefInDDLExport
Description	Configurable DDL parameters
Introduction	true: Provide ExtendedDataClassRef value in Dem_DDL.csv file false: Not Provide ExtendedDataClassRef value in Dem_DDL.csv file
Multiplicity	0..1
Default Value	true
Type	ECUC-BOOLEAN-PARAM-DEF

3.3.1.144 Dem/DemGeneral/DemRbGeneral/DemRbDocumentation/DemRb-DDL_CSV/DemRbIncludeFreezeFrameClassRefInDDLExport

Short Name	DemRbIncludeFreezeFrameClassRefInDDLExport
Description	Configurable DDL parameters
Introduction	true: Provide FreezeFrameClassRef value in Dem_DDL.csv file false: Not Provide FreezeFrameClassRef value in Dem_DDL.csv file
Multiplicity	0..1
Default Value	true
Type	ECUC-BOOLEAN-PARAM-DEF

3.3.1.145 Dem/DemGeneral/DemRbGeneral/DemRbDocumentation/DemRb-DDL_CSV/DemRbIncludeIsReplacementEventOfStocoInDDLExport

Short Name	DemRbIncludeIsReplacementEventOfStocoInDDLExport
Description	Configurable DDL parameters
Introduction	true: Provide IsReplacementEventOfStoco value in Dem_DDL.csv file false: Not Provide IsReplacementEventOfStoco value in Dem_DDL.csv file
Multiplicity	0..1

Default Value	true
Type	ECUC-BOOLEAN-PARAM-DEF

3.3.1.146 Dem/DemGeneral/DemRbGeneral/DemRbDocumentation/DemRb-DDL_CSV/DemRbIncludeNodeRefInDDLExport

Short Name	DemRbIncludeNodeRefInDDLExport
Description	Configurable DDL parameters
Introduction	true: Provide NodeRef value in Dem_DDL.csv file false: Not Provide NodeRef value in Dem_DDL.csv file
Multiplicity	0..1
Default Value	true
Type	ECUC-BOOLEAN-PARAM-DEF

3.3.1.147 Dem/DemGeneral/DemRbGeneral/DemRbDocumentation/DemRb-DDL_CSV/DemRbIncludeOperationCycleInDDLExport

Short Name	DemRbIncludeOperationCycleInDDLExport
Description	Configurable DDL parameters
Introduction	true: Provide OperationCycle value in Dem_DDL.csv file false: Not Provide OperationCycle value in Dem_DDL.csv file
Multiplicity	0..1
Default Value	true
Type	ECUC-BOOLEAN-PARAM-DEF

3.3.1.148 Dem/DemGeneral/DemRbGeneral/DemRbDocumentation/DemRb-DDL_CSV/DemRbIncludeRecoverableInDDLExport

Short Name	DemRbIncludeRecoverableInDDLExport
Description	Configurable DDL parameters
Introduction	true: Provide Recoverable value in Dem_DDL.csv file false: Not Provide Recoverable value in Dem_DDL.csv file
Multiplicity	0..1
Default Value	true
Type	ECUC-BOOLEAN-PARAM-DEF

3.3.1.149 Dem/DemGeneral/DemRbGeneral/DemRbDocumentation/DemRb-DDL_CSV/DemRbIncludeShiftViewInDDLExport

Short Name	DemRbIncludeShiftViewInDDLExport
Description	Configurable DDL parameters

Introduction	true: Provide DemRbShiftView value in Dem_DDL.csv file false: Not Provide DemRbShiftView value in Dem_DDL.csv file
Multiplicity	0..1
Default Value	true
Type	ECUC-BOOLEAN-PARAM-DEF

3.3.1.150 Dem/DemGeneral/DemRbGeneral/DemRbDocumentation/DemRb-DDL_CSV/DemRbIncludeStorageConditionsInDDLExport

Short Name	DemRbIncludeStorageConditionsInDDLExport
Description	Configurable DDL parameters
Introduction	true: Provide StorageConditions value in Dem_DDL.csv file false: Not Provide StorageConditions value in Dem_DDL.csv file
Multiplicity	0..1
Default Value	true
Type	ECUC-BOOLEAN-PARAM-DEF

3.3.1.151 Dem/DemGeneral/DemRbGeneral/DemRbEcucPartitionRef

Short Name	DemRbEcucPartitionRef
Description	Maps the Dem to one or multiple ECUC partitions to make the Dem_SetEventStatus() API available in the respective partition
Multiplicity	0..65535
Default Value	-
Type	ECUC-SYMBOLIC-NAME-REFERENCE-DEF

3.3.1.152 Dem/DemGeneral/DemRbGeneral/DemRbEnableDemNvmBlock-sAutoConfiguration

Short Name	DemRbEnableDemNvmBlocksAutoConfiguration
Description	This configuration switch defines, whether the needed Dem Blocks shall be added automatically in NvM or not. If it is enabled, then the expected NvM Blocks are automatically configured in the NVRAM Manager and respectively referenced in Dem.
Multiplicity	0..1
Default Value	false
Type	ECUC-BOOLEAN-PARAM-DEF

3.3.1.153 Dem/DemGeneral/DemRbGeneral/DemRbEnableNvmBlockLength-Checks

Short Name	DemRbEnableNvmBlockLengthChecks
Description	This configuration switch defines, whether the static assertion check for the NvM Block length is enabled or not. The assertion checks ensure that the configured size of the NvM blocks used by Dem are large enough to hold the data structures that is persisted in them. By enabling the assertion checks, it is enforced that the configured NvM block sizes are exactly equal to the size of the data structures. The compilation is aborted in case the assertion check fails.
Introduction	Note: The parameter is not relevant if the "Auto configuration of NvM blocks" feature is enabled in Dem (see DemRbEnableDemNvmBlocksAutoConfiguration) or if the NvM is configured to calculate the block size automatically by setting NvMRbAutoBlockLength.
Multiplicity	0..1
Default Value	true
Type	ECUC-BOOLEAN-PARAM-DEF

3.3.1.154 Dem/DemGeneral/DemRbGeneral/DemRbEnableReportingChronologicalOrderHooks

Short Name	DemRbEnableReportingChronologicalOrderHooks
Description	This switch controls the availability of hooks for implementing Project specific DTC filter and reporting.(like report DTCs in Chronological order). The declarations are available in Dem_ChronoSort.h file. Project has to provide their implementation in template file Dem_PrjChronoSorting.c_tpl.
Multiplicity	0..1
Default Value	false
Type	ECUC-BOOLEAN-PARAM-DEF

3.3.1.155 Dem/DemGeneral/DemRbGeneral/DemRbEvMemProjectExtension

Short Name	DemRbEvMemProjectExtension

Description	Activate project specific implementation and calculation in evmem (e.g. cycle counters for environmental data). On activating the feature, the struct 'Dem_EvMemProjectExtension' is included in each EvMem location and the project specific calculations 'Dem_EvMemProjectExtension(Init|Main|UnRobust|Failed|StartOpCycle)' are triggered. The struct and the functions have to be implemented in the project based on the available templates 'Dem_PrjEvmemProjectExtension.*_tpl'. Additionally data items can be defined in the environmental data by specifying appropriate dataelement (refer to DemInternalDataElement: DEM_PRJ_SPECIFIC_DATA_ON_RETRIEVE or DEM_PRJ_SPECIFIC_DATA_ON_REPORT).
Multiplicity	0..1
Default Value	false
Type	ECUC-BOOLEAN-PARAM-DEF

3.3.1.156 Dem/DemGeneral/DemRbGeneral/DemRbEventCategory

Short Name	DemRbEventCategory
Description	Specification of the event categories known in ESP-Systems; used to categorize failure for analysis in RB plants. This parameter is also used as the priority of RB failure memory records.
Multiplicity	0..255
Default Value	-
Type	ECUC-PARAM-CONF-CONTAINER-DEF

3.3.1.157 Dem/DemGeneral/DemRbGeneral/DemRbEventCategory/DemRbEventCategoryId

Short Name	DemRbEventCategoryId
Description	This contains unique event category id. The values “0” and “255” are reserved for internal use. In the RB failure memory “0” is the highest priority and “255” is the lowest one.
Multiplicity	1..1
Value Range	1..254
Default Value	254
Type	ECUC-INTEGER-PARAM-DEF

3.3.1.158 Dem/DemGeneral/DemRbGeneral/DemRbEventCategory/DemRbEventCategoryPriorityId

Short Name	DemRbEventCategoryPriorityId

Description	The parameter is used to configure event category priority. Least the value higher the priority.
Multiplicity	0..1
Value Range	1..254
Default Value	254
Type	ECUC-INTEGER-PARAM-DEF

3.3.1.159 Dem/DemGeneral/DemRbGeneral/DemRbEventCombinationOnlyAllEventsPassed

Short Name	DemRbEventCombinationOnlyAllEventsPassed
Description	Changes the processing of passed for combined DTCs. If set to “true”: On OperationCycle change a DTC will be processed as PASSED only if all combined Events of this DTC have reported PASSED (and never FAILED) in this operation cycle If ser to “false”: On OperationCycle change the passed for the DTC will be processed if one of the combined events reported passed Default: “false” (old behavior)
Multiplicity	0..1
Default Value	false
Type	ECUC-BOOLEAN-PARAM-DEF

3.3.1.160 Dem/DemGeneral/DemRbGeneral/DemRbEventMemoryGenericSupported

Short Name	DemRbEventMemoryGenericSupported
Description	Enables/Disables support for the generic event memory services Dem_GetDTCByOccurrenceTime and Dem_GetEventMemoryOverflow
Multiplicity	0..1
Default Value	true
Type	ECUC-BOOLEAN-PARAM-DEF

3.3.1.161 Dem/DemGeneral/DemRbGeneral/DemRbEvtProjectExtension

Short Name	DemRbEvtProjectExtension
Description	Enables a project specific callback on each setEventStatus. May be used to calculate project specific status information (and e.g. store it as customer specific attributes; see DemRbDefaultEventCustAttribute).
Multiplicity	0..1
Default Value	false

Type	ECUC-BOOLEAN-PARAM-DEF

3.3.1.162 Dem/DemGeneral/DemRbGeneral/DemRbExternalProcessing

Short Name	DemRbExternalProcessing
Description	This container contains the configuration (parameter) for processing the function externally.
Multiplicity	0..1
Default Value	-
Type	ECUC-PARAM-CONF-CONTAINER-DEF

3.3.1.163 Dem/DemGeneral/DemRbGeneral/DemRbExternalProcessing/DemRbDtcFilter

Short Name	DemRbDtcFilter
Description	This configuration switch defines whether the DTC filter function shall be processed inside Dem_Mainfunction or it shall be processed when it is called from the external function
Introduction	true: External processing is enabled false: External Processing is disabled
Multiplicity	0..1
Default Value	false
Type	ECUC-BOOLEAN-PARAM-DEF

3.3.1.164 Dem/DemGeneral/DemRbGeneral/DemRbFailureDependencyRecheckLimit

Short Name	DemRbFailureDependencyRecheckLimit
Description	Also known as “Recheck Causality”: With failure dependency structure enabled, this parameter configures the number of events which are checked per main-cycle. The check addresses the situation, where an event is TestFailed, but not yet considered as beeing causal. This might happen for example, if an event is reported as FAILED while the storagecondition is not fulfilled or a higher priority fault is already reported and later on, the storagecondition is fulfilled again or the other failure disappears.
Multiplicity	0..1
Value Range	1..65535
Default Value	80
Type	ECUC-INTEGER-PARAM-DEF

3.3.1.165 Dem/DemGeneral/DemRbGeneral/DemRbGetEventAvailableBe-

haviourDemRbGetEventAvailableBehaviour

Short Name	DemRbGetEventAvailableBehaviour
Description	This parameter defines how Dem_GetEventAvailable() API assigns value to the out parameter "AvailableStatus". (1) AUTOSAR_COMPLIANT: If chosen, TRUE is assigned to the out parameter if event is available (Not suppressed) and FALSE is assigned to the out parameter if event is not available (suppressed) (2) LEGACY: If chosen, FALSE is assigned to the out parameter if event is available (Not suppressed) and TRUE is assigned to the out parameter if event is not available (suppressed) Note: It is the responsibility of the user that all callers of the Dem_GetEventAvailable() API in the project rely on the same type of behaviour and that specific behaviour is configured here.
Multiplicity	1..1
Default Value	-
Enumeration Literals	AUTOSAR_COMPLIANT, LEGACY
Type	ECUC-ENUMERATION-PARAM-DEF

3.3.1.166 Dem/DemGeneral/DemRbGeneral/DemRbGlobalCallback-ClearEventAllowedFnc

Short Name	DemRbGlobalCallbackClearEventAllowedFnc
Description	This parameter specifies a global callback, which is called as a "permission gate" before deleting an event. The behavior is identical to DemCallbackClearEventAllowed which is configured within DemEventParameter. In cases when DemGlobalCallbackClearEventAllowedFnc is configured, for the events which have DemCallbackClearEventAllowedFnc configured, the configured callback per Event shall have the precidence and thus the global callback shall not be invoked. Thus, the global callback is restricted to be invoked for all the Events which does not have container DemCallbackClearEventAllowed configured with DemEventParameter.
Multiplicity	0..1
Default Value	-
Type	ECUC-FUNCTION-NAME-DEF

3.3.1.167 Dem/DemGeneral/DemRbGeneral/DemRbGlobalCallbackEventStatusChanged

Short Name	DemRbGlobalCallbackEventStatusChanged

Description	The presence of this container indicates, that the Dem has access to a “DemRbGlobalCallbackEventStatusChangedFnc” callback, which the Dem will call for every event status changed to notify other components about the change in the status of a event. Note: All callbacks configured in this container shall be called in a single function.
Introduction	if DemRbGlobalCallbackEventStatusChangedFnc is configured, this parameter defines the name of the function that the Dem will call.
Multiplicity	0..255
Default Value	-
Type	ECUC-PARAM-CONF-CONTAINER-DEF

3.3.1.168 Dem/DemGeneral/DemRbGeneral/DemRbGlobalCallbackEventStatusChanged/DemRbGlobalCallbackEventStatusChangedFnc

Short Name	DemRbGlobalCallbackEventStatusChangedFnc
Description	Function name of prototype “GlobalEventStatusChanged”.
Multiplicity	1..1
Default Value	-
Type	ECUC-FUNCTION-NAME-DEF

3.3.1.169 Dem/DemGeneral/DemRbGeneral/DemRbGlobalCallbackGetFaultDetectionCounter

Short Name	DemRbGlobalCallbackGetFaultDetectionCounter
Description	This parameter specifies a global callback for getting the fault detection counter. The function returns 127 if the event is failed and -128 if it is passed . In all other cases it returns 0.
Multiplicity	0..1
Default Value	-
Type	ECUC-FUNCTION-NAME-DEF

3.3.1.170 Dem/DemGeneral/DemRbGeneral/DemRbGlobalCallbackInitMForE

Short Name	DemRbGlobalCallbackInitMForE
Description	The presence of this container indicates, that the Dem has access to an “DemRbGlobalInitMonitorForEvent” callback, which the Dem will call to initialize all monitors.

Introduction	In case the container has a DemRbGlobalCallbackInitM-ForEFnc, this parameter defines the name of the function that the Dem will call. In case there is no DemRbGlobal CallbackInitMForEFnc, the Dem will not call the GLobal Callback to notify the reset monitor reasons to the application.
Multiplicity	0..1
Default Value	-
Type	ECUC-PARAM-CONF-CONTAINER-DEF

3.3.1.171 Dem/DemGeneral/DemRbGeneral/DemRbGlobalCallbackInitMForE/DemRbGlobalCallbackInitMForEFnc

Short Name	DemRbGlobalCallbackInitMForEFnc
Description	Function name of prototype “GlobalInitMonitorForEvent”.
Multiplicity	0..1
Default Value	-
Type	ECUC-FUNCTION-NAME-DEF

3.3.1.172 Dem/DemGeneral/DemRbGeneral/DemRbHistoryStatusSupported

Short Name	DemRbHistoryStatusSupported
Description	AllowHistoryStatus is triggered or not
Multiplicity	0..1
Default Value	false
Type	ECUC-BOOLEAN-PARAM-DEF

3.3.1.173 Dem/DemGeneral/DemRbGeneral/DemRbLockAllFailureInfoSupported

Short Name	DemRbLockAllFailureInfoSupported
Description	LockAllFailureInformation is supported or not
Multiplicity	0..1
Default Value	false
Type	ECUC-BOOLEAN-PARAM-DEF

3.3.1.174 Dem/DemGeneral/DemRbGeneral/DemRbMasterEcucPartitionRef

Short Name	DemRbMasterEcucPartitionRef
Description	Maps the Dem to one ECUC partition to assign the master functionality to certain core. The partition referenced is a subset of the ECUC partition where the Dem is mapped to.

Multiplicity	0..1
Default Value	-
Type	ECUC-SYMBOLIC-NAME-REFERENCE-DEF

3.3.1.175 Dem/DemGeneral/DemRbGeneral/DemRbMirrorMemoryDisplacmentStrategyType

Short Name	DemRbMirrorMemoryDisplacmentStrategyType
Description	This parameter defines the type of Displacement Strategy used for the Mirror Memory (1) DEM_DISPLACEMENT_FULL (priority based) (2) DEM_DISPLACEMENT_RBA_FIFO (3) DEM_DISPLACEMENT_RBA_CUST
Multiplicity	0..1
Default Value	DEM_DISPLACEMENT_FULL
Enumeration Literals	DEM_DISPLACEMENT_FULL, DEM_DISPLACEMENT_RBA_CUST, DEM_DISPLACEMENT_RBA_FIFO
Type	ECUC-ENUMERATION-PARAM-DEF

3.3.1.176 Dem/DemGeneral/DemRbGeneral/DemRbMirrorMemoryDtcStatusStored

Short Name	DemRbMirrorMemoryDtcStatusStored
Description	[NON-AUTOSAR] This parameter defines that the DTC Status is additionally stored in Mirror Memory when a Mirror Memory location is written. If a DTC Status (with parameter Origin = MIRROR) is requested the stored DTC status is reported instead of the current DTC Status
Multiplicity	0..1
Default Value	false
Type	ECUC-BOOLEAN-PARAM-DEF

3.3.1.177 Dem/DemGeneral/DemRbGeneral/DemRbMirrorMemoryType

Short Name	DemRbMirrorMemoryType
Description	[NON-AUTOSAR] This parameter defines the type of Mirror Memory: (1) Store in Mirror Memory (On Failed) (2) Store in Mirror Memory (On Clear)
Multiplicity	0..1
Default Value	MIRROR_ON_CLEAR
Enumeration Literals	MIRROR_ON_CLEAR, MIRROR_ON_FAILED

Type	ECUC-ENUMERATION-PARAM-DEF

3.3.1.178 Dem/DemGeneral/DemRbGeneral/DemRbMonitorDataBeforeInit

Short Name	DemRbMonitorDataBeforeInit
Description	DemRbMonitorDataBeforeInit (i.e monitor data data support between Preinit and Init) is supported or not
Multiplicity	0..1
Default Value	true
Type	ECUC-BOOLEAN-PARAM-DEF

3.3.1.179 Dem/DemGeneral/DemRbGeneral/DemRbMonitorDataForTime-BasedDebouncing

Short Name	DemRbMonitorDataForTimeBasedDebouncing
Description	With this it is possible to store the monitordata reported with the prepassed/-failed and process it after the timebaseddebouncing qualified the report. CAREFULL: this leads to additional memory consumption (MonitorDataSize * 2 * countOfTimeBasedDebouncedEvents)
Multiplicity	0..1
Default Value	false
Type	ECUC-BOOLEAN-PARAM-DEF

3.3.1.180 Dem/DemGeneral/DemRbGeneral/DemRbNvMBlockValidationAsError

Short Name	DemRbNvMBlockValidationAsError
Description	With this switch the validation error for missing NvM blocks can be turned into a warning instead of error. This is to support the usecase of not using the NvM together with the Dem.
Multiplicity	0..1
Default Value	true
Type	ECUC-BOOLEAN-PARAM-DEF

3.3.1.181 Dem/DemGeneral/DemRbGeneral/DemRbNvMNumberOfStorage-Buffer

Short Name	DemRbNvMNumberOfStorageBuffer

Description	This parameter defines how many storagebuffer are available for copying to NVRAM. Storagebuffer are needed for all DemNvMBlocks which don't support callbacks and therefore need to store the content in a sperate buffer for data consistency. A higher number of storagebuffer means faster copying of multiple blocks but also more RAM consumption.
Multiplicity	0..1
Value Range	1..255
Default Value	2
Type	ECUC-INTEGER-PARAM-DEF

3.3.1.182 Dem/DemGeneral/DemRbGeneral/DemRbOccurrenceCounterType

Short Name	DemRbOccurrenceCounterType
Description	[NON-AUTOSAR] This parameter defines the Type of Occurrence Counter. (1) Not Supported (2) AUTOSAR
Multiplicity	0..1
Default Value	AUTOSAR
Enumeration Literals	AUTOSAR, NONE
Type	ECUC-ENUMERATION-PARAM-DEF

3.3.1.183 Dem/DemGeneral/DemRbGeneral/DemRbOperationCycleStatusStorage

Short Name	DemRbOperationCycleStatusStorage
Description	Defines if the operation cycle state is available over the power cycle (stored non-volatile) or not.
Introduction	true: the operation cycle state is stored non-volatile false: the operation cycle state is only stored volatile
Multiplicity	1..1
Default Value	-
Type	ECUC-BOOLEAN-PARAM-DEF

3.3.1.184 Dem/DemGeneral/DemRbGeneral/DemRbOptimization

Short Name	DemRbOptimization
Description	This container contains the configuration (parameters) introduced for the optimization purpose.
Multiplicity	1..1
Default Value	-

Type	ECUC-PARAM-CONF-CONTAINER-DEF

3.3.1.185 Dem/DemGeneral/DemRbGeneral/DemRbOptimization/DemRbAdvanceOperationCycleAffectedEventsPerLock

Short Name	DemRbAdvanceOperationCycleAffectedEventsPerLock
Description	On restart of an operation cycle, all events are checked and the events affected by the operation cycle transition are processed concerning their eventstatus. This parameter defines the maximum number of events that are affected by the operation cycle transition and are processed within a single critical section. This parameter considers only those events that are affected by the operation cycle transition. See also DemRbAdvanceOperationCycleEventsPerLock. If DemRbAdvanceOperationCycleAffectedEventsPerLock is not configured or is configured with a value larger than that of DemRbAdvanceOperationCycleEventsPerLock, then only the limit specified by DemRbAdvanceOperationCycleEventsPerLock is effective..
Multiplicity	0..1
Value Range	1..65535
Default Value	65535
Type	ECUC-INTEGER-PARAM-DEF

3.3.1.186 Dem/DemGeneral/DemRbGeneral/DemRbOptimization/DemRbAdvanceOperationCycleEventsPerLock

Short Name	DemRbAdvanceOperationCycleEventsPerLock
Description	On restart of an operation cycle, all events are checked and the events affected by the operation cycle transition are processed concerning their eventstatus. This parameter defines the maximum number of events that are processed within a single critical section during the operation cycle change. This parameter considers all events, independent of whether an event is affected by the operation cycle transition or not. See also DemRbAdvanceOperationCycleAffectedEventsPerLock.
Multiplicity	0..1
Value Range	1..65535
Default Value	16
Type	ECUC-INTEGER-PARAM-DEF

3.3.1.187 Dem/DemGeneral/DemRbGeneral/DemRbOptimization/DemRbClearDTCNumberOfEventsPerCycle

Short Name	DemRbClearDTCNumberOfEventsPerCycle
Description	Configures the number of events to be processed per DemMainCycle when processing a clear command. Be aware, the time to clear an individual event is highly dependent on ecu speed, task timing, and configuration like clearEventAllowed callbacks and eventStatusChange callbacks. →If this parameter is not configured, all events are processed in one call.
Multiplicity	0..1
Value Range	1..65535
Default Value	-
Type	ECUC-INTEGER-PARAM-DEF

3.3.1.188 Dem/DemGeneral/DemRbGeneral/DemRbOptimization/DemRbDtcFilterNumberOfEventsPerCycle

Short Name	DemRbDtcFilterNumberOfEventsPerCycle
Description	Defines how many events are considered in processing of a DTCFilter function in the Dem task cycle. If a DTC has more events than this parameter it will be processed anyway if it is the first DTC of this Dem task cycle (so that at least one dtc will be processed per cycle).
Multiplicity	0..1
Value Range	1..65535
Default Value	25
Type	ECUC-INTEGER-PARAM-DEF

3.3.1.189 Dem/DemGeneral/DemRbGeneral/DemRbOptimization/DemRbDtcFilterRetrieveNumberOfDTCsPerCycle

Short Name	DemRbDtcFilterRetrieveNumberOfDTCsPerCycle
Description	This parameter defines how many DTC's are considered to search matching DTC while retrieval in processing of one GetNextFilteredDTC call. Increasing this value will increase runtime marginally as we look for more DTC's if matching DTC's are not present. Refer BSWDOC for further details.
Multiplicity	0..1

Value Range	1..255
Default Value	50
Type	ECUC-INTEGER-PARAM-DEF

3.3.1.190 Dem/DemGeneral/DemRbGeneral/DemRbOptimization/DemRbDtcStateManagerNumberOfDTCsPerCycle

Short Name	DemRbDtcStateManagerNumberOfDTCsPerCycle
Description	Defines the namuber of DTCs to be processed in DTCStateManager function per cycle
Multiplicity	0..1
Value Range	1..65535
Default Value	10
Type	ECUC-INTEGER-PARAM-DEF

3.3.1.191 Dem/DemGeneral/DemRbGeneral/DemRbOptimization/DemRbNumberOfDTCsToProcessPerLOCK

Short Name	DemRbNumberOfDTCsToProcessPerLOCK
Description	The number of DTCs to be allowed for processing per LOCK while Control DTC Setting Service is called. Projects should take care that LOCK might take more time in two cases, (i) when DEM_CFG_CUSTOMIZABLEDTCSTATUSBYTE is switched ON(and some project specific functionlaity is done inside the Customer Status Byte Function), (ii) When Event combination is used. → If this parameter is not configured, by default, it takes the value 1 such that it is safe to have LOCK per DTC irrespective of Project configuration.
Multiplicity	0..1
Value Range	1..65535
Default Value	1
Type	ECUC-INTEGER-PARAM-DEF

3.3.1.192 Dem/DemGeneral/DemRbGeneral/DemRbOptimization/DemRbNumberOfDemRatiosPerCycle

Short Name	DemRbNumberOfDemRatiosPerCycle

Description	Defines the number of DemRatios processed within one main function cycle. If it is not configured or is configured with a value larger than the number of DemRatios, then all ratios are processed once within each main function cycle. The value configured shall be such that the ratios are processed fast enough to comply with any applicable legal regulations. In case of restarting the driving cycle all DemRatios will be processed at once.
Multiplicity	0..1
Value Range	1..65535
Default Value	65535
Type	ECUC-INTEGER-PARAM-DEF

3.3.1.193 Dem/DemGeneral/DemRbGeneral/DemRbOptimization/DemRbProcessNumberOfComponentsPerLock

Short Name	DemRbProcessNumberOfComponentsPerLock
Description	Defines the number of Components to be processed in one lock time execution. Default Configuration Hint: The current DemComponents size will be used as reference for the final value generation.
Multiplicity	0..1
Value Range	1..65535
Default Value	16
Type	ECUC-INTEGER-PARAM-DEF

3.3.1.194 Dem/DemGeneral/DemRbGeneral/DemRbPrestoreFreezeFrameNonVolatile

Short Name	DemRbPrestoreFreezeFrameNonVolatile
Description	[NON-AUTOSAR] If enabled, prestored freeze frame will be stored non volatile and will be still available after restarting the ECU.
Multiplicity	0..1
Default Value	false
Type	ECUC-BOOLEAN-PARAM-DEF

3.3.1.195 Dem/DemGeneral/DemRbGeneral/DemRbProjectSpecificDebOptimization

Short Name	DemRbProjectSpecificDebOptimization

Description	Controls call-out point for optimising debouncing of events. If enabled (TRUE), a call-out point will be available in which the run-time optimzation logic needs to be implemented as per the template file Dem_PrjSpecificDebOptimization.h_tpl.
Multiplicity	0..1
Default Value	false
Type	ECUC-BOOLEAN-PARAM-DEF

3.3.1.196 Dem/DemGeneral/DemRbGeneral/DemRbProjectSpecificEventStatusHandling

Short Name	DemRbProjectSpecificEventStatusHandling
Description	Controls project-specific handling of EventStatus. If enabled (TRUE), the event handling functions have to be implemented in the project based on the template file Dem_PrjEventStatus.h_tpl.
Multiplicity	0..1
Default Value	false
Type	ECUC-BOOLEAN-PARAM-DEF

3.3.1.197 Dem/DemGeneral/DemRbGeneral/DemRbProjectSpecificWarningIndicatorHandling

Short Name	DemRbProjectSpecificWarningIndicatorHandling
Description	Enable for project specific implementation of WIR bit handling. This parameter has to be enabled to use the IndicatorLiteforCCPS sub-module. Along with this parameter, DemEventParameter/DemRbEventClass/DemRbCustEventAtrributes has to be Configured with short name “DeAct-Trigger”.
Multiplicity	0..1
Default Value	false
Type	ECUC-BOOLEAN-PARAM-DEF

3.3.1.198 Dem/DemGeneral/DemRbGeneral/DemRbReadEventRelatedDataFromDifferentOsTask

Short Name	DemRbReadEventRelatedDataFromDifferentOsTask

Description	Needs to be enabled if event-related data (extended data or freeze frames) is read synchronously from a task different than the one running the Dem_MainFunction() (e.g. via the Dem_GetEventExtendedDataRecord* or Dem_GetEventFreezeFrameData* functions). The asynchronous Dcm interfaces Dem_GetNextExtended-DataRecord() and Dem_GetNextFreezeFrameData() are not affected. Enables additional synchronisation mechanisms which may incur some overhead (memory and runtime). In particular it needs to be enabled in OBD-relevant systems where the Dcm_MainFunction() runs in a different task than the Dem_MainFunction(). Non-OBD-relevant systems generally do not need to enable it unless the event-related data is accessed from outside the Dcm.
Multiplicity	0..1
Default Value	false
Type	ECUC-BOOLEAN-PARAM-DEF

3.3.1.199 Dem/DemGeneral/DemRbGeneral/DemRbRuntimeErrorDetect

Short Name	DemRbRuntimeErrorDetect
Description	This parameter is deprecated and not used anymore. Runtime errors are always on and cannot be turned off.
Multiplicity	0..1
Default Value	false
Type	ECUC-BOOLEAN-PARAM-DEF

3.3.1.200 Dem/DemGeneral/DemRbGeneral/DemRbShadowEntriesVisible

Short Name	DemRbShadowEntriesVisible
Description	This switch enables the visibility of deleted event memory entries (cleared or aged) in API services reporting extended data records and freeze frame data records to DCM.
Multiplicity	0..1
Default Value	false
Type	ECUC-BOOLEAN-PARAM-DEF

3.3.1.201 Dem/DemGeneral/DemRbGeneral/DemRbShadowMemory

Short Name	DemRbShadowMemory

Description	[NON-AUTOSAR] This parameter defines if the virtual deleted entries (Shadow memory) in the primary memory shall be mapped to the MIRROR Memory. This setting requires that Mirror Memory is not configured!
Multiplicity	0..1
Default Value	false
Type	ECUC-BOOLEAN-PARAM-DEF

3.3.1.202 Dem/DemGeneral/DemRbGeneral/DemRbStartAgingWhen-HealingIsCompleted

Short Name	DemRbStartAgingWhenHealingIsCompleted
Description	This configuration switch defines whether the process of aging starts when healing is completed or starts independent of healing
Introduction	TRUE: The process of aging starts when healing is completed. FALSE: The process of aging starts independent of healing.
Multiplicity	0..1
Default Value	false
Type	ECUC-BOOLEAN-PARAM-DEF

3.3.1.203 Dem/DemGeneral/DemRbGeneral/DemRbStatusByteConsisten-cyPreference

Short Name	DemRbStatusByteConsistencyPreference
Description	Select which source to prefer if the event status bits stored in the event status byte Nvm block and the event memory Nvm block are inconsistent. Currently this only affects the TestFailed status bit. Options: STATUS_BYTE_BLOCK: Prefer the status bits restored from the status byte block. EVENT_MEMORY: Prefer the status bits restored from the event memory.
Multiplicity	0..1
Default Value	STATUS_BYTE_BLOCK
Enumeration Literals	EVENT_MEMORY, STATUS_BYTE_BLOCK
Type	ECUC-ENUMERATION-PARAM-DEF

3.3.1.204 Dem/DemGeneral/DemRbGeneral/DemRbStorageCondition-BlocksPassedReports

Short Name	DemRbStorageConditionBlocksPassedReports

Description	Control whether StorageCondition will block the storage of Passed reports. TRUE : Storage Condition blocks storage on both Passed and Failed reports. FALSE : Storage Condition blocks only storage on Failed reports and allows storage on Passed reports.
Multiplicity	0..1
Default Value	false
Type	ECUC-BOOLEAN-PARAM-DEF

3.3.1.205 Dem/DemGeneral/DemRbGeneral/DemRbStoreWaitingForMonBufferDelayTime

Short Name	DemRbStoreWaitingForMonBufferDelayTime
Description	This parameter defines the number of Dem main function cycles for which the storing of an event is delayed. During that time the event is kept in the event buffer and not written to the storage. If demRbStoreWaitingForMonitoringEvent is set to FALSE: If the event reports FAILED within this time then it will be moved to the Dem event memory and BFM as a FAILED event. If it reports PASSED within this time it will be discarded from buffer i.e. no storage. If demRbStoreWaitingForMonitoringEvent is set to TRUE: Even if there is no report within this time (either PASSED or FAILED) the event will be stored in the Dem event memory and BFM as a FAILED event after this time. Please refer demRbStoreWaitingForMonitoringEvent for further details. Note: If configured as 0, the event will be directly written to storage as soon as it enters the event buffer via recheck logic.[NON-AUTOSAR EXTENSION]
Multiplicity	0..1
Value Range	0..255
Default Value	255
Type	ECUC-INTEGER-PARAM-DEF

3.3.1.206 Dem/DemGeneral/DemRbGeneral/DemRbStoreWaitingForMonitoringEvent

Short Name	DemRbStoreWaitingForMonitoringEvent
Multiplicity	0..1
Default Value	false

Type	ECUC-BOOLEAN-PARAM-DEF

Description: This switch controls whether the event in WAITING_FOR_MONITORING state to be stored as FAILED event in Event memory and BFM, even though no report until waiting time.

Consider the situation, an event was affected by Storage condition (Storage filtered) or sequential fault (Parent or higher prio event is FAILED) and now it becomes causal due to storage condition removed or parent/high prio fault healed. Then the event will be in WAITING_FOR_MONITORING state i.e. it will be in the event buffer for a definite time (DemTaskTime * DemRbStoreWaitingForMonBufferDelayTime) expecting for a new report again either PASSED or FAILED.

If it reports FAILED then it will be stored in Event memory and BFM.

If PASSED then it will be discarded from buffer itself.

If there is no report after time elapsed, then the event storage is dependant on this switch.

TRUE: Event will be stored as FAILED in Event memory and BFM even though no report after time elapsed.

FALSE: Event will be discarded and no storage in Event memory and BFM as there is no report after time elapsed.

3.3.1.207 Dem/DemGeneral/DemRbGeneral/DemRbSupportAgingCounterOnlyWhenAgingAllowed

Short Name	DemRbSupportAgingCounterOnlyWhenAgingAllowed
Description	This configuration switch defines whether Aging counter has to be supported for all DTCs or only for DTCs which have DemAgingAllowed configured as TRUE
Introduction	TRUE: DTCs which have DemAgingAllowed configured as TRUE - will have an Aging counter, and for DTCs with DemAgingAllowed configured as FALSE - the corresponding aging counter will be set to a Default value 255. ; FALSE : All DTCs will have aging counters independent of the configuration value of DemAgingAllowed. ; Note : If this parameter is configured as TRUE, the Aging counter behaviour will be compliant to AUTOSAR 42.3. If this parameter is not configured or if it is configured as FALSE, then the behaviour will be not as per AUTOSAR(it will be compliant to previous versions of DEM) - If additionally DemAgingCycleCounterThreshold is not configured, the default value generated will be '1' and Aging counter will be updated independent of the configuration value of DemAgingAllowed.
Multiplicity	0..1
Default Value	false

Type	ECUC-BOOLEAN-PARAM-DEF

3.3.1.208 Dem/DemGeneral/DemRbGeneral/DemRbSupportDebouncerResetAfterStorageCondition

Short Name	DemRbSupportDebouncerResetAfterStorageCondition
Description	[NON-AUTOSAR] This Parameter Enables or Disables the resetting of the debouncer after storagecondition. (1) OFF : do not reset the debouncer after StorageCondition (2) ACCORDING_DEBOUNCEBEHAVIOR : resetting after StorageCondition is re-enebled based on the debounce behavior
Multiplicity	0..1
Default Value	OFF
Enumeration Literals	ACCORDING_DEBOUNCEBEHAVIOR, OFF
Type	ECUC-ENUMERATION-PARAM-DEF

3.3.1.209 Dem/DemGeneral/DemRbGeneral/DemRbSupportEraseAllNvMData

Short Name	DemRbSupportEraseAllNvMData
Description	Allows the project to erase all nvm data, e.g. in case of flashing. (Templatefile Dem_PrjEraseAllNvMDataHandling.c needs to be used, project specific callback “Dem_PrjEraseAllNvMDataHandling” to be implemented to trigger erase all.) 1.) ERASE_ALL_OFF - Erase all Nvm data functionality disabled. 2.) ERASE_ALL_WITHOUT_HASHID_CHECK - Erase all is enabled without having hash id check. 3.) ERASE_ALL_WITH_HASHID_CHECK - Erase all is enabled with hash id check, i.e result of HashId check to detect any change in Order of Events or Number of Events or Event ID or Event Shortname is provided as argument in callback. (Refer section 12.13 in Dem_BswDoc for more details)
Multiplicity	0..1
Default Value	ERASE_ALL_OFF
Enumeration Literals	ERASE_ALL_OFF, ERASE_ALL_WITH_HASHID_CHECK, ERASE_ALL_WITHOUT_HASHID_CHECK
Type	ECUC-ENUMERATION-PARAM-DEF

3.3.1.210 Dem/DemGeneral/DemRbGeneral/DemRbSupportEventCategory

Short Name	DemRbSupportEventCategory
Description	This is used e.g. for prioritization of monitoring in the RB Failure Memory
Multiplicity	0..1
Default Value	false
Type	ECUC-BOOLEAN-PARAM-DEF

3.3.1.211 Dem/DemGeneral/DemRbGeneral/DemRbSupportEventFDC-ThresholdReached

Short Name	DemRbSupportEventFDCThresholdReached
Description	[NON-AUTOSAR] Switch to enable or disable the availability to querry whether the Event FDC value has reached FDC threshold configured or not. The querying can be done through the interface "Dem_GetEventFdcThresholdReached( )".
Multiplicity	0..1
Default Value	false
Type	ECUC-BOOLEAN-PARAM-DEF

3.3.1.212 Dem/DemGeneral/DemRbGeneral/DemRbSupportEventMemo-ryEntryOnFDCThreshold

Short Name	DemRbSupportEventMemoryEntryOnFDCThreshold
Description	[NON-AUTOSAR] Switch to enable or disable the Robustness feature
Multiplicity	0..1
Default Value	false
Type	ECUC-BOOLEAN-PARAM-DEF

3.3.1.213 Dem/DemGeneral/DemRbGeneral/DemRbSupportProjectSpecific-ConsistencyCheck

Short Name	DemRbSupportProjectSpecificConsistencyCheck
Description	Enables or Disables Project Specific Consistency Checks.The Project specific function have to be implemented based on the project specific templates available in Dem_PrjSpecificConsistencyCheck*.tpl.
Multiplicity	0..1
Default Value	CONSISTENCY_CHECK_OFF
Enumeration Literals	CONSISTENCY_CHECK_OFF, CONSISTENCY_CHECK_ON

Type	ECUC-ENUMERATION-PARAM-DEF

3.3.1.214 Dem/DemGeneral/DemRbGeneral/DemRbSupportProjectSpecificFailureMemory

Short Name	DemRbSupportProjectSpecificFailureMemory
Description	Activates Project Specific Failure Memory Data handling. On activation, a new variable DemRbProjectSpecificFailureMemoryType DemRbProjectSpecificFailureMemoryData will be available in Dem_EvBuffEvent structure. The Project specific data type DemRbProjectSpecificFailureMemoryType and functions have to be implemented based on the project specific templates available in Dem_PrjSpecificFailureMemory*.tpl.
Multiplicity	0..1
Default Value	false
Type	ECUC-BOOLEAN-PARAM-DEF

3.3.1.215 Dem/DemGeneral/DemRbGeneral/DemRbSupportQueueingUnrobustEvent

Short Name	DemRbSupportQueueingUnrobustEvent
Description	[NON-AUTOSAR] Switch to enable or disable the queueing of events reporting unrobust before the Dem has been initialized.
Multiplicity	0..1
Default Value	false
Type	ECUC-BOOLEAN-PARAM-DEF

3.3.1.216 Dem/DemGeneral/DemRbGeneral/DemRbSupportRecoveryLimitation

Short Name	DemRbSupportRecoveryLimitation
Description	The number of possible recoveries of failure words may be limited. This function switch configures the availability of a feature to configure a failure recovery limit for each node in the failure graph
Multiplicity	0..1
Default Value	false
Type	ECUC-BOOLEAN-PARAM-DEF

3.3.1.217 Dem/DemGeneral/DemRbGeneral/DemRbSupportSuspicious

Short Name	DemRbSupportSuspicious
Description	This configuration switch defines, event and component suspicious feature support is enabled or not.
Introduction	true: event and component suspicious feature support is enabled false: event and component suspicious feature support is disabled
Multiplicity	0..1
Default Value	false
Type	ECUC-BOOLEAN-PARAM-DEF

3.3.1.218 Dem/DemGeneral/DemRbGeneral/DemRbTestModeSupported

Short Name	DemRbTestModeSupported
Description	This configuration switch defines, whether Test Mode reporting is supported for events or not. If Test mode is enabled only test reporting is accepted (through Dem_SetEventStatus_TestMode, DemSetEventStatusWithMonitorData_TestMode). Normal reporting is deactivated
Introduction	true: Test Mode Reporting for event is enabled; false: Test Mode Reporting for event is disabled
Multiplicity	0..1
Default Value	false
Type	ECUC-BOOLEAN-PARAM-DEF

3.3.1.219 Dem/DemGeneral/DemRbGeneral/DemRbTriggerToStoreNvMSupported

Short Name	DemRbTriggerToStoreNvMSupported
Description	This configuration switch defines, whether the API to trigger immediate storage to NVM (API : Dem_TriggerStorageToNvm) is available.
Multiplicity	0..1
Default Value	false
Type	ECUC-BOOLEAN-PARAM-DEF

3.3.1.220 Dem/DemGeneral/DemRbGeneral/DemRbUserDefinedAgingCounter

Short Name	DemRbUserDefinedAgingCounter
Description	[NON-AUTOSAR] This parameter defines the custamizable Aging Counter. (1) true (2) false

Introduction	If this parameter is True, then the Aging counter value can be obtained from the Data-element DEM_AGINGCTR_UPCNT, but it will not be ensured that the data-element DEM_AGINGCTR_DOWNCNT gives correct value of Aging counter. False: No effect.
Multiplicity	0..1
Default Value	false
Type	ECUC-BOOLEAN-PARAM-DEF

3.3.1.221 Dem/DemGeneral/DemRbGeneral/DemRbUserDefinedEventMemoryStorageStrategyEnabled

Short Name	DemRbUserDefinedEventMemoryStorageStrategyEnabled
Description	Enables/Disables support for a user defined storage and displacement strategy in primary and secondary event memory. This strategy replaces the AUTOSAR defined methods. When user-defined storage strategy is enabled the user needs to implement the function Dem_EvMemGetEventMemoryStorageLocation. The header file Dem.h is used to include the function prototyp
Multiplicity	0..1
Default Value	false
Type	ECUC-BOOLEAN-PARAM-DEF

3.3.1.222 Dem/DemGeneral/DemRbGeneral/DemRbUserDefinedMirrorMemoryStorageStrategyEnabled

Short Name	DemRbUserDefinedMirrorMemoryStorageStrategyEnabled
Description	Enables/Disables support for a user defined function to implement storage and displacement strategy in mirror memory. If default behavior is not overwritten here, the mirror memory is using the same strategy as the primary and secondary memory. When user-defined storage strategy is enabled the user needs to implement the function Dem_EvMemGetMirrorMemoryStorageLocation. The header file Dem.h is used to include the function prototyp
Multiplicity	0..1
Default Value	false
Type	ECUC-BOOLEAN-PARAM-DEF

3.3.1.223 Dem/DemGeneral/DemRbGeneral/DemRbValidationMode

Short Name	DemRbValidationMode
Description	This configuration switch defines,whether the executed validations shall be specially enhanced for RTA-BSW Events or whether - in standard mode - the standard project validations shall be used.
Multiplicity	0..1
Default Value	Dem_ValidationMode_Standard
Enumeration Literals	Dem_ValidationMode_ExtendedEventChecks, Dem_ValidationMode_Standard
Type	ECUC-ENUMERATION-PARAM-DEF

3.3.1.224 Dem/DemGeneral/DemRbHeaderFileInclusion

Short Name	DemRbHeaderFileInclusion
Description	Name of the header file(s) to be included by the Dem module containing the used C-callback declarations.
Multiplicity	0..*
Default Value	-
Type	ECUC-STRING-PARAM-DEF

3.3.2 Additional Services

The following API are available in version v12.7.0 in addition to those defined in AR v22-11

Syntax	`Dem_AllowHistoryStatus`
Description	Trigger function to set the history status upon events reporting as failed.

Syntax	`Dem_ClearDisturbanceMemory`
Description	Interface to clear events that are stored in disturbance memory.

Syntax	`Dem_DebSwitchToParameters`
Parameter In	`CounterParamSetId` Identification of a Paramset. 1. Use the symbolic name as ParamsetId with the naming convention of DemConf_DemConf_DemRbDebounceCounterSet_xx (available in Dem_Cfg_Deb.h) to switch to new paramset. 2. If you need to switch to standard parameters, use DemConf_DemRbDebounceCounterSet_Standard.
Description	Switch between different debounce parameter sets.

Syntax	`Dem_EnableTestMode`

Parameter In	`TestModeStatus` Status to Enable (TRUE) or Disable (FALSE) Test Mode reporting.
Description	Allows activation or de-activation of Test Mode Reporting. If Test Mode reporting is active then Normal reporting will be disabled, and vice versa.

Syntax	`Dem_EvtGetLastReportedEventStatus`
Parameter In	`EventId` ID of the Event for which the Last Reported Status is needed.
Description	Function to query the Last Reported Event Status.

Syntax	`Dem_GetComponentAreAncestorsInitialized`
Parameter In	`ComponentId` Identifier of the component.
Parameter Out	`ComponentAreAncestorsInitialized`: TRUE - All ancestor components are initialized. FALSE - Any ancestor component is not initialized.
Description	Interface to query whether a Component's (Node's) ancestors (parents) are initialized. At startup Components may not be initialized.
Warning	Don't use this function to query whether a Component can be used. Use Dem_GetComponentUsable() or Dem_GetComponentRestrictedUsable() instead.

Syntax	`Dem_GetComponentAvailable`
Parameter In	`ComponentId` Identifier of the component.
Parameter Out	`ComponentAvailable`: TRUE - The component is available. FALSE - The component is not available.
Description	Interface to query whether a Component (Node) is available. A Component may be unavailable due to a variant code configuration or other conditions.
Warning	Don't use this function to query whether a Component can be used. Use Dem_GetComponentUsable() or Dem_GetComponentRestrictedUsable() instead.

Syntax	`Dem_GetComponentFailedItself`
Parameter In	`ComponentId` Identifier of the component.
Parameter Out	`ComponentFailedItself`: TRUE - The component itself is failed. FALSE - The component is valid. No event reported the component as failed.
Description	Interface to query whether a Component(Node) is invalid/failed by itself. That means that monitoring functions at this Component did report failed.

Warning	Don't use this function to query whether a Component can be used. Use Dem_GetComponentUsable() or Dem_GetComponentRestrictedUsable() instead.

Syntax	`Dem_GetComponentInitialized`
Parameter In	`ComponentId` Identifier of the component.
Parameter Out	`ComponentInitialized`: TRUE - Component is initialized. FALSE - Component is not initialized.
Description	Interface to query whether a Component (Node) is initialized. At startup Components (Nodes) may not be initialized.
Warning	Don't use this function to query whether a Component can be used. Use Dem_GetComponentUsable() or Dem_GetComponentRestrictedUsable() instead.

Syntax	`Dem_GetComponentRestrictedUsable`
Parameter In	`ComponentId` Identifier of the component.
Parameter Out	`ComponentRestrictedUsable`: TRUE - Component may be restrictedly used. FALSE - Component may not be used.
Description	Interface to query whether a Component (Node) is restricted usable. The function returns TRUE, if the Component may restrictedly be used (Component is not failed, and Component is initialized, and Component is available, but Component may be suspicious). If the function returns FALSE, the Component may not be used (one of the 3 criteria is not met).

Syntax	`Dem_GetComponentSuspicious`
Parameter In	`ComponentId` Identifier of the component.
Parameter Out	`ComponentSuspicious`: TRUE - Component is suspicious. FALSE - Component is not suspicious.
Description	Interface to get the suspicious status of the Component (Node).

Syntax	`Dem_GetComponentSuspiciousItself`
Parameter In	`ComponentId` Identifier of the component.

Parameter Out	`ComponentSuspiciousItself`: TRUE - The component is suspicious. FALSE - The component is valid. No event reported the component as suspicious.
Description	Interface to query whether a Component (Node) is suspicious itself. A Component may be marked as suspicious by monitoring functions. If a Component is marked as failed, it isn't marked automatically as suspicious.
Warning	Don't use this function to query whether a Component can be used. Use Dem_GetComponentUsable() or Dem_GetComponentRestrictedUsable() instead.

Syntax	`Dem_GetComponentUsable`
Parameter In	`ComponentId` Identifier of the component.
Parameter Out	`ComponentUsable`: TRUE - Component is usable. FALSE - Component is not usable.
Description	Interface to query whether a Component (Node) is usable. This function returns E_OK if the Component may be used (Component is not failed, and Component is not suspicious, and Component is initialized, and Component is available). If the function returns E_NOT_OK the Component may not be used (one of the 4 criteria is not met).

Syntax	`Dem_GetCycleQualified`
Parameter In	`OperationCycleId` Identification of a configured DemOperationCycle.
Parameter Out	`isQualified` Qualification flag of cycle.
Description	Returns the qualification state of the operation cycle.

Syntax	`Dem_GetDtcFaultDetectionCounter`
Parameter In	`dtcCode` Defines the mapping dtcCode from which a DTC Id shall be retrieved.
Parameter Out	`FaultDetectionCounter` The Fault Detection Counter for the specified dtcCode.
Description	Gets the Fault detection counter of the specified dtcCode.
Return Values	E_OK - When a valid dtcCode is passed, and also when the pointer argument is not a null pointer. E_NOT_OK - When either an invalid dtcCode is passed or when the pointer argument is a null pointer.

Syntax	`Dem_GetDtcKindOfEvent`
Parameter In	`EventId` Defines the event for which the DTC kind shall be retrieved.
Parameter Out	`DtcKind` The DTCKind.
Description	Returns the DTCKind of a specified event.
Return Values	E_OK - If DTCKind is returned correctly. E_NOT_OK - Invalid eventId or no DTC assigned.

Syntax	`Dem_GetDTCSelectionResult`
Parameter In	`ClientId` Defines the Unique client id, assigned to the instance of the calling module.
Description	Provides information if the last call to Dem_SelectDTC has selected a valid DTC or group of DTCs.

Syntax	`Dem_GetDTCSelectionResultForClearDTC`
Parameter In	`ClientId` Defines the unique client id assigned to the instance of the calling module.
Description	Provides information if the last call to Dem_SelectDTC has selected a valid DTC or group of DTCs, respecting the settings if the Dem shall clear only all DTCs.

Syntax	`Dem_GetDTCSuppression`
Parameter In	`ClientID` Identifier of the client.
Parameter Out	`SuppressionStatus` Defines whether the respective DTC is suppressed (TRUE) or enabled (FALSE)
Description	Get the suppression status of a specific DTC.

Syntax	`Dem_GetEnableCondition`
Parameter In	`EnableConditionID` Identifies the Enable condition.
Parameter Out	`ConditionFulfilled` Specifies whether the enable condition assigned to the EnableConditionID is fulfilled (TRUE) or not fulfilled (FALSE).
Description	Gets an Enable condition.

Syntax	`Dem_GetEraseAllNvMDataResult`
Parameter In	`EraseResult` Pointer.
Description	Interface to query the Result of Erase all Data.

Syntax	`Dem_GetEventAvailable`

Parameter In	`EventId` Identification of an event by assigned EventId.
Parameter Out	`AvailableStatus` Specifies whether the respective Event is available (FALSE) or not (TRUE).
Description	Get the available status of a specific Event.
Return Values	E_OK - Operation was successful. E_NOT_OK - Specified EventId is invalid or AvailableStatus is a NULL_PTR.

Syntax	`Dem_GetEventCategory`
Parameter In	`EventId` Defines the event for which the category shall be retrieved.
Parameter Out	`EventCategory` The category of the event.
Description	Return the category of a specified event.

Syntax	`Dem_GetEventFdcThresholdReached`
Parameter In	`EventId` Identifier of event.
Parameter Out	`FdcThresholdReached`: TRUE - Event FDC value has reached the FDCThreshold. FALSE - Event FDC value not reached the FDCThreshold.
Description	Ascertains whether the event FDC value has reached the FDCThreshold.

Syntax	`Dem_GetEventIdCausingLastDetError`
Parameter Out	`EventId` The EventId which has caused last DET error
Description	Returns the event ID used in Dem service which has caused the last DET error.

Syntax	`Dem_GetEventSuspicious`
Parameter In	`EventId` Identifier of event.
Parameter Out	`EventSuspicious`: TRUE - Event is suspicious. FALSE - Event is not suspicious.
Description	Interface to get the suspicious status of an event.

Syntax	`Dem_GetEvMemLock`
Description	Interface to get the Event Memory lock.

Syntax	`Dem_GetHistoryStatus`
Parameter In	`EventId` for which the HistoryStatus is needed.
Parameter Out	`HistoryStatus` for the specified event.

Description	Interface to get the HistoryStatus.

Syntax	`Dem_GetNextDisturbanceData`
Parameter InOut	`BufSize` When the function is called, this parameter contains the maximum number of data bytes that can be written to the buffer.
Parameter Out	`EventId` The identifier of the calling event. `DestBuffer` Contains a byte pointer that points to the buffer.
Description	Interface to get the actual data from the disturbance memory. Returns the actual number of written data bytes in this parameter.
Return Values	E_OK - Returned next element. E_NOT_OK - No further element found.

Syntax	`Dem_GetNextEventFromMemory`
Parameter In	`ReadEventsFromMemory` Read event, and event related information, from memory.
Parameter Out	`EventId` The next found eventId is stored in the variable, but only if the return value is DEM_READEVENT_OK. `LocId` The Location Id to read further event related information (e.g. Occurrence Counter)
Description	Reads eventids and Location Ids from event memory after the read was started with Dem_ReadEvents-FromMemory. Returns whether reading was successful or no event was found (NO_MATCH-ING_ELEMENT). In concurrently running systems, the eventID returned by this interface may subsequently be removed from event memory (e.g. due to displace-ment or event combination).

Syntax	`Dem_GetStorageCondition`
Parameter In	`StorageConditionID` Identifies the storage condition.
Parameter Out	`ConditionFulfilled` Specifies whether the storage condition assigned to the StorageConditionID is fulfilled (TRUE) or not fulfilled (FALSE)
Description	Gets a storage condition.

Syntax	`Dem_IsAnyInitMonitorForEventRequested`
Parameter InOut	`localCounter` Pointer to the local copy of the global counter.
Parameter Out	`modified` Contains the result of the comparison (TRUE = monitoring should reinitialize, FALSE = reinitialization is not necessary)

Description	Global good check function which checks if there have been modifications to a global counter that indicate the need to initialize monitorings again. You have to pass a buffer for the local copy of the global counter and a parameter that will contain the result of the comparison. The local copy and the global counter are compared, and the local copy is updated if necessary.
Return Values	E_OK - You can trust the content of the parameter "modified". E_NOT_OK - Invalid pointers have been passed.

Syntax	`Dem_IsInitMonitorForEventRequested`
Parameter In	`EventId` The identifier of the calling event.
Parameter Out	`InitMonitorReason` Specific (re-)initialization reason evaluated from the monitor to identify the type of initialization to be performed.
Description	Checks if the diagnostic monitor of a specific event shall be initialized. This interface can be used as an alternative to the AUTOSAR-defined callback mechanism.

Syntax	`Dem_J1939DcmDTCFilterMainFunction`
Parameter Out	`J1939DTC` Receives the J1939DTC value. If the return value of the function is other than DEM_FILTERED_OK, this parameter does not contain valid data.
Parameter Out	`OccurenceCounter` Receives the corresponding occurrence counter. If the return value of the function call is other than DEM_FILTERED_OK, this parameter does not contain valid data.
Description	Gets the next filtered J1939 DTC. Returns the status of the operation to retrieve a DTC from the Dem.

Syntax	`Dem_LockAllFailureInfo`
Description	Interface to lock the FailureInfo.

Syntax	`Dem_LockDisturbanceMemory`
Parameter In	`Lock` Describes whether the disturbance memory is locked or unlocked.
Description	Interface to lock the disturbance memory.
Return Values	E_OK - Accessing any of the functions is possible. E_NOT_OK - Only read and clear functions are accessible. Accessing other functions is not possible since the memory is locked for any insertion of events.

Syntax	`Dem_LockEventMemory`
Parameter In	`Lock` Describes whether the event memory is locked or unlocked.

Description	Interface to lock the event memory.
Return Values	E_OK - Accessing any of the functions is possible. E_NOT_OK - Only read and clear functions are accessible. Accessing other functions is not possible since the memory is locked for any insertion of events.

Syntax	`Dem_ManipulateEventDebounceStatus`
Parameter In	`EventId` The identifier of the calling event. `DebugData0` Free defineable debug data. `DebugData1` Free defineable debug data. `manipulationValue` Specifies the value which is to be added to the current debounce level.
Description	Interface to manipulate the dem-internal debouncing.

Syntax	`Dem_OverwriteWIRStatus`
Parameter In	`EventId` The event id where the status of the WIR bit will be overwritten. `WIRStatus` The status that will be written (TRUE=set, FALSE=reset).
Description	Overwrites the WIR (warning indicator) bit of an event's ISO14229 byte.

Syntax	`Dem_PrestoreFreezeFrameWithEnvData`
Parameter In	`EventId` Identification of Event by ID. `debug0` First envData information. `debug1` Second envData information.
Description	Stores a prestored freeze frame of a specific event along with the debug information.

Syntax	`Dem_ReadDisturbanceMemory`
Description	Interface to start reading the disturbance memory.

Syntax	`Dem_ReadEventsFromMemory`
Parameter In	`ReadEventsFromMemory` Reads event, and event related information, from memory. `origin` Specify the event memory that will be read.
Description	Start reading the eventIds from the event memory. Returns whether a valid origin was requested.

Syntax	`Dem_SetCycleQualified`
Parameter In	`OperationCycleId` Identification of a configured DemOperationCycle.
Description	Sets an operation cycle qualified so that it may be processed along with its leading cycle.

Syntax	`Dem_SetEventStatus_TestMode`
Parameter In	`EventId` The identifier of the calling event. `EventStatus` Status of the calling event.
Description	Set the status of an event for Test Mode Reporting. If the Test Mode is active then only the event is set, otherwise no action is performed.

Syntax	`Dem_SetEventSuppressionByDTC`
Parameter In	`DTC` The Diagnostic Trouble Code. `DTCFormat` Defines the output format of the requested DTC value (currently only UDS format is supported). `SuppressionStatus` Specifies whether the events associated with the DTC shall be suppressed (TRUE) or not (FALSE).
Description	Sets the suppression status of all events assigned to a specific DTC. This API is supported only when Event suppression is enabled (DemGeneral/DemAvailabilitySupport set to DEM_EVENT_AVAILABILITY). Events that have a status bit set of TestFailed, Pending, Confirmed or WarningIndicatorRequested will not be suppressed (they are skipped).
Return Values	E_OK - Operation was successful. E_NOT_OK - Event suppression support is disabled, or the operation failed due to DTCFormat != DEM_DTC_FORMAT_UDS, or DTC is invalid, or one or more of its Events could not be suppressed (e.g. because it has the TestFailed, Pending, Confirmed or WIR status bit set).
CAUTION	It's possible that some events are suppressed and some left unsuppressed, if E_NOT OK is returned. To overcome this, call the API again with suppression status "FALSE" to make all the events available.

Syntax	`Dem_SetEventSuspicion`
Parameter In	`EventId` The identifier of the calling monitoring. `suspicion` Specifies whether to set or reset a failure suspicion. TRUE = set suspicious, FALSE = reset suspicious.
Description	This function is used by a monitoring to report the failure suspicion state.

Syntax	`Dem_SetIndicatorStatusForEvent`
Parameter In	`EventId` Identification of an event by assigned EventId. `IndicatorId` Number of indicator. `IndicatorActivation` TRUE/FALSE: set the indicator status according to the configuration.

Description	Sets the indicator status for a particular event.

Syntax	`Dem_SwitchToAlternativeDtc`
Description	Function to Enable Alternative DtcCode during startup.

Syntax	`Dem_TimeBasedDebounceMainFunction`
Description	An optional Interface to be called periodically when a reference cyclic task time for Time-Based Debounce method is chosen other than Dem_MainFunction cyclic time. This API is only exported if it's configured.

Syntax	`Dem_TriggerStorageToNvm`
Description	Trigger the storage to NVM for all persistently stored DEM-internal data.

Syntax	`Dem_GetEnableCondition`
Parameter In	`EnableConditionID` Identifies the Enable condition.
Parameter Out	`ConditionFulfilled` Specifies whether or not the enable condition assigned to the EnableConditionID is fulfilled (TRUE/FALSE).
Description	Gets an Enable condition.

Syntax	`Dem_GetEventIdCausingLastDetError`
Parameter Out	`EventId` Returns the EventId which has caused last DET error.
Description	Returns the event ID used in Dem service which has caused the last DET error.

Syntax	`Dem_GetDtcFaultDetectionCounter`
Parameter In	`dtcCode` Defines the mapping DtcCode from which a DTC Id shall be retrieved.
Parameter Out	`FaultDetectionCounter` Fault Detection Counter for the specified dtcCode.
Description	Gets the Fault detection counter of the specified dtcCode.

Syntax	`Dem_SetEventStatusWithMonitorData_TestMode`
Parameters In	`EventId` Identifier of the calling event. `EventStatus` Status of the calling event. `monitorData0` Free defineable debug/monitor data. `monitorData1` Free defineable debug/monitor data.
Description	Sets the status of an event and allows specification of environmental data, which may be stored to memory within snapshot data when Test mode reporting is active.

3.3.3 AUTOSAR 4.2 APIs

The following APIs were in the AR42 specification but have been removed in AR45. For this reason, they are now regarded as extensions to the AR45 baseline.

- Dem_GetEventFailed
- Dem_GetEventTested
- Dem_GetEventExtendedDataRecord
- Dem_GetEventFreezeFrameData
- Dem_DltGetAllExtendedDataRecords
- Dem_DltGetMostRecentFreezeFrameRecordData

3.4 Limitations

Version v12.7.0 has following limitations.

3.4.1 DEM_E_UNINIT

AUTOSAR (R21-11) states that if development error detection is enabled, and any instance calls any Dem API before the Dem has been fully initialized (excluding Dem_ReportErrorStatus), Dem should set the error code DEM_E_UNINIT.

Not all Dem API services currently support this requirement, and so the system architecture must ensure that Dem services (excluding Dem_ReportErrorStatus) are not called before Dem is fully initialized.

3.4.2 FDC Threshold Reached

The FDC Threshold Reached feature is enabled by configuring DemEventMemoryEntryFdcThresholdStorageValue. The following limitations/considerations should be noted:

- FDC Threshold Reached is not currently supported for events that use the Autosar Counter-Based Debounce algorithm, and it is only supported for events using Monitor-Internal algorithm methods.
- The handling of FDC Threshold Reached before Dem is initialized is not supported. The queue buffer size is limited, and so handling FDC Threshold Reached events would block the queue, potentially preventing failed events from getting entered into the queue. Events that reach FDC Threshold after Dem initialisation are correctly considered as FDC Reached.
- If, when reporting FDC_Threshold_REACHED for an event, another higher priority fault has already been reported (e.g. TestFailed), the FDC_Threshold_REACHED is ignored as it is secondary to the other failure (and it won't be stored in the buffer/memory).
- If DemFreezeFrameRecordTrigger or DemExtendedDataRecordTrigger is set to DEM_TRIGGER_ON_FDC_THRESHOLD, and the event reports DEM_EVENT_STATUS_FDC_THRESHOLD_REACHED, then the algorithm will try to insert the event into the Event Buffer and capture the Environmental

data, and distribute it to the Event Memory, only once per operation cycle. There is no retry in the same operation cycle in the event of overflow or displacement in the Event Buffer or Event Memory.

3.4.3 Specifying Additional Event Attributes

3.4.4 Limitations on J1939-Related Functionalities

DM31 is not implemented.

3.4.5 DemEventCombinationSupport

When DemEventCombinationSupport is configured as "DEM_EVCOMB_ONSTORAGE", there's a limitation currently in the processing of PASSED report of event(s) related to a DTC (which has its Status as "Passed") in that it is allowed to Evmem irrespective of the TestCompleteTOC flag (it sets DEM_EVMEM_STSMASK_TESTCOMPLETE_TFC and DEM_EVMEM_STSMASK_TESTCOMPLETE_TAC in EvMem).

3.4.6 Non-Volatile Data

Dem stores data in several non-volatile data blocks which, in contrast to storing a single non-volatile data block, allows for the partial updating in some, but not all, of the non-volatile blocks instead of always having to update the complete set of non-volatile data.

This is done to reduce the overall amount of data that needs to be written to non-volatile memory, but it can potentially result in inconsistencies if not all of the changed non-volatile data blocks are safely written. This situation might occur, for example, in an under-voltage situation. For this reason, Dem performs data plausibility checks to detect and resolve conflicting data.

3.4.7 Reinitilization of Dem after Software Programming

The feature of re-initializing Dem after Software Programming (Erase All) is currently supported only for OBD Projects (without BFM).

3.4.8 Miscellaneous

- Some of the synchronous API calls defined within the Dem might take more time to complete than a software component or basic software component is assigned to run. Thus, the calling instance has to ensure that the blocking caused by the execution of the Dem API call is handled appropriately.
- Dcm should only be used when the vehicle is in a safe state. Typically the Safe State for Dcm usage is when the vehicle is stationary and not capable of movement, consistent with it being in a test or maintenance environment.
- The callbacks for EventStatusChanged and DTCStatusChanged may not be deterministic in rare cases, therefore they should not be relied on for safety-relevant use-cases. Also, according to ISO-14229-1:2013 (section 9.9.1), it is not recommended to rely on the DTC status for failsafe strategies.

- The Last Reported Event Status is the actual status reported for events that have been configured with Time-based and Monitoring-Internal debouncing. For events configured with Counter-based debouncing, it's the calculated status based on a debouncing algorithm. For example, if "Pre-Failed" is reported until the Threshold is reached, the last reported event status will be stored as "Pre-Failed". At the point of reaching the Threshold level, the calculated event status will become "Failed"', and the last reported event status will be stored as "Failed".

- Re-enabling a node makes all events inside the node available again, regardless of the previous usage of the single event suppression. As soon as a node is set to not available, single event enabling in the suppressed node is restricted.

3.5 Exclusions

The following features from AR v22-11 are not supported in version v12.7.0.

API functions not currently implemented:

- `Dem_SetEventFailureCycleCounterThreshold`

The following configuration parameters from AR v22-11 are not supported in version v12.7.0.

- Dem/DemConfigSet/DemDebounceCounterBasedClass/DemCounterBasedFdcThresholdStorageValue
- Dem/DemConfigSet/DemDebounceCounterBasedClass/DemDebounceCounterStorage
- Dem/DemConfigSet/DemDebounceTimeBaseClass/DemTimeBasedFdcThresholdStorageValue
- Dem/DemConfigSet/DemDebounceTimeBasedClass
- Dem/DemConfigSet/DemEventParameter/DemDebounceAlgorithmClass/DemDebounceTimeBased
- Dem/DemConfigSet/DemEventParameter/DemDebounceAlgorithmClass/DemDebounceTimeBased/DemDebounceTimeBasedClassRef
- Dem/DemConfigSet/DemEventParameter/DemEventReportingType
- Dem/DemConfigSet/DemEventParameter/DemFFPrestorageInNvm
- Dem/DemConfigSet/DemMultiEventTriggering
- Dem/DemConfigSet/DemMultiEventTriggering/DemMultiEventTriggeringMasterEventRef
- Dem/DemConfigSet/DemMultiEventTriggering/DemMultiEventTriggeringSlaveEventRef
- Dem/DemConfigSet/DemObdDTC/DemConsiderPtoStatus
- Dem/DemGeneral/DemAgingRequiresTestedCycle

- Dem/DemGeneral/DemClearEventsWithoutDTCEventMemoryRef
- Dem/DemGeneral/DemClient/DemClientFunctionality
- Dem/DemGeneral/DemDataElementClass/DemExternalSRDataElement-Class/DemDiagnosisScaling/DemAlternativeDataInterface
- Dem/DemGeneral/DemDataElementClass/DemExternalSRDataElement-Class/DemDiagnosisScaling/DemAlternativeDataInterface/DemDataElement
- Dem/DemGeneral/DemDataElementClass/DemExternalSRDataElement-Class/DemDiagnosisScaling/DemAlternativeDataInterface/DemPortInterfaceMapping
- Dem/DemGeneral/DemDataElementClass/DemExternalSRDataElement-Class/DemDiagnosisScaling/DemAlternativeDataType/DemTextTableMapping
- Dem/DemGeneral/DemDataElementClass/DemExternalSRDataElement-Class/DemDiagnosisScaling/DemAlternativeDataType/DemTextTableMapping/DemDiagnosisRepresentationDataValue
- Dem/DemGeneral/DemDataElementClass/DemExternalSRDataElement-Class/DemDiagnosisScaling/DemAlternativeDataType/DemTextTableMapping/DemInternalDataValue
- Dem/DemGeneral/DemDataElementClass/DemExternalSRDataElement-Class/DemDiagnosisScaling/DemAlternativeDiagnosticDataElement
- Dem/DemGeneral/DemDataElementClass/DemExternalSRDataElement-Class/DemDiagnosisScaling/DemAlternativeDiagnosticDataElement/DemDiagnosticDataElementRef
- Dem/DemGeneral/DemDataElementClass/DemExternalSRDataElement-Class/DemSRDataElementClass
- Dem/DemGeneral/DemDataElementClass/DemExternalSRDataElement-Class/DemSRDataElementClass/DemDataElementInstance
- Dem/DemGeneral/DemDataElementClass/DemExternalSRDataElement-Class/DemSRDataElementClass/DemDataElementInstance/DemDataElementInstanceRef
- Dem/DemGeneral/DemDataElementClass/DemExternalSRDataElement-Class/DemSRDataElementClass/DemSubElementInDataElementInstance
- Dem/DemGeneral/DemDataElementClass/DemExternalSRDataElement-Class/DemSRDataElementClass/DemSubElementInDataElementInstance/DemSubElementInDataElementInstanceRef
- Dem/DemGeneral/DemDataElementClass/DemExternalSRDataElement-Class/DemSRDataElementClass/DemSubElementInImplDataElementInstance
- Dem/DemGeneral/DemDataElementClass/DemExternalSRDataElement-Class/DemSRDataElementClass/DemSubElementInImplDataElementInstance/DemSubElementInImplDataElementInstanceRef
- Dem/DemGeneral/DemEventMemorySet/DemPrimaryMemory/DemEvent-

MemoryEntryStorageTriggerDemEventMemoryEntryStorageTrigger

- Dem/DemGeneral/DemEventMemorySet/DemUserDefinedMemory/DemEventMemoryEntryStorageTrigger
- Dem/DemGeneral/DemGeneralJ1939/DemCallbackJ1939DTCStatusChanged
- Dem/DemGeneral/DemGeneralOBD/DemCallbackOBDDTCStatusChanged
- Dem/DemGeneral/DemGeneralOBD/DemOBDDelayedDCYConfirmedAndMIL
- Dem/DemGeneral/DemGeneralOBD/DemOBDInputAcceleratorPedalInformation
- Dem/DemGeneral/DemGeneralOBD/DemOBDInputAmbientPressure
- Dem/DemGeneral/DemGeneralOBD/DemOBDInputAmbientTemperature
- Dem/DemGeneral/DemGeneralOBD/DemOBDInputEngineSpeed
- Dem/DemGeneral/DemGeneralOBD/DemOBDInputEngineTemperature
- Dem/DemGeneral/DemGeneralOBD/DemOBDInputProgrammingEvent
- Dem/DemGeneral/DemGeneralOBD/DemOBDInputVehicleSpeed
- Dem/DemGeneral/DemOBDEventMemorySetRef
- Dem/DemGeneral/DemPTOSupport

3.6 Deviations

3.6.1 Configuration Deviations

Configuration parameters which differ from the AR v22-11 specification are listed below.

3.6.1.1 Dem/DemConfigSet/DemDTC

[RTA-BSW] Lower Multiplicity	1
[AUTOSAR] Lower Multiplicity	0

3.6.1.2 Dem/DemConfigSet/DemDTC/DemDtcValue

[RTA-BSW] Description	Unique Diagnostic Trouble Code value for UDS
[AUTOSAR] Description	3 Byte value of the UDS Diagnostic Trouble Code (DTC). The value is unique within a fault memory.
[RTA-BSW] Introduction	(Range: 0x000000 and 0xFFFFFF are reserved for DTC groups by ISO 14229-1)
[AUTOSAR] Introduction	The ranges: 0x000000-0x0000FF and 0xFFFFFF are reserved by ISO 14229-1.
[RTA-BSW] Minimum Value	1

[AUTOSAR] Minimum Value	256

3.6.1.3 Dem/DemConfigSet/DemDTC/DemNvStorageStrategy

[RTA-BSW] Description	This parameter defines when a specific event memory entry is allowed to be stored in the NVRAM.
[AUTOSAR] Description	This parameter defines when the event related data of the DTC is stored in the NVRAM.
[ADDED] Introduction	If DemNvStorageStrategy is configured with RB_IMMEDIATE_LIMITED_BY_OCCCTR, and DemRbImmediateNvStorageLimit is not configured for the relevant memory destination, then immediate storage will not be bound by the occurrence counter value for such DTC. This can have an impact on the life cycle of the memory media used for persistent storage!
[RTA-BSW] Enumeration Literals	DURING_SHUTDOWN, IMMEDIATE_AT_FIRST_OCCURRENCE, RB_IMMEDIATE_LIMITED_BY_OCCCTR
[AUTOSAR] Enumeration Literals	DURING_SHUTDOWN, IMMEDIATE_AT_FIRST_OCCURRENCE

3.6.1.4 Dem/DemConfigSet/DemDTCAttributes/DemAgingCycleCounterThresholdForTFSLC

[RTA-BSW] Description	Number of aging cycles needed to reset the testFailedSinceLastClear Bit.When it is not configured, aging of TFSLC is not performed. Value has to be greater than 'DemAgingCycleCounterThreshold'. The parameter DemStatusBitHandlingTestFailedSinceLastClear has to be configured as "DEM_STATUS_BIT_AGING_AND_DISPLACEMENT". AgingCycleCounterThresholdForTFSLC is not supported if:→ Event is configured for Fast aging (demAgingCycleCounterThreshold = 0).→ No DTC mapped to the Event.
[AUTOSAR] Description	Number of aging cycles needed to reset the testFailedSinceLastClear Bit.

3.6.1.5 Dem/DemConfigSet/DemDTCAttributes/DemDTCPriority

[RTA-BSW] Minimum Value	0
[AUTOSAR] Minimum Value	1

3.6.1.6 Dem/DemConfigSet/DemDTCAttributes/DemMemoryDestinationRef

[RTA-BSW] Description	The memory destination assigns DTCs to one or two memory destinations. If more than one memory destination is assigned to a specific DTC, the DTC can be present in the corresponding event memories. In this case one of the references has to be DemMirrorMemory (SWS_Dem_CONSTR_6104).
[AUTOSAR] Description	The memory destination assigns DTCs to one memory destination.
[RTA-BSW] Upper Multiplicity	2
[AUTOSAR] Upper Multiplicity	1

3.6.1.7 Dem/DemConfigSet/DemDTCAttributes/DemUDSOBDFreezeFrameClassRef

[RTA-BSW] Description	This reference defines the link to the freeze frame for emission related diagnostics over UDS as defined by ISO 27145-3 (WWHOBD). This is relevant in case DemOBDCompliancy is configured with the value 0x15.
[AUTOSAR] Description	This reference defines the link to the freeze frame for emission related diagnostics over UDS.

3.6.1.8 Dem/DemConfigSet/DemDebounceCounterBasedClass/DemDebounceBehavior

[REMOVED] Default Value	DEM_DEBOUNCE_RESET

3.6.1.9 Dem/DemConfigSet/DemDebounceCounterBasedClass/DemDebounceCounterFailedThreshold

[RTA-BSW] Lower Multiplicity	0
[AUTOSAR] Lower Multiplicity	1
[ADDED] Default Value	32767

3.6.1.10 Dem/DemConfigSet/DemDebounceCounterBasedClass/DemDebounceCounterJumpDown

[RTA-BSW] Lower Multiplicity	0
[AUTOSAR] Lower Multiplicity	1
[ADDED] Default Value	true

3.6.1.11 Dem/DemConfigSet/DemDebounceCounterBasedClass/DemDe-

bounceCounterJumpDownValueDemDebounceCounterJump-DownValue

[RTA-BSW] Lower Multiplicity	0
[AUTOSAR] Lower Multiplicity	1

3.6.1.12 Dem/DemConfigSet/DemDebounceCounterBasedClass/DemDebounceCounterJumpUp

[RTA-BSW] Lower Multiplicity	0
[AUTOSAR] Lower Multiplicity	1
[ADDED] Default Value	true

3.6.1.13 Dem/DemConfigSet/DemDebounceCounterBasedClass/DemDebounceCounterJumpUpValue

[RTA-BSW] Lower Multiplicity	0
[AUTOSAR] Lower Multiplicity	1

3.6.1.14 Dem/DemConfigSet/DemDebounceCounterBasedClass/DemDebounceCounterPassedThreshold

[RTA-BSW] Lower Multiplicity	0
[AUTOSAR] Lower Multiplicity	1
[ADDED] Default Value	-32768

3.6.1.15 Dem/DemConfigSet/DemDebounceTimeBaseClass/DemDebounceBehavior

[RTA-BSW] Parent Container	Dem/DemConfigSet/DemDebounceTimeBaseClass/DemDebounceBehavior
[AUTOSAR] Parent Container	Dem/DemConfigSet/DemDebounceTimeBasedClass/DemDebounceBehavior
[REMOVED] Default Value	DEM_DEBOUNCE_RESET

3.6.1.16 Dem/DemConfigSet/DemDebounceTimeBaseClass/DemDebounceTimeFailedThreshold

[RTA-BSW] Parent Container	Dem/DemConfigSet/DemDebounceTimeBaseClass/DemDebounceTimeFailedThreshold
[AUTOSAR] Parent Container	Dem/DemConfigSet/DemDebounceTimeBasedClass/DemDebounceTimeFailedThreshold

3.6.1.17 Dem/DemConfigSet/DemDebounceTimeBaseClass/DemDe-

bounceTimePassedThresholdDemDebounceTimePassedThreshold

[RTA-BSW] Parent Container	Dem/DemConfigSet/DemDebounceTimeBaseClass/DemDebounceTimePassedThreshold
[AUTOSAR] Parent Container	Dem/DemConfigSet/DemDebounceTimeBasedClass/DemDebounceTimePassedThreshold

3.6.1.18 Dem/DemConfigSet/DemEventParameter/DemCallbackClearEventAllowed/DemCallbackClearEventAllowedFnc

[RTA-BSW] Description	Function name of prototype “ClearEventAllowed”. For further details refer Dem563. Hint: It has presedence (Higher Prio) over DemRbGlobalCallbackClearEventAllowedFnc.
[AUTOSAR] Description	Function name of prototype “ClearEventAllowed”.

3.6.1.19 Dem/DemConfigSet/DemEventParameter/DemCallbackClearEventAllowed/DemClearEventAllowedBehavior

[RTA-BSW] Description	Defines the resulting UDS status byte for the related event, which must not be cleared according to the ClearEventAllowed callback. Note : If DemClearEventAllowedBehavior is configured as DEM_ONLY_THIS_CYCLE_AND_READINESS for an event, the aging and confirmation process could speed up for that Event. Currently any event which is linked to an OBD Dtc shall not be configured as DemClearEventAllowedBehavior to be set to DEM_ONLY_THIS_CYCLE_AND_READINESS(An error will be thrown for this scenario by OAW check)
[AUTOSAR] Description	Defines the resulting UDS status byte for the related event, which must not be cleared according to the ClearEventAllowed callback.

3.6.1.20 Dem/DemConfigSet/DemEventParameter/DemCallbackEventDataChanged

[RTA-BSW] Description	The presence of this container indicates that the Dem has access to an “EventDataChanged” callback. (Callback on any change in Event related data i.e ENV DATA→ Ext data or freeze frame)
[AUTOSAR] Description	The presence of this container indicates that the Dem has access to an “EventDataChanged” callback.

3.6.1.21 Dem/DemConfigSet/DemEventParameter/DemCallbackEventDataChanged/DemCallbackEventDataChangedFnc

[RTA-BSW] Description	Function name of prototype “EventDataChanged”. Refer [SWS_Dem_00562] for more details

[AUTOSAR] Description	Function name of prototype “EventDataChanged”

3.6.1.22 Dem/DemConfigSet/DemEventParameter/DemCallbackInitMForE/DemCallbackInitMForEFnc

[RTA-BSW] Description	Function name of prototype “InitMonitorForEvent”. For further details refer [SWS_Dem_00256].
[AUTOSAR] Description	Function name of prototype “InitMonitorForEvent”.

3.6.1.23 Dem/DemConfigSet/DemEventParameter/DemDebounceAlgorithmClass/DemDebounceMonitorInternal/DemCallbackGetFDC/DemCallbackGetFDCFnc

[RTA-BSW] Description	This parameter defines the name of the function that the Dem will call to retrieve the fault-detection counter value from a complex driver. Refer [SWS_Dem_00263] for further details. Hint:The function name must not be same as any Dem API’s
[AUTOSAR] Description	This parameter defines the name of the function that the Dem will call to retrieve the fault-detection counter value from a complex driver.

3.6.1.24 Dem/DemConfigSet/DemEventParameter/DemEventConfirmationThresholdCounterAdaptable

[RTA-BSW] Description	Indicates whether the events confirmation Cycle threshold can be adapted by Dem_SetEventConfirmationThresholdCounter
[AUTOSAR] Description	Indicates whether the events confirmation Cycle threshold can be adapted by Dem_SetEventConfirmationThresholdCounter.

3.6.1.25 Dem/DemConfigSet/DemEventParameter/DemEventId

[RTA-BSW] Introduction	If DemRbDisableAutomaticEventIds is false, then this parameter should not be changed by the user, because the Id is generated by Dem itself to prevent gaps and multiple use of an Id. If DemRbDisableAutomaticEventIds is true, then the user has to ensure that the events are sequentially ordered beginning with 1 and no gaps in between.
[AUTOSAR] Introduction	This parameter should not be changeable by user, because the Id should be generated by Dem itself to prevent gaps and multiple use of an Id. The events should be sequentially ordered beginning with 1 and no gaps in between.

3.6.1.26 Dem/DemConfigSet/DemEventParameter/DemEventKind

[REMOVED] Default Value	DEM_EVENT_KIND_SWC

3.6.1.27 Dem/DemConfigSet/DemEventParameter/DemFFPrestorageSup-

portedDemFFPrestorageSupported

[RTA-BSW] Lower Multiplicity	0
[AUTOSAR] Lower Multiplicity	1

3.6.1.28 Dem/DemConfigSet/DemEventParameter/DemIndicatorAttribute/DemIndicatorBehaviour

[RTA-BSW] Enumeration Literals	DEM_INDICATOR_BLINKING, DEM_INDICATOR_BLINK_CONT, DEM_INDICATOR_CONTINUOUS, DEM_INDICATOR_FAST_FLASH, DEM_INDICATOR_ON_DEMAND, DEM_INDICATOR_SHORT, DEM_INDICATOR_SLOW_FLASH
[AUTOSAR] Enumeration Literals	DEM_INDICATOR_BLINKING, DEM_INDICATOR_BLINK_CONT, DEM_INDICATOR_CONTINUOUS, DEM_INDICATOR_FAST_FLASH, DEM_INDICATOR_SLOW_FLASH

3.6.1.29 Dem/DemConfigSet/DemEventParameter/DemReportBehavior

[RTA-BSW] Description	Indicates the reporting behavior of the BSW Module (DemEventKind == DEM_EVENT_KIND_BSW) in order to determine the size of the reporting queue. Hint: This configuration is necessary in case of DemBswErrorBufferSize in DemGeneral is not configured
[AUTOSAR] Description	Indicates the reporting behavior of the BSW Module (DemEventKind == DEM_EVENT_KIND_BSW) in order to determine the size of the reporting queue.

3.6.1.30 Dem/DemConfigSet/DemObdDTC/DemJ1939DTCValue

[RTA-BSW] Description	Unique Diagnostic Trouble Code value for J1939 (consisting of SPN and FMI). The value has to be configured as per SAEJ1939 SPN conversion method. Dem or J1939Dcm will not perform any conversion.
[AUTOSAR] Description	Unique Diagnostic Trouble Code value for J1939 (consisting of SPN and FMI).

3.6.1.31 Dem/DemGeneral/DemBswErrorBufferSize

[RTA-BSW] Description	Maximum number of elements in buffer for handling of BSW errors (ref. to SWS_Dem_00207). If this is not configured, the internal buffer queue size will be equal to the number of events configured with DemEventkind = DEM_EVENT_KIND_BSW and DemReportBehavior == REPORT_BEFORE_INIT.
[AUTOSAR] Description	Maximum number of elements in buffer for handling of BSW errors (ref. to SWS_Dem_00207).

3.6.1.32 Dem/DemGeneral/DemDataElementClass/DemExternalCSDataElementClass/DemDataElementDataType

[RTA-BSW] Lower Multiplicity	0
[AUTOSAR] Lower Multiplicity	1
[ADDED] Default Value	UINT8_N

3.6.1.33 Dem/DemGeneral/DemDataElementClass/DemExternalCSDataElementClass/DemDataElementFncSignature

[RTA-BSW] Description	If the parameter is set to DATA_FNC_NORMAL the function omits DataLength and Dcm_NegativeResponseCodeType parameters. If the parameter is set to DATA_FNC_PROXY the generated function call to retrieve the data element uses the signature for Dcm proxy with DataLength and Dcm_NegativeResponseCodeType parameters.
[AUTOSAR] Description	If the parameter is set to DATA_FNC_NORMAL the function omits DataLength and Dcm_NegativeResponseCodeType parameters. Tags: atp.Status=draft
[RTA-BSW] Introduction	Please note that the Dcm proxy is commonly handling the diagnostic data connections for the Dcm and Dem module
[AUTOSAR] Introduction	If the parameter is set to DATA_FNC_PROXY the generated function call to retrieve the data element uses the signature for Dcm proxy with DataLength and Dcm_NegativeResponseCodeType parameters. Please note that the Dcm proxy is commonly handling the diagnostic data connections for the Dcm and Dem module.

3.6.1.34 Dem/DemGeneral/DemDataElementClass/DemExternalCSDataElementClass/DemDataElementProvideMonitorData

[RTA-BSW] Description	If the parameter is set to True, the generated function call to retrieve the data element has the monitorData0 as additional first parameter.
[AUTOSAR] Description	If the parameter is set to true, the generated function call to retrieve the data element has the monitorData0 as additional first parameter.

3.6.1.35 Dem/DemGeneral/DemDataElementClass/DemExternalSRDataElementClass/DemDataElementDataType

[RTA-BSW] Lower Multiplicity	0
[AUTOSAR] Lower Multiplicity	1
[ADDED] Default Value	UINT8_N

3.6.1.36 Dem/DemGeneral/DemDataElementClass/DemInternalDataEle-

mentClass/DemDataElementDataSize

[ADDED] Introduction	Note: This parameter shall not be configured by the user in case of the internal data element DEM_DTR, as the size could be dynamic. Dem itself forwards a dummy value of 1 in this case for ease of internal implementation.

3.6.1.37 Dem/DemGeneral/DemDataElementClass/DemInternalDataElementClass/DemInternalDataElement

[RTA-BSW] Enumeration Literals: DEM_AGINGCTR_DOWNCNT, DEM_AGINGCTR_UPCNT, DEM_AGINGCTR_UPCNT_FIRST_ACTIVE, DEM_CLR_DIST, DEM_CLR_TIME, DEM_CURRENT_FDC, DEM_CYCLES_SINCE_FIRST_FAILED, DEM_CYCLES_SINCE_LAST_FAILED, DEM_DTC_PRIORITY, DEM_FAILED_CYCLES, DEM_J1939LAMP_STATUS, DEM_MAX_FDC_DURING_CURRENT_CYCLE, DEM_MAX_FDC_SINCE_LAST_CLEAR, DEM_MIL_DIST, DEM_MIL_TIME, DEM_MONITOR_DATA_0, DEM_MONITOR_DATA_1, DEM_OCCCTR, DEM_OVFLIND, DEM_SIGNIFICANCE, DEM_WARM_UPS, DEM_BFM_OP_CYCLE, DEM_BFM_ABS_OP_TIME, DEM_BFM_REL_OP_TIME, DEM_BFM_TIMESTAMP, DEM_BFM_UNIQUE_ID, DEM_CYCLES_SINCE_LAST_FAILED_EXCLUDING_TNC, DEM_EVENT_STATUS_BYTE, DEM_EVENT_ID, DEM_PRJ_SPECIFIC_DATA_ON_RETRIEVE, DEM_PRJ_SPECIFIC_DATA_ON_REPORT, DEM_DTR, DEM_EVENT_ASSOCIATED_IDENTIFICATION, DEM_IUMPR, DEM_MONITOR_ACTIVITY_DATA, DEM_RB_B1_COUNTER

[AUTOSAR] Enumeration Literals: DEM_AGINGCTR_DOWNCNT, DEM_AGINGCTR_UPCNT, DEM_AGINGCTR_UPCNT_FIRST_ACTIVE, DEM_CLR_DIST, DEM_CLR_TIME, DEM_CURRENT_FDC, DEM_CYCLES_SINCE_FIRST_FAILED, DEM_CYCLES_SINCE_LAST_FAILED, DEM_DTC_PRIORITY, DEM_DTR, DEM_EVENT_ASSOCIATED_IDENTIFICATION, DEM_FAILED_CYCLES, DEM_IUMPR, DEM_J1939LAMP_STATUS, DEM_MAX_FDC_DURING_CURRENT_CYCLE, DEM_MAX_FDC_SINCE_LAST_CLEAR, DEM_MIL_DIST, DEM_MIL_TIME, DEM_MONITOR_ACTIVITY_DATA, DEM_MONITOR_DATA_0, DEM_MONITOR_DATA_1, DEM_OCCCTR, DEM_OVFLIND, DEM_SIGNIFICANCE, DEM_WARM_UPS

3.6.1.38 Dem/DemGeneral/DemDebounceTimeBasedSupport

[RTA-BSW] Introduction	true: time based debouncing support is enabled. In case it is enabled and has to run in a different cyclic task time (refer to: DemRbDebounceTimeBasedTaskTime) then the API Dem_TimeBasedDebounceMainFunction() must be scheduled in appropriate cycle task. false: time based debouncing support is disabled
[AUTOSAR] Introduction	true: time based debouncing support is enabled false: time based debouncing support is disabled

3.6.1.39 Dem/DemGeneral/DemEventMemorySet

[RTA-BSW] Upper Multiplicity	1

[AUTOSAR] Upper Multiplicity	255

3.6.1.40 Dem/DemGeneral/DemEventMemorySet/DemClearDTCNotification

[ADDED] Introduction	As per AUTOSAR definition, Dem shall pass the DTC as an argument when invoking the callback. But in case of J1939 ClearDTC requests (DM03 / DM11) that use the Dem API Dem_J1939DcmClearDTC(), the DTC information is not available and hence, the callback is also not invoked.

3.6.1.41 Dem/DemGeneral/DemEventMemorySet/DemIndicator/DemIndicatorID

[RTA-BSW] Minimum Value	1
[AUTOSAR] Minimum Value	0

3.6.1.42 Dem/DemGeneral/DemEventMemorySet/DemMILIndicatorRef

[RTA-BSW] Description	This parameter defines the indicator representing the MIL. This parameter may be used for ECUs supporting J1939.
[AUTOSAR] Description	This parameter defines the indicator representing the MIL.

3.6.1.43 Dem/DemGeneral/DemEventMemorySet/DemMaxNumberEventEntryPermanent

[RTA-BSW] Description	Maximum number of events which can be stored in the permanent memory.Please consider that some legislations may require a certain minimum size for the permanent memory
[AUTOSAR] Description	Maximum number of events which can be stored in the permanent memory.

3.6.1.44 Dem/DemGeneral/DemEventMemorySet/DemPrimaryMemory/DemEventDisplacementStrategy

[RTA-BSW] Enumeration Literals	DEM_DISPLACEMENT_FULL, DEM_DISPLACEMENT_NONE, DEM_DISPLACEMENT_PRIO_OCC, DEM_DISPLACEMENT_RBA_FIFO, DEM_DISPLACEMENT_RBA_CUST
[AUTOSAR] Enumeration Literals	DEM_DISPLACEMENT_FULL, DEM_DISPLACEMENT_NONE, DEM_DISPLACEMENT_PRIO_OCC

3.6.1.45 Dem/DemGeneral/DemEventMemorySet/DemPrimaryMemory/DemTypeOfFreezeFrameRecordNumeration

[RTA-BSW] Description	This parameter defines the type of assigning freeze frame record numbers for event-specific freeze frame records. Hint: For CALCULATED, the numeration will start from 1 and increase by 1 for each subsequent freezeframe record
[AUTOSAR] Description	This parameter defines the type of assigning freeze frame record numbers for event-specific freeze frame records.

3.6.1.46 Dem/DemGeneral/DemEventMemorySet/DemUserDefinedMemory/DemEventDisplacementStrategy

[RTA-BSW] Enumeration Literals	DEM_DISPLACEMENT_FULL, DEM_DISPLACEMENT_NONE, DEM_DISPLACEMENT_PRIO_OCC, DEM_DISPLACEMENT_RBA_FIFO, DEM_DISPLACEMENT_RBA_CUST
[AUTOSAR] Enumeration Literals	DEM_DISPLACEMENT_FULL, DEM_DISPLACEMENT_NONE, DEM_DISPLACEMENT_PRIO_OCC

3.6.1.47 Dem/DemGeneral/DemExtendedDataClass

[RTA-BSW] Lower Multiplicity	1
[AUTOSAR] Lower Multiplicity	0

3.6.1.48 Dem/DemGeneral/DemExtendedDataRecordClass/DemExtendedDataRecordTrigger

[RTA-BSW] Enumeration Literals	DEM_TRIGGER_ON_CONFIRMED, DEM_TRIGGER_ON_FDC_THRESHOLD, DEM_TRIGGER_ON_MIRROR, DEM_TRIGGER_ON_PASSED, DEM_TRIGGER_ON_PENDING, DEM_TRIGGER_ON_TEST_FAILED
[AUTOSAR] Enumeration Literals	DEM_TRIGGER_ON_CONFIRMED, DEM_TRIGGER_ON_FDC_THRESHOLD, DEM_TRIGGER_ON_PASSED, DEM_TRIGGER_ON_PENDING, DEM_TRIGGER_ON_TEST_FAILED

3.6.1.49 Dem/DemGeneral/DemFreezeFrameClass

[RTA-BSW] Description	This container contains the combinations of DIDs for a non OBD2 and WWH-OBD relevant freeze frame class.
[AUTOSAR] Description	This container contains the combinations of DIDs for freeze frames of UDS and emission related diagnostics over UDS.

3.6.1.50 Dem/DemGeneral/DemFreezeFrameRecordClass/DemFreeze-FrameRecordNumber

[RTA-BSW] Description	This parameter defines a record number for a freeze frame record. This record number is unique per freeze frame record number class.
[AUTOSAR] Description	This parameter defines the snapshot record number for a freeze frame. This record number is unique per freeze frame record number class. The number needs to be unique within a fault memory.

3.6.1.51 Dem/DemGeneral/DemFreezeFrameRecordClass/DemFreeze-FrameRecordTrigger

[RTA-BSW] Description	Defines the trigger to store the FreezeFrameRecordTrigger. Tags: atp.Status=draft
[AUTOSAR] Description	Defines the trigger to store the FreezeFrameRecord. Tags: atp.Status=draft

3.6.1.52 Dem/DemGeneral/DemFreezeFrameRecordClass/DemFreeze-FrameRecordUpdate

[RTA-BSW] Description	This parameter defines the case, when the freeze frame record is stored/updated. In case of DEM_TRIGGER_ON_EVERY_TEST_FAILED this parameter should be set to DEM_UPDATE_RECORD_YES
[AUTOSAR] Description	This parameter defines the case, when the freeze frame record is stored/updated.

3.6.1.53 Dem/DemGeneral/DemGeneralInterfaceSupport

[RTA-BSW] Description	The interfaces GeneralEvtInfo, GeneralCallbackEventDataChanged, GeneralCallbackMonitorStatusChanged and GeneralCallbackEvent-tUdsStatusChanged are provided if DemGeneralInterfaceSupport is equal to true.
[AUTOSAR] Description	The interfaces GeneralEvtInfo, GeneralCallbackEventDataChanged, GeneralCallbackMonitorStatusChanged and GeneralCallbackEvent-tUdsStatusChange are provided if DemGeneralInterfaceSupport is equal to true.

3.6.1.54 Dem/DemGeneral/DemGeneralOBD/DemOBDEventDisplacement

[RTA-BSW] Lower Multiplicity	0
[AUTOSAR] Lower Multiplicity	1

[ADDED] Default Value	true

3.6.1.55 Dem/DemGeneral/DemRatio/DemFunctionIdRef

[RTA-BSW] Upper Multiplicity	5
[AUTOSAR] Upper Multiplicity	*

3.6.1.56 Dem/DemGeneral/DemRatio/DemIUMPRDenGroup

[RTA-BSW] Description	This parameter specifies the assigned denominator type.
[AUTOSAR] Description	This parameter specifies the assigned denominator type

3.6.1.57 Dem/DemGeneral/DemResetPendingBitOnOverflow

[RTA-BSW] Description	This configuration switch defines, whether the pending bit is reset or not while an event memory entry will be displaced.
[AUTOSAR] Description	This configuration switch defines, whether the pending bit is reset or
[REMOVED] Introduction	not while an event memory entry will be displaced.

3.6.1.58 Dem/DemGeneral/DemStatusBitStorageTestFailed

[RTA-BSW] Description	Activate/Deactivate the permanent storage of the “TestFailed” status bits. If this is activated, the failure of this event is stored to the next power cycle.
[AUTOSAR] Description	Activate/Deactivate the permanent storage of the “TestFailed” status bits.

3.6.1.59 Dem/DemGeneral/DemStorageConditionGroup/DemStorageConditionRef

[RTA-BSW] Description	References a storage condition.
[AUTOSAR] Description	References an enable condition.

3.7 API Definition

Table 3.338.Dem Supported API functions

`Dem_Init`	Initialises the Dem module.
`Dem_PreInit`	Initializes the internal states necessary to process events reported by BSW-modules.
`Dem_Shutdown`	Shuts down the Dem module.
`Dem_MainFunction`	Processes all non-event-based Dem internal functions.

Dem_GetVersionInfo	Returns the version information of the Dem module.
Dem_ClearDTC	Clears single DTCs and groups of DTCs.
Dem_ClearPrestoredFreezeFrame	Clears a prestored freeze frame of a specific event.
Dem_GetComponentFailed	Gets the failed status of a DemComponent.
Dem_GetDebouncingOfEvent	Gets the debouncing status of an event.
Dem_GetDTCOfEvent	Gets the DTC of an event.
Dem_GetDTCSelectionResult	Provides information if the last call to Dem_SelectDTC has selected a valid DTC or group of DTCs.
Dem_GetDTCSelectionResultForClearDTC	Provides information if the last call to Dem_SelectDTC has selected a valid DTC or group of DTCs respecting the settings if Dem clears only all DTCs.
Dem_GetDTCSuppression	Returns the suppression status of a specific DTC.
Dem_GetEventExtendedDataRecordEx	Gets the data of an extended data record by event.
Dem_GetEventFreezeFrameDataEx	Gets the data of a freeze frame by event.
Dem_GetEventMemoryOverflow	Gets the event memory overflow indication status.
Dem_GetEventUdsStatus	Gets the current UDS status byte assigned to the DTC for the event.
Dem_GetFaultDetectionCounter	Gets the fault detection counter of an event.
Dem_GetIndicatorStatus	Gets the indicator status derived from the event status.
Dem_GetMonitorStatus	Gets the current monitor status for an event.
Dem_GetNextFreezeFrameData	Gets freeze frame data by the DTC selected by Dem_SelectFreezeFrameData.
Dem_GetNumberOfEventMemoryEntries	Returns the number of entries currently stored in the requested event memory.
Dem_GetDTCStatusAvailabilityMask	Gets the DTC Status availability mask.
Dem_GetSizeOfExtendedData RecordSelection	Gets the size of Extended Data Record by DTC selected by the call of Dem_SelectExtended-DataRecord.
Dem_GetSizeOfFreezeFrameSelection	Gets the size of freeze frame data by DTC selected by the call of Dem_SelectFreezeFrame-Data.
Dem_PrestoreFreezeFrame	Captures the freeze frame data for a specific event.

Dem_ResetEventDebounceStatus	Controls the internal debounce counter/timer by BSW modules and SWCs.
Dem_ResetEventStatus	Resets the event failed status.
Dem_ResetMonitorStatus	Resets the monitor status for an event.
Dem_RestartOperationCycle	Sets an operation cycle state.
Dem_SelectDTC	Selects a DTC or DTC group as target for further operations.
Dem_SelectExtendedDataRecord	Sets the filter to be used by Dem_GetNextExtendedDataRecord and Dem_GetSizeOfExtendedDataRecordSelection.
Dem_SelectFreezeFrameData	Sets the filter to be used by Dem_GetNextFreezeFrameData and Dem_GetSizeOfFreezeFrameSelection.
Dem_SetComponentAvailable	Set the availability of a specific DemComponent.
Dem_SetDTCFilter	Sets the DTC Filter.
Dem_SetDTCSuppression	Set the suppression status of a specific DTC.
Dem_SetEnableCondition	Sets an enable condition.
Dem_SetEventAvailable	Set the available status of a specific event.
Dem_SetEventStatus	Processes the events reported by SW-Cs via RTE.
Dem_SetEventStatusWithMonitorData	Processes the events reported by SW-Cs via RTE with Monitor data.
Dem_SetStorageCondition	Sets a storage condition.
Dem_SetWIRStatus	Sets the WIR status bit via failsafe SW-Cs.
Dem_GetDTCByOccurrenceTime	Gets the DTC by occurrence time.
Dem_GetFunctionalUnitOfDTC	Gets the functional unit of the requested DTC.
Dem_GetNextExtendedDataRecord	Gets extended data record for the DTC selected by Dem_SelectExtendedDataRecord.
Dem_GetNextFilteredDTC	Gets the next filtered DTC matching the filter criteria.
Dem_GetNextFilteredDTCAndFDC	Gets the next filtered DTC and its associated Fault Detection Counter (FDC) matching the filter criteria.
Dem_GetNextFilteredDTCAndSeverity	Gets the next filtered DTC and its associated Severity matching the filter criteria.
Dem_GetNextFilteredRecord	Gets the next freeze frame record number and its associated DTC stored in the event memory.
Dem_GetNumberOfFilteredDTC	Gets the number of a filtered DTC.
Dem_GetNumberOfFreezeFrameRecords	Returns the number of freeze frame records currently stored in the primary event memory.
Dem_GetSeverityOfDTC	Gets the severity of the requested DTC.

Dem_GetStatusOfDTC	Gets the status of a DTC.
Dem_GetTranslationType	Gets the supported DTC formats of the ECU.
Dem_SetFreezeFrameRecordFilter	Sets a freeze frame record filter
Dem_DisableDTCRecordUpdate	Disables the event memory update of a specific DTC (only one at one time).
Dem_EnableDTCRecordUpdate	Enables the event memory update of the DTC previously disabled by Dem_DcmDisableDTCRecordUpdate().
Dem_DisableDTCSetting	Disables the DTC setting for a DTC group.
Dem_EnableDTCSetting	Enables the DTC setting for a DTC group.
Dem_J1939DcmFirstDTCwithLampStatus	Sets the filter to the first applicable DTC for the DM31 response for a specific node.
Dem_J1939DcmGetNextDTCwithLampStatus	Gets the next filtered J1939DTC for DM31 including current LampStatus.
Dem_J1939DcmGetNextFilteredDTC	Gets the next filtered J1939 DTC.
Dem_J1939DcmGetNumberOfFilteredDTC	Gets the number of currently filtered DTCs set by the function Dem_J1939DcmSetDTCFilter.
Dem_J1939DcmSetDTCFilter	Sets the DTC filter for a specific node and returns the composite lamp status of the filtered DTCs.
Dem_J1939DcmClearDTC	Clears active DTCs as well as previously active DTCs.
Dem_J1939DcmGetNextFreezeFrame	Gets next freeze frame data.
Dem_J1939DcmGetNextSPNInFreezeFrame	Gets next SPN.
Dem_J1939DcmSetFreezeFrameFilter	Sets the FreezeFrame filter for a specific node.
Dem_J1939DcmGetNextFilteredRatio	Gets the next filtered Ratio.
Dem_J1939DcmReadDiagnosticReadiness1	Service to report the value of Diagnostic Readiness 1 (DM05) computed by the Dem.
Dem_J1939DcmReadDiagnosticReadiness2	Service to report the value of Diagnostic Readiness 2 (DM21) computed by the Dem.
Dem_J1939DcmReadDiagnosticReadiness3	Service to report the value of Diagnostic Readiness 3 (DM26) computed by the Dem.
Dem_J1939DcmSetRatioFilter	Sets the Ratio filter for a specific node and returns the corresponding Ignition Cycle Counter and General Denominator.

For further information on these functions, please refer to AUTOSAR_SWS_DiagnosticEventManager.pdf

3.8 Configuration Advice

> **NOTE**
>
> Before configuring this module, please note that some of the default values used by ConfGen to produce the default ECU Configuration can be changed, and this is done by either adding a Conf-Gen Enhancer Mapping (CEM) or by adding a rule to an *algo.properties* file. For more information, see the "Configuration Generation" section of the RTA-CAR User Guide.

3.8.1 Interrupt Protection Locks

Dem uses Exclusive-Areas (implemented by Interrupt Locks) to support a multi-task system architecture.

The following lock is always required because the monitorings are usually running in different tasks from Dem, and may therefore interrupt the processing of Dem at any time:

- SchM_Enter_Dem_Monitoring()
- SchM_Exit_Dem_Monitoring()

These locks are required if Dem and either NvM or Dcm might interrupt each other, for example if both components are running in different tasks on a single CPU:

- SchM_Enter_Dem_Nvm()
- SchM_Exit_Dem_Nvm()
- SchM_Enter_Dem_Dcm()
- SchM_Exit_Dem_Dcm()

Dem requires these Interrupt Locks to implement atomic operations on Dem-internal data structures. Atomic executions of long sequences of data operations would affect the real-time behaviour of the Operating System, so Dem avoids them.

In a multi-task/multi-core system the considerations below should be observed in order to ensure atomic Dem data processing.

3.8.1.1 Init/Shutdown

Dem_Init and Dem_Shutdown are not allowed to interrupt any other running Dem API function (e.g. Dem_MainFunction, Dem_SetEventStatus). This can be ensured by only calling Dem_Init and Dem_Shutdown in a task that has a task priority lower or equal to the task priorities of all pre-emptive scheduled tasks that are using Dem API functions. Using the lowest priority task, or the idle task, is recommended.

3.8.1.2 Dcm and Dlt APIs

Unless the DemRbReadEventRelatedDataFromDifferentOsTask configuration switch is enabled, the system architecture has to ensure that the Dcm or Dlt that is calling a Dem API, and the Dem_MainFunction, are not pre-empting each other. This can be ensured by having Dem and Dcm/Dlt scheduled in the same task, or within cooperative tasks.

Affected Dtc interfaces:

- Dem_GetStatusOfDTC (only for optional stored status in mirror memory)
- Dem_SetFreezeFrameRecordFilter
- Dem_GetNextFilteredRecord
- Dem_DcmReadDataOfOBDFreezeFrame

Affected Dlt interfaces:

- Dem_DltGetAllExtendedDataRecords
- Dem_DltGetMostRecentFreezeFrameRecordData

3.8.1.3 SWC APIs

Unless DemRbReadEventRelatedDataFromDifferentOsTask is enabled, the Dem and the SWC should be scheduled in the same task, or within cooperative tasks.

Affected SWC interfaces:

- Dem_GetEventExtendedDataRecord
- Dem_GetEventFreezeFrameData

3.8.1.4 Dem_ClearDTC (Sequential locking strategy)

When the event status of an event is reported, and this event is currently being cleared, the event status report will be ignored.

Furthermore, if the processing of the event clearing in the Dem_MainFunction might pre-empt or run in parallel with this lower-priority monitoring function which reported either PASSED or FAILED, this may result in an event memory entry for that event (with the PendingDTC bit in the event status byte set) although the rest of the status byte is cleared.

This situation does not occur if Dem_MainFunction is not pre-empting, or if it runs in parallel with any monitor. This is valid only if the system is a single-core system, or all monitors and the Dem_MainFunction run on the same core and the Dem task has the lowest priority, or is a cooperative task and any other lower priority task using/calling Dem API functions is also cooperative.

3.8.2 Non-Volatile Memory Blocks

The configuration of NvM blocks can be done in one of two ways -

- **Manual Configuration:** In case of manual configuration (i.e., when DemR-

bEnableDemNvmBlocksAutoConfiguration= false), the user has to configure all the required NvM blocks and their attibutes manually. The tables below can be used as reference to configure the different attributes of the NvM blocks. The attributes marked with an asterisk (*) must be configured exactly as specified in the table. Other parameters are only recommendations and may be configured differently by the user.

- **Auto Configuration:** In case of auto configuration (i.e., when DemRbEnableDemNvmBlocksAutoConfiguration = true), Dem automatically configures the required NvM blocks and several attributes of the corresponding NvMBlockDescriptor container.

The following attributes are automatically configured with the values mentioned in the tables below:

- NvMBlockUseSyncMechanism
- NvMRbAutoBlockLength
- NvMRbNvBlockLengthString
- NvMWriteBlockOnce
- NvMBlockWriteProt
- NvMReadRamBlockFromNvCallback
- NvMWriteRamBlockToNvCallback
- NvMSelectBlockForReadAll
- NvMSelectBlockForWriteAll
- NvMResistantToChangedSw
- NvMBlockJobPriority
- NvMRomBlockNum

Note: There are also some mandatory attributes that are not supported by the current NvM implemention in RTA-CAR, and these attributes are also automatically configured with dummy values, to simplify the configuration process.

- NvMStaticBlockIDCheck
- NvMWriteVerification
- NvMWriteVerificationDataSize.

Additionally, some of the attributes with recommended values mentioned in the tables below are also automatically configured, providing that these have not been configured manually. Note: The NvMNvBlockNum must also be manually configured, if NvMBlockManagementType is manually configured and vice-versa.

- NvMBlockManagementType
- NvMNvBlockNum
- NvMMaxNumOfReadRetries
- NvMMaxNumOfWriteRetries

- NvMBlockUseCrc

Note: In case of auto configuration, the names of the NvM blocks mentioned below are not recommendations, and must be named exactly as shown. This is important to note because some of the attributes of the NvM blocks (e.g., NvMNvramDeviceId) must be configured manually, even when auto configuration is used.

3.8.3 NvM Configuration Proposal

3.8.3.1 Configuration of DemNvBlock

Event Status Byte	mandatory
Event Indicator Attribute	conditional "one indicator configured per at least one event"
History Status Byte	conditional "DemHistoryStatusSupported ==true"
Event Memory Locations	mandatory
Generic Data	mandatory
Disturbance Memory	conditional "DemDisturbanceMemorySupported== true"
Pre-stored FF Memory Locations	conditional"(DemRbPrestoreFreezeFrameNonVolatile == true)&&(demMaxNumberPrestoredFF>0)"
Hash ID	conditional "(DemRbEraseAllNvMDataSupported== ERASE_ALL_WITH_HASHID_CHECK)"

3.8.3.2 Use of Indexed DemNvBlocks

Event Status Byte	no
Event Indicator Attribute	no
History Status Byte	no
Event Memory Locations	yes, number of locations/indices: DemMaxNumberEventEntryPrimary + DemMaxNumberEventEntrySecondary + DemMaxNumberEventEntryMirror (first index is zero).
Generic Data	no
Disturbance Memory	no
Pre-stored FF Memory Locations	yes, number of locations/indices: demMaxNumberPrestoredFF (first index is zero)

Hash ID	no

3.8.3.3 Dem/DemGeneral/DemNvRamBlockId

ShortName

Event Status Byte	DEM_NVM_ID_EVT_STATUSBYTE
Event Indicator Attribute	DEM_NVM_ID_INDICATOR_ATTRIBUTES
History Status Byte	DEM_NVM_ID_HISTORY_STATUS_BYTE
Event Memory Locations	DEM_NVM_ID_EVMEM_LOC_<index>
Generic Data	DEM_NVM_ID_DEM_GENERIC_NV_DATA
Disturbance Memory	DEM_NVM_ID_DISTURBANCE_MEMORY
Pre-stored FF Memory Locations	DEM_NVM_ID_PRESTORED_FREEZEFRAME_LOC_<index>
Hash ID	DEM_NVM_ID_DEM_HASH_ID

DemNvRamBlockIdRef

Event Status Byte	Ref to NvM/NvMBlockDescriptor
Event Indicator Attribute	Ref to NvM/NvMBlockDescriptor
History Status Byte	Ref to NvM/NvMBlockDescriptor
Event Memory Locations	Ref to NvM/NvMBlockDescriptor
Generic Data	Ref to NvM/NvMBlockDescriptor
Disturbance Memory	Ref to NvM/NvMBlockDescriptor
Pre-stored FF Memory Locations	Ref to NvM/NvMBlockDescriptor
Hash ID	Ref to NvM/NvMBlockDescriptor

3.8.3.4 NvM/NvMBlockDescriptor

ShortName

Event Status Byte	Cust (Recommended: NVM_ID_EVT_STATUSBYTE)
Event Indicator Attribute	Cust (Recommended: NVM_ID_INDICATOR_ATTRIBUTES)
History Status Byte	Cust (Recommended: NVM_ID_HISTORY_STATUS_BYTE)
Event Memory Locations	Cust (Recommended: NVM_ID_EVMEM_LOC_<index>)
Generic Data	Cust (Recommended: NVM_ID_DEM_GENERIC_NV_DATA)
Disturbance Memory	Cust (Recommended: NVM_ID_DISTURBANCE_MEMORY)
Pre-stored FF Memory Locations	Cust (Recommended: NVM_ID_PRESTORED_FREEZE-FRAME_LOC_<index>)

Hash ID	Cust (Recommended: NVM_ID_DEM_HASH_ID)

NvMReadRamBlockFromNvCallback*

Event Status Byte	Dem_EventStatusByteReadRamBlockFromNvCallback
Event Indicator Attribute	Dem_EventIndicatorReadRamBlockFromNvCallback
History Status Byte	Dem_HistoryStatusByteReadRamBlockFromNvCallback
Event Memory Locations	Dem_EvMemNvmReadRamBlockFromNvCallback<index>
Generic Data	Dem_GenericNVDataReadRamBlockFromNvCallback
Disturbance Memory	Dem_DistMemReadRamBlockFromNvCallback
Pre-stored FF Memory Locations	Dem_PreStoredFFReadRamBlockFromNvCallback<index>
Hash ID	Dem_HashIdReadRamBlockFromNvCallback

NvMWriteRamBlockToNvCallback*

Event Status Byte	Dem_EventStatusByteWriteRamBlockToNvCallback
Event Indicator Attribute	Dem_EventIndicatorWriteRamBlockToNvCallback
History Status Byte	Dem_HistoryStatusByteWriteRamBlockToNvCallback
Event Memory Locations	Dem_EvMemNvMWriteRamBlockToNvCallback<index>
Generic Data	Dem_GenericNVDataWriteRamBlockToNvCallback
Disturbance Memory	Dem_DistMemWriteRamBlockToNvCallback
Pre-stored FF Memory Locations	Dem_PreStoredFFWriteRamBlockToNvCallback<index>
Hash ID	Dem_HashIdWriteRamBlockToNvCallback

The settings for these parameters are the same across all options...

NvMBlockManagementType	Cust (Recommended: NVM_BLOCK_REDUNDANT)
NvMBlockUseCrc	Cust (Recommended: True)
NvMBlockUseSyncMechanism*	TRUE
NvMBlockWriteProt*	FALSE
NvMInitBlockCallback*	empty
NvMMaxNumOfReadRetries	0
NvMMaxNumOfWriteRetries	0
NvMRamBlockDataAddress*	empty
NvMResistantToChangedSw*	FALSE
NvMRomBlockDataAddress*	empty
NvMSelectBlockForReadAll*	TRUE
NvMSelectBlockForWriteAll*	TRUE
NvMWriteBlockOnce*	FALSE

NvMCalcRamBlockCrc	Cust
NvMRbAutoBlockLength (1)	Cust
NvMBlockHeaderInclude (2)	Cust
NvMBlockCrcType	Cust
NvMBlockJobPriority*	1
NvMBlockUseAutoValidation	Cust
NvMBlockUseCRCCompMechanism	Cust
NvMBlockUsePort*	empty
NvMNvBlockNum	Cust (Recommended: 2)
NvMNvramBlockIdentifier	Cust
NvMRomBlockNum*	0
NvMSingleBlockCallback*	empty

1. NvMRbAutoBlockLength

If "NvMRbAutoBlockLength" is set to true, then the block size must be configured using "NvMRbNvBlockLengthString", else using NvMNvBlockLength. Please refer the documentation of NvM for more details.

2. NvMBlockHeaderInclude

If the size is configured using "NvMRbNvBlockLengthString", then "NvMBlock-HeaderInclude" should be configured such that the Dem header file Dem_Fwd-ToNvm.h is included by NvM, this is to allow automatic calculation of the block size.

If the auto-configuration feature is enabled in Dem by setting "DemRbEn-ableDemNvmBlocksAutoConfigurationas" to true, then Dem makes use of "NvM-RbNvBlockLengthString" to configure the block size.

Additional Hint for Event Status Byte

For "NvMNvBlockLength" it is possible to configure a value bigger than size of (Dem_AllEventsStatusByte). This might be used to reduce NVM-layout changes during the project development phase. The correct value must be configured within the project before release.

3.8.4 Memory Mapping

Dem currently only supports the following memory mapping sections:

- DEM_START_SEC_VAR_CLEARED
- DEM_STOP_SEC_VAR_CLEARED
- DEM_START_SEC_CODE
- DEM_STOP_SEC_CODE
- DEM_START_SEC_CONST
- DEM_STOP_SEC_CONST
- DEM_START_SEC_VAR_INIT

- DEM_STOP_SEC_VAR_INIT
- DEM_START_SEC_VAR_SAVED_ZONE
- DEM_STOP_SEC_VAR_SAVED_ZONE
- DEM_START_SEC_CONST_UNSPECIFIED
- DEM_STOP_SEC_CONST_UNSPECIFIED
- DEM_START_SEC_VAR_CLEARED_UNSPECIFIED
- DEM_STOP_SEC_VAR_CLEARED_UNSPECIFIED
- DEM_START_SEC_VAR_CLEARED_BOOLEAN
- DEM_STOP_SEC_VAR_CLEARED_BOOLEAN
- DEM_START_SEC_VAR_INIT_UNSPECIFIED
- DEM_STOP_SEC_VAR_INIT_UNSPECIFIED
- DEM_START_SEC_VAR_SAVED_ZONE_8
- DEM_STOP_SEC_VAR_SAVED_ZONE_8
- DEM_START_SEC_VAR_SAVED_ZONE_UNSPECIFIED
- DEM_STOP_SEC_VAR_SAVED_ZONE_UNSPECIFIED
- DEM_START_SEC_VAR_CLEARED_8
- DEM_STOP_SEC_VAR_CLEARED_8
- DEM_START_SEC_CONST_8
- DEM_STOP_SEC_CONST_8
- DEM_START_SEC_VAR_INIT_BOOLEAN
- DEM_STOP_SEC_VAR_INIT_BOOLEAN
- DEM_START_SEC_VAR_INIT_8
- DEM_STOP_SEC_VAR_INIT_8
- DEM_START_SEC_VAR_CLEARED_16
- DEM_STOP_SEC_VAR_CLEARED_16
- DEM_START_SEC_CONST_16
- DEM_STOP_SEC_CONST_16

Instead of using e.g. SAVED_ZONE, the immediate NvM-storage trigger functionality can be used.

3.8.5 Multiple Clear requests

Multiple clear operations might influence each other (e.g. ClearDTC is triggered by Dcm, parallel clear of BFM), so only one clear should be performed at a time, otherwise the following effects might occur:

- One block will not be able to be cleared (either a block addressed by the DCM clear, or by the BFM clear) and both operations will return Clear Failed.
- The time for clearing may be increased because Clear Finished is not

returned until all blocks marked for clear are cleared.

3.8.6 Re-entrancy of functions

AUTOSAR (R21-11) defines the re-entrancy of services that are updating the stored events in Dem as “Re-entrant” for different EventIds and “Non re-entrant” for the same EventId.

The system architecture should ensure that none of the services listed below are pre-empting each other for the same EventId. This can be achieved for each individual EventId if its API services are only called within the same task, or within cooperative tasks. Affected interfaces:

- Dem_ManipulateEventDebounceStatus

3.8.7 Environment Data configuration

In order to avoid compiling issues, at least one Freeze Frame Class and one Extended Data Class must be defined. If Disturbance Memory is supported, a DemRbDisturbanceMemoryExtendedDataClassRef (in DemRbGeneral) must be defined.

3.8.8 J1939Nm Node IDs

To optimize access to J1939Nm NodeIds, it’s assumed that the Ids are arranged in consecutive order, beginning with 0.

3.8.9 DemRbStoreWaitingForMonitoringEvent

The extension parameter DemRbStoreWaitingForMonitoringEvent should not be used for emission-related events (OBDrelevant).

3.8.10 DemRbGetEventAvailableBehaviour

The mandatory parameter DemRbGetEventAvailableBehaviour must be configured appropriately, depending on the behavior needed for the Dem_GetEventAvailable API. If required, the callers of the Dem_GetEventAvailable in the project must be adopted, so that all of them expect the same behavior from the API. The parameter DemRbGetEventAvailableBehaviour can then be configured accordingly.

3.8.11 DemRbAgingCounterType

The DemRbAgingCounterType extension parameter in RTA-BSW can be used (instead of the AUTOSAR-defined DemAgingRequiresTestedCycle) to define the type of Aging Counter. If DemRbAgingCounterType is set to COUNT_UP, the aging counter value is processed when the respective aging cycle ends/restarts. If DemRbAgingCounterType is set to COUNT_UP_TC, the aging counter value is processed when the respective aging cycle ends/restarts and the UDS status bit 6 (TestNotCompleteThisOperationCycle) is set to zero.

3.8.12 Additional Configuration Advice

3.8.12.1 Manual Configuration

The following parameters must be manually configured in the ISOLAR-AB application after running the Confgen tool.

- Dem/DemGeneral/DemNvRamBlockId
- Dem/DemInternalDataElementClass/DemDataElementDataSize

3.8.12.2 Dem & RTE Generation

To avoid build errors when running the RTE Generation for projects that include Dem, the following files should be unselected.

- Dem_Cfg_DtcMcSupportData.arxml
- Dem_Cfg_McSupportData.arxml

3.8.12.3 Manual Configuration of DemeventId (DemEventParameter)

The DemeventId parameter (in Dem/DemConfigSet/DemEventParameter) can be configured manually, and if you choose to do so, the following rules must be observed:

- If one DemeventId is manually configured, all DemeventIds in the DemEventParameters list must be configured.
- The value configured for each DemeventId must be unique, and it must not be larger than the number of DemEventParameters in the DemEventParameters list.
- The failure dependency feature must be disabled, so demRbSupportFailureDependency must be configured as FALSE. If failure dependency is left enabled (configured as TRUE), the RTA-BSW code generation script will be based on the "graph" of DemComponents to generate the value for DemeventId.

3.8.12.4 DemComponent configuration

DEM_CFG_DEPENDENCY will be defined as DEM_CFG_DEPENDENCY_ON when there is any demComponent configured. For a DemComponent, if DemComponent/DemComponentFailedCallbackUsePort is set to false, DemComponentFailedCallbackFnc should be configured. If DemComponent/DemComponentFailedCallbackUsePort is set to true, Rte_Call_Dem_CBStatusComp_<component.shortName>_ComponentStatusChanged is required to be generated by RTE. Then the below actions need to be completed:

- Create the relevant ClientServer Provided Port and connect it to the RPort CBStatusComp_<Component.shortName>generated in Dem_Cfg_SWCD.arxml. Provide the Rte_Call_Dem_CBStatusComp_<component.shortName>_ComponentStatusChanged implementation in the

Application software

- Because of a limitation, Rte_Call_Dem_CBStatusComp_<component.shortName>_ComponentStatusChanged is not externed in RTE. To overcome this issue, Add DemAppl.h in Dem/DemGeneral/DemHeaderFileInclusion, run RTA-BSW CodeGen again; then create DemAppl.h in Dem/integration, add the extern for these Rte_Call_Dem_CBStatusComp_<component.shortName>_ComponentStatusChanged api.

3.8.12.5 Configuring of FreezeFrame Captured in Pending status

From Dem module, in order to have FreezeFrame captured when event in Pending status, two parameters need to be configured:

- DemRbOBDFreezeFrameCapture == DEM_TRIGGER_PENDINGDTC
- DemRbOBDFreezeFrameVisibility == DEM_OBDFF_PENDING_AND_CONFIRMED_DTCS

3.8.12.6 Configuring of Dem's parameters in case of CanSMRbDynCreate_CANSM_E_BUS_OFF is set to Standard

List of parameters below are generated automatically with default value in case of existence of Dem module and CanSMRbDynCreate_CANSM_E_BUS_OFF is set to Standard:

- Dem/DemConfigSet/DemDebounceCounterBasedClass/ShortName: CANSM_E_DemDebounceCounterBasedClass_NETWORK_0
- Dem/DemConfigSet/DemDebounceCounterBasedClass/DemDebounceCounterJumpDownValue
- Dem/DemConfigSet/DemDebounceCounterBasedClass/DemDebounceCounterJumpUpValue
- Dem/DemConfigSet/DemDebounceCounterBasedClass/DemDebounceCounterStorage
- Dem/DemConfigSet/DemDTCAttributess/DemDTCAttributes/ShortName: CANSM_E_DemDTCAttributes_NETWORK_0
- Dem/DemConfigSet/DemDTCAttributess/DemDTCAttributes/DemAgingAllowed
- Dem/DemConfigSet/DemDTCAttributess/DemDTCAttributes/DemAgingCycleCounterThreshold
- Dem/DemConfigSet/DemDTCAttributess/DemDTCAttributes/DemDTCPriority
- Dem/DemConfigSet/DemDTCAttributess/DemDTCAttributes/DemImmediateNvStorage
- Dem/DemEventParameters/DemEventParameter/ShortName: CANSM_E_BUS_OFF_NETWORK_0
- Dem/DemEventParameters/DemEventParameter/DemEventAvailable

- Dem/DemEventParameters/DemEventParameter/DemEventFailureCycleCounterThreshold
- Dem/DemEventParameters/DemEventParameter/DemFFPrestorageSupported
- Dem/DemEventParameters/DemEventParameter/DemReportBehavior
- Dem/DemEventParameters/DemEventParameter/DemDebounceAlgorithmClass/DemDebounceCounterBased
- Dem/DemEventParameters/DemEventParameter/DemDebounceAlgorithmClass/DemDebounceCounterBasedClassRef

For value of parameters DemAgingCycleRef and DemOperationCycleRef should be configured in algo.properties file as user project in order to generated successfully

- Dem/DemConfigSet/DemDTCAttributess/DemDTCAttributes/DemAgingCycleRef
- Dem/DemEventParameters/DemEventParameter/DemOperationCycleRef

Example:

- manprop_DemDTCAttributes_ALL=DemAgingCycleRef: DemOpCycle_IGNITION
- manprop_DemEventParameter_ALL=DemOperationCycleRef: DemOpCycle_POWER

3.8.12.7 Enabling and Disabling Calibration for Dem Structure Elements

Calibration for the structure elements can be enabled or disabled by setting the child parameters in demGeneral.demRbGeneral.demRbCalibParameter to true or false.

From Dem module, in order for the calibration to be enabled for MaxNumberFreezeFrameRecords, EventPriority, FailureConfirmationThreshold, AgingThreshold, AgingAllowed, Significance, IsEventOBDRelevant and IsEventAvailableInVariant, the ten parameters need to be configured:

- DemRbCalibAgingCycleCounterThreshold == true
- DemRbCalibEventConfirmationThreshold == true
- DemRbCalibEventPriority == true
- DemRbCalibMaxNumberFreezeFrameRecords == true
- DemRbCalibSeverity == true
- DemRbCalibDTCKind == true
- DemRbCalibEventAvailability == true
- DemRbCalibAgingAllowed == true
- DemRbCalibSignificance == true

- DemRbCalibOBDReadiness == true

3.8.12.8 OBD Readiness

The Dem module is responsible for the computation of readiness relevent PIDs $01, $41 and $F501. This allows the user to assign an OBD relevant DTC to a specific readiness group via the configuration parameter DemEventOBDReadinessGroup under the DemObdDTC container. If required, the user may get the list of DTCs assigned to specific readiness group number (RGID) via the UDS service 0x19 56 (reportDTCInformationByReadinessGroupNumber).

The Dem API Dem_SetDTCFilterByReadinessGroup(), that is called by Dcm when 0x19 56 is requested, is functional only when OBDonUDS is enabled (via the parameter DemRbOBDProtocol) or if there is any DTC assigned to a readiness group other than DEM_OBD_RDY_CMPRCMPT or DEM_OBD_RDY_NONE. The same applies also for the API Dem_DcmReadDataOfPIDF501() that is used to get the value of PID$F501. It is done so to optimize the resource consumption for primary ECUs, since service 0x19 56 is usually necessary only in case of OBDonUDS.

The following table shows how the AUTOSAR configuration is internally mapped to the SAE defined readiness groups in PID $01/$41/$F501 as well as to the ReadinessGroupIDs used by service 0x19 56.

AUTOSAR Readiness Group to Readiness PIDs and 0x19 56 RGID Mapping

AUTOSAR Readiness Group [1]	Engine Type [2]	PID $01 / $41	PID $F501	RGID (0x19 56)
DEM_OBD_RDY_AC	S/C	N/A	N/A	N/A
DEM_OBD_RDY_BOOSTPR	C	Boost Pressure System Monitoring	Boost Pressure System Monitoring	0x12
DEM_OBD_RDY_CAT	S	Catalyst Monitoring	Catalyst Monitoring	0x01
DEM_OBD_RDY_CMPRCMPT	S/C	Comprehensive Component Monitoring	Comprehensive Component Monitoring	0x0E
DEM_OBD_RDY_CSER	S/C	N/A	CSER System Monitoring	0x0B
DEM_OBD_RDY_DOR	S/C	N/A	DOR Monitoring	0x0D
DEM_OBD_RDY_ECS	S/C	N/A	Engine Cooling System Monitoring	0x0A

DEM_OBD_RDY_EGR	S/C	EGR and+or VVT System Monitoring	EGR System Monitoring	0x08
DEM_OBD_RDY_EGSENS	C	Exhaust Gas Sensor Monitoring	Exhaust Gas Sensor Monitoring	0x07
DEM_OBD_RDY_EVAP	S	Evaporative System Monitoring	Evaporative System Monitoring	0x04
DEM_OBD_RDY_FLSYS [3]	S/C	Fuel System Monitoring	Fuel System Monitoring	0x06
DEM_OBD_RDY_FLSYS_NONCONT [3]	S/C	Fuel System Monitoring	Fuel System Monitoring	0x06
DEM_OBD_RDY_GPF	S	Gasoline Particulate Filter Monitoring	Particulate Matter Filter Monitoring	0x14
DEM_OBD_RDY_HCCAT	C	NMHC Catalyst Monitoring	NMHC Catalyst Monitoring	0x10
DEM_OBD_RDY_HTCAT	S	Heated Catalyst Monitoring	Heated Catalyst Monitoring	0x02
DEM_OBD_RDY_MISF	S/C	Misfire Monitoring	Misfire Monitoring	0x03
DEM_OBD_RDY_NONE [4]	S/C	N/A	N/A	N/A
DEM_OBD_RDY_NOXADSORB	C	NOx Aftertreatment Monitoring	NOx Adsorber Monitoring	0x13
DEM_OBD_RDY_NOXCAT	C	NOx Aftertreatment Monitoring	NOx Catalyst Monitoring	0x11
DEM_OBD_RDY_O2SENS	S	Oxygen Sensor Monitoring	Exhaust Gas Sensor Monitoring	0x07
DEM_OBD_RDY_O2SENSHT	S	Oxygen Sensor Heater Monitoring	Exhaust Gas Sensor Monitoring	0x07
DEM_OBD_RDY_OTHER	S/C	N/A	“Other” System Monitoring	0x0F
DEM_OBD_RDY_PCV	S/C	N/A	CV System Monitoring	0x09
DEM_OBD_RDY_PMFLT	C	PM Filter Monitoring	Particulate Matter Filter Monitoring	0x14

DEM_OBD_RDY_SECAIR	S	Secondary Air System Monitoring	Secondary Air System Monitoring	0x05
DEM_OBD_RDY_VVT	S/C	EGR and+or VVT System Monitoring	VVT System Monitoring	0x0C

[1] AUTOSAR Readiness Group corresponds to values of the configuration parameter DemEventOBDReadinessGroup.

[2] Engine Type corresponds to the value of the configuration parameter DemOBDEngineType. In the table, S = DEM_IGNITION_SPARK, C = DEM_IGNITION_COMPRESSION.

[3] Continuous fuel system monitoring affects only the enabled/support status bit but not the complete/ready status bit in the PIDs.

[4] The SAE J1979-DA defines the RGID 0x00 as corresponding to "No Readiness Group". But in the current Dem implementation, RGID 0x00 is not supported and is treated as an invalid RGID.

3.9 Integration Advice

3.9.1 Init Functions

Dem_PreInit must be called first.

Dem_Init has to be called during the startup phase of the ECU, after NvM has finished the restore of NVRAM data.

Dem allows some interfaces to be called before its own initialization is completed. This allows proper processing of the events that are reported and stored to the event queue before Dem's own initialization. The following interfaces can be called before Dem_Init:

- Dem_ReportErrorStatus
- Dem_RestartOperationCycle
- Dem_SetEventAvailable
- Dem_SetStorageCondition
- Dem_SetEnableCondition
- Dem_SetComponentAvailable
- Dem_GetFaultDetectionCounter
- Dem_ResetEventDebounceStatus
- Dem_GetDebouncingOfEvent
- Dem_SetDTCSuppression

3.9.1.1 DemRbLockAllFailureInfoSupported

If DemRbLockAllFailureInfoSupported is set to True, then Dem_LockAllFailureInfo

must be called before Dem_Init. If it is set to False, then the Dem initialization sequence is described above.

3.9.2 DeInit Functions

After the shutdown of Dem, no calls to Dem_SetEventStatus are allowed.

Dem_TriggerStorageToNvm can be used to trigger the storage (to NvM) for all persistently-stored DEM-internal data. This might be necessary if the ECU is unplugged from power directly at shutdown. Dem_TriggerStorageToNvm can be called shortly before the power (and data) is lost. This feature is only supported when the configuration parameter "DemTriggerToStoreNvMSupported" is set to TRUE.

3.9.3 Scheduled Functions

Dem_MainFunction must be scheduled cyclically. The calling period has to be the same as configured in the Dem EcuC-values file (DemTaskTime).

3.9.4 operationCycle

The operationCycle of an event has to be a sub-part of the failureCycle, so if failureCycle is started, operationCycle must also be started at the same time.

3.9.5 DemRatio

The following configuration of the Diag_Extract file is required in order to ensure a full and proper generation of the DemRatio container.

- **Diagnosticlumpr:**
    - Reference to the DemEvent
    - AdminData/Sdg: Gid = RB:DEXT
    - AdminData/Sdg/SdgContents/<sds>Sd: Gid = RATIO-KIND, Value = API/OBSERVER
- **DiagnosticlumprGroup:**
    - Give DemIUMPRGroup value as defined by AUTOSAR
    - Reference to the Diagnosticlumpr within the same group
- **DiagnosticlumprDenominatorGroup:**
    - Refer to Diagnosticlumpr with the same Denominator Group
    - Give the Category to be one of the following:
        - IUMPR_DENOMINATOR_500_MILES
        - IUMPR_DENOMINATOR_COLDSTART
        - IUMPR_DENOMINATOR_EVAP
        - IUMPR_DENOMINATOR_NONE
        - IUMPR_DENOMINATOR_PHYSICAL_API

- **DiagnosticCommonElementRefConditional:**
    - Reference to all of above configuration

3.9.6 RTE Configuration

3.9.7 Required Integration for Dem TEXT-VALUE-SPECIFICATION

Due to a limitation in Dem, Rte may raise the following error or similar:

E002205: file: .\BSW\ecu_config\bsw\gen\swcd\Dem_Cfg_SWCD.arxml, line: 4864

Unable to identify an enumeration value for the text label 'DemConf_DemEventParameter_CANSM_E_BUS_OFF_NETWORK_0' specified in a ValueSpecification as this did not match any CompuConst.vt, shortLabel or symbol of any CompuScale in any relevant CompuMethod.

Extra integration is required to successfully run RTEGen. Add the following inclusions into Rte_UserCfg.h:

- #include "Dem_Cfg_Client.h"
- #include "Dem_Cfg_EventId.h"
- #include "Dem_Cfg_OperationCycle.h"

And add the following as a command during RTE Gen:

- –text-value-spec-policy=symbolic-pdav-always

3.9.8 Multi-Partition Handling

The purpose of a multi-partition setup is to primarily reduce the load of the master partition. Therefore the following APIs can be mapped to specific Ecuc partitions:

- Dem_SetEventStatus
- Dem_SetEventDisabled
- Dem_ResetEventStatus
- Dem_PrestoreFreezeFrame
- Dem_ClearPrestoredFreezeFrame
- Dem_ResetMonitorStatus
- Dem_ResetEventDebounceStatus

In order to use the multi-partition option, additional integration steps are required inside of Dem.

Ecuc Assignment

In the EcucPartitionCollection the partitions can be added and configured.

Valid Ecuc Assignment

To enable the multi-partition feature, all the valid Dem partitions are required to be configured in the *DemRbEcucPartitionRef*, as the *DemRbEcucPartitionRef* is a sub-set of the EcucPartitionCollection.

Master Partition Assignment

The master partition is set using *DemRbMasterEcucPartitionRef*.

The configured value of *DemRbMasterEcucPartitionRef* needs to be a valid subset of the *DemRbEcucPartitionRef*.

Event Assignment

Each event can be mapped to one partition using the *DemRbEventEcucPartitionRef*.

The configured value of *DemRbEcucPartitionRef* needs to be a valid subset of the *DemRbEcucPartitionRef*.

If an event is not mapped to a partition, then this will be automatically assigned to the master partition.

3.9.8.1 OS Resources

The number of the active OS resources, and their usage, depends on the user configuration of the AUTOSAR System and Basic Software. It is the user's responsibility to ensure that the `ENTER` and `EXIT` resource API are implemented to disable/enable interrupts as required. Alternatively they must be configured in the OS with the correct task/ISR allocation.

The following is a list of all exclusive areas that this module might use regardless of the user configuration. It is the job of the integrator to ensure that the implementation of these exclusive areas is correctly defined. This implementation can be done using the RTE exclusive area configuration wizard or by manual implementation of the corresponding BSW API calls to ensure exclusivity.

The Dem module uses the following OS resources:

- Dem_Monitoring

3.10 Hardware Requirements

Table 3.348.Dem Hardware Requirements

Support	Usage
FLASH	29562 bytes
RAM	5912 bytes

Usage calculated based on a sample configuration.

4 Diagnostic Log and Trace

Module name	Dlt
Package	RTA-DIAG
AUTOSAR Module ID	55
AUTOSAR Version	23-11

4.1 Introduction

The Diagnostic Log and Trace module (Dlt) is located in the system services abstraction layer, forming part of the diagnostic (RTA-DIAG) stack.

Dlt is a basic software module that handles and stores log and trace messages, which are generated by calling APIs that it provides.

Dlt receives log information from DET, DEM and SW-Cs, and trace information from the RTE. It transmits this data via communication busses (through a PduR interface) to make this information visible outside the ECU.

4.1.1 Interaction/Dependencies

The diagram in the RTA-DIAG Introduction section shows the position of Dlt (and the other RTA-DIAG modules) in the Autosar layered architecture. The interaction/dependencies of Dlt with other modules is summarised here.

Run-Time Environment (RTE)

- The RTE (including the VFB and the BSW Scheduler) is used to interact with SW-Cs to generate Log and Trace messages and to cyclically call the Dlt module's Tx function.

Default Error Tracer (Det)

- Dlt interacts with Det in order to report default errors and to forward DET errors to the communication bus. Dlt's interaction with Det is optional.

Diagnostic Event Manager (Dem)

- Dlt's interaction with Dem was removed in AR 4.4.0.

NVRAM Manager (Nvm)

- The NvM module can be optionally used to persistently store updated filter settings and/or Log Channel assignments of the Dlt module. This enables the ECU to transmit log/trace information with the desired level without the need of an explicit setup request coming from the communication bus (via a logging tool) at every ECU startup.

Protocol Data Unit Router (PduR)

- Dlt interacts with the PDU Router to transmit Dlt messages on the communication bus.

Diagnostic Communication Manager (Dcm)

- Dlt uses Dcm as a communication interface. The diagnostic services ReadDataByIdentifier(), WriteDataByIdentifier() and ResponseOnEvent() are used. Dlt uses the security mechanisms of Dcm.

General Purpose Timer (GPT)

- The Gpt module can be optionally used to derive a time stamp.

rba_DiagLib

- The rba_Diaglib provides common functionality and utils for all RTA-DIAG modules, including Dlt.

The following diagram summarises Dlt's interaction and dependencies with other modules.

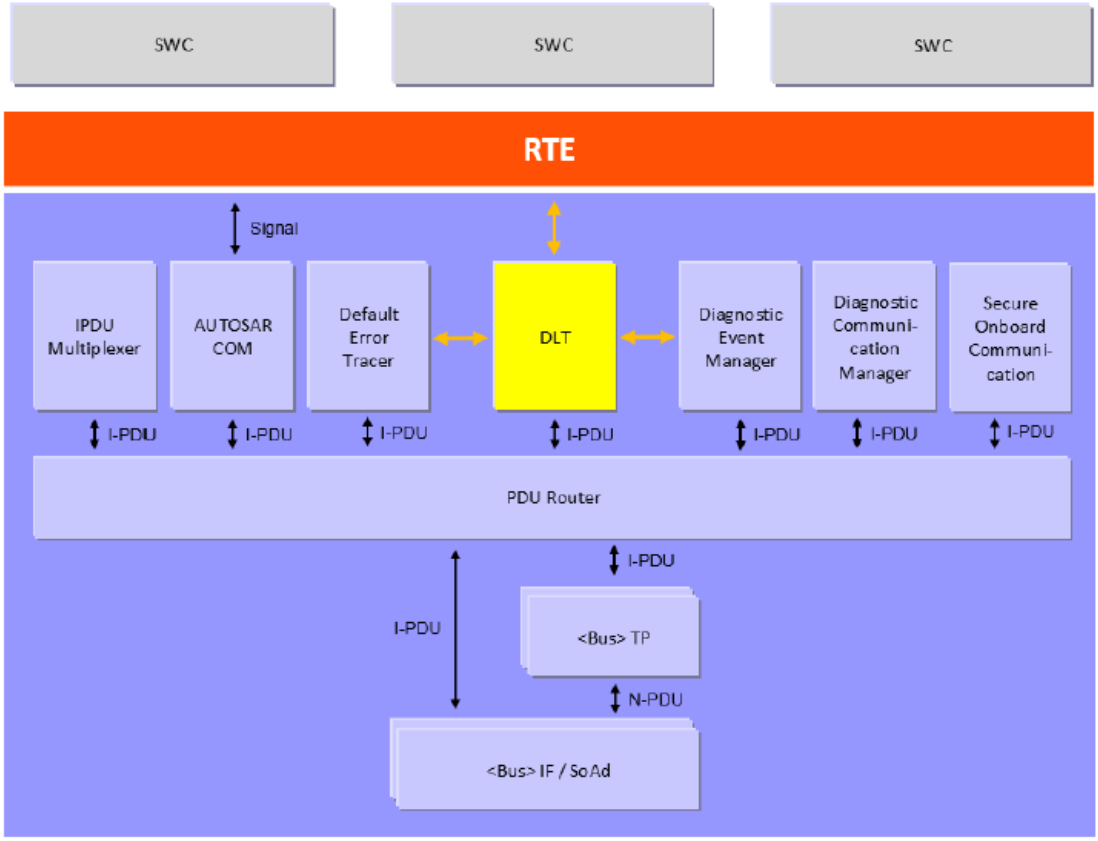


Figure 4.1.Dlt Interaction and Dependencies

4.1.2 Autosar 4.3

The Dlt module is based on Autosar 4.3 - this is an AR43 implementation that is embedded in a AR42 scheme.

Please use the AR43 specification of Dlt as the reference document for this module. Requirements, Design, Code and Tests of Dlt AR42 have been completely replaced by this new AR43 Dlt version. This also includes the configuration, as AR43 Paramdef is used but the schema stays AR42 for compatibility.

This implementation does not yet fully support the feature set of Dlt AR43, see the Limitations section for more details.

Module Configuration

This module is not currently supported by the RTA-BSW Configuration Generator (ConfGen) and must be manually configured.

4.2 Supported Features

Dlt provides the following functionalities:

1. Logging

   - The logging of messages from Det and Dem, and the logging of errors, warnings and info messages from SWCs.
   - Log levels define a classification for the severity grade of each Log Message.
   - A log channel (physical communication bus) is used to transmit Dlt messages.

2. Tracing

   - Receipt of trace messages from SW-Cs and trace information from the RTE.

3. Dlt commands

   - The Dlt Protocol specifies various Dlt Commands which are identified by unique Service IDs. The Dlt Commands are used to modify the behaviour of the Dlt module at runtime, e.g., fetching information about the current Dlt configuration or altering filter settings.

4.3 Extensions

4.3.1 Configuration Extensions

The following configuration parameters are available in version v12.7.0 in addition to those defined in AR v22-11

4.3.1.1 Dlt/DltConfigSet/DltLogLevelSetting/DltRbLogLevelThreshold

Short Name	DltRbLogLevelThreshold
Description	This container contains a preconfiguration of ApplicationId / ContextId pairs and their assigned log level.
Multiplicity	0..*

Default Value	-
Type	ECUC-PARAM-CONF-CONTAINER-DEF

4.3.1.2 Dlt/DltConfigSet/DltLogLevelSetting/DltRbLogLevelThreshold/DltRbLogLevelThresholdSwcContextRef

Short Name	DltRbLogLevelThresholdSwcContextRef
Description	Reference to an ApplicationId/ContextId pair to which a DltLogLevelThreshold is assigned.
Multiplicity	1..1
Default Value	-
Type	ECUC-REFERENCE-DEF

4.3.1.3 Dlt/DltConfigSet/DltLogLevelSetting/DltRbLogLevelThreshold/DltRbThreshold

Short Name	DltRbThreshold
Description	This is the effective log level used for given AppicationId and ContextId
Multiplicity	1..1
Default Value	-
Enumeration Literals	DLT_LOG_DEBUG, DLT_LOG_ERROR, DLT_LOG_FATAL, DLT_LOG_INFO, DLT_LOG_OFF, DLT_LOG_VERBOSE, DLT_LOG_WARN
Type	ECUC-ENUMERATION-PARAM-DEF

4.3.1.4 Dlt/DltGeneral/DltRbGeneral

Short Name	DltRbGeneral
Description	This container contains RB specific configuration parameters.
Multiplicity	1..1
Default Value	-
Type	ECUC-PARAM-CONF-CONTAINER-DEF

4.3.1.5 Dlt/DltGeneral/DltRbGeneral/DltRbComSupportSoADPduHeaderEnable

Short Name	DltRbComSupportSoADPduHeaderEnable
Description	This parameter is used to indicate whether PDU Header is enabled or not when SoAD is used as transport protocol. Set to true if PDU header is enabled, Set to false when PDU header is disabled e.g, when DltViewer is used.
Multiplicity	0..1

Default Value	true
Type	ECUC-BOOLEAN-PARAM-DEF

4.3.1.6 Dlt/DltGeneral/DltRbGeneral/DltRbImplementVfbTrace

Short Name	DltRbImplementVfbTrace
Description	This parameter is used to indicate whether VFB trace is used or not.It is used to enable the VFB trace functionality
Multiplicity	0..1
Default Value	false
Type	ECUC-BOOLEAN-PARAM-DEF

4.3.1.7 Dlt/DltGeneral/DltRbGeneral/DltRbMaxControlSize

Short Name	DltRbMaxControlSize
Description	This RB-specific configuration parameter specifies the maximum size in bytes of a control message that DLT can receive. This parameter shall only be updated if the project uses DLT viewer and Trace message is also sent along with Log Message.
Multiplicity	0..1
Value Range	0..254
Default Value	43
Type	ECUC-INTEGER-PARAM-DEF

4.3.1.8 Dlt/DltGeneral/DltRbGeneral/DltRbTaskTime

Short Name	DltRbTaskTime
Description	This RB-specific configuration parameter specifies Dlt task time. It allows to configure the time for the periodic cyclic task. Please note: this configuration value shall be equal to the value in the Basic Software Scheduler configuration of the RTE module. The AUTOSAR configuration standard is to use SI units, so this parameter is defined as float value in seconds. Exactly one task time must be specified per configuration.
Multiplicity	1..1
Value Range	0.001..0.1
Default Value	-
Type	ECUC-FLOAT-PARAM-DEF

4.3.1.9 Dlt/DltGeneral/DltRbGeneral/DltRbVFBHooks_ArrayLength

Short Name	DltRbVFBHooks_ArrayLength
Description	Size of buffer in Byte. Hint: To be able to trace the Hook Parameters, an array is needed, The length of this array which is common for all hooks, Is The maximum value between all the hook functions parameters Length. Example Hoohfunction1(Parameter1,Parameter2), hookfunction2(Parameter3, Parameter4) DltVFBHooks_ArrayLength = MAX of((-Sizeof(Parameter1) + Sizeof(Parameter2)) and ((Sizeof(Parameter1) + Sizeof(Parameter2)))), Sizeof(Parameter1) = 10 , Sizeof(Parameter2) = 5 , Sizeof(Parameter3)=10 , Sizeof(Parameter4)=6 then DltVFBHooks_ArrayLength = 16 .
Multiplicity	0..1
Value Range	1..65535
Default Value	1
Type	ECUC-INTEGER-PARAM-DEF

4.3.1.10 Dlt/DltGeneral/DltRbGeneral/DltRbVfbTrace

Short Name	DltRbVfbTrace
Description	All functions to trace from the VFB by the Dlt.
Multiplicity	0..*
Default Value	-
Type	ECUC-PARAM-CONF-CONTAINER-DEF

4.3.1.11 Dlt/DltGeneral/DltRbGeneral/DltRbVfbTrace/Activate

Short Name	Activate
Description	Active or deactive the hook function.
Multiplicity	1..1
Default Value	-
Type	ECUC-BOOLEAN-PARAM-DEF

4.3.1.12 Dlt/DltGeneral/DltRbGeneral/DltRbVfbTrace/ArgumentParameter

Short Name	ArgumentParameter
Description	Hook functions parameter

Introduction	Note that this container definition does not explicitly define a symbolic name parameter. Instead, the short name of the container will be used in the Ecu Configuration Description to specify the symbolic name of the SessionId Parameter.
Multiplicity	0..20
Default Value	-
Type	ECUC-PARAM-CONF-CONTAINER-DEF

4.3.1.13 Dlt/DltGeneral/DltRbGeneral/DltRbVfbTrace/ArgumentParameter/ArrayLength

Short Name	ArrayLength
Description	Lenght of the Array .
Introduction	Unit: byte
Multiplicity	0..1
Value Range	0..32767
Default Value	-
Type	ECUC-INTEGER-PARAM-DEF

4.3.1.14 Dlt/DltGeneral/DltRbGeneral/DltRbVfbTrace/ArgumentParameter/Direction

Short Name	Direction
Description	Determines the Direction of the parameter.
Multiplicity	1..1
Default Value	-
Enumeration Literals	IN, OUT, INOUT
Type	ECUC-ENUMERATION-PARAM-DEF

4.3.1.15 Dlt/DltGeneral/DltRbGeneral/DltRbVfbTrace/ArgumentParameter/ImplementationDataType

Short Name	ImplementationDataType
Description	Type Of the Parameter e.g. unit8, uint16, uint32 etc.
Multiplicity	1..1
Default Value	-
Type	ECUC-STRING-PARAM-DEF

4.3.1.16 Dlt/DltGeneral/DltRbGeneral/DltRbVfbTrace/ArgumentParameter/ImplementationDataTypeCategory

Short Name	ImplementationDataTypeCategory
Description	Implementation Type Category.
Multiplicity	1..1
Default Value	-
Type	ECUC-STRING-PARAM-DEF

4.3.1.17 Dlt/DltGeneral/DltRbGeneral/DltRbVfbTrace/DltRbBswModuleEntryRef

Short Name	DltRbBswModuleEntryRef
Description	Foreign reference to the BSWModuleEntry describing the trace function implementation.
Multiplicity	0..1
Default Value	-
Type	ECUC-FOREIGN-REFERENCE-DEF

4.3.1.18 Dlt/DltGeneral/DltRbGeneral/DltRbVfbTrace/DltRbVfbHookContexID

Short Name	DltRbVfbHookContexID
Description	4 letter as ContexId, this ContextID will be send to the Viewr and ApplicationId is always “VFBT”.
Introduction	Example: “ESP1”
Multiplicity	0..1
Default Value	-
Type	ECUC-FUNCTION-NAME-DEF

4.3.1.19 Dlt/DltGeneral/DltRbGeneral/DltRbVfbTrace/DltRbVfbTraceFunction

Short Name	DltRbVfbTraceFunction
Description	The Dlt generator shall enable VFB tracing for a given hook function when there is a #define in the Dlt-VFB configuration header file for the hook function name and tracing is globally enabled.

Introduction	Example: #define Dlt_Rte WriteHook i1 p1 a Start. Also the corresponding function shall be generated. The exact argument description for the function is to take from the provieded BSWModudulDescription from the RTE module.
Multiplicity	1..1
Default Value	-
Type	ECUC-FUNCTION-NAME-DEF

4.3.2 Additional Services

The following API are available in version v12.7.0 in addition to those defined in AR v22-11

Syntax	`DltMainFunction`
Description	Replaces the Autosar Dlt_TxFunction.
Note	All requirements that apply to Dlt_TxFunction are instead covered by Dlt_Main-Function.

4.4 Limitations

- Interaction with NvM for persistent storing (via the DltImplementNVRam-Storage parameter) is not currently supported.

- Currently only a subset of control messages is implemented. Control Message GetLogInfo is currently implemented in such way that it returns log information for a requested AppId/ContextId tuple. ECU and Dlt Viewer must have the same endianess for correct interpretation of control messages.

- Because of missing control messages, runtime configuration is not fully supported.

- Currently callbacks to inform SW-C about status changes are not implemented.

- Traffic Shaping is not supported.

- The Dem interface that triggers Dlt on changes of the event related data in the event memory (Dlt_DemTriggerOnEventData) is not supported.

- Context Registration/Unregistration (via the Dlt_RegisterContext and Dlt_UnregisterContext functions) is not supported.

- The CANIF interface (Tx data Patch) is not supported because Dlt log messages are always bigger than 8 bytes.

- Currently VFB-Trace is implemented but the session ID being passed to the hook functions is kept as zero, because the session ID is a port generated value and cannot be extracted seperately within the hook functions.

- The Dlt Injection feature is currently tested and verified for only Non -RTE projects

4.5 Exclusions

The following features from AR v22-11 are not supported in version v12.7.0.

API functions not currently implemented:

- `Dlt_RegisterContext`
- `Dlt_StoreConfiguration`
- `Dlt_TxFunction`
- `Dlt_UnregisterContext`
- `Dlt_InjectCall_<SESSION>`
- `Dlt_SendLogMessageWithAttributes`
- `Dlt_SendTraceMessageWithAttributes`

The following configuration parameters from AR v22-11 are not supported in version v12.7.0.

- Dlt/DltConfigSet/DltLogLevelSetting/DltLogLevelThreshold
- Dlt/DltConfigSet/DltLogLevelSetting/DltLogLevelThreshold/DltLogLevelThresholdSwcContextRef
- Dlt/DltConfigSet/DltLogLevelSetting/DltLogLevelThreshold/DltThreshold
- Dlt/DltConfigSet/DltLogOutput/DltLogChannel/DltLogChannelBufferOverflowTimer
- Dlt/DltConfigSet/DltLogOutput/DltLogChannel/DltLogChannelSegmentationSupported
- Dlt/DltConfigSet/DltLogOutput/DltLogChannel/DltLogChannelTrafficShapingBandwidth
- Dlt/DltConfigSet/DltLogOutput/DltLogChannel/DltTxPdu/DltTxPduUsesTp
- Dlt/DltConfigSet/DltRxPdu/DltRxPduUsesTp
- Dlt/DltGeneral/DltGeneralStartUpDelayTimer
- Dlt/DltGeneral/DltGeneralTimeStampSupport
- Dlt/DltGeneral/DltGeneralTrafficShapingSupport
- Dlt/DltGeneral/DltMaxNumberOfChannels
- Dlt/DltGeneral/DltProtocolVersion

4.6 Deviations

4.6.1 Autosar Deviations

The following deviations should be noted with respect to the stated Autosar requirements.

4.6.1.1 Log level and log threshold handling

AUTOSAR specifies that messages with a log level lower than the value of a log level threshold shall be discarded, but the list of log levels given in [SWS_Dlt_00230] indicates that the highest priority log levels have the lowest value. Therefore -

- If DLT_LOG_FATAL messages need to be seen, setting the log level threshold to DLT_LOG_FATAL would always also show informational messages.
- DLT_LOG_OFF would not be possible to reach, as all other log levels would still be active.

For these reasons, only those messages that have a log level value that is lower than the value of the log level threshold are logged. This will result in a proper DLT_LOG_OFF, and if you are only interested in DLT_LOG_ERROR, no log messages other than DLT_LOG_ERROR and DLT_LOG_FATAL will be shown.

4.6.1.2 Handling of messages that can't be copied completely into a buffer

[SWS_Dlt_00670] and [SWS_Dlt_00672] state that DLT_E_NO_BUFFER is returned if no assigned LogChannel buffer is available to store the complete message, and if at least one LogChannel buffer is available, then E_OK is returned.

It's unlikely that a situation will occur where no buffer is available to store the message and, in any event, the method of returning E_OK even though an overflow occurred is not robust.

For this reason, Dlt always returns DLT_E_NO_BUFFER if at least one assigned LogChannel buffer is unable to store the complete message, and it returns E_OK only if the message has been successfully copied to all assigned LogChannel buffers.

4.6.1.3 Message counters

[SWS_Dlt_00671] states that the message counter shall be increased after the message is copied to an assigned LogChannel buffer. If more than one LogChannel buffer is in use, this will result in gaps because a message can be copied to several buffers.

To avoid such situations, the message counter is set when the header is assembled. This will happen before a message is copied to any buffer. So, independent of how many buffers exist, each buffer contains messages with a consistent incremented counter. This has the benefit that the message counter can be used

to identify a message regardless of the buffer in which it is contained.

The transmit counter described in [SWS_Dlt_00677] is not implemented because the use case is not clear, and because there is no information given about ranges or fault behavior (e.g. retry).

4.6.1.4 Verbose / Non-Verbose

The UEH bit described in [SWS_Dlt_00678] cannot be handled because setting the bit to 1 or 0 is also influenced by the caller of Dlt_SendLogMessage, and it's not valid to simply set the bit to 0 if DltUseExtHeaderInNonVerbMode is set to FALSE.

Instead, Dlt will always set the UEH bit to 1 if DltUseExtHeaderInNonVerbMode is configured TRUE to use an extended header for non-verbose mode messages, but if DltUseExtHeaderInNonVerbMode is set to FALSE, Dlt will decide depending on how the UEH bit is set by the caller.

Other than specified by [SWS_Dlt_00692] the VERB bit is not modified by the configuration parameter DltUseExtHeaderInNonVerbMode.

4.6.2 Configuration Deviations

Configuration parameters which differ from the AR v22-11 specification are listed below.

4.6.2.1 Dlt

[REMOVED] Description	Configuration of the Dlt (Log&Trace) module.

4.6.2.2 Dlt/DltConfigSet/DltLogOutput/DltLogChannel/DltLogChannel-BufferSize

[RTA-BSW] Maximum Value	65534
[AUTOSAR] Maximum Value	4294967295

4.6.2.3 Dlt/DltConfigSet/DltLogOutput/DltLogChannel/DltLogChannel-MaxMessageLength

[RTA-BSW] Maximum Value	65534
[AUTOSAR] Maximum Value	65535

4.6.2.4 Dlt/DltConfigSet/DltLogOutput/DltLogChannel/DltLogChannel-MaxNumOfRetries

[RTA-BSW] Description	The maximum length of a Dlt log or trace message.
[AUTOSAR] Description	The maximum amount of retries for sending a Dlt log or trace message.

4.6.2.5 Dlt/DltConfigSet/DltLogOutput/DltLogChannel/DltLogTraceSta-

tusFlagDltLogTraceStatusFlag

[RTA-BSW] Description	Parameter to turn on/off on this LogChannel completely.
[AUTOSAR] Description	Parameter to turn on/off tracing on this LogChannel completely.

4.6.2.6 Dlt/DltConfigSet/DltProtocol/DltHeaderUseEcuId

[RTA-BSW] Introduction	If the parameter DltGeneralNvRamRef is used this parameter defines the initial value for the corresponding NVRam entry. If the parameter DltGeneralNvRamRef is not set, Link-Time or Post-Build configuration shall be used.
[AUTOSAR] Introduction	If DltGeneralNvRAMSupport is enabled the value of the parameter defined here is also the initial value for the corresponding NvRam entry. If DltGeneralNvRAMSupport is not set, Link-Time or Post-Build configuration shall be used.

4.6.2.7 Dlt/DltConfigSet/DltProtocol/DltHeaderUseSessionID

[RTA-BSW] Introduction	If the parameter DltGeneralNvRamRef is used this parameter defines the initial value for the corresponding NVRam entry. If the parameter DltGeneralNvRamRef is not set, Link-Time or Post-Build configuration shall be used.
[AUTOSAR] Introduction	If DltGeneralNvRAMSupport is enabled the value of the parameter defined here is also the initial value for the corresponding NvRam entry. If DltGeneralNvRAMSupport is not set, Link-Time or Post-Build configuration shall be used.

4.6.2.8 Dlt/DltConfigSet/DltProtocol/DltHeaderUseTimestamp

[RTA-BSW] Introduction	If the parameter DltGeneralNvRamRef is used this parameter defines the initial value for the corresponding NVRam entry. If the parameter DltGeneralNvRamRef is not set, Link-Time or Post-Build configuration shall be used.
[AUTOSAR] Introduction	If DltGeneralNvRAMSupport is enabled the value of the parameter defined here is also the initial value for the corresponding NvRam entry. If DltGeneralNvRAMSupport is not set, Link-Time or Post-Build configuration shall be used.

4.6.2.9 Dlt/DltConfigSet/DltProtocol/DltUseExtHeaderInNonVerbMode

[RTA-BSW] Introduction	If DltGeneralNvRAMSupport is enabled this parameter is the initial value for the corresponding NVRam entry. If DltGeneralNvRAMSupport is not set, Link-Time or Post-Build configuration shall be used.

[AUTOSAR] Introduction	If DltGeneralNvRAMSupport is enabled the value of the parameter defined here is also the initial value for the corresponding NvRam entry. If DltGeneralNvRAMSupport is not set, Link-Time or Post-Build configuration shall be used.

4.6.2.10 Dlt/DltConfigSet/DltRxPdu/DltRxPduId

[RTA-BSW] Description	The numerical value used as the ID of this I-PDU. The DltRxPduHandleId is required by the API calls Dlt_RxIndication, Dlt_TpRxIndication, Dlt_StartOfReception and Dlt_CopyRxData to receive I-PDUs from the PduR.
[AUTOSAR] Description	The numerical value used as the ID of this I-PDU. The DltRxPduId is required by the API calls Dlt_RxIndication, Dlt_TpRxIndication, Dlt_StartOfReception and Dlt_CopyRxData to receive I-PDUs from the PduR.

4.6.2.11 Dlt/DltGeneral/DltGeneralNvRAMSupport

[ADDED] Introduction	Set to TRUE for enabling the NvRAM support feature or it shall be set to FALSE.

4.6.2.12 Dlt/DltSwc/DltSwcContext/DltSwcApplicationId

[REMOVED] Value Range	0..4294967295

4.6.2.13 Dlt/DltSwc/DltSwcContext/DltSwcContextId

[REMOVED] Value Range	0..4294967295

4.6.2.14 Dlt/DltSwc/MaxSwcLogMessageLength

[RTA-BSW] Description	Defines the maximum allowed length (uint16) for LogMessages. The upper limit for the range of this parameter is currently defined by the range of the data type. The actual upper limit for the range of this parameter is identical to the maximum length of all configured Dlt log or trace messages, which is known when all log or trace messages are configured.
[AUTOSAR] Description	Defines the maximum allowed length (uint16) for LogMessages.
[REMOVED] Introduction	The upper limit for the range of this parameter is currently defined by the range of the data type. The actual upper limit for the range of this parameter is identical to the maximum length of all configured Dlt log or trace messages, which is known when all log or trace messages are configured.

4.6.2.15 Dlt/DltSwc/MaxSwcTraceMessageLength

[RTA-BSW] Description	Defines the maximum allowed length (uint16) for TraceMessages.The upper limit for the range of this parameter is currently defined by the range of the data type. The actual upper limit for the range of this parameter is identical to the maximum length of all configured Dlt log or trace messages, which is known when all log or trace messages are configured.
[AUTOSAR] Description	Defines the maximum allowed length (uint16) for TraceMessages.
[REMOVED] Introduction	The upper limit for the range of this parameter is currently defined by the range of the data type. The actual upper limit for the range of this parameter is identical to the maximum length of all configured Dlt log or trace messages, which is known when all log or trace messages are configured.

4.7 API Definition

Table 4.37.Dlt Supported API functions

`Dlt_Init`	Initialises the Dlt module.
`Dlt_CopyRxData`	Provides the received data of an I-PDU segment (N-PDU) to the upper layer.
`Dlt_CopyTxData`	Called to acquire the transmit data of an I-PDU segment (N-PDU).
`Dlt_GetDefaultLogLevel`	Returns the Default Log Level currently used by the Dlt module.
`Dlt_GetLogChannelNames`	Returns all configured LogChannel names.
`Dlt_GetLogChannelThreshold`	Returns the filter threshold for the given LogChannel.
`Dlt_GetLogInfo`	Called to request information about registered Application Ids and their ContextIds and the corresponding log level.
`Dlt_GetVersionInfo`	Returns the version information of this module.
`Dlt_ResetToFactoryDefault`	Sets the LogLevel and TraceStatus back to the persistently stored default values.
`Dlt_RxIndication`	Indication of a received PDU from a lower layer communication interface module.
`Dlt_SendLogMessage`	Used by basic software modules or by software component to send log messages.
`Dlt_SetDefaultLogLevel`	Modifies the pass-through range for Log Messages for all not explicit set ContextIds.
`Dlt_SetLogChannelAssignment`	Adds/removes the addressed tuple ApplicationId/-ContextId to/from the addressed LogChannel.
`Dlt_SetLogChannelThreshold`	Sets the filter threshold for the given LogChannel.

`Dlt_SetLogLevel`	Changes the LogLevel for the given tuple of ApplicationID/ContextID.
`Dlt_SetMessageFiltering`	Switches on/off the message filtering functionality of the Dlt module.
`Dlt_StartOfReception`	Called at the start of receiving an N-SDU.
`Dlt_TpRxIndication`	Called after an I-PDU has been received via the TP API and the result indicates whether the transmission was successful or not.
`Dlt_TpTxConfirmation`	Called after the I-PDU has been transmitted on its network and the result indicates whether the transmission was successful or not.
`Dlt_TriggerTransmit`	The upper layer module (called module) checks whether the available data fits into the buffer size reported by PduInfoPtr→SduLength.
`Dlt_TxConfirmation`	The lower layer communication interface module confirms the transmission of a PDU or the failure to transmit a PD.
`Dlt_DetForwardErrorTrace`	Service to forward error reports from Det to Dlt.
`Dlt_SendTraceMessage`	Service interface used by basic software modules or by software components to trace parameters.
`Dlt_SetTraceStatus`	Sets the trace status for a specific tuple of ApplicationID and ContextID.
`Dlt_SetDefaultTraceStatus`	Called to enable or disable trace messages for all not explicitly set ContextIds.
`Dlt_GetDefaultTraceStatus`	Returns the current Trace Status of the addressed LogChannel.
`Dlt_GetTraceStatus`	Returns the current Trace Status for a given tuple ApplicationId/ContextId.

For further information on these functions, please refer to AUTOSAR_SWS_DiagnosticLogAndTrace.pdf

4.8 Configuration Advice

NOTE

Before configuring this module, please note that some of the default values used by ConfGen to produce the default ECU Configuration can be changed, and this is done by either adding a Conf-Gen Enhancer Mapping (CEM) or by adding a rule to an *algo.properties* file. For more information, see the “Configuration Generation” section of the RTA-CAR User Guide.

4.8.1 Wildcard configuration

The Ecuc parameter DltSwcContextId should be configured as 0000 to configure

wildcard contextid.

4.8.2 GPT configuration

GPT must be configured with channel frequency 10KHZ to get 0.1ms resolution. Currently only 24bit or 32bit GPT timer is supported.

4.8.3 Post-Build

Post build is not supported in Dlt. PduR related configuration must be done in the Common Variant.

4.9 Integration Advice

4.9.1 Init Functions

The Dlt module includes several configuration parameters that can be reconfigured during runtime via API calls or via Dlt control messages. Related data structures have to be loaded into runtime variables during the startup of the Dlt module.

It's possible that SW-Cs and/or BSW modules are already generating log and trace data even though the Dlt module itself has not yet been initialized. In this scenario, the Dlt module can optionally buffer this data until it is fully initialized.

4.9.2 DeInit Functions

No specific handling necessary.

4.9.3 Scheduled Functions

Dlt_MainFunction has to be scheduled cyclically. The calling period has to be the same as configured in the Dlt EcuC-values file (DltRbTaskTime).

4.9.4 Dlt_TxFunction

The cyclically triggered Dlt_TxFunction is permanently replaced with Dlt_MainFunction. All requirements that apply to Dlt_TxFunction are instead covered by Dlt_MainFunction.

4.9.4.1 OS Resources

The number of the active OS resources, and their usage, depends on the user configuration of the AUTOSAR System and Basic Software. It is the user's responsibility to ensure that the `ENTER` and `EXIT` resource API are implemented to disable/enable interrupts as required. Alternatively they must be configured in the OS with the correct task/ISR allocation.

The following is a list of all exclusive areas that this module might use regardless of the user configuration. It is the job of the integrator to ensure that the implementation of these exclusive areas is correctly defined. This implementation can be done using the RTE exclusive area configuration wizard or by manual implementation of the corresponding BSW API calls to ensure exclusivity.

The Dlt module uses the following OS resources:

- Dlt_Buffer
- Dlt_ServiceDispatcher

4.10 Hardware Requirements

Table 4.38.Dlt Hardware Requirements

Support	Usage
FLASH	9496 bytes
RAM	14254 bytes

Usage calculated based on a sample configuration.

5 Function Inhibition Manager

Module name	Fim
Package	RTA-DIAG
AUTOSAR Module ID	11
AUTOSAR Version	23-11

5.1 Introduction

The Function Inhibition Manager (FiM) is located in the system services abstraction layer, forming part of the diagnostic (RTA-DIAG) stack.

FiM provides a control mechanism for functionality within software components. In this context, a “functionality” can be built up of the contents of one, several or parts of runnable entities with the same set of permission/inhibit conditions.

Through FiM, the inhibiting (i.e. the enabling/disabling) of functionalities can be configured and even modified during runtime (in post-built configuration).

Although FiM’s focus is on functionalities in SW-Cs, the functionalities of the BSW can also use FiM services.

A functionality is assigned to a Function Identifier (FID) along with the inhibit conditions for that particular identifier. Functionalities poll for the permission state of their respective FIDs before execution. If an inhibit condition is TRUE for a particular identifier, the corresponding functionality will be disabled.

5.1.1 Interaction/Dependencies

The diagram in the RTA-DIAG Introduction section shows the position of FiM (and the other RTA-DIAG modules) in the Autosar layered architecture. The interaction/dependencies of FiM with other modules is summarised here.

Diagnostic Event Manager (Dem)

- Dem handles detected malfunctions (events) and informs and updates FiM on changes of the event status. This is done in order to stop or release functionalities according to assigned dependencies.

SW-Components (SW-C)

- Software components with a FID interface will query for permission to execute certain functionality, as identified by a FID. The FIDs have to be provided by the SW components. The diagram below shows the interaction between FiM, Dem and SW-Cs.

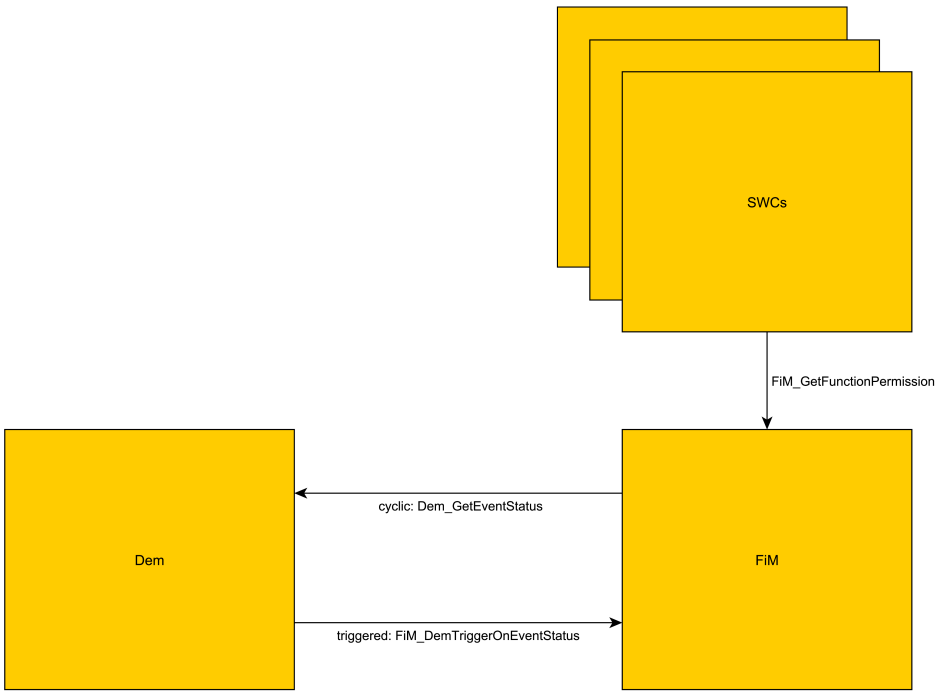


Figure 5.1.FiM Interaction

BSW modules

- Any BSW modules that are to have functionalities inhibited by FiM will use the same FiM interface as mentioned above for SW-Cs. These modules have to provide the corresponding configuration data (EventID, FID, Inhibition Mask relation) at configuration time.

rba_DiagLib

- The rba_Diaglib provides common functionality and utils for all RTA-DIAG modules, including FiM.

Module Configuration

This module is not currently supported by the RTA-BSW Configuration Generator (ConfGen) and must be manually configured.

5.2 Supported Features

- Provides an inhibition mechanism for functionalities within SW-Cs.
- Support also for inhibition of BSW functionalities.
- Runtime configuration/modification of inhibition (in post-built configuration).
- Choice of Triggered Operation mode or Cyclic Operation mode.
- Multi-partition feature to split the FiM computation load on multiple partitions.

5.3 Extensions

5.3.1 Configuration Extensions

The following configuration parameters are available in version v12.7.0 in addition to those defined in AR v22-11

5.3.1.1 FiM/FiMConfigSet/FiMFID/FiMRbDoc

Short Name	FiMRbDoc
Long Name	Documentation
Description	This container contains the documentation for the FiMFID.
Multiplicity	0..1
Default Value	-
Type	ECUC-PARAM-CONF-CONTAINER-DEF

5.3.1.2 FiM/FiMConfigSet/FiMFID/FiMRbDoc/FiMRbDocSubstReaction

Short Name	FiMRbDocSubstReaction
Long Name	Substitute function
Description	Verbal description of the substitute function.
Multiplicity	0..1
Default Value	-
Type	ECUC-ADD-INFO-PARAM-DEF

5.3.1.3 FiM/FiMConfigSet/FiMFID/FiMRbFIDDescription

Short Name	FiMRbFIDDescription
Description	The purpose of the specified FiMFID shall be described.
Multiplicity	0..1
Default Value	-

Type	ECUC-STRING-PARAM-DEF

5.3.1.4 FiM/FiMConfigSet/FiMFID/FiMRbFIdSequence

Short Name	FiMRbFIdSequence
Description	Sequence Number of a FimFunctionId. This configuration specifies the sequence of FID's and must be unique. This must be configured if FiMRbFid-SortMethod is set as Sequence number.
Multiplicity	0..1
Value Range	1..65535
Default Value	-
Type	ECUC-INTEGER-PARAM-DEF

5.3.1.5 FiM/FiMConfigSet/FiMFID/FiMRbFunctionIdEcucPartitionRef

Short Name	FiMRbFunctionIdEcucPartitionRef
Description	Maps this FunctionId to one EcuC partition.
Multiplicity	0..1
Default Value	-
Type	ECUC-SYMBOLIC-NAME-REFERENCE-DEF

5.3.1.6 FiM/FiMConfigSet/FiMFID/FiMRbProvidingRef

Short Name	FiMRbProvidingRef
Description	This parameter is added for DGS FiM. It refereces to the Dem events defined by this FiMFunction.
Multiplicity	0..*
Default Value	-
Type	ECUC-REFERENCE-DEF

5.3.1.7 FiM/FiMConfigSet/FiMFID/FiMRbShiftView

Short Name	FiMRbShiftView

Description	This parameter is added for DGS FiM. It specifies where to arrange calibration parameters of the FiMFID in asap file (shiftview). DGS DSM creates diagnostic calibration parameters for fids. If your module has own calibration parameters then give your assigned function name in a2l-file and DGS-DSM will shift the diagnostic calibration parameters to this a2l function. Otherwise define a function name for the diagnostic calibration, only.
Multiplicity	0..1
Default Value	-
Type	ECUC-STRING-PARAM-DEF

5.3.1.8 FiM/FiMConfigSet/FiMSummaryEvent/FiMRbSummaryEventDescription

Short Name	FiMRbSummaryEventDescription
Description	The purpose of the specified FiMSummaryEvent shall be described.
Multiplicity	0..1
Default Value	-
Type	ECUC-STRING-PARAM-DEF

5.3.1.9 FiM/FiMConfigSet/FiMSummaryEvent/FiMRbSummaryEventShiftView

Short Name	FiMRbSummaryEventShiftView
Description	This parameter is added for DGS FiM for documentation purpose.
Multiplicity	0..1
Default Value	-
Type	ECUC-STRING-PARAM-DEF

5.3.1.10 FiM/FiMGeneral/FiMRbGeneral

Short Name	FiMRbGeneral
Description	This container contains the RB General configuration parameters.
Multiplicity	1..1
Default Value	-
Type	ECUC-PARAM-CONF-CONTAINER-DEF

5.3.1.11 FiM/FiMGeneral/FiMRbGeneral/FiMRbCallbackClass

Short Name	FiMRbCallbackClass

Description	This container contains the configuration (parameters) for an internal/ external callback class.
Multiplicity	0..1
Default Value	-
Type	ECUC-PARAM-CONF-CONTAINER-DEF

5.3.1.12 FiM/FiMGeneral/FiMRbGeneral/FiMRbCallbackClass/FiMRbExternalCSCallbackClass

Short Name	FiMRbExternalCSCallbackClass
Description	This container contains the configuration (parameters) for an external client/server based callback class. It defines, how the FiM can obtain the callback function from either a SW-C or another BSW module. Whether a client/ server port or a C function-call is used, is defined by FiMRbCallbackUsePort.
Multiplicity	1..1
Default Value	-
Type	ECUC-PARAM-CONF-CONTAINER-DEF

5.3.1.13 FiM/FiMGeneral/FiMRbGeneral/FiMRbCallbackClass/FiMRbExternalCSCallbackClass/FiMRbCallbackReadFnc

Short Name	FiMRbCallbackReadFnc
Description	In case of FiMCallbackUsePort is false, this parameter defines the prototype of the C function
Multiplicity	0..1
Default Value	-
Type	ECUC-FUNCTION-NAME-DEF

5.3.1.14 FiM/FiMGeneral/FiMRbGeneral/FiMRbCallbackClass/FiMRbExternalCSCallbackClass/FiMRbCallbackUsePort

Short Name	FiMRbCallbackUsePort
Description	If the parameter is set to True, a R-Port is generated to obtain the callback. If the parameter is set to False, the information is obtained by C function-call specified by the parameter FiMRbCallbackReadFnc.
Multiplicity	1..1
Default Value	-
Type	ECUC-BOOLEAN-PARAM-DEF

5.3.1.15 FiM/FiMGeneral/FiMRbGeneral/FiMRbEcucPartition

Short Name	FiMRbEcucPartition
Description	Contains references to the EcuC Partition. The EcuC partition referenced are a subset of the EcuC partitions where the FiM is mapped to.
Multiplicity	0..65535
Default Value	-
Type	ECUC-PARAM-CONF-CONTAINER-DEF

5.3.1.16 FiM/FiMGeneral/FiMRbGeneral/FiMRbEcucPartition/FiMRbEcucPartitionRef

Short Name	FiMRbEcucPartitionRef
Description	Maps the FiM to one or multiple EcuC partitions to make its C-APIs available in the according partition.
Multiplicity	1..1
Default Value	-
Type	ECUC-SYMBOLIC-NAME-REFERENCE-DEF

5.3.1.17 FiM/FiMGeneral/FiMRbGeneral/FiMRbEventsPerMonitorCycle

Short Name	FiMRbEventsPerMonitorCycle
Description	This RB-specific configuration parameter specifies the number of Dem events to be processed in a scheduling cycle during monitoring and cyclic event evaluation (in the phase CHECK). If the Parameter FiMEventUpdateTriggeredByDem is set to TRUE, this value is ignored. A value of 0xFFFF indicates that all Dem events shall be processed in one scheduling cycle. The number of events to be processed controls the runtime of the FiM_MainFunction() process. Reasonable values may be 50.100 events per cycle.
Multiplicity	0..1
Value Range	1..65535
Default Value	-
Type	ECUC-INTEGER-PARAM-DEF

5.3.1.18 FiM/FiMGeneral/FiMRbGeneral/FiMRbFidSortMethod

Short Name	FiMRbFidSortMethod

Description	Provides the sorting method for FunctionId. If the FIDs are to be sorted by shortname ,then the SortMethod must be selected as SHORTNAME. If the FIDs are to be sorted by sequence number ,then the SortMethod must be selected as SEQUENCE_NUMBER. If this parameter is not configured,the default value will be taken as SHORTNAME.
Multiplicity	0..1
Default Value	SHORTNAME
Enumeration Literals	SHORTNAME, SEQUENCE_NUMBER
Type	ECUC-ENUMERATION-PARAM-DEF

5.3.1.19 FiM/FiMGeneral/FiMRbGeneral/FiMRbFidsPerMonitorCycle

Short Name	FiMRbFidsPerMonitorCycle
Description	This RB-specific configuration parameter specifies the number of FIds to be processed in a scheduling cycle during monitoring and cyclic event evaluation (in the phase INITIALIZATION and EVALUATION). If the Parameter FiMEventUpdateTriggeredByDem is set to TRUE, this value is ignored. A value of 0xFFFF indicates that all FIds shall be processed in one scheduling cycle. The number of FIds to be processed controls the runtime of the FiM_MainFunction() process. Reasonable values may be 50..100 FIds per cycle
Multiplicity	0..1
Value Range	1..65535
Default Value	-
Type	ECUC-INTEGER-PARAM-DEF

5.3.1.20 FiM/FiMGeneral/FiMRbGeneral/FiMRbFunctionIdType

Short Name	FiMRbFunctionIdType
Description	Provide the implementation data type belonging to a FunctionId. If the number of configured FiM Function IDs is more than 255 ,then the type must be selected as datatype uint16. If the number of configured FiM Function IDs is less than or equal to 255 ,then the data type should be selected as uint8. If this parameter is not configured,the default value will be taken as uint16.
Multiplicity	0..1
Default Value	uint16

Enumeration Literals	UINT8, UINT16
Type	ECUC-ENUMERATION-PARAM-DEF

5.3.1.21 FiM/FiMGeneral/FiMRbGeneral/FiMRbHeaderFileInclusion

Short Name	FiMRbHeaderFileInclusion
Description	Name of the header file(s) to be included by the FiM module containing the used C-callback declarations.
Multiplicity	0..*
Default Value	-
Type	ECUC-STRING-PARAM-DEF

5.3.1.22 FiM/FiMGeneral/FiMRbGeneral/FiMRbMasterEcucPartitionRef

Short Name	FiMRbMasterEcucPartitionRef
Description	Maps the FiM master to one of the EcuC partition to assign the master functionality to a certain core.
Multiplicity	0..1
Default Value	-
Type	ECUC-SYMBOLIC-NAME-REFERENCE-DEF

5.3.1.23 FiM/FiMGeneral/FiMRbGeneral/FiMRbObdServ07VisCalc

Short Name	FiMRbObdServ07VisCalc
Description	This RB-specific configuration parameter gives an option to enable or disable the OBD service $07 visibility calculation for each FID. If enabled, the API FiM_GetService07Visibility will be available.
Multiplicity	0..1
Default Value	false
Type	ECUC-BOOLEAN-PARAM-DEF

5.3.1.24 FiM/FiMGeneral/FiMRbGeneral/FiMRbRteInUse

Short Name	FiMRbRteInUse

Description	The RTE generates some of the APIs which are already provided by FiM itself. This results in redundant redeclarations of these functions, which is wanted to ensure consistency between generated and normal APIs. Important:This parameter needs to be configured ,only if global RTE is on and still component does not support RTE due to project specific reasons. TRUE If this parameter is not set, the default value TRUE is taken into consideration. FALSE If the Rte is not used for the component ,then this parameter needs to be FALSE.
Multiplicity	0..1
Default Value	true
Type	ECUC-BOOLEAN-PARAM-DEF

5.3.2 Additional Services

The following API are available in version v12.7.0 in addition to those defined in AR v22-11

Syntax	`FiM_GetService07Visibility`
Parameters In	`FID` The function ID that is going to be checked for its service $07 visibility. `Visibility` A pointer to a variable that will contain the result of the service07 visibility counter.
Description	Reports the service $07 visible entry for the FID.

5.4 Limitations

- Support of IUMPR handling is not yet available in FiM.
- Post build loadable is not supported, only Post build selectable is supported.
- The FiM_DemTriggerOnComponentStatus function has been modified according to the Dem SWS requirement [SWS_Dem_01116]. Instead, therefore, Dem_GetComponentFailed will be used to fetch the component status.

5.5 Exclusions

The following configuration parameters from AR v22-11 are not supported in version v12.7.0.

- FiM/FiMGeneral/FiMAvailabilitySupport
- FiM/FiMGeneral/FiMMaxInputEventsPerSummaryEvents

- FiM/FiMGeneral/FiMMaxSumEventsPerFidInhibitionConfiguration
- FiM/FiMGeneral/FiMMaxSummaryEvents

5.6 Deviations

5.6.1 Configuration Deviations

Configuration parameters which differ from the AR v22-11 specification are listed below.

5.6.1.1 FiM/FiMConfigSet/FiMInhibitionConfiguration/FiMInhComponentRef

[ADDED] Introduction	At least one FiMInhSumRef or FiMInhEventRef or FiMInhComponentRef needs to be configured.

5.6.1.2 FiM/FiMConfigSet/FiMInhibitionConfiguration/FiMInhEventRef

[ADDED] Introduction	At least one FiMInhSumRef or FiMInhEventRef or FiMInhComponentRef needs to be configured.

5.6.1.3 FiM/FiMConfigSet/FiMInhibitionConfiguration/FiMInhFunctionIdRef

[ADDED] Description	Reference to a FiM Function ID for this inhibition configuration.

5.6.1.4 FiM/FiMConfigSet/FiMInhibitionConfiguration/FiMInhInhibitionMask

[RTA-BSW] Enumeration Literals	FIM_FAILED_OR_NOT_TESTED, FIM_LAST_FAILED, FIM_NOT_TESTED, FIM_TESTED, FIM_TESTED_AND_FAILED
[AUTOSAR] Enumeration Literals	FIM_LAST_FAILED, FIM_NOT_TESTED, FIM_TESTED, FIM_TESTED_AND_FAILED

5.6.1.5 FiM/FiMConfigSet/FiMInhibitionConfiguration/FiMInhSumRef

[ADDED] Introduction	At least one FiMInhSumRef or FiMInhEventRef or FiMInhComponentRef needs to be configured.

5.6.1.6 FiM/FiMGeneral

[ADDED] Description	This container contains the general configuration parameters.

5.6.1.7 FiM/FiMGeneral/FiMDevErrorDetect

[RTA-BSW] Description	Switches the Default Error Tracer (Det) detection and notification ON or OFF.

[AUTOSAR] Description	Switches the development error detection and notification on or off.
[RTA-BSW] Introduction	* true: enabled (ON). * false: disabled (OFF).
[AUTOSAR] Introduction	* true: detection and notification is enabled. * false: detection and notification is disabled.
[REMOVED] Default Value	false

5.6.1.8 FiM/FiMGeneral/FiMEventUpdateTriggeredByDem

[RTA-BSW] Introduction	TRUE: the DEM informs FIM about changes of event status, FALSE: the FIM polls event status from the DEM module either cyclically or on demand.
[AUTOSAR] Introduction	TRUE: the DEM informs FIM about changes of monitor status, FALSE: the FIM polls monitor status from the DEM module either cyclically or on demand.

5.6.1.9 FiM/FiMGeneral/FiMMainFunctionPeriod

[RTA-BSW] Description	Allow to configure the time for the periodic cyclic task. Please note: This configuration value shall be equal to the value in the Basic Software Scheduler configuration of the RTE module.
[AUTOSAR] Description	Allow to configure the time for the periodic cyclic task.
[RTA-BSW] Introduction	The AUTOSAR configuration standard is to use SI units, so this parameter is defined as float value in seconds. FiM configuration tools must convert this float value to the appropriate value format for the use in the software implementation of DEM. min: A negative value is not allowed. max: FID must be set after a maximal time of 100ms after FiM status is set. upperMultiplicity: Exactly one TaskTime must be specified per configuration. lowerMultiplicity: Exactly one TaskTime must be specified per configuration. Note: It is recommended to configure it as a non zero value and less than 0.1
[AUTOSAR] Introduction	Please note: This configuration value shall be equal to the value in the Basic Software Scheduler configuration of the RTE module. The AUTOSAR configuration standard is to use SI units, so this parameter is defined as float value in seconds. FiM configuration tools shall convert this float value to the appropriate value format for the use in the software implementation of FiM.
[RTA-BSW] Maximum Value	Inf
[AUTOSAR] Maximum Value	INF

5.6.1.10 FiM/FiMGeneral/FiMMaxEventsPerFidInhibitionConfiguration

[RTA-BSW] Description	This configuration parameter specifies the total maximum number of inhibiting events in a FiMInhibitionConfiguration. RB:Parameter can be used in case Postbuild is enabled.
[AUTOSAR] Description	This configuration parameter specifies the total maximum number of inhibiting events in a FiMInhibitionConfiguration.

5.6.1.11 FiM/FiMGeneral/FiMMaxFiMInhibitionConfigurations

[RTA-BSW] Description	This configuration parameter specifies the total maximum number of FiMInhibitionConfigurations. RB:Parameter can be used in case Postbuild is enabled.
[AUTOSAR] Description	This configuration parameter specifies the total maximum number of FiMInhibitionConfigurations.

5.6.1.12 FiM/FiMGeneral/FiMVersionInfoApi

[REMOVED] Default Value	false

5.7 API Definition

Table 5.39.Fim Supported API functions

`FiM_Init`	Initialises the FiM module.
`FiM_GetVersionInfo`	Returns the version information of the FiM module.
`FiM_MainFunction`	FiM main function.
`FiM_GetFunctionPermission`	Reports the permission state to the functionality.
`FiM_DemTriggerOnComponentStatus`	Triggers on changes of the component failed status.
`FiM_DemInit`	Re-initializes the FiM.
`FiM_DemTriggerOnMonitorStatus`	Called by Dem in order to inform the Fim about monitor status changes.

For further information on these functions, please refer to AUTOSAR_SWS_FunctionInhibitionManager.pdf

5.8 Configuration Advice

> **NOTE**
>
> Before configuring this module, please note that some of the default values used by ConfGen to produce the default ECU Configuration can be changed, and this is done by either adding a Conf-Gen Enhancer Mapping (CEM) or by adding a rule to an *algo.properties* file. For more information, see the "Configuration Generation" section of the RTA-CAR User Guide.

5.8.1 Summarized Events

FiM supports the configuration of summarized events, but the summarized events exist only in the configuration model. The generators forward new inhibit relations into the existing inhibit structure based on the summarized events and their associated Dem events.

5.8.2 FiMEventUpdateTriggeredByDem

A new parameter FiMEventUpdateTriggeredByDem (TRUE/FALSE) has replaced the parameter FiMCyclicEventEvaluation (FALSE/TRUE) with inversion.

5.8.3 FiMFunctionId

FiMFunctionId is generated by CodeGen. By default, the FiMFunctionIds are generated in short name order. However, if FiMRbFidSortMethod is set to SEQUENCE_NUMBER, then the order is determined by the FiMRbFIdSequence parameter instead.

5.8.3.1 Code Generation based on configuration class

FiM supports 3 kinds of configuration class

- Pre-compile
- Post-build Loadable
- Post-build Selectable

If either of the two Post-Build options is to be used for FiM, the following two configuration steps must first be completed.

Step 1 Select the Configuration Class

Go to the FiM module configuration, and then the "properties" tab.

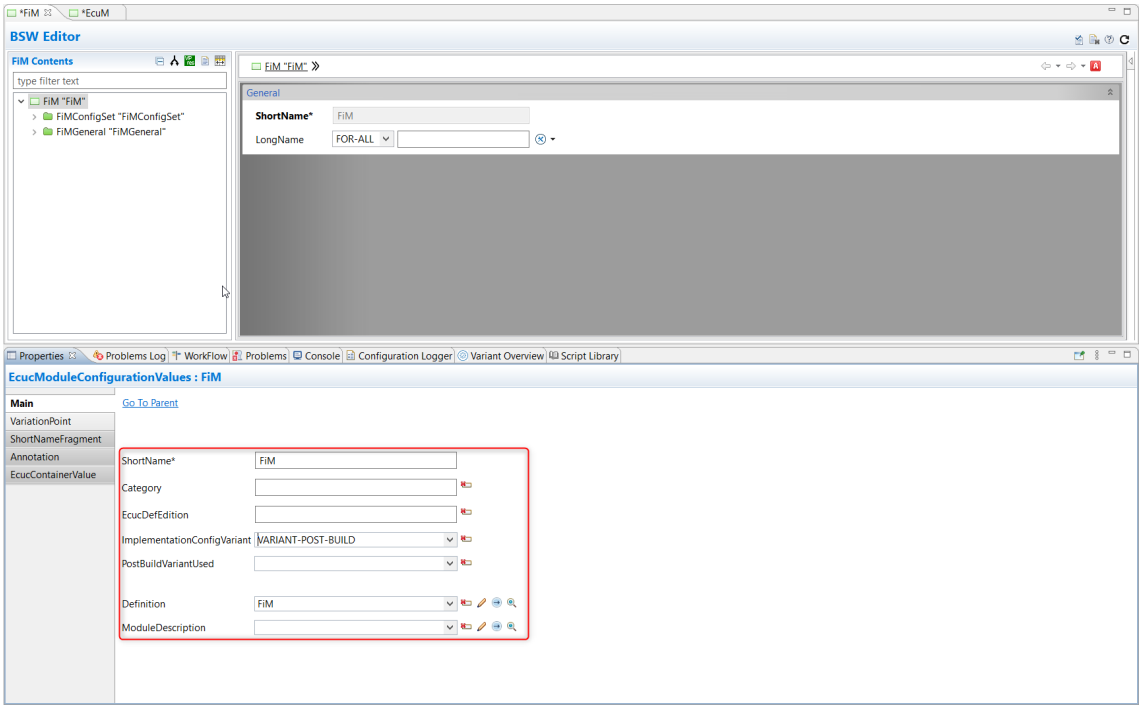


Figure 5.2.Select which configuration class of FiM

ImplementationConfigVariant will be set to “variant-pre-compile” by default, but if either of the two Post-Build options will be used for FiM, you must select the generic “variant-post-build” option (and not the specific “variant-post-build-loadable” or “variant-post-build-selectable” options).

Step 2 EcuM Configuration

If ImplementationConfigVariant is set to “variant-post-build”, the EcuM configuration for FiM and all its dependent modules must be configured with POST-BUILD_PTR.

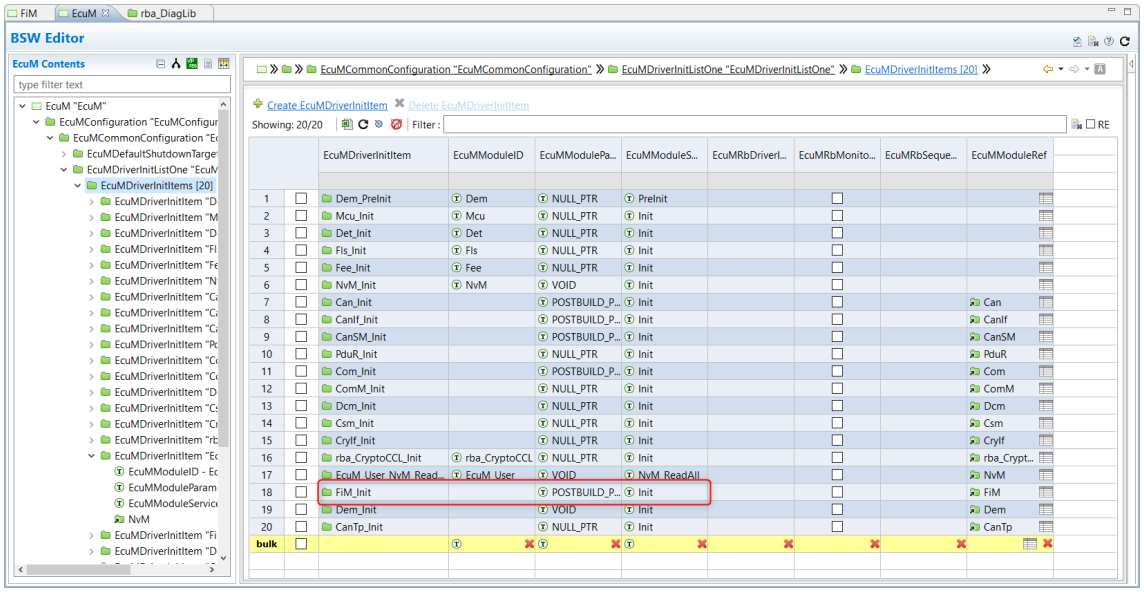


Figure 5.3.EcuM configuration

5.8.4 Code Generation Options

Depending on the above setting of ImplementationConfigVariant, the following options can be deployed at Code Generation.

Pre-Compile

If this option is chosen, all ECUs using the generated BSW will have the same configuration and there is no way to change this configuration without re-generating and re-compiling the whole of the ECU's software.

As shown below, select the "FiM_PreCompile" option for the FiM module. The other options on the right can be set at your discretion.

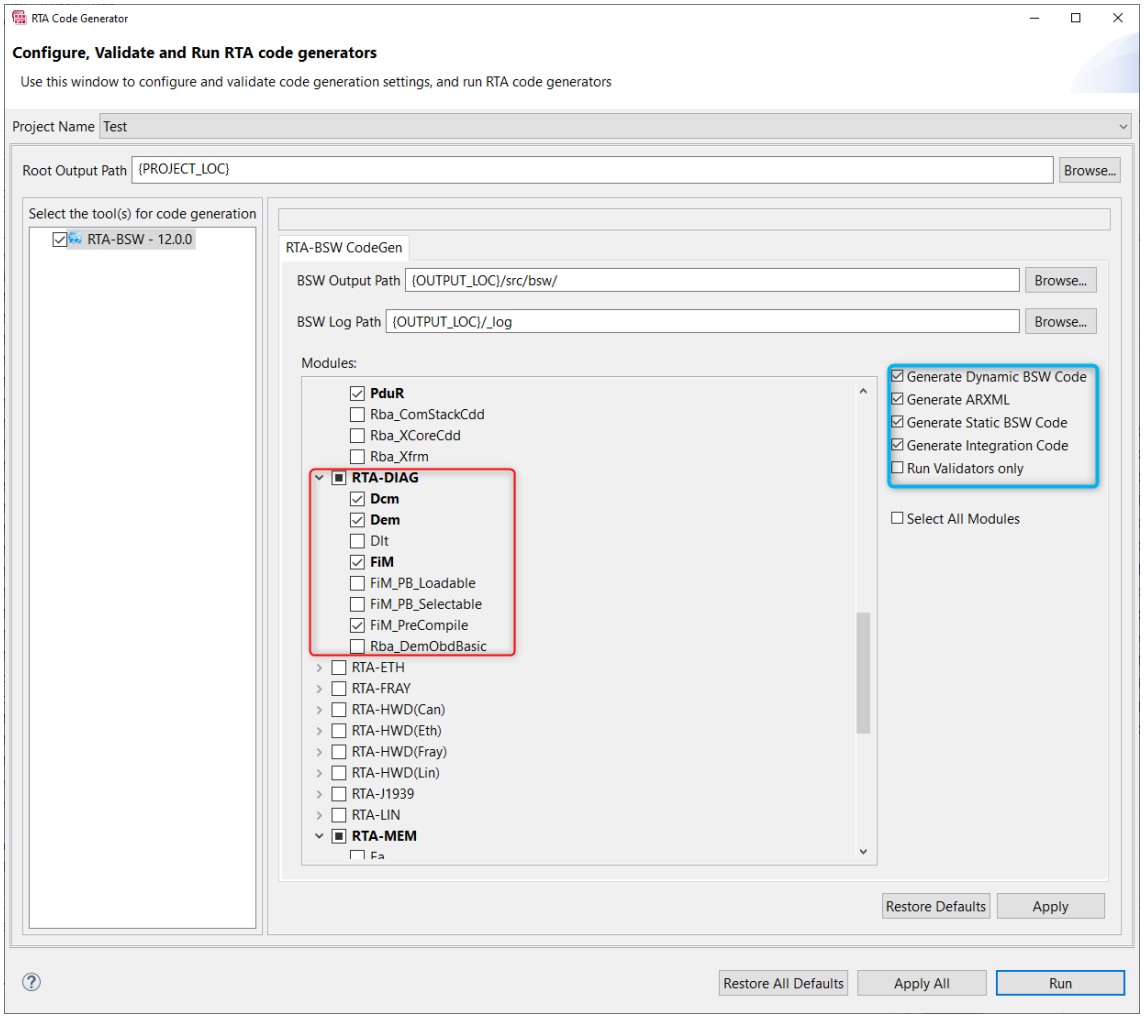


Figure 5.4.Generate FiM with Configuration Class as Pre-compile

Post-Build Loadable

This option results in the generation of only the configuration for the FiM module, and it does not generate any of the other code.

Post-Build Loadable is only available after the FiM module's main code has initially been fully generated in a Post-Build configurable form, and this is achieved with a one-time Code Generation using Post-Build Selectable, as shown here.

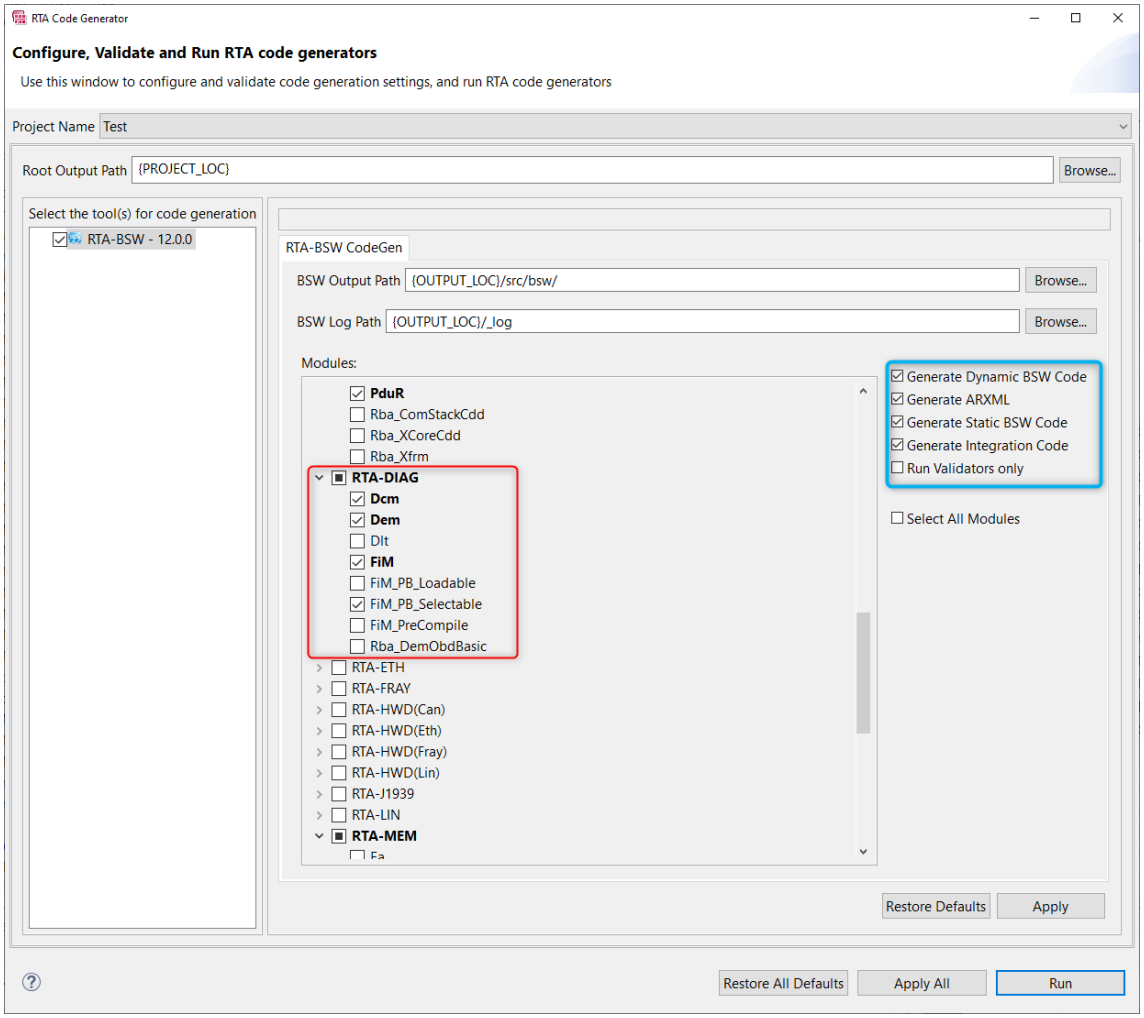


Figure 5.5.Full Code Generation with Post-Build

Then, if the module's configuration is subsequently changed, as shown below the Post-Build Loadable option can be used so that only the configuration files are re-generated (the main FiM code will not be re-generated).

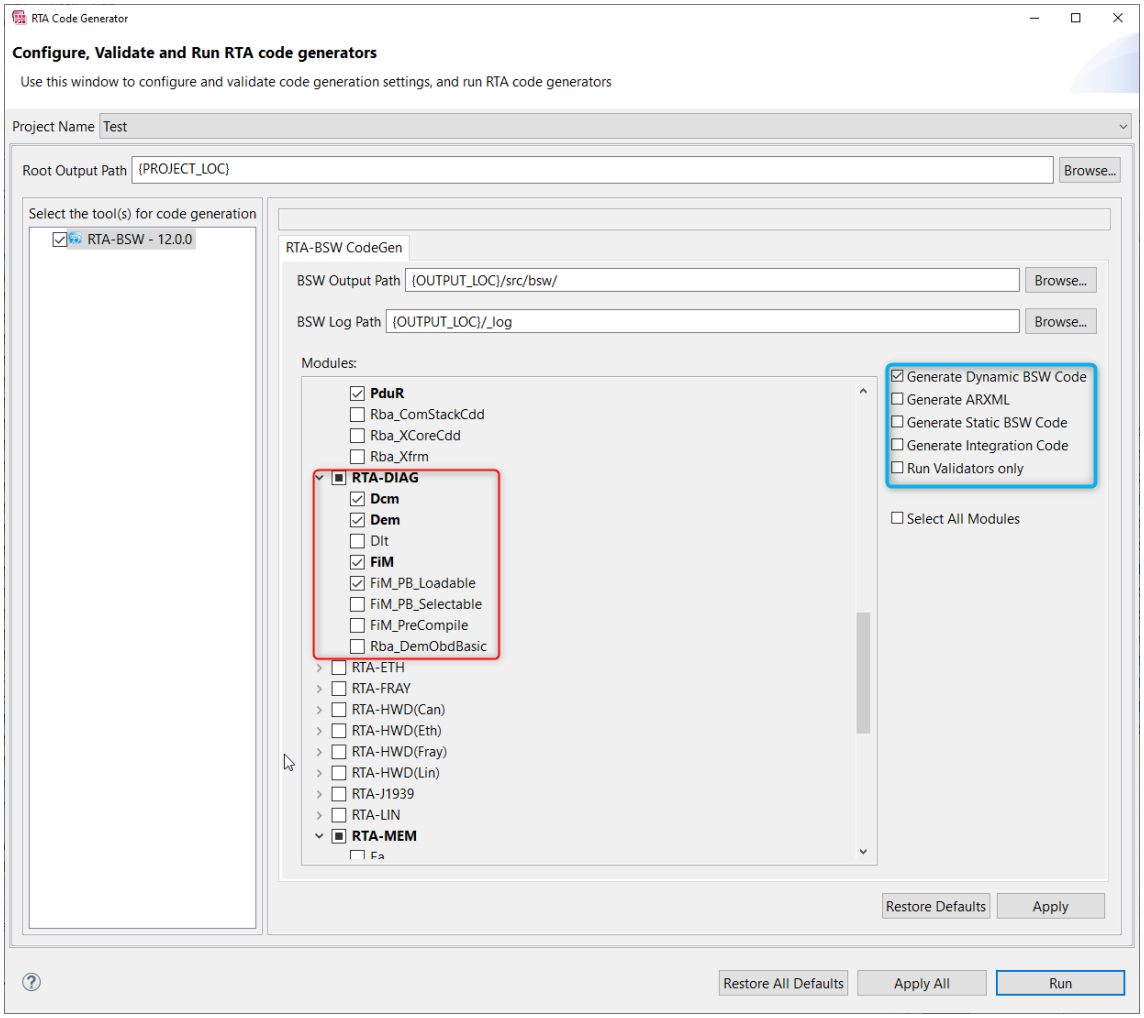


Figure 5.6.Regeneration for Post-Build

Note that the “Delete Existing Output Folder” option must be un-ticked when using Post-Build Loadable. This results in the loadable parts being allocated into a different folder from the full generation that was previously created by the initial Post-Build Selectable.

Post-Build Selectable

This option is used when multiple configurations for FiM will be created, and the required configuration can be selected at run time.

As shown below, the FiM_PB_Selectable option is ticked.

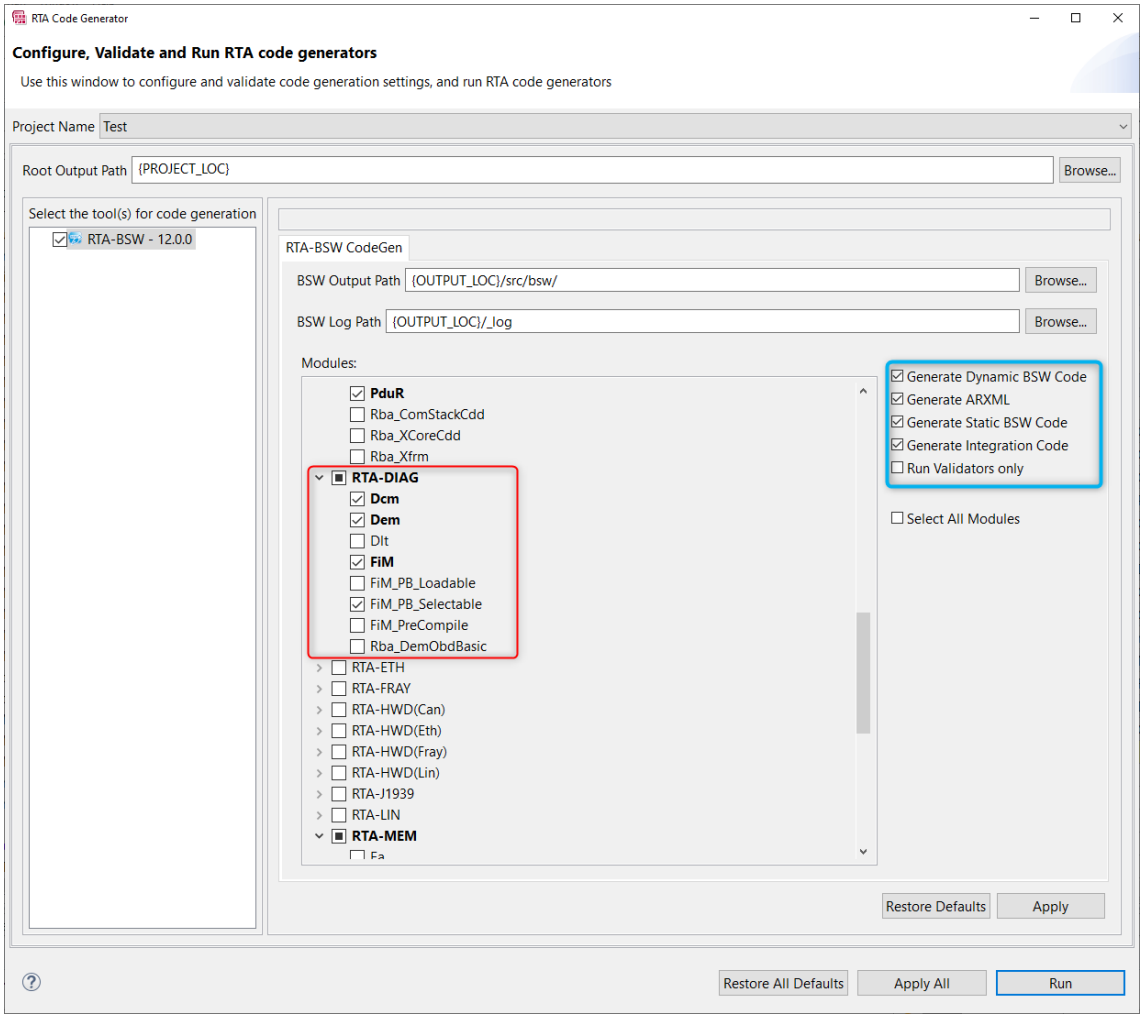


Figure 5.7.Code Generation with Selectable Post-Build

5.8.5 Multi-Partitioning

FIDs can be mapped to specific EcuC partitions, enabling the execution of the C/S operation `GetFunctionPermission` (which is a member of the *FunctionInhibition* interface) on the respective partition.

In order to use the multi-partition feature, additional steps are required:

EcuC Partition Configuration

EcuC partitions can be added and configured in the `EcucPartitionCollection` container.

EcuC Partitions Relevant For FiM

The `FiMRbEcucPartition` container contains references to all the FiM-relevant EcuC partitions. `FiMRbEcucPartition` is a (sub-)set of `EcucPartitionCollection`.

Master Partition Assignment

The FiM Master Partition is defined by `FiMRbMasterEcucPartitionRef`. This parti-

tionsupports the general functionality of FiM and the `GetFunctionPermission` calls for all FIDs that are mapped to the Master Partition.

FID Assignment To Partition

Each FID can be mapped to one partition in `FiMRbEcucPartition` using `FiMRbFunctionIdEcucPartitionRef`. The API `GetFunctionPermission` is executed on the partition the respective FID is mapped to.

If a FID is not mapped to any partition, it gets automatically assigned to the Master Partition.

5.9 Integration Advice

5.9.1 Init Functions

FiM must be initialized before Dem, via FiM_Init().

During Dem_Init(), Dem can reinit/resync FiM by calling the service FiM_DemInit().

5.9.2 Fim Operation Modes

FiM can be configured to run exclusively in "Triggered Operation" mode (i.e. triggered by Dem), or it can run with its own main function in "Cyclic Operation" mode. The table below provides a comparison of the two modes of operation.

Header 1	Triggered Operation Mode	Cyclic Operation Mode
Main Function	No main function	Main function required
Responsiveness	Synchronous to Dem	Asynchronous to Dem
RAM	1 or 2 byte counter per FID	2*1bit (dirty flag) per FID
ROM	Small code	Larger code
Runtime (1)	Minimum runtime	Depends on number of events
Runtime (2)	In Dem / SWC Task context	Possible in another task
Reliability	Doesn't miss event status change	Can miss toggling event status changes

5.9.3 Scheduled Functions

(Cyclic Operation Mode)

If the configuration switch FIM_CFG_DEM_TRIGGERFIMREPORTS is set to OFF (FIM_CFG_OFF), the following applies:

- The function FiM_MainFunction() needs to be called on a regular basis. It depends on the amount of EventIds and FIDs and the configuration settings how often the function should be triggered.
- The time required calculate all inhibition relations can be estimated with the following formula:

complete time = (2*ceiling(FIM_CFG_NUMBEROFFIDS / FIM_CFG_FIDS_PER_-MONITORCYCLE_ECUC) + ceiling(FIM_CFG_NUMBEROFDEMEVENTIDS /

FIM_CFG_EVENTS_PER_MONITORCYCLE_ECUC)) * cycle time of the task

If FIM_CFG_DEM_TRIGGERFIMREPORTS is set to ON (FIM_CFG_ON), no harm is done if the main function is called, but this should be avoided.

5.9.4 DeInit Functions

No specific handling necessary.

5.9.4.1 OS Resources

The number of the active OS resources, and their usage, depends on the user configuration of the AUTOSAR System and Basic Software. It is the user's responsibility to ensure that the `ENTER` and `EXIT` resource API are implemented to disable/enable interrupts as required. Alternatively they must be configured in the OS with the correct task/ISR allocation.

The following is a list of all exclusive areas that this module might use regardless of the user configuration. It is the job of the integrator to ensure that the implementation of these exclusive areas is correctly defined. This implementation can be done using the RTE exclusive area configuration wizard or by manual implementation of the corresponding BSW API calls to ensure exclusivity.

The FiM module uses the following OS resources:

- FiM_Monitor_NoNest
- FiM_Status_NoNest

5.9.4.2 Post Build Selectable

When Post-build Selectable is used with multiple variants, target variant can be selected by using user defined API - EcuM_DeterminePbConfiguration from EcuM integration. Please refer relevant integration advice from ECU State Management for more detail information.

5.10 Hardware Requirements

Table 5.41.FiM Hardware Requirements

Support	Usage
FLASH	1343 bytes
RAM	17 bytes

Usage calculated based on a sample configuration.

6 rba_DemObdBasic

Module name	rba_DemObdBasic
Package	RTA-DIAG
AUTOSAR Module ID	N/A
AUTOSAR Version	23-11

6.1 Introduction

The rba_DemObdBasic component is an extension to the RTA-DIAG Dem module, and it provides the basic functionality for OBD diagnostic data, as described in the CARB OBD law.

The diagram in the RTA-DIAG Introduction section shows the position of rba_DemObdBasic (and the other RTA-DIAG modules) in the system services abstraction layer of the Autosar layered architecture.

IMPORTANT: Any project using rba_DemObdBasic should ensure that the specific legal requirements for the target market are fulfilled.

Module Configuration

This module does not have configuration parameters.

6.2 Supported Features

Rba_DemObdBasic provides various functions for OBD-relevant ECUs, including:

- Primary and Secondary ECUs fully supported.
- Master ECUs features supported with alternative configurations.
- Locking/Releasing a denominator of a specific monitor.
- Reporting faults based on fulfilment of all conditions.
- Forwarding condition status to a Dem of a particular ECU.
- Reporting value of PID $01, $1C, $21, $30, $31, $41, $4D and $4E computed by Dem.
- Reporting an event as disabled to the Dem for the PID $41 computation.
- Requesting IUMPR data according to InfoTypes $08 and $0B.
- Getting DTC by freeze frame record number, and data element per PID and index of the most important freeze frame being selected for the output of service $02.

6.3 Extensions

6.3.1 Configuration Extensions

The following configuration parameters are available in version v12.7.0 in addition to those defined in AR v22-11

6.3.2 Additional Services

The following API are available in version v12.7.0 in addition to those defined in AR v22-11

- rba_DemObdBasic_Clear
- rba_DemObdBasic_Dtc_GetCode
- rba_DemObdBasic_Dtc_GetNextFilteredDTC
- rba_DemObdBasic_Dtc_DtcFilterInit
- rba_DemObdBasic_Dtc_SetDTCFilter
- rba_DemObdBasic_Dtc_DTCFilterMatches
- rba_DemObdBasic_Event_IsRequestingMil
- rba_DemObdBasic_EvMem_GetLocationOfOBDFreezeFrame
- rba_DemObdBasic_EvMem_IsTriggerObdFreezeFrame
- rba_DemObdBasic_EvMem_CopyFreezeFrame
- rba_DemObdBasic_EvMem_CommonMilDebModifyFailureCounter
- rba_DemObdBasic_EvMem_IsAgingOfLocationAllowed
- rba_DemObdBasic_EvMem_IsDisplaceEventMemoryLocAllowed
- rba_DemObdBasic_EvMem_IsDtcConfirmed
- rba_DemObdBasic_EvMem_HandleImmediateAgeing
- rba_DemObdBasic_EvMem_StatusNotificationPending
- rba_DemObdBasic_EvMem_StatusNotificationThresholdReached
- rba_DemObdBasic_EvMem_StatusNotificationConfirmed
- rba_DemObdBasic_FF_CaptureFF
- rba_DemObdBasic_FF_IsObdFFAvailabl
- rba_DemObdBasic_FF_RetrievePidData
- rba_DemObdBasic_FF_SetObdFFAvailable
- rba_DemObdBasic_Init
- rba_DemObdBasic_InitCheckNvm
- rba_DemObdBasic_InitCheckPlausability
- rba_DemObdBasic_MainFunction
- rba_DemObdBasic_Mil_GetIndicatorStatusExternal
- rba_DemObdBasic_NvMInit
- rba_DemObdBasic_NvMShutdown
- rba_DemObdBasic_SetCycleQualified
- rba_DemObdBasic_Shutdown
- rba_DemObdBasic_StartOperationCycle

- `Dem_SignalEndOfOBDIgnitionCycle`
- `Dem_IsOBDRelevantNvmBlockNotReadCorrectly`
- `Dem_IsOBDFullClearPossible`
- `Dem_GetIUMPRRatioData`

6.4 Limitations

Rba_DemObdBasic does not currently provide the following functionality:

- Dem does not currently re-set the DTR data (using the reference to each individual event per DTR) when the event is affected by the fault clear command. ([SWS_Dem_00763]. This is because, in projects that support OBD, only Clear All is allowed during Clear. This means that DemClearDT-CLimitation (in /DemGeneral/) must be configured to DEM_ONLY_-CLEAR_ALL_DTCS.

6.5 Exclusions

The following features from AR v22-11 are not supported in version v12.7.0.

API functions not currently implemented:

- `Dem_DcmReadDataOfPID91`
- `Dem_ReadDataOfPID01`
- `Dem_SetDTR`
- `Dem_SetPtoStatus`

The following configuration parameters from AR v22-11 are not supported in version v12.7.0.

6.6 Deviations

6.6.1 Configuration Deviations

Configuration parameters which differ from the AR v22-11 specification are listed below.

6.7 API Definition

Table 6.2.Rba_DemObdBasic Supported API functions

`Dem_DcmGetDTCOfOBDFreezeFrame`	Gets DTC by freeze frame record number.
`Dem_DcmGetInfoTypeValue08`	Used for requesting IUMPR data according to Info-Type $08.
`Dem_DcmGetInfoTypeValue0B`	Used for requesting IUMPR data according to Info-Type $0B.
`Dem_DcmReadDataOfOBDFreezeFrame`	Gets data element per PID and index of the most important freeze frame being selected for the output of service $02.

Dem_DcmReadDataOfPID01	Service to report the value of PID $01 computed by the Dem.
Dem_DcmReadDataOfPID1C	Service to report the value of PID $1C computed by the Dem.
Dem_DcmReadDataOfPID21	Reports the value of PID $21 computed by the Dem.
Dem_DcmReadDataOfPID30	Reports the value of PID $30 computed by the Dem.
Dem_DcmReadDataOfPID31	Reports the value of PID $31 computed by the Dem.
Dem_DcmReadDataOfPID41	Reports the value of PID $41 computed by the Dem.
Dem_DcmReadDataOfPID4D	Reports the value of PID $4D computed by the Dem.
Dem_DcmReadDataOfPID4E	Reports the value of PID $4E computed by the Dem.
Dem_GetCycleQualified	Returns the qualification state of the dependent operation cycle.
Dem_GetDTCSeverityAvailabilityMask	Gets the DTC Severity availability mask.
Dem_GetIUMPRDenCondition	Gets the current condition status.
Dem_RepIUMPRDenRelease	Releases a denominator of a specific monitor.
Dem_RepIUMPRFaultDetect	Reports that faults are possibly found because all conditions are fullfilled.
Dem_SetCycleQualified	Sets a dependent operation cycle as qualified so it may be processed along with its leading cycle.
Dem_SetEventDisabled	Reports the event as disabled to the Dem for the PID $41 computation.
Dem_SetIUMPRDenCondition	Forwards the condition status to a Dem of a particular ECU.
Dem_GetB1Counter	Service to report the value of the B1 counter computed by the Dem.
Dem_DcmGetAvailableOBDMIDs	Reports the value of a requested “availability-OBDMID” to the DCM upon a Service $06 request.
Dem_GetDataOfPID21	Service to get the value of PID $21 from the Dem by a software component.
Dem_DcmGetDTRData	Reports a DTR data along with TID-value / UaSID and test result with lower and upper limit.
Dem_DcmGetNumTIDsOfOBDMID	Gets the number of TIDs per (functional) OBDMID.
Dem_SetDataOfPID21	Service to set the value of PID $21 in the Dem by a software component.
Dem_SetDataOfPID31	Service to set the value of PID $31 in the Dem by a software component.
Dem_SetDataOfPID4D	Service to set the value of PID $4D in the Dem by a software component.
Dem_SetDataOfPID4E	Service to set the value of PID $4E in the Dem by a software component.

For further information on these functions, please refer to AUTOSAR_SWS_Diag-

nosticEventManager.pdfAUTOSAR_SWS_DiagnosticEventManager.pdf

6.8 Configuration Advice

> **NOTE**
>
> Before configuring this module, please note that some of the default values used by ConfGen to produce the default ECU Configuration can be changed, and this is done by either adding a Conf-Gen Enhancer Mapping (CEM) or by adding a rule to an *algo.properties* file. For more information, see the "Configuration Generation" section of the RTA-CAR User Guide.

6.8.1 Non-Volatile Memory Blocks

The configuration of NvM blocks can be done in one of two ways -

- Manual Configuration. Follow the proposed NvM configuration below.
- Automatic Configuration. Enable the demRbEnableDemNvmBlocksAutoConfiguration configuration switch. All the configuration shall be forwarded, Fee shall be used as defauts, NvMRbNvBlockLengthString shall be used instead of NvMNvBlockLength to store the NvM Block Length. If you want to switch to Ea, you will need to create NvmBlockDescriptor (see manual configuration), change NvMNvramDeviceId to 1 onwards, you need to supply the NvMNvBlockLength (because NvMRbNvBlockLengthString is not supported for Ea) - the other information in NvmBlockDescriptor will be forwarded.

6.8.2 Configuration Changes

If `((Dem.DemGeneral.DemOBDSupport != DEM_OBD_NO_OBD_SUPPORT) && (Dem.DemGeneral.DemRbGeneral.DemRbEnableDemNvmBlocksAutoConfiguration != True))`, then a new NvM block must be configured and the same must be referenced in Dem by defining a container of type Dem.DemGeneral.DemNvRamBlockId with the shortname DEM_NVM_ID_OBD_GENOBD_DATA.

6.8.3 NvM Configuration Proposal

6.8.3.1 Configuration of DemNvBlock

OBD-Permanent Memory Location: conditional "DemObdSupport != DEM_OBD_NO_OBD_SUPPORT"
OBD IUMPR Data: conditional "demRatio.size > 0"
OBD DTR Data: conditional "demDtr.size > 0"
OBD SMAD Data: An extendedDataRecord is referencing the internalData element DEM_MONITOR_ACTIVITY_DATA
OBD Readiness Data: conditional ("DemObdSupport != DEM_OBD_NO_OBD_SUPPORT") && ("DemRbOBDProtocol != DEM_RB_J1979")
OBD GeneralOBD Data: conditional "DemObdSupport != DEM_OBD_NO_OBD_SUPPORT"

6.8.3.2 Dem/DemGeneral/DemNvBlockId

ShortName
OBD-Permanent Memory Location: DEM_NVM_ID_OBD_PERMANENT_MEMORY
OBD IUMPR Data: DEM_NVM_ID_OBD_IUMPR_DATA
OBD DTR Data: DEM_NVM_ID_OBD_DTR_DATA
OBD SMAD Data: DEM_NVM_ID_OBD_SMAD_DATA
OBD Readiness Data: DEM_NVM_ID_OBD_READINESS_DATA
OBD GeneralOBD Data: DEM_NVM_ID_OBD_GENOBD_DATA

DemNvRamBlockIdRef
OBD-Permanent Memory Location: Ref to NvM/NvMBlockDescriptor
OBD IUMPR Data: Ref to NvM/NvMBlockDescriptor
OBD DTR Data: Ref to NvM/NvMBlockDescriptor
OBD SMAD Data: Ref to NvM/NvMBlockDescriptor
OBD Readiness Data: Ref to NvM/NvMBlockDescriptor
OBD GeneralOBD Data: Ref to NvM/NvMBlockDescriptor

DemNvRamImmediateStoragePriority
OBD-Permanent Memory Location: highest
OBD IUMPR Data: highest
OBD DTR Data: highest
OBD SMAD Data: highest
OBD Readiness Data: highest
OBD GeneralOBD Data: highest

6.8.3.3 NvM/NvMBlockDescriptor

ShortName
OBD-Permanent Memory Location: Cust (Recommended: CustNVM_ID_OBD_PERMANENT_MEMORY)
OBD IUMPR Data: Cust (Recommended: NVM_ID_OBD_IUMPR_DATA)
OBD DTR Data: Cust (Recommended: NVM_ID_OBD_DTR_DATA)
OBD SMAD Data: Cust (Recommended: NVM_ID_OBD_SMAD_DATA)
OBD Readiness Data: Cust (Recommended: NVM_ID_OBD_READINESS_DATA)
OBD GeneralOBD Data: Cust (Recommended: NVM_ID_OBD_DATA)

NvMBlockManagementType
OBD-Permanent Memory Location: Cust (Recommended: NVM_BLOCK_REDUNDANT)
OBD IUMPR Data: Cust (Recommended: NVM_BLOCK_REDUNDANT)
OBD DTR Data: Cust (Recommended: NVM_BLOCK_REDUNDANT)
OBD SMAD Data: Cust (Recommended: NVM_BLOCK_REDUNDANT)
OBD Readiness Data: Cust (Recommended: NVM_BLOCK_REDUNDANT)
OBD GeneralOBD Data: Cust (Recommended: NVM_BLOCK_REDUNDANT)

NvMBlockUseCrc
OBD-Permanent Memory Location: Cust (Recommended: True)
OBD IUMPR Data: Cust (Recommended: True)
OBD DTR Data: Cust (Recommended: True)
OBD SMAD Data: Cust (Recommended: True)

OBD Readiness Data: Cust (Recommended: True)
OBD GeneralOBD Data: Cust (Recommended: True)

NvMBlockUseSyncMechanism
OBD-Permanent Memory Location: True
OBD IUMPR Data: True
OBD DTR Data: True
OBD SMAD Data: True
OBD Readiness Data: True
OBD GeneralOBD Data: True

NvMBlockWriteProt
OBD-Permanent Memory Location: False
OBD IUMPR Data: False
OBD DTR Data: False
OBD SMAD Data: False
OBD Readiness Data: False
OBD GeneralOBD Data: False

NvMInitBlockCallback
OBD-Permanent Memory Location: empty
OBD IUMPR Data: empty
OBD DTR Data: empty
OBD SMAD Data: empty
OBD Readiness Data: empty
OBD GeneralOBD Data: empty

NvMNvBlockBaseNumber
OBD-Permanent Memory Location: empty
OBD IUMPR Data: empty
OBD DTR Data: empty
OBD SMAD Data: empty
OBD Readiness Data: empty
OBD GeneralOBD Data:empty

NvMRbAutoBlockLength
OBD-Permanent Memory Location: Cust
OBD IUMPR Data: Cust
OBD DTR Data: Cust
OBD SMAD Data: Cust
OBD Readiness Data: Cust
OBD GeneralOBD Data: Cust

Additional hint: If `NvMRbAutoBlockLength` is set to *TRUE*, then the block size must be configured using `NvMRbNvBlockLengthString`, else using `NvMNvBlockLength`. (Please refer to the NvM chapter in *RTA-Mem_Stack_Reference_Guide_EN.pdf* for more details.) If the size is configured using `NvMRbNvBlockLengthString`, then `NvMBlockHeaderInclude` should be configured such that the Dem header file *Dem_FwdToNvm.h* is included by NvM, to allow automatic calculation of the block size. If the auto-configuration feature is enabled in Dem by setting `DemRbEnableDemNvmBlocksAutoConfiguration` to *TRUE*, then Dem makes use of `NvMRbNvBlockLengthString` to configure the block size.

NvMNvBlockLength
OBD-Permanent Memory Location: empty
OBD IUMPR Data: empty
OBD DTR Data: empty
OBD SMAD Data: empty
OBD Readiness Data: empty
OBD GeneralOBD Data: empty

NvMRbNvBlockLengthString
OBD-Permanent Memory Location: sizeof(rba_DemObdBasic_PdtcMem)
OBD IUMPR Data: sizeof(rba_DemObdBasic_IumprNvDataType)
OBD DTR Data: sizeof(rba_DemObdBasic_DtrDataType)
OBD SMAD Data: sizeof(rba_DemObdBasic_SmadNvDataType)
OBD Readiness Data: sizeof(rba_DemObdBasic_ReadinessNvDataType)
OBD GeneralOBD Data: sizeof(rba_DemObdBasic_AllDenomCondStatus)

NvMRamBlockDataAddress
OBD-Permanent Memory Location: empty
OBD IUMPR Data: empty
OBD DTR Data: empty
OBD SMAD Data: empty
OBD Readiness Data: empty
OBD GeneralOBD Data: empty

NvMReadRamBlockFromNvCallback
OBD-Permanent Memory Location: rba_DemObdBasic_PdtcMemReadRamBlock-FromNvCallback
OBD IUMPR Data: rba_DemObdBasic_IumprReadRamBlockFromNvCallback
OBD DTR Data: rba_DemObdBasic_DtrReadRamBlockFromNvCallback
OBD SMAD Data: rba_DemObdBasic_SmadReadRamBlockFromNvCallback
OBD Readiness Data: rba_DemObdBasic_ReadinessReadRamBlockFromNvCall-back
OBD GeneralOBD Data: rba_DemObdBasic_GenObdDataReadRamBlockFromN-vCallback

NvMResistantToChangedSw
OBD-Permanent Memory Location: False
OBD IUMPR Data: False
OBD DTR Data: False
OBD SMAD Data: False
OBD Readiness Data: False
OBD GeneralOBD Data: False

NvMRomBlockDataAddress
OBD-Permanent Memory Location: empty
OBD IUMPR Data: empty
OBD DTR Data: empty
OBD SMAD Data: empty
OBD Readiness Data: empty
OBD GeneralOBD Data: empty

NvMRomBlockNum

OBD-Permanent Memory Location: empty
OBD IUMPR Data: empty
OBD DTR Data: empty
OBD SMAD Data: empty
OBD Readiness Data: empty
OBD GeneralOBD Data: empty

NvMSelectBlockForReadAll
OBD-Permanent Memory Location: True
OBD IUMPR Data: True
OBD DTR Data: True
OBD SMAD Data: True
OBD Readiness Data: True
OBD GeneralOBD Data: True

NvMSelectBlockForWriteAll
OBD-Permanent Memory Location: True
OBD IUMPR Data: True
OBD DTR Data: True
OBD SMAD Data: True
OBD Readiness Data: True
OBD GeneralOBD Data: True

NvMSingleBlockCallback
OBD-Permanent Memory Location: empty
OBD IUMPR Data: empty
OBD DTR Data: empty
OBD SMAD Data: empty
OBD Readiness Data: empty
OBD GeneralOBD Data: empty

NvMWriteBlockOnce
OBD-Permanent Memory Location: False
OBD IUMPR Data: False
OBD DTR Data: False
OBD SMAD Data: False
OBD Readiness Data: False
OBD GeneralOBD Data: False

NvMWriteRamBlockToNvCallback
OBD-Permanent Memory Location: rba_DemObdBasic_PdtcMemWriteRamBlock-ToNvCallback
OBD IUMPR Data: rba_DemObdBasic_IumprWriteRamBlockToNvCallback
OBD DTR Data: rba_DemObdBasic_DtrWriteRamBlockToNvCallback
OBD SMAD Data: rba_DemObdBasic_SmadWriteRamBlockToNvCallback
OBD Readiness Data: rba_DemObdBasic_ReadinessWriteRamBlockToNvCallback
OBD GeneralOBD Data: rba_DemObdBasic_GenObdDataWriteRamBlockToN-vCallback

NvMBlockCrcType
OBD-Permanent Memory Location: Cust
OBD IUMPR Data: Cust

OBD DTR Data: empty
OBD SMAD Data: Cust
OBD Readiness Data: Cust
OBD GeneralOBD Data: Cust

NvMBlockJobPriority
OBD-Permanent Memory Location: Cust
OBD IUMPR Data: Cust
OBD DTR Data: empty
OBD SMAD Data: Cust
OBD Readiness Data: Cust
OBD GeneralOBD Data: Cust

NvMCalcRamBlockCrc
OBD-Permanent Memory Location: empty
OBD IUMPR Data: empty
OBD DTR Data: empty
OBD SMAD Data: empty
OBD Readiness Data: empty
OBD GeneralOBD Data: empty

NvMInitBlockCallback
OBD-Permanent Memory Location: empty
OBD IUMPR Data: empty
OBD DTR Data: empty
OBD SMAD Data: empty
OBD Readiness Data: empty
OBD GeneralOBD Data: empty

NvMMaxNumOfReadRetries
OBD-Permanent Memory Location: Cust (Recommended: 1)
OBD IUMPR Data: Cust (Recommended: 1)
OBD DTR Data: empty
OBD SMAD Data: Cust (Recommended: 1)
OBD Readiness Data: Cust (Recommended: 1)
OBD GeneralOBD Data: Cust (Recommended: 1)

NvMMaxNumOfWriteRetries
OBD-Permanent Memory Location: Cust (Recommended: 1)
OBD IUMPR Data: Cust (Recommended: 1)
OBD DTR Data: empty
OBD SMAD Data: Cust (Recommended: 1)
OBD Readiness Data: Cust (Recommended: 1)
OBD GeneralOBD Data: Cust (Recommended: 1)

NvMNvBlockNum
OBD-Permanent Memory Location: empty
OBD IUMPR Data: empty
OBD DTR Data: empty
OBD SMAD Data: empty
OBD Readiness Data: empty
OBD GeneralOBD Data: empty

NvMNvramBlockIdentifier
OBD-Permanent Memory Location: empty
OBD IUMPR Data: empty
OBD DTR Data: empty
OBD SMAD Data: empty
OBD Readiness Data: empty
OBD GeneralOBD Data: empty

NvMNvramDeviceId
OBD-Permanent Memory Location: empty
OBD IUMPR Data: empty
OBD DTR Data: empty
OBD SMAD Data: empty
OBD Readiness Data: empty
OBD GeneralOBD Data: empty

NvMStaticBlockIDCheck
OBD-Permanent Memory Location: Cust
OBD IUMPR Data: Cust
OBD DTR Data: empty
OBD SMAD Data: Cust
OBD Readiness Data: Cust
OBD GeneralOBD Data: Cust

NvMWriteVerification
OBD-Permanent Memory Location: Cust (Recommended: False)
OBD IUMPR Data: Cust (Recommended: False)
OBD DTR Data: empty
OBD SMAD Data: Cust (Recommended: False)
OBD Readiness Data: Cust (Recommended: False)
OBD GeneralOBD Data: Cust (Recommended: False)

NvMWriteVerificationDataSize
OBD-Permanent Memory Location: Cust
OBD IUMPR Data: Cust
OBD DTR Data: empty
OBD SMAD Data: Cust
OBD Readiness Data: Cust
OBD GeneralOBD Data: Cust

NvMCommon
OBD-Permanent Memory Location: empty
OBD IUMPR Data: empty
OBD DTR Data: empty
OBD SMAD Data: empty
OBD Readiness Data: empty
OBD GeneralOBD Data: empty

NvMSetRamBlockStatusApi
OBD-Permanent Memory Location: True
OBD IUMPR Data: True
OBD DTR Data: empty

OBD SMAD Data: True
OBD Readiness Data: True
OBD GeneralOBD Data: True

Trigger NvM Storage
OBD-Permanent Memory Location: conditional “DemTriggerToStoreNvMSupported == True”
OBD IUMPR Data: conditional “DemTriggerToStoreNvMSupported == True”
OBD DTR Data: empty
OBD SMAD Data: conditional “DemTriggerToStoreNvMSupported == True”
OBD Readiness Data: conditional “DemTriggerToStoreNvMSupported == True”
OBD GeneralOBD Data: conditional “DemTriggerToStoreNvMSupported == True”

Additional hint: For “NvMNvBlockLength” it is possible to configure a value bigger than sizeof(Dem_AllEventsStatusByte). This might be used to reduce NVM-layout changes during project development phase. The project is responsible to configure the correct value before the project release.

6.8.4 Centralized PID $21 / $31 / $4D / $4E handling

6.8.4.1 PID calculation

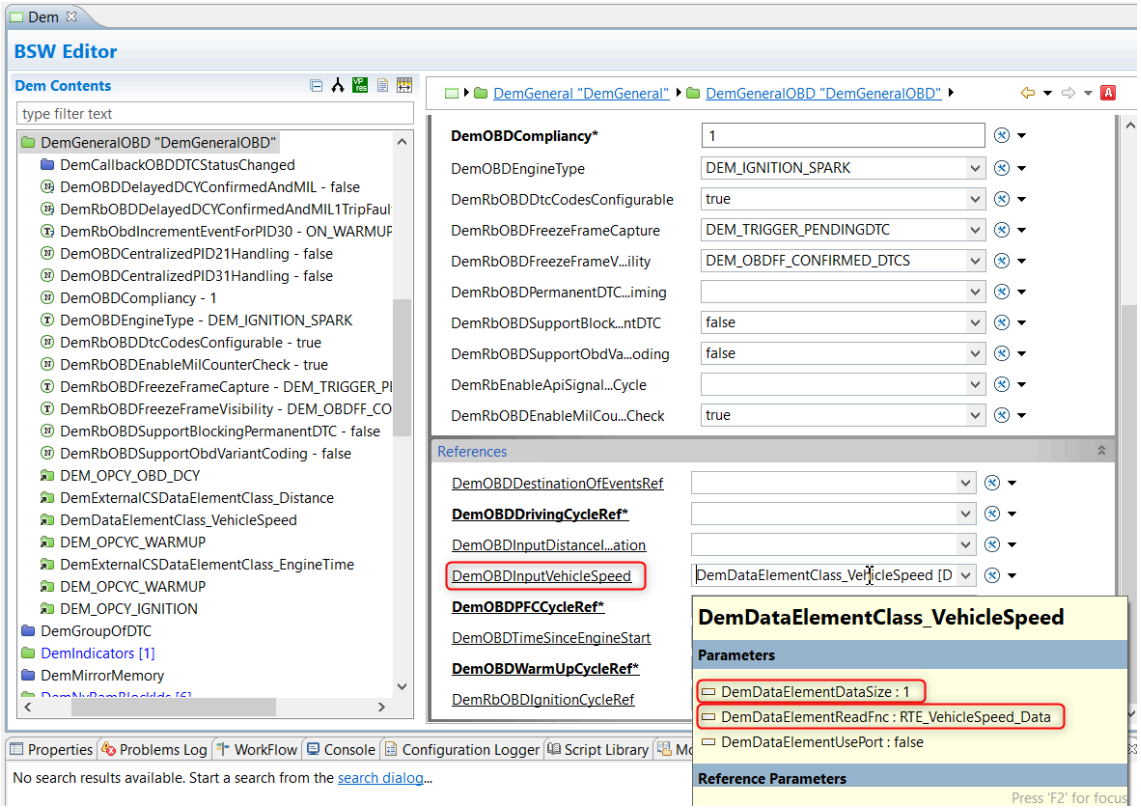
The Dem of the Master ECU supports the calculation of information for these PIDs, which requires the following input information managed by the Master ECU:

Vehicle speed. Distance information. Time since engine start.

For this reason, reference information for these PIDs must be configured through the following parameters:

DemOBDInputVehicleSpeed with data size as 1 bytes DemOBDInputDistanceInformation with data size as 4 bytes DemOBDTimeSinceEngineStart with data size as 2 bytes

Below is a sample configuration for DemOBDInputVehicleSpeed:



Vehicle speed should be provided through the implementation of function RTE_VehicleSpeed_Data.

6.8.4.2 PID Interfaces

Centralized handling allows the calculatation and reporting of data for these PIDs from Dem, based on the input distance configured in DemOBDInputDistanceInformation. To enable this feature, the following parameters must be disabled:

DemOBDCentralizedPID21Handling DemOBDCentralizedPID31Handling DemRbOBDCentralizedPID4DHandling DemRbOBDCentralizedPID4EHandling

With each parameter disabled, the ECU can access calculated PID data through a Dem_GetDataOfPIDxx interface (where xx can be: 21, 31, 4D or 4E).

6.9 Integration Advice

6.9.1 Init Functions

No specific handling necessary.

6.9.2 DeInit Functions

No specific handling necessary.

6.9.3 Scheduled Functions

There are no scheduled functions.

6.9.4 OBD Mode $06 - Diagnostic Test result support

6.9.4.1 DTR Central Handing support

After CodeGeneration, a ProvidedPort named 'DTRCentralReport_'+DTRShortName is generated for each DTR. A RequiredPort should also be created for each DTR in order to report the test result values to Dem.

6.9.4.2 Consistency of Availability OBDMID and OBDMID Test Values responses

- As defined in [SWS_Dem_01108], the Dem_DcmGetDTRData API returns E_NOT_OK when the referenced event is set to unavailable. This leads to no data being reported for the TIDs of requested OBDMID Test Values.
- In order to ensure the consistency between Availability OBDMID and OBDMID Test Values responses, when the referenced event is unavailable, Dem_SetDTR can be called with a "control value" of DEM_DTR_CTL_INVISIBLE. This will mark the DTR as invisible, making it appear from the external (tester) view as if it is not integrated.

6.9.4.3 DTR Calibration

Turn off an OBDMID:

- Clear the OBDMID's inidicate bit in rba_DemObdBasic_ObdmidMask
- Clear the OBDMID's inidicate bit in rba_DemObdBasic_DtrData.ObdmidMaskRunTime
- Turn off all of related OBDTIDs

NOTE: Update/Calibrate rba_DemObdBasic_DtrData.ObdmidMaskRunTime[x] and rba_DemObdBasic_DtrConfig[x] after the ECU is started (after rba_DemObdBasic_Dtr_InitCheckNvm() has been called).

- This will ensure that the value of ObdmidMaskRunTime is updated from the calibration.
- This will only required for the first calibration because rba_DemObdBasic_Dtr_Shutdown will be called (during ECU shutdown) to indicate to the software to update the new value to the flash memory.

Turn off an OBDTID:

- Set OBDTID's rba_DemObdBasic_DtrConfig.ObdTid_u8 to zero.
- Clear OBDTID's datas (rba_DemObdBasic_DtrData.TestValue_u16, LowLimitValue_u16, UppLimitValue_u16, DtrStatus_u8) to their

default value.

- If all an OBDMID's OBDTIDs are off, you should also turn off the OBDMID.

6.9.5 OS Resources

The number of the active OS resources, and their usage, depends on the user configuration of the AUTOSAR System and Basic Software. It is the user's responsibility to ensure that the `ENTER` and `EXIT` resource API are implemented to disable/enable interrupts as required. Alternatively they must be configured in the OS with the correct task/ISR allocation.

The Rba_DemObdBasic module does not use any exclusive OS resources.

6.10 Hardware Requirements

Information relating to the hardware resources required by this module is not currently available.

7 Glossary

CAN
Controller Area Network

CARB
California Air Resources Board

CDD
Complex Device Driver

CRC
Cyclic Redundancy Check

DCM
Diagnostic Communication Manager

DEM
Diagnostic Event Manager

DET
Default Error Tracer

DID Data Identifier

DIt Diagnostic Log and Trace

DSD
Diagnostic Service Dispatcher (submodule of the DCM module)

DSL
Diagnostic Session Layer (submodule of the DCM module)

DSP
Diagnostic Service Processing (submodule of the DCM module)

DTC
Diagnostic Trouble Codes

DTR
Diagnostic Test Result

DYC
Driving Cycle

FDC
Fault Detection Counter

FID Function Identifier

FiM Function Inhibition Manager

FMI Failure Mode Indicator

FTB
Failure Type Byte

ID Identifier

IUMPR
In Use Monitoring Performance Ratio

KWP
Keyword Protocol

LIN Local Interconnect Network

LSB
Least Significant Bit

MCU
Micro-Controller Unit

MIL Malfunction Indicator Light

MOST
Media Orientated System Transport

MSB
Most Significant Bit

NRC
Negative Response Code

NVRAM
Non volatile RAM

OBD
On-Board Diagnosis

OBDII
Emission-related On-board Diagnostics

OC Occurrence Count

ODX
Open Diagnostic Data Exchange

OSI Open Systems Interconnection

PDU
Protocol Data Unit

PID Parameter Identifier

PTO
Power Take Off

ROE
ResponseOnEvent

SAP
Service Access Point

SDU
Service Data Unit

SID Service Identifier

SPN
Suspect Parameter Number

SSCP
synchronous server call point

UDS
Unified Diagnostic Services

VOBD
Vehicle On-Board-Diagnostic

WIR
Warning Indicator Request

WUC
Warm up cycle

WWH-OBD
World Wide Harmonized On-Board-Diagnostic

XCP
Universal Measurement and Calibration Protocol